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Computation of Multi-Particle Phase-Space Integrals and Their Applications in High-Energy Physics

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Resumo

A precisão alcançada pelas medições realizadas no Large Hadron Collider (LHC) impõe exigências cada vez mais rigorosas às previsões teóricas. Em Cromodinâmica Quântica (QCD) perturbativa, a precisão destas previsões é sistematicamente melhorada através da inclusão de correções de ordem superior na constante de acoplamento forte, α_s. Contudo, aumentar a precisão de um cálculo não corresponde apenas a acrescentar diagramas de Feynman: exige controlar uma estrutura complexa de contribuições de emissão real e virtual, bem como as divergências que surgem em cada uma delas.

As divergências infravermelhas constituem uma das principais dificuldades dos cálculos perturbativos em QCD. As contribuições de emissão real tornam-se singulares quando um partão é emitido com energia muito baixa, o limite *soft*, ou quando dois partões se tornam colineares. Estas singularidades estão associadas à emissão de radiação não resolvida, isto é, de partões cuja energia ou separação angular é demasiado pequena para serem distinguidos como estados finais distintos. Paralelamente, as amplitudes que incluem correções virtuais de loop apresentam polos infravermelhos associados às mesmas regiões cinemáticas. Embora estas divergências se cancelem em observáveis físicos suficientemente inclusivos, as componentes individuais do cálculo não são finitas. É, por isso, necessário reorganizar a previsão teórica de modo a isolar as singularidades e a efetuar o seu cancelamento de forma analítica e controlada.

O formalismo de subtração por antenas fornece um enquadramento adequado para esse objetivo. Neste contexto, uma antena é formada por dois partões ligados por cor, designados por radiadores, entre os quais ocorre a emissão de radiação. A radiação não resolvida é então descrita em termos destas antenas, permitindo tratar de forma sistemática as regiões singulares do espaço de fases. As funções de antena reproduzem os limites *soft* e colineares relevantes de forma local no espaço de fase. Quando são subtraídas às contribuições reais, eliminam as singularidades antes da integração numérica. Depois de integradas analiticamente sobre o espaço de fase da radiação não resolvida, originam polos em regularização dimensional que cancelam os polos das contribuições virtuais.

Esta dissertação apresenta o desenvolvimento de *AntCalc*, um pacote simbólico para a construção e integração automatizadas de funções de antena em QCD. O projeto procura automatizar etapas que requerem implementações específicas para cada antena, reunindo num mesmo fluxo as convenções de normalização, construção e integração. O trabalho centra-se nas antenas quark–antiquark sem massa, considerando a configuração cinemática final-final, na qual ambos os radiadores da antena são partículas do estado final, até à ordem next-to-next-to-leading order (NNLO). Para além deste objetivo, é explorada uma extensão ao regime de quarks massivos, através da implementação da antena $A_3^0(Q, g, \bar{Q})$, que descreve a emissão de um gluão entre um quark e um antiquark massivos.

A base teórica da dissertação inclui uma introdução à QCD perturbativa, aos campos de quarks e gluões e à construção de amplitudes a partir de diagramas de Feynman. São discutidas as divergências ultravioleta, tratadas através da renormalização, e as divergências infravermelhas, que se cancelam mediante a combinação das diferentes contribuições perturbativas reais e virtuais à mesma ordem. Utiliza-se regularização dimensional, na qual o número de dimensões do espaço-tempo é tomado como

$d = 4 - 2\epsilon$. Neste enquadramento, as divergências manifestam-se como polos em $1/\epsilon$, e as quantidades integradas são expressas por séries de Laurent.

É também apresentada a estrutura das funções de antena e a sua decomposição em cor. As antenas são obtidas a partir de amplitudes físicas normalizadas e organizadas de acordo com os radiadores e as partículas emitidas. A fatorização do espaço de fase é essencial neste processo, pois permite separar o espaço de fase reduzido, associado aos radiadores depois de removida a emissão não resolvida, do espaço de fase correspondente à própria emissão. A integração pode então ser efetuada apenas sobre o espaço de fase da radiação não resolvida, independentemente do processo duro em que a antena venha a ser inserida.

A integração analítica das funções de antena sobre o correspondente espaço de fases de antena constitui uma das etapas mais exigentes na obtenção das antenas integradas. Para realizar esta integração, as integrais de espaço de fases são reescritas como integrais de loop através da técnica de reverse unitarity. Estas são então reduzidas, através de identidades de integração por partes (IBP) e do algoritmo de Laporta, a um conjunto finito de integrais mestres. Em casos adequados, recorre-se também à redução de Passarino–Veltman. Uma vez identificados os integrais mestres e substituídas as respetivas expressões conhecidas, obtém-se a forma integrada da antena como uma expansão em ϵ.

O pacote *AntCalc* foi implementado no ambiente *Wolfram Mathematica* e integra ferramentas especializadas para as diferentes fases do cálculo. *FeynArts* é utilizado para gerar os diagramas de Feynman relevantes, enquanto *FeynCalc* e *FeynHelpers* apoiam a construção e a manipulação das amplitudes. A redução IBP é realizada com *LiteRed2*. A arquitetura do pacote é orientada por perfis, isto é, por descrições estruturadas que especificam o processo, as amplitudes, as famílias de integrais, os fatores de normalização e os integrais mestres necessários. A informação específica de cada antena fica, assim, concentrada no respetivo perfil, enquanto as rotinas de construção e integração permanecem comuns aos diferentes casos.

O fluxo de trabalho divide-se em duas fases principais: construção e integração. Na fase de construção, o utilizador fornece a descrição do processo e o pacote gera os diagramas e amplitudes correspondentes. São depois efetuadas as manipulações algébricas necessárias para extrair a função de antena não integrada, ou as suas componentes de cor quando aplicável. Na fase de integração, essa expressão é convertida para uma forma apropriada à redução, associada à família de integrais correta e reduzida aos integrais mestres. Finalmente, a substituição dos resultados dos integrais mestres e a expansão em torno de $\epsilon = 0$ produzem a antena integrada.

Como exemplo detalhado, é apresentada a derivação da antena A_3^0, que descreve a emissão de um glúão entre um quark e um antiquark sem massa a *tree-level* (onde *tree-level* corresponde a contribuições de diagramas de Feynman sem loops). Este caso ilustra todo o percurso, desde a geração da amplitude até à expressão integrada. A validação local da antena é efetuada através dos seus limites infravermelhos. No limite *soft*, verifica-se que a expressão reproduz o fator eikonal esperado; nos limites colineares, confirma-se a recuperação da função de desdobramento (*splitting function*) de Altarelli–Parisi apropriada.

Para além de A_3^0, *AntCalc* é aplicado a um conjunto de antenas sem massa até NNLO. São obtidas contribuições de dois, três e quatro partões, incluindo os conjuntos A_2^2, A_3^1 e A_4^0, bem como antenas com outras estruturas de cor, tais como B_4^0 e C_4^0. Os resultados não integrados e integrados são comparados com resultados estabelecidos na literatura, verificando-se concordância.

A validação global é realizada através do cálculo do rácio hadrónico R, no limite de quarks sem massa, até NNLO em QCD, isto é, até $O(\alpha_s^2)$. Este observável é definido como o rácio entre a secção eficaz inclusiva de produção de hadrões em aniquilação e^+e^- e a secção eficaz para a produção de um

par de muões,

$$R(s) = \frac{\sigma(e^+e^- \to \text{hadrões})}{\sigma(e^+e^- \to \mu^+\mu^-)}.$$

Por ser inclusivo, constitui um teste rigoroso da consistência do formalismo: os polos infravermelhos provenientes das contribuições reais e virtuais devem cancelar exatamente. O cálculo do R-ratio a partir dos objetos integrados produzidos por *AntCalc* confirma esse cancelamento e reproduz o resultado finito conhecido até NNLO.

A dissertação explora ainda a extensão ao caso massivo pela construção e integração da antena massiva A_3^0. A presença de quarks massivos nesta antena altera a cinemática, a estrutura do espaço de fase e os limites infravermelhos, eliminando, em particular, algumas singularidades colineares existentes no caso sem massa. Este cálculo fornece uma primeira implementação do fluxo massivo em *AntCalc* e identifica as alterações necessárias à construção e integração de futuras antenas massivas.

Em síntese, o trabalho desenvolvido mostra que as etapas de construção, redução e integração das funções de antena consideradas podem ser reunidas num único procedimento automatizado. O pacote *AntCalc* reproduz resultados conhecidos para antenas sem massa até NNLO, valida os limites infravermelhos locais e recupera corretamente o R-ratio hadrónico. A implementação obtida pode servir de base à inclusão de novas classes de antenas sem exigir que todo o procedimento de construção e integração seja novamente implementado de raiz.

Palavras-chave: QCD; Subtração por Antenas; NNLO; Singularidades Infravermelhas; *Mathematica*

Abstract

The precision achieved by measurements at the Large Hadron Collider (LHC) places increasingly stringent demands on theoretical predictions. In perturbative Quantum Chromodynamics (QCD), predictions are systematically improved by including higher-order corrections in the strong coupling constant, α_s. Beyond leading order, however, real and virtual contributions develop infrared singularities associated with soft and collinear parton radiation. These singularities cancel in infrared-safe observables, but the real and virtual terms are separately divergent and are defined over phase spaces of different multiplicities. Their numerical evaluation therefore requires a procedure that isolates the singular behaviour and implements its cancellation explicitly.

Antenna subtraction does this systematically: universal antenna functions describe unresolved radiation between colour-connected hard partons; they are subtracted locally from the real-emission contribution and integrated analytically in dimensional regularisation, making the cancellation of infrared poles against the virtual contributions explicit.

This thesis develops *AntCalc*, a symbolic framework in *Wolfram Mathematica* for constructing and integrating QCD antenna functions. The package combines *FeynArts*, *FeynCalc*, *FeynHelpers*, and *LiteRed2*, keeping process-dependent information separate from the general workflow. It generates the required amplitudes, constructs unintegrated antennae or colour components, maps the corresponding phase-space integrals onto integral families, and reduces them to master integrals through integration-by-parts identities.

The implementation reproduces the massless final-final quark-antiquark antennae through next-to-next-to-leading order. The tree-level three-parton antenna A_3^0 is used as a detailed example and is checked in its soft and collinear limits. The integrated antennae agree with existing results in the literature, and their combined consistency is validated through the massless hadronic R-ratio up to NNLO, $O(\alpha_s^2)$. The framework is also extended beyond the massless case through the implementation of the massive A_3^0 antenna. Future work will extend *AntCalc* to further massive antennae, additional partonic configurations, and a more modular multi-language architecture.

Keywords: QCD; Antenna Subtraction; NNLO; Infrared Singularities; *Mathematica*

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List of Abbreviations and Symbols

Abbreviations and Acronyms

Term	Meaning
QCD	Quantum Chromodynamics
LO / NLO / NNLO	Leading / Next-to-Leading / Next-to-Next-to-Leading Order
IR / UV	Infrared / Ultraviolet
MI	Master Integral(s)
RR / RV / VV	double-Real, Real-Virtual and double-Virtual NNLO contributions
AP	Altarelli–Parisi (splitting function)
CDR	Conventional Dimensional Regularisation
KLN	Kinoshita–Lee–Nauenberg (theorem)
SM	Standard Model of Particle Physics
MC	Monte Carlo
IBP	Integration by Parts

Antenna and Regularisation Notation

Symbol	Meaning
X_n^l	Generic antenna symbol: n is the final-state multiplicity, l the loop order
A, B, C	Quark-antiquark antenna families (implemented)
D, E	Quark-gluon antenna families (not implemented in the present work)
F, G, H	Gluon-gluon antenna families (not implemented in the present work)
X (undecorated)	Leading-colour component (from the raw "Lead" interference term)
$\tilde{X}$	Subleading-colour component (raw "Sublead" interference term)
$\hat{X}$	Quark-loop component (raw "QL" interference term)
$\check{X}$	One-loop-squared component (NNLO A_2^2 only)
S_ϵ	$(4\pi)^\epsilon e^{-\epsilon\gamma_E}$, the $\overline{\text{MS}}$ normalisation factor

Symbol	Meaning
Λ_l	$(8\pi^2)^l$, loop-order normalisation
G_k	$(8\pi^2)^k$, perturbative-order scale normalisation
$C(\epsilon, k)$	G_k / S_ϵ^k, combined normalisation factor
$\mathcal{N}(\epsilon, k)$	Full integration normalisation, $C(\epsilon, k)/\Phi_2$
$\mathcal{T}$-objects	Process-wide colour- and kinematics-conscious objects assembled from antennae, used in the R-ratio construction
Φ_n	d-dimensional n-body phase-space volume

General Kinematics

Symbol	Meaning
α_s	Strong coupling constant
ϵ, $d = 4 - 2\epsilon$	Dimensional-regularisation parameter; spacetime dimension
s_{ij}	Mandelstam invariant, $2p_i \cdot p_j$
μ	Renormalisation/regularisation scale
q^2	Hard-process invariant mass squared
N	Number of colours (= 3)
N_f	Number of active quark flavours
C_F, C_A	Quadratic Casimir invariants (fundamental / adjoint representation)
g_s	Strong coupling, $g_s = \sqrt{4\pi\alpha_s}$

Chapter 1

Introduction

High-energy collider experiments, in particular proton-proton collisions at the Large Hadron Collider (LHC) at CERN, provide increasingly precise tests of the strong interaction and of the Standard Model of particle physics. Matching this experimental precision requires theoretical predictions with correspondingly high accuracy, making higher-order perturbative calculations in Quantum Chromodynamics (QCD) essential.

QCD is the quantum field theory describing the strong interaction. It is a non-Abelian gauge theory based on the $SU(3)_c$ colour symmetry, with quarks as fermionic matter fields and gluons as the corresponding gauge bosons. Quarks occur in six flavours and transform in the fundamental representation of $SU(3)_c$, while gluons transform in the adjoint representation and themselves carry colour charge. The quarks and gluons within the Standard Model particle content are shown in Fig. 1.1.

At sufficiently large momentum-transfer scales, the strong coupling α_s becomes small as a conse-

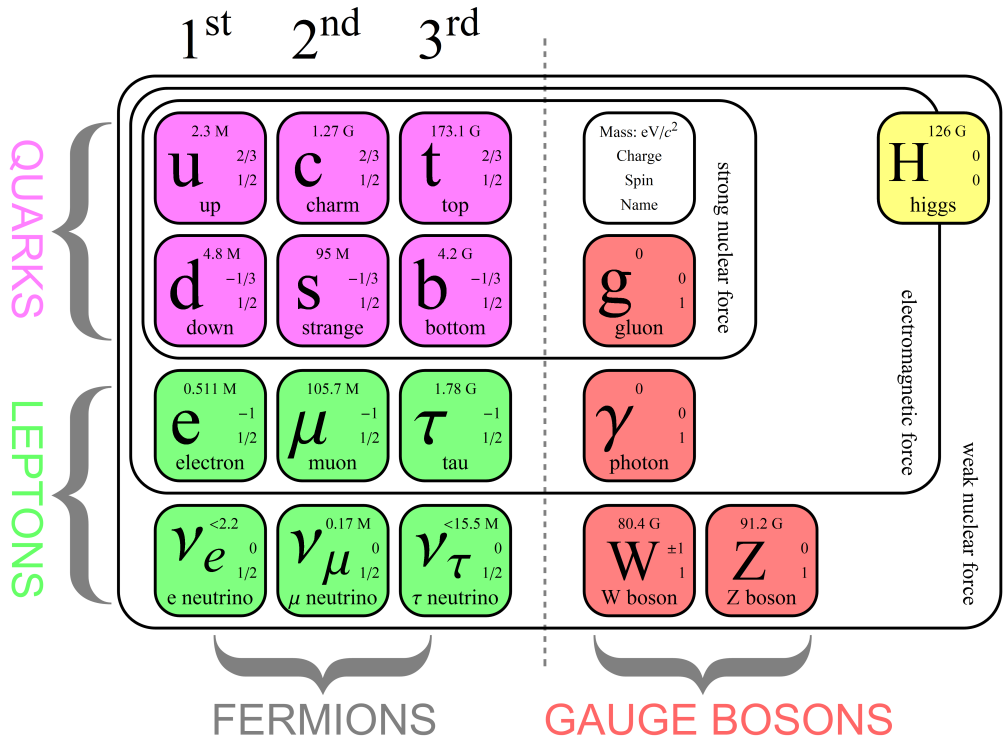


Figure 1.1: A diagram representing the Standard Model of Particle Physics. In it, the six quark flavours are visible on the top left in purple, among the fermions, and the gluon is shown in red, as the topmost of the gauge bosons. Image by Matic Lubej [1].

quence of asymptotic freedom. This allows QCD observables involving a hard scale Q to be calculated perturbatively as an expansion in powers of $\alpha_s(Q)$. Successive perturbative orders—leading order (LO), next-to-leading order (NLO), next-to-next-to-leading order (NNLO), and beyond—generally improve the accuracy of theoretical predictions and reduce their residual dependence on unphysical renormalisation and factorisation scales. This improvement, however, comes with a rapid increase in the complexity of the required calculations. Obtaining higher-order perturbative predictions is not simply a matter of calculating additional Feynman diagrams. Real-emission and virtual-loop contributions develop infrared singularities in kinematic configurations involving soft (low energy) or collinear partons. These individual contributions are divergent, and their infrared singularities must be extracted and cancelled before numerical integration can be performed. For infrared-safe observables, such as the hadronic R-ratio defined in Chapter 5, the singularities cancel between the real and virtual contributions, leaving a finite physical result.

The antenna subtraction formalism provides a framework for extracting and cancelling these infrared singularities, rendering the individual contributions suitable for numerical integration. The formalism describes unresolved radiation between pairs of colour-connected hard partons through universal antenna functions. These functions reproduce the relevant infrared limits locally and, after analytic integration over the unresolved phase space in dimensional regularisation, $d = 4 - 2\epsilon$, yield Laurent series whose ϵ-pole structure cancels that of the virtual contributions. Their systematic construction and integration are therefore central to higher-order QCD calculations within the antenna subtraction framework.

This thesis develops and validates *AntCalc*, a symbolic framework for the automated construction and integration of QCD antenna functions. The work focuses on massless final-final quark-antiquark antennae through next-to-next-to-leading order (NNLO), combining symbolic amplitude generation with reduction methods for phase-space integrals. The resulting implementation is tested globally through the calculation of the massless hadronic R-ratio, a well-known inclusive observable in e^+e^- annihilation that provides a benchmark for the complete implementation, up to NNLO. Locally, individual antenna functions are tested, where appropriate, by verifying that they reproduce the expected eikonal factors and Altarelli–Parisi (AP) splitting functions in the relevant unresolved limits.

The remainder of the thesis is organised as follows. Chapter 2 introduces the theoretical foundations required throughout this work, including perturbative QCD, infrared singularities, dimensional regularisation, and the antenna subtraction formalism. It defines the massless antenna functions and discusses their colour structure, phase-space factorisation, and role in fixed-order calculations. The chapter also introduces the normalisation conventions used throughout. Chapter 3 describes *AntCalc*'s framework, including its architecture, antenna construction and integration workflows, reduction procedures, and the conventions and validation tools used to obtain the results. Chapter 4 presents the antenna functions obtained using *AntCalc*, organised by final-state multiplicity, with the master integrals required for each antenna set presented after the corresponding antennae. The A_3^0 antenna is discussed in detail, from its unintegrated construction to its integrated form and the validation of its soft and collinear limits. Selected NLO and NNLO antenna results are then presented more concisely. The chapter concludes with an exploratory extension of the A_3^0 antenna to the massive case. Chapter 5 provides a global validation of the framework through the calculation of the massless hadronic R-ratio through NNLO. The relevant antenna contributions are combined, demonstrating the cancellation of infrared poles and agreement with the known finite result. Finally, Chapter 6 presents the conclusions of the thesis and discusses future directions for *AntCalc*, including more efficient symbolic and reduction backends, completion of the massive antenna sector, further antenna families and kinematic configurations, and extensions to higher perturbative orders, particularly $\mathrm{N^3LO}$.

Chapter 2

Theoretical Foundations

2.1 QCD: Fields, Interactions and Perturbative Amplitudes

QCD is the quantum field theory of the strong interaction, whose elementary particles are the quarks and the gluons. Quarks are fermions that carry colour charge and interact through the exchange of gluons. Owing to the non-Abelian gauge structure of QCD, gluons themselves also carry colour charge and, therefore, undergo self-interaction.

At sufficiently high energy scales, the strong coupling, α_s, becomes sufficiently small for physical observables to be calculated perturbatively, as an expansion in powers of α_s. In this framework, the interaction structure of QCD determines the amplitudes for quark and gluon radiation that are studied throughout this thesis.

2.1.1 Quark and Gluon Fields

Quarks are spin-1/2 fields carrying flavour and a three-valued colour charge. They are QCD's matter fields. Gluons are the eight gauge fields associated with local $SU(3)_c$ colour symmetry. The QCD Lagrangian describes their propagation and interaction. It is written as follows,

$$\mathcal{L}_{\text{QCD}} = -\frac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \sum_f \bar{\psi}_f \left(i\gamma^\mu D_\mu - m_f \right) \psi_f . \tag{2.1}$$

In this expression, $G^a_{\mu\nu}$ is the non-Abelian field-strength tensor, D_μ is the covariant derivative, ψ_f is the quark field and m_f is the quark mass. The flavour label f is summed over the quark species included in the theory, while the adjoint colour index a runs from one to eight. This Lagrangian is cleanly divided into two pieces, the quark and the gluon pieces.

Defining the covariant derivative as,

$$D_\mu = \partial_\mu - i g_s T^a A^a_\mu \tag{2.2}$$

where T^a represents the generators of the fundamental representation of $SU(3)_c$ and A^a_μ is the gluon gauge field, the quark Lagrangian term may be expanded to,

$$\mathcal{L}_q = \sum_f \left[\bar{\psi}_f \left(i\gamma^\mu \partial_\mu - m_f \right) \psi_f + g_s \bar{\psi}_f \gamma^\mu T^a \psi_f A^a_\mu \right] . \tag{2.3}$$

Here g_s is the strong coupling. In the expanded quark term, the first piece describes free quark propagation and the second quark-gluon interaction. This second piece is also responsible for generating the $q\bar{q}g$

vertex.

We now also define the totally antisymmetric structure constants of the $SU(3)_c$ Lie algebra f^{abc} as,

$$\left[T^a, T^b\right] = i f^{abc} T^c. \tag{2.4}$$

They may be used to define the non-Abelian field-strength tensor $G^a_{\mu\nu}$ as,

$$G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g_s f^{abc} A^b_\mu A^c_\nu. \tag{2.5}$$

The first two terms of the field-strength tensor are Abelian-like and can be rewritten using the linear part of the non-Abelian field-strength tensor $F^a_{\mu\nu}$, as,

$$G^a_{\mu\nu} = F^a_{\mu\nu} + g_s f^{abc} A^b_\mu A^c_\nu, \quad F^a_{\mu\nu} \equiv \partial_\mu A^a_\nu - \partial_\nu A^a_\mu. \tag{2.6}$$

The second term arises because, as shown in Eq. (2.4), the $SU(3)_c$ generators do not commute. The non-Abelian field-strength tensor is used to write the gluon Lagrangian term. After expanding it becomes,

$$\mathcal{L}_g = -\frac{1}{4}\left(F^a_{\mu\nu}\right)^2 - \frac{g_s}{2} f^{abc} F^{a\,\mu\nu} A^b_\mu A^c_\nu - \frac{g_s^2}{4} f^{abc} f^{ade} A^b_\mu A^c_\nu A^{d\,\mu} A^{e\,\nu}. \tag{2.7}$$

Here, the gluon Lagrangian is divided into three terms. The first term is quadratic in the gluon field and, after gauge-fixing, describes free gluon propagation as [2],

$$\Delta_{ab,\mu\nu}(k) = \delta_{ab} \frac{i}{k^2} \left(-g_{\mu\nu} + (1-\lambda)\frac{k_\mu k_\nu}{k^2}\right), \quad \mathcal{L}_{\text{gf}} = -\frac{1}{2\lambda}\left(\partial^\mu A^a_\mu\right)^2 \tag{2.8}$$

where $\mathcal{L}_{\text{gf}}$ is the gauge-fixing Lagrangian, δ_{ab} is the Kronecker delta, and λ is the gauge parameter. Throughout this text the 't Hooft-Feynman convention is used, which makes $\lambda = 1$, following the default convention from *FeynArts* [3]. It is important to note, however, that physical observables are independent of the gauge choice. In addition, the last two terms in Eq. (2.7) contain cubic and quartic terms in the gluon fields that generate, respectively, the three- and four-gluon self-interaction vertices, ggg and $gggg$.

The Feynman gauge adopted throughout this work, corresponding to $\lambda = 1$, is a particular choice within the class of covariant gauges. In a non-Abelian gauge theory such as QCD, covariant gauge fixing introduces Faddeev–Popov ghost fields. Their dynamics and interactions with the gluon field are described by the ghost Lagrangian [2],

$$\mathcal{L}_{\text{ghost}} = \partial_\mu \eta^{a\,\dagger} \left(D^\mu_{ab} \eta^b\right). \tag{2.9}$$

This is the default ghost sector as used in *FeynArts* [3] and, as for the $\lambda = 1$ choice above, is the one used for the duration of the present text and in *AntCalc*. These ghost fields cancel the unphysical timelike and longitudinal gluon polarisations that would, otherwise, propagate in covariant-gauge loop diagrams. This ensures that only the two physical transverse polarisation states contribute to physical amplitudes [2].

Together, the two pieces, $\mathcal{L}_q$ and $\mathcal{L}_g$, constitute the gauge-invariant core of QCD. They generate the three elementary interaction vertices among the quark and gluon fields: $q\bar{q}g$, ggg and $gggg$ [2, 4].

2.1.2 From the QCD Lagrangian to Feynman Diagrams

The interaction terms in $\mathcal{L}_q$ and $\mathcal{L}_g$ determine the interaction vertices of QCD, which, together with the corresponding propagators, form the Feynman rules we use to compute scattering amplitudes perturbatively. Feynman diagrams are graphical representations of the individual contributions to the perturbative expansion of an amplitude. Each process may require more than one diagram to be fully

represented; in that case, the complete amplitude is given as the sum of the amplitudes associated with all contributing diagrams. Fig. 2.1 depicts the two Feynman diagrams contributing to the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$, where a virtual photon splits into a quark-antiquark pair, with the gluon emitted from either the quark or antiquark line.



(a) Gluon emitted from the quark line.

(b) Gluon emitted from the antiquark line.

Figure 2.1: Tree-level diagrams for the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

The final amplitude in this case would then follow,

$$\mathcal{M}^{(0)}_{\gamma^* \to qg\bar{q}} = \mathcal{M}^{(0)}_{q\text{-emission}} + \mathcal{M}^{(0)}_{\bar{q}\text{-emission}}. \tag{2.10}$$

In the diagrams in Fig. 2.1, straight, curly, and wavy lines denote fermions (quarks), gluons and photons, respectively. These diagrams represent the reduced hadronic-current process with γ^* as an off-shell and colourless source. In the full process, $e^+e^- \to \gamma^* \to qg\bar{q}$, the photon is internal between the lepton (electron-positron) and hadron (quark-antiquark) currents. Apart from the single internal quark propagator, the remaining lines are external, meaning that they represent outgoing asymptotic partons, the quarks and gluons. Each of the diagrams features two vertices, where the lines meet. The vertices represent the interactions between the particles, which are determined by their respective Lagrangians. Each diagram contains an electromagnetic $\gamma^* q\bar{q}$ current insertion and a QCD $q\bar{q}g$ interaction vertex. The former follows from the QED sector, whereas the latter is generated by $\mathcal{L}_{\text{QCD}}$. A rate or cross section often depends on the squared amplitude $|\mathcal{M}|^2$, given as,

$$|\mathcal{M}|^2 = \sum_{D,D'} \mathcal{M}_D \mathcal{M}^*_{D'}, \tag{2.11}$$

where D and D' index the Feynman diagrams contributing to the amplitude, so that the sum includes both the diagonal $D = D'$ terms and the interference between distinct diagrams.

An amplitude $\mathcal{M}$ is computed from a Feynman diagram using Feynman rules. This ruleset is given in Section B.1.

2.2 Divergences in Perturbative QCD

2.2.1 Infrared Divergences

Beyond leading order in perturbation theory, QCD predictions receive contributions from both real and virtual corrections. Real corrections correspond to processes with additional parton emission in the final state, while virtual corrections arise from loop contributions to the amplitude. Both are required at a given perturbative order and can develop infrared singularities, which must ultimately cancel in infrared-safe physical observables.

The kinematic conditions for the two QCD infrared limits, soft and collinear, are obtained as follows. We start by defining the physical system under study: consider two massless final-state partons i and j with four-momenta p_i and p_j. By momentum conservation, the four-momentum of the internal virtual propagator is $P = p_i + p_j$. The matrix element contains this propagator and in the squared amplitude it appears as a $1/(P^2)^2$ factor. Expanding it and applying the massless condition ($p^2 = 0$), we get $P^2 = 2p_i \cdot p_j = 2E_iE_j(1 - \cos\theta_{ij})$ [2]. Importantly, this quantity defines the Mandelstam invariants as $s_{ij} = 2p_i \cdot p_j = P^2$, in the massless regime, which will serve as the natural integration variable throughout the work. The limits are then apparent: the soft limit is hit when one of the energies goes to zero and the collinear limit, when θ_{ij} does. In both cases,

$$\lim_{E_i \to 0} s_{ij} = 0, \quad \lim_{\theta_{ij} \to 0} s_{ij} = 0. \tag{2.12}$$

At NLO, the process $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$ illustrates the two types of single-unresolved infrared limits. The gluon becomes unresolved either when it is soft, $k_3 \to 0$, or when it becomes collinear with one of the neighbouring quarks, $q \parallel g$ or $g \parallel \bar{q}$. The quarks can also participate in these collinear limits, but a single soft quark does not generate an independent infrared singularity. This distinction is not determined solely by the vanishing of an invariant s_{ij}, but by the soft and collinear factorisation properties of the matrix element; in particular, soft-gluon emission exhibits an eikonal singularity for which there is no analogue in the single-soft-quark limit.

At NNLO, two partons can become unresolved simultaneously, giving rise to additional infrared configurations. For a secondary $q'\bar{q}'$ pair produced through gluon splitting, these include the double-soft limit in which q' and $\bar{q}'$ become simultaneously soft, the collinear limit $q' \parallel \bar{q}'$, and triple-collinear configurations such as $q \parallel q' \parallel \bar{q}'$, in which the secondary pair becomes collinear with a hard parton. The double-soft $q'\bar{q}'$ limit is singular despite the absence of a single-soft-quark singularity, as it is associated with the soft behaviour of the parent gluon. These and the other unresolved limits relevant to this thesis are summarised in Table 2.1.

Table 2.1: Infrared limits of the real-radiation matrix elements for $\gamma^* \to q\bar{q}$ through NNLO, listed by final-state parton configuration. Each configuration is colour-ordered, with the hard quark and antiquark first and last; p_i is the momentum of parton i, $\parallel$ denotes a collinear pair, and $s_{i\ldots k} = (p_i + \ldots + p_k)^2$. Limits related by $q \leftrightarrow \bar{q}$ or by gluon relabelling are not listed separately. The subtraction counterterm of Eq. (2.13) must reproduce the matrix element in each of these limits [5]. Complete analytic expressions for the universal soft and collinear factors listed in this table are given in [5], Sections 8.1 and 8.2.2.

Unresolved limit	**Kinematic condition**	**Universal IR factor**
$(1_q, 3_g, 2_{\bar{q}})$ — NLO real radiation		
Soft gluon	$p_3 \to 0$	S_{132}
Collinear	$1_q \parallel 3_g$	$P_{qg\to Q}(z)$
$(1_q, 3_g, 4_g, 2_{\bar{q}})$ — NNLO real radiation		
Soft gluon	$p_3 \to 0$	S_{134}
Collinear (qg)	$1_q \parallel 3_g$	$P_{qg\to Q}(z)$
Collinear (gg)	$3_g \parallel 4_g$	$P_{gg\to G}(z)$
Double soft	$p_3,\, p_4 \to 0$	S_{1342}
Soft–collinear	$p_3 \to 0,\ 4_g \parallel 2_{\bar{q}}$	$S_{1;342}(z)\, P_{qg\to Q}(z)$
Triple collinear	$1_q \parallel 3_g \parallel 4_g$	$P_{134\to Q}(w, z, y)$
Double single collinear	$1_q \parallel 3_g,\ 2_{\bar{q}} \parallel 4_g$	$P_{qg\to Q}(z)\, P_{\bar{q}g\to \bar{Q}}(y)$
$(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}})$ — NNLO real radiation		

Unresolved limit	Kinematic condition	Universal IR factor
Soft $q'\bar{q}'$ pair	$p_3,\, p_4 \to 0$	$S_{12}(3,4)$ (double pole)
Collinear $(q'\bar{q}')$	$3_{q'} \parallel 4_{\bar{q}'}$	$P_{q\bar{q}\to G}(z)$
Triple collinear	$1_q \parallel 3_{q'} \parallel 4_{\bar{q}'}$	$P^{\text{non-ident.}}_{134\to Q}(x,y)$
$(1_q,\, 3_q,\, 4_{\bar{q}},\, 2_{\bar{q}})$ — NNLO real radiation, identical flavour		
Triple collinear	$2_{\bar{q}} \parallel 3_q \parallel 4_{\bar{q}}$	$\frac{1}{2}P^{\text{ident.}}_{234\to\bar{Q}}(w,x,y)$

From an experimental point of view, a real emission that becomes soft or collinear may be unresolved, meaning that it cannot be distinguished from a configuration in which that emission is absent. For an infrared-safe observable, the two configurations yield the same value of the observable in the corresponding limit.

The virtual corrections contain infrared singularities related to those arising from unresolved real radiation. According to the Kinoshita–Lee–Nauenberg (KLN) theorem, these singularities cancel once all degenerate contributions to an infrared-safe physical observable are appropriately combined [6, 7]. The resulting observable is therefore finite, despite the presence of infrared divergences in the individual real and virtual contributions.

In practice, however, this cancellation cannot be performed directly at the integrand level, since the real and virtual contributions are separately divergent and are defined on phase spaces with different particle multiplicities. At NLO, the real contribution contains one additional final-state parton with respect to the LO n-parton process, and is integrated over the $(n+1)$-particle phase space, whereas the virtual contribution is integrated over the n-particle phase space. A subtraction method resolves this mismatch by introducing a local counterterm $d\sigma^S$ that reproduces the singular behaviour of the real contribution $d\sigma^R$ in all unresolved limits. The counterterm can then be subtracted from the real contribution, rendering their difference finite and suitable for numerical integration. The same counterterm is analytically integrated over the unresolved one-particle phase space and added to the virtual contribution, where its explicit infrared poles cancel those of the virtual correction. Schematically, at NLO, this is expressed as,

$$\sigma = \int_{n+1} \left(d\sigma^R - d\sigma^S \right) + \int_n \left(d\sigma^V + \int_1 d\sigma^S \right) \tag{2.13}$$

where $d\sigma^R$ represents the differential real radiation cross-section, $d\sigma^V$ the differential virtual corrections cross-section and $d\sigma^S$ the differential subtraction counterterm [8]. Importantly, $d\sigma^S$ must reproduce $d\sigma^R$'s exact singular behaviour in every unresolved limit, such that the divergences cancel locally.

2.2.2 Ultraviolet Divergences

Infrared divergences are not the only kind of divergences we find in QCD. In loop diagrams, the loop momentum l is integrated over the whole of momentum space. Regions in which l becomes soft or collinear to an external massless momentum can give rise to infrared singularities, which contribute to the singular structure of $d\sigma^V$. In addition, loop integrals may become divergent when the magnitude of the loop momentum becomes arbitrarily large, $|l| \to \infty$. These are referred to as ultraviolet (UV) divergences.

The UV behaviour of a loop integral can be understood through simple power counting. In d dimensions, the loop integration measure can be written in spherical coordinates as,

$$d^d l = l^{d-1} dl\, d\Omega_{d-1}. \tag{2.14}$$

Whether the integral diverges or not depends on whether the integrand converges fast enough to counter the l^{d-1} measure term. Since each propagator contributes $\sim 1/l^2$ to the integrand, schematically, a diagram with n_p propagators in the loop and no extra powers of l in the numerator becomes,

$$\int^{\infty} l^{d-1} \cdot \frac{1}{l^{2n_p}} = \int^{\infty} l^{d-1-2n_p} \, dl \tag{2.15}$$

which is UV divergent whenever $d - 1 - 2n_p \geq -1 \Rightarrow d \geq 2n_p$ [4, 9].

Ultraviolet divergences arising in loop corrections require the parameters and fields appearing in the bare QCD Lagrangian to be renormalised. In particular, the bare strong coupling is related to the renormalised coupling $\alpha_s(\mu)$ through a coupling renormalisation constant, which absorbs the corresponding ultraviolet divergences. The introduction of the renormalisation scale μ leads to a scale-dependent renormalised coupling, whose evolution is governed by the QCD beta function. The coupling renormalisation relevant to the one-loop antenna functions considered in this work is discussed explicitly in Subsection 3.3.2.

2.3 Dimensional Regularisation

The subtraction scheme from Eq. (2.13) separates the NLO cross-section into contributions that can ultimately be evaluated numerically. The difference $d\sigma^R - d\sigma^S$ is finite in the unresolved regions and can therefore be integrated numerically in four dimensions. The subtraction term must, however, still be integrated analytically over the unresolved one-parton phase space,

$$\int_1 d\sigma^S. \tag{2.16}$$

Since $d\sigma^S$ is designed explicitly to carry over all of $d\sigma^R$'s divergent structures, this integral diverges when integrated in four dimensions, as does the virtual contribution $d\sigma^V$, whose infrared divergences originate from the loop integration, as explored in Subsection 2.2.2. To regulate these divergent contributions we continue the number of space-time dimensions away from four, via,

$$d = 4 - 2\epsilon \tag{2.17}$$

so that infrared divergences appear as explicit analytic poles in $1/\epsilon$. After analytic integration over the unresolved phase space, this leaves us with two ϵ-dependent expressions, for the virtual corrections and for the now unresolved-parton-integrated subtraction cross-section,

$$\int_n \left(d\sigma^V + d\sigma_1^S \right). \tag{2.18}$$

From the KLN theorem, as described in Subsection 2.2.1, we know that these poles must cancel. This combination is therefore finite as $\epsilon \to 0$, allowing the remaining phase-space integration to be performed numerically in four dimensions.

Though the underlying principle is always the same, dimensional regularisation can be implemented using different schemes. The most common are 't Hooft-Veltman [10], Four-Dimensional Helicity [11] and Conventional Dimensional Regularisation (CDR) [2]. In CDR, all momenta and polarisation states are treated in $d = 4-2\epsilon$ dimensions. This is the scheme used throughout the present work, since *AntCalc*'s primary objective is the simplification and democratisation of the antenna framework established in [5], which uses CDR throughout.

While using CDR, the integrated and unintegrated antennae considered in this work are normalised using the following convention. First, the antennae are expressed in the $\overline{\text{MS}}$ (modified minimal subtraction)-convention using $S_\epsilon = (4\pi)^\epsilon e^{-\epsilon\gamma_E}$ [5]. Second, they are also normalised in relation to the strong coupling constant expansion. For an antenna with n final-state partons and l loops, we define,

$$k = (n-2) + l, \tag{2.19}$$

the perturbative-order index, which is zero for LO, one for NLO and two for NNLO, and introduce the normalisation factor

$$G_k = \left(8\pi^2\right)^{n-2} \Lambda_l = \left(8\pi^2\right)^{n-2} \left(8\pi^2\right)^l. \tag{2.20}$$

These two normalisations are then bundled into a general convention used throughout, which is written as,

$$\mathcal{N}(\epsilon, k) = \frac{C(\epsilon, k)}{\Phi_2}, \quad C(\epsilon, k) = \frac{G_k}{S_\epsilon^k} \tag{2.21}$$

using the d-dimensional two-particle phase-space volume Φ_2. This is the general representation of the work's normalisation convention.

As discussed in Subsection 2.2.2, ultraviolet divergences are removed by expressing the bare strong coupling α_0 in terms of the renormalised coupling $\alpha_s(\mu)$. In the $\overline{\text{MS}}$ scheme, the relation between the bare and renormalised couplings is given by [5],

$$\begin{gathered}\alpha_0 \mu_0^{2\epsilon} S_\epsilon = \alpha_s \mu^{2\epsilon} \left[1 - \frac{\beta_0}{\epsilon}\frac{\alpha_s}{2\pi} + \left(\frac{\beta_0^2}{\epsilon^2} - \frac{\beta_1}{2\epsilon}\right)\left(\frac{\alpha_s}{2\pi}\right)^2 + O(\alpha_s^3)\right], \\ \beta_0 = \frac{11N - 2N_f}{6}, \, \beta_1 = \frac{34N^3 - 13N^2 N_f + 3N_f}{12N}\end{gathered} \tag{2.22}$$

where S_ϵ denotes the conventional dimensional-regularisation factor, while μ_0 and μ are the dimensional-regularisation and renormalisation scales associated with the bare and renormalised couplings, respectively. The coefficients β_0 and β_1 are the first two coefficients of the QCD beta function, with $N = 3$ the number of colours and N_f the number of active massless quark flavours [2, 5]. For the massless antenna functions considered here, the renormalisation scale is chosen as $\mu^2 = q^2$, with q^2 the invariant mass squared of the antenna system.

2.4 The Antenna Subtraction Formalism

In perturbative QCD, computing the perturbative corrections to a certain jet observable $J_m^{(n)}$ requires summing over all partonic multiplicity channels up to the desired order.

At LO, the approximation to the m-jet cross section is computed as the integral over the appropriate phase space,

$$d\sigma_{\text{LO}} = \int_m d\sigma^B, \quad d\sigma^B = d\Phi_m(q : p_1, ..., p_m) \frac{1}{S_m} |\mathcal{M}_m^0(p_1, ..., p_m)|^2 J_m^{(m)}(p_1, ..., p_m) \tag{2.23}$$

where $d\sigma^B$ is the tree-level m-parton contribution to the m-jet cross section, built from the squared tree-level m-parton amplitude $|\mathcal{M}_m^0(p_1, ..., p_m)|^2$. At LO every final-state particle forms its own jet, so the jet function carries m jets for m final-state particles, $J_m^{(m)}$. S_m is a symmetry factor for identical partons in the final state, and Φ_m is the m-parton final-state phase space in d dimensions [5],

$$d\Phi_m(q : p_1, ..., p_m) = \frac{d^{d-1}p_1}{2E_1(2\pi)^{d-1}} ... \frac{d^{d-1}p_m}{2E_m(2\pi)^{d-1}} (2\pi)^d \delta^{(d)}(q - p_1 - ... - p_m). \tag{2.24}$$

At NLO, the m-jet cross section receives a real-emission and a virtual contribution, built from an $(m+1)$-parton tree-level amplitude and from the interference between the m-parton tree-level and one-loop amplitudes, respectively, for a generic process,

$$\begin{aligned} d\sigma^{R}_{\mathrm{NLO}} &= d\Phi_{m+1}(q : p_1, ..., p_{m+1}) \frac{1}{S_{m+1}} |\mathcal{M}^{0}_{m+1}(p_1, ..., p_{m+1})|^2 J^{(m+1)}_{m}(p_1, ..., p_{m+1}), \\ d\sigma^{V}_{\mathrm{NLO}} &= d\Phi_{m}(q : p_1, ..., p_{m}) \frac{1}{S_{m}} 2\,\mathrm{Re}\Big(\mathcal{M}^{1}_{m}(p_1, ..., p_{m}) \mathcal{M}^{0\,*}_{m}(p_1, ..., p_{m})\Big) J^{(m)}_{m}(p_1, ..., p_{m}). \end{aligned} \tag{2.25}$$

These contributions are summed to reach,

$$d\sigma_{\mathrm{NLO}} = \int_{m+1} d\sigma^{R}_{\mathrm{NLO}} + \int_{m} d\sigma^{V}_{\mathrm{NLO}}. \tag{2.26}$$

At NNLO, the m-jet cross section receives three contributions: a double-real contribution, built from the $(m+2)$-parton tree-level amplitude; a real-virtual contribution, built from the interference between the $(m+1)$-parton tree-level and one-loop amplitudes; and a double-virtual contribution, built from the interference between the m-parton tree-level and two-loop amplitudes and from the squared m-parton one-loop amplitude, for a generic process,

$$\begin{aligned} d\sigma^{RR}_{\mathrm{NNLO}} &= d\Phi_{m+2}(q : p_1, ..., p_{m+2}) \frac{1}{S_{m+2}} |\mathcal{M}^{0}_{m+2}(p_1, ..., p_{m+2})|^2 J^{(m+2)}_{m}(p_1, ..., p_{m+2}), \\ d\sigma^{RV}_{\mathrm{NNLO}} &= d\Phi_{m+1}(q : p_1, ..., p_{m+1}) \frac{1}{S_{m+1}} 2\,\mathrm{Re}\Big(\mathcal{M}^{1}_{m+1}(p_1, ..., p_{m+1}) \mathcal{M}^{0\,*}_{m+1}(p_1, ..., p_{m+1})\Big) \\ &\qquad \times J^{(m+1)}_{m}(p_1, ..., p_{m+1}), \\ d\sigma^{VV}_{\mathrm{NNLO}} &= d\Phi_{m}(q : p_1, ..., p_{m}) \frac{1}{S_{m}} \Big[2\,\mathrm{Re}\Big(\mathcal{M}^{2}_{m}(p_1, ..., p_{m}) \mathcal{M}^{0\,*}_{m}(p_1, ..., p_{m})\Big) \\ &\qquad + |\mathcal{M}^{1}_{m}(p_1, ..., p_{m})|^2\Big] J^{(m)}_{m}(p_1, ..., p_{m}). \end{aligned} \tag{2.27}$$

These contributions are summed to reach,

$$d\sigma_{\mathrm{NNLO}} = \int_{m+2} d\sigma^{RR}_{\mathrm{NNLO}} + \int_{m+1} d\sigma^{RV}_{\mathrm{NNLO}} + \int_{m} d\sigma^{VV}_{\mathrm{NNLO}}. \tag{2.28}$$

2.4.1 Subtraction Terms

The contributions that form part of the NLO and NNLO m-jet cross sections, given in Eq. (2.26) and Eq. (2.28) respectively, are separately IR-divergent. Consequently, a direct numerical evaluation of the individual contributions is not possible, as the corresponding integrals are divergent. The standard solution to this problem is to introduce subtraction terms that locally reproduce the infrared singular behaviour of the corresponding contributions throughout the relevant phase space.

2.4.1.1 NLO

At NLO, a subtraction term, $d\sigma^{S}_{\mathrm{NLO}}$, is introduced to locally reproduce the infrared singular behaviour of the real-emission contribution $d\sigma^{R}_{\mathrm{NLO}}$. The m-jet cross section can then be written as,

$$d\sigma_{\mathrm{NLO}} = \int_{m+1} \Big(d\sigma^{R}_{\mathrm{NLO}} - d\sigma^{S}_{\mathrm{NLO}}\Big) + \int_{m} \left(d\sigma^{V}_{\mathrm{NLO}} + \int_{1} d\sigma^{S}_{\mathrm{NLO}}\right). \tag{2.29}$$

The subtraction term must reproduce the real-emission contribution in all single-unresolved limits, while remaining local in phase space and introducing no additional singularities. Consequently, the difference $d\sigma^{R}_{\mathrm{NLO}} - d\sigma^{S}_{\mathrm{NLO}}$ is finite in these limits and can be integrated numerically in four dimensions. At the same

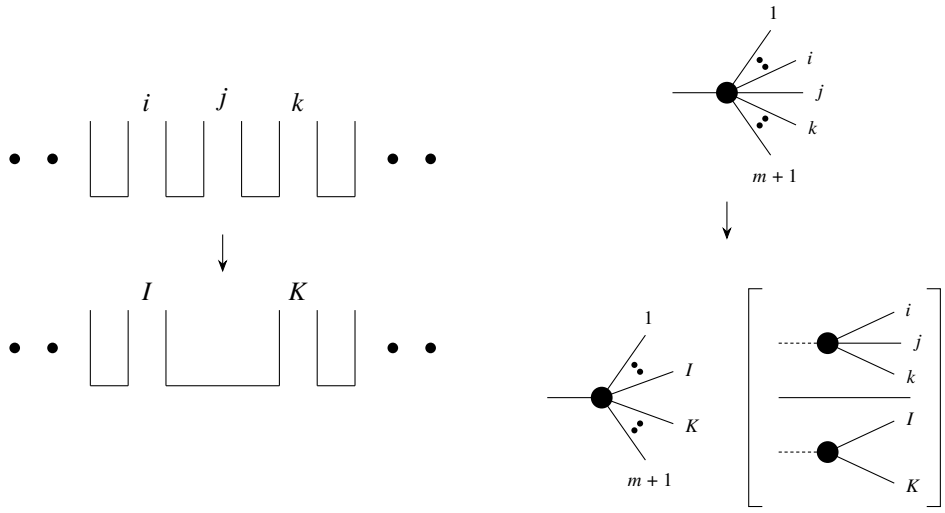


Figure 2.2: The colour connections in between parent and daughter partons before and after an emission, accompanied also by the corresponding amplitude-level factorisation, using the momenta labels.

time, $d\sigma^S_{\rm NLO}$ must be analytically integrable over the unresolved one-parton phase space. Its integrated form contains explicit poles in ϵ that cancel those of the virtual contribution $d\sigma^V_{\rm NLO}$.

Within the antenna subtraction formalism, $d\sigma^S_{\rm NLO}$ is constructed from three-parton tree-level antenna functions X^0_{ijk}, which capture the unresolved radiation emitted between a pair of colour-connected hard radiators. It takes the schematic form,

$$d\sigma^S_{\rm NLO} = \sum_{m+1} d\Phi_{m+1}(q : p_1, ..., p_{m+1}) \frac{1}{S_{m+1}} \sum_j X^0_{ijk} |\mathcal{M}^0_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1})|^2 J^{(m)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}). \tag{2.30}$$

Here, i and k denote the hard radiators and j the unresolved parton. The momenta p_i, p_j, p_k are mapped onto the reduced on-shell momenta $\tilde{p}_I$ and $\tilde{p}_K$, such that the reduced matrix element has the appropriate m-parton kinematics. For a $q\bar{q}g$ antenna, i and k correspond to the quark and antiquark radiators, while j corresponds to the unresolved gluon. Parent partons I and K are colour connected. Daughter partons i and j, and j and k are also connected, but i and k are not. A diagram of these relationships is shown in Fig. 2.2.

The jet function $J^{(m)}_m$ in the antenna subtraction term above does not depend on the individual momenta p_i, p_j and p_k, but only on $\tilde{p}_I$ and $\tilde{p}_K$. One can therefore carry out the integration over the unresolved antenna phase space appropriate to p_i, p_j and p_k analytically, exploiting the factorisation of the phase space,

$$d\Phi_{m+1}(q : p_1, ..., p_{m+1}) = d\Phi_m(q : p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}) d\Phi_{X_{ijk}}(q = \tilde{p}_I + \tilde{p}_K : p_i, p_j, p_k). \tag{2.31}$$

The integrated counterterm $\int_1 d\sigma^S_{\rm NLO}$ then takes the form,

$$\int_1 d\sigma^S_{\rm NLO} = \sum_m d\Phi_m(q : p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}) \frac{1}{S_{m+1}} \sum_j \left(\int d\Phi_{X_{ijk}} X^0_{ijk} \right) |\mathcal{M}^0_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1})|^2 J^{(m)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}). \tag{2.32}$$

Table 2.2: NNLO unresolved configurations and their matching subtraction term, part of the $d\sigma^S_{\text{NNLO}}$ and $d\sigma^T_{\text{NNLO}}$ subtraction counterterms.

Configuration	Subtraction Term
RR subtraction counterterm – $d\sigma^S_{\text{NNLO}}$	
Single unresolved parton	$d\sigma^{S,a}$
Two colour-connected unresolved partons	$d\sigma^{S,b}$
Two almost colour-unconnected unresolved partons	$d\sigma^{S,c}$
Two fully colour-unconnected unresolved partons	$d\sigma^{S,d}$
Soft-gluon emission at large angles	$d\sigma^{S,e}$
RV subtraction counterterm – $d\sigma^T_{\text{NNLO}}$	
Explicit infrared poles from the one-loop matrix elements	$d\sigma^{T,a}$
One of the partons becomes soft or collinear	$d\sigma^{T,b}$
Spurious poles introduced by the other two subterms	$d\sigma^{T,c}$

The integral $\int d\Phi_{X_{ijk}}\, X^0_{ijk}$ defines the integrated antenna, which is then combined with the virtual contribution, rendering its contribution IR-finite. Setting $m = 2$ in Eq. (2.31), we can see that the NLO antenna phase space $d\Phi_{X_{ijk}}$ is proportional to the three-particle phase space,

$$d\Phi_3 = \Phi_2\, d\Phi_{X_{ijk}}, \quad \Phi_2 = \int d\Phi_2. \tag{2.33}$$

2.4.1.2 NNLO

Analogously, at NNLO, two subtraction counterterms, $d\sigma^S_{\text{NNLO}}$ and $d\sigma^T_{\text{NNLO}}$, are introduced to reproduce the infrared singular behaviour of the double-real and real–virtual contributions, $d\sigma^{RR}_{\text{NNLO}}$ and $d\sigma^{RV}_{\text{NNLO}}$, respectively. The m-jet cross section can then be written as,

$$\begin{aligned} d\sigma_{\text{NNLO}} = \int_{m+2} \left(d\sigma^{RR}_{\text{NNLO}} - d\sigma^S_{\text{NNLO}} \right) + \int_{m+1} \left(d\sigma^{RV}_{\text{NNLO}} - d\sigma^T_{\text{NNLO}} \right) \\ + \int_m \left(d\sigma^{VV}_{\text{NNLO}} + \int_2 d\sigma^S_{\text{NNLO}} + \int_1 d\sigma^T_{\text{NNLO}} \right). \end{aligned} \tag{2.34}$$

At NNLO, the subtraction structure becomes more involved due to the presence of several single- and double-unresolved infrared configurations. The double-real and real–virtual subtraction counterterms are therefore decomposed into several contributions,

$$\begin{aligned} d\sigma^S_{\text{NNLO}} &= d\sigma^{S,a} + d\sigma^{S,b} + d\sigma^{S,c} + d\sigma^{S,d} + d\sigma^{S,e} \\ d\sigma^T_{\text{NNLO}} &= d\sigma^{T,a} + d\sigma^{T,b} + d\sigma^{T,c}. \end{aligned} \tag{2.35}$$

Each contribution accounts for a specific infrared singular configuration of the double-real or real–virtual matrix elements, as summarised in Table 2.2. For the double-real contribution, the different colour connections associated with the unresolved partons are illustrated schematically in Fig. 2.3.

The individual contributions to the subtraction terms are constructed from antenna functions. At NNLO, in addition to the tree-level three-parton antennae X^0_{ijk} already required at NLO, tree-level four-parton antennae X^0_{ijkl} and one-loop three-parton antennae X^1_{ijk} are also required. The contributions to

the double-real subtraction counterterm are given by,

$$
\begin{aligned}
d\sigma^{S,a} &= \sum_{m+2} d\Phi_{m+2}(q : p_1, ..., p_{m+2}) \frac{1}{S_{m+2}} \\
&\qquad \sum_j X^0_{ijk} |\mathcal{M}^0_{m+1}(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+2})|^2 J^{(m+1)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+2}), \\
d\sigma^{S,b} &= \sum_{m+2} d\Phi_{m+2}(q : p_1, ..., p_{m+2}) \frac{1}{S_{m+2}} \\
&\qquad \sum_{j,k} \left(X^0_{ijkl} - X^0_{ijk} X^0_{IKl} - X^0_{jkl} X^0_{iJL} \right) |\mathcal{M}^0_m(p_1, ..., \tilde{p}_I, \tilde{p}_L, ..., p_{m+2})|^2 \\
&\qquad J^{(m)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_L, ..., p_{m+2}),
\end{aligned}
\tag{2.36}
$$

and for the real-virtual counterterm,

$$
\begin{aligned}
d\sigma^{T,a} &= \sum_{m+1} d\Phi_{m+1}(q : p_1, ..., p_{m+1}) \frac{1}{S_{m+1}} \\
&\qquad \sum_{i,k} -\mathcal{X}^0_{ijk}(s_{ik}) |\mathcal{M}^0_{m+1}(p_1, ..., p_i, p_k, ..., p_{m+1})|^2 J^{(m+1)}_m(p_1, ..., p_i, p_k, ..., p_{m+1}), \\
&\qquad \text{with } \mathcal{X}^0_{ijk} = C(\epsilon, 1) \int d\Phi_{X_{ijk}} X^0_{ijk}, \\
d\sigma^{T,b} &= \sum_{m+1} d\Phi_{m+1}(q : p_1, ..., p_{m+1}) \frac{1}{S_{m+1}} \\
&\qquad \sum_j \Big[X^0_{ijk} |\mathcal{M}^1_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1})|^2 J^{(m)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}) \\
&\qquad + X^1_{ijk} |\mathcal{M}^0_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1})|^2 J^{(m)}_m(p_1, ..., \tilde{p}_I, \tilde{p}_K, ..., p_{m+1}) \Big], \\
&\qquad \text{with } X^1_{ijk} \to X^1_{ijk} + \frac{\beta_0}{\epsilon} \frac{(4\pi)^\epsilon e^{-\epsilon\gamma_E}}{8\pi^2} X^0_{ijk} \left(s_{ijk}^{-\epsilon} - (\mu^2)^{-\epsilon} \right),
\end{aligned}
\tag{2.37}
$$

The terms $d\sigma^{S,c}$, $d\sigma^{S,d}$, $d\sigma^{S,e}$, and $d\sigma^{T,c}$ arise only when the underlying reduced hard process contains at least three resolved partons. Since the configurations considered in this work contain only the two hard $q\bar{q}$ radiators, these terms are not required.

For the double-real subtraction terms, the corresponding phase-space factorisation generalises the NLO $3 \to 2$ mapping to the four-parton antenna configuration,

$$
d\Phi_{m+2}(q : p_1, ..., p_{m+2}) = d\Phi_m(q : p_1, ..., \tilde{p}_I, \tilde{p}_L, ..., p_{m+2}) d\Phi_{X_{ijkl}}(q = \tilde{p}_I + \tilde{p}_L : p_i, p_j, p_k, p_l). \tag{2.38}
$$

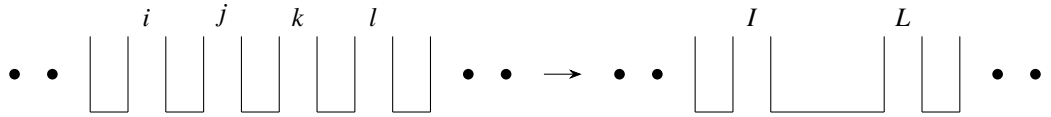
For a two-parton reduced configuration, this becomes

$$
d\Phi_4 = \Phi_2 \, d\Phi_{X_{ijkl}}, \tag{2.39}
$$

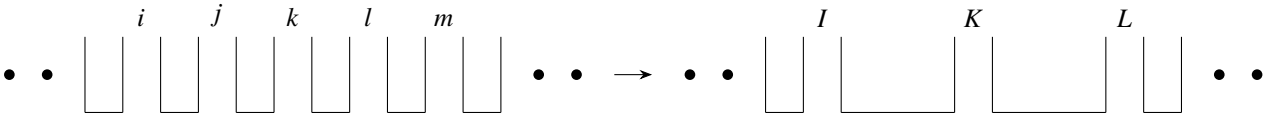
so that the four-parton antenna phase space $d\Phi_{X_{ijkl}}$ provides the integration measure for the tree-level four-parton antennae X^0_{ijkl}.

2.4.2 Antenna Functions

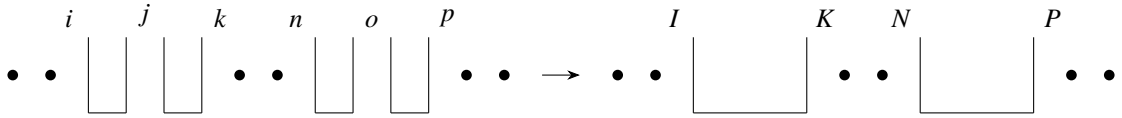
In this thesis, the complete set of massless quark–antiquark final–final antenna functions required up to NNLO is considered. All antennae share the same underlying hard $q\bar{q}$ radiator pair, with configurations containing up to one unresolved parton at NLO and up to two unresolved partons at NNLO. In addition,



(a) The colour connection of the partons for the double unresolved antenna.



(b) The colour connection of the partons for two adjacent single unresolved antennae.



(c) The colour connections of the partons for two disconnected single unresolved antennae.

Figure 2.3: The three characteristic double-unresolved colour-connection configurations at NNLO: (a) two colour-connected unresolved partons, corresponding to $d\sigma^{S,b}$; (b) two adjacent single-unresolved antennae, corresponding to $d\sigma^{S,c}$; and (c) two disconnected single-unresolved antennae, corresponding to $d\sigma^{S,d}$.

the corresponding two-parton virtual contributions, derived from the quark–antiquark form factor, are included. Although these two-parton objects do not describe unresolved real radiation, they enter the antenna framework alongside the real-radiation antenna functions and will, for convenience, be referred to as antenna functions throughout this thesis.

Each antenna is generally determined by its external state and the hard parton pair it collapses to. In this thesis, the hard parton pair is always $q\bar{q}$. Up to NNLO, quark-antiquark final-final antennae are organised into three families, which differ in the types of particles emitted by the hard parton pair. A-type antennae are characterised by gluon emissions, leading to final states like $qgg\bar{q}$ at NNLO or $qg\bar{q}$ at NLO; C-type antennae are characterised by a final-state quark-antiquark pair with the original hard pair's flavour, as $q\bar{q}q\bar{q}$; and B-type by a quark-antiquark pair of a different flavour, notated as $q\bar{q}q'\bar{q}'$. Following QCD Feynman rules, described in Section B.1, A-type antennae may exist at any order, while B- and C-type antennae first appear at NNLO.

Throughout the remainder of this thesis, a generic X-type antenna is denoted by X_n^l, where n is the number of final-state partons and l the number of loops. Where applicable, antennae are decomposed into leading- and subleading-colour components, denoted by X_n^l and $\tilde{X}_n^l$, respectively. For loop antennae, an additional closed-quark-loop contribution, denoted by $\hat{X}_n^l$, may be present. Here, "leading" and "subleading" refer to the corresponding colour structures in the expansion in the number of colours N.

For a given antenna type X_n^l, the collection of its different colour components is referred to as the X_n^l antenna set. Depending on the antenna, some of these components may vanish; in particular, not all antenna sets contain non-zero subleading-colour or, at loop level, closed-quark-loop contributions. The A_2^2 set requires an additional contribution, denoted by $\breve{A}_2^2$. This additional term does not represent a new colour component. Rather, it distinguishes the one-loop-squared contribution from the interference between the two-loop and tree-level amplitudes. Both contributions have the same two-parton final state and enter at NNLO, and are therefore grouped within the A_2^2 set for convenience. The antenna functions (and $q\bar{q}$ form factors) considered are organised by order, loop number, and final state in Table 2.3.

Antenna functions are constructed from physical, colour-ordered matrix elements (shown in Table 2.3)

Table 2.3: Quark-antiquark antenna families used in this work, including the $q\bar{q}$ form factors [5]. Final states are ordered with the two hard radiators listed first and last, following that paper's convention.

Final state	Tree level	One loop	Two loop
$\gamma^* \to q\bar{q}$	A_2^0	A_2^1	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$
$\gamma^* \to qg\bar{q}$	A_3^0	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$	—
$\gamma^* \to qgg\bar{q}$	A_4^0, $\tilde{A}_4^0$	—	—
$\gamma^* \to qq'\bar{q}'\bar{q}$	B_4^0	—	—
$\gamma^* \to qq\bar{q}\bar{q}$	C_4^0	—	—

and are normalised to the corresponding two-parton matrix element of the underlying hard radiator pair. To define the notation used below, the matrix-element contributions at successive perturbative orders are written as,

$$\begin{aligned} |\mathcal{M}_n^0|^2 &= \mathcal{M}_n^{0\,\dagger}\mathcal{M}_n^0 \\ |\mathcal{M}_n^1|^2 &= 2\,\mathrm{Re}\left(\mathcal{M}_n^{0\,\dagger}\mathcal{M}_n^1\right) \\ |\mathcal{M}_n^2|^2 &= 2\,\mathrm{Re}\left(\mathcal{M}_n^{0\,\dagger}\mathcal{M}_n^2\right) + \mathcal{M}_n^{1\,\dagger}\mathcal{M}_n^1. \end{aligned} \tag{2.40}$$

Here, $\mathcal{M}_n^l$ denotes the l-loop amplitude for an n-parton final state. Thus, $|\mathcal{M}_n^1|^2$ denotes the tree–one-loop interference, while $|\mathcal{M}_n^2|^2$ contains both the tree–two-loop interference and the one-loop-squared contribution.

At tree level ($l = 0$), the three- and four-parton antenna functions are defined as [5],

$$\begin{aligned} X_{ijk}^0 &= S_{ijk,IK}\frac{|\mathcal{M}_{ijk}^0|^2}{|\mathcal{M}_{IK}^0|^2} \\ X_{ijkl}^0 &= S_{ijkl,IL}\frac{|\mathcal{M}_{ijkl}^0|^2}{|\mathcal{M}_{IL}^0|^2} \end{aligned} \tag{2.41}$$

Here, i, j, k and i, j, k, l label the partons in the three- and four-parton configurations, respectively, while I, K and I, L denote the corresponding hard radiators in the reduced two-parton configuration and S denotes the symmetry factor associated to the antenna, which accounts both for potential identical particle symmetries and for the presence of more than one antenna in the basic two-parton process. For all $q\bar{q}$ antenna topologies considered in this work, the corresponding symmetry factors are equal to unity.

At one loop ($l = 1$), the two- and three-parton antenna functions are defined as [5],

$$\begin{aligned} X_{ij}^1 &= S_{ij,IJ}\frac{|\mathcal{M}_{ij}^1|^2}{|\mathcal{M}_{IJ}^0|^2} \\ X_{ijk}^1 &= S_{ijk,IK}\frac{|\mathcal{M}_{ijk}^1|^2}{|\mathcal{M}_{IK}^0|^2} - X_{ijk}^0\frac{|\mathcal{M}_{IK}^1|^2}{|\mathcal{M}_{IK}^0|^2}. \end{aligned} \tag{2.42}$$

where the second term removes the one-loop correction associated with the reduced two-parton process.

At two loops ($l = 2$), the two-parton antenna considered in this work contributes to the A_2^2 antenna set and is defined by

$$X_{ij}^2 = S_{ij,IJ}\frac{|\mathcal{M}_{ij}^2|^2}{|\mathcal{M}_{IJ}^0|^2}. \tag{2.43}$$

Loop antennae are additionally coupling-renormalised according to Eq. (2.22) whenever the corresponding reduced process carries powers of the strong coupling. This procedure is discussed further in Subsection 2.4.4. Finally, all antenna functions are normalised to the two-parton Born squared matrix

element, $|\mathcal{M}_2^0|^2$, such that the Born-level two-parton antenna satisfies $A_2^0 = 1$.

For antennae beyond A_2^0, the matrix-element ratios in Eqs. (2.41)–(2.43) are understood with the overall strong-coupling factor, g_s^{2k} with k as in Eq. (2.19), and the $2C_F$ colour factor associated with the $q\bar{q}$ radiator pair divided out, so that antenna functions are purely kinematic and colour-stripped. The same convention applies to the process-level contributions of Eq. (2.44) when they are decomposed into antennae in Subsection 2.4.4. $2C_F$ is restored when the antennae are combined with the Born contribution, as in Chapter 5.

In addition to the colour-ordered amplitudes used to define individual antenna functions, it is useful to introduce notation for the complete Born-normalised matrix-element contribution associated with a given physical final state. Such a contribution may contain several colour orderings, flavour assignments, or coupling configurations, together with the corresponding interference terms. This notation will be particularly useful below when decomposing the complete matrix elements into their individual antenna contributions. It is defined as,

$$|M^l_{p_1 p_2 \ldots p_n}|^2 = \frac{1}{|\mathcal{M}_2^0|^2} \sum_{r=0}^{l} \mathcal{M}_n^{r\,\dagger}(p_1, ..., p_n)\mathcal{M}_n^{l-r}(p_1, ..., p_n), \tag{2.44}$$

where more than one colour ordering, flavour assignment, or coupling configuration contributes to the specified final state, the right-hand side is understood to include the complete sum over the corresponding amplitude products, including any interference terms. Thus, $|M^l_{p_1 \ldots p_n}|^2$ denotes the complete process-level contribution, whereas $\mathcal{M}^l_n(p_1, \ldots, p_n)$ denotes an amplitude with a specified ordering or assignment from which the individual antenna contributions are subsequently identified.

The complete matrix-element contributions in Eq. (2.44) also contain the colour factors associated with the underlying QCD process. After the relevant amplitude products are formed, the colour indices must therefore be contracted and summed. The resulting colour algebra separates the matrix element into distinct colour structures, whose coefficients are used to identify the corresponding antenna colour components. The colour identities and decomposition required for this procedure are introduced in the following subsection.

2.4.3 Colour Algebra and the Antenna Colour Decomposition

The interactions of quarks and gluons are determined by the local $SU(3)_c$ gauge symmetry of QCD. For the colour algebra relevant here, quarks transform in the fundamental representation and carry a colour index $i = 1, \ldots, N$, while gluons transform in the adjoint representation and carry an index $a = 1, \ldots, N^2 - 1$, with $N = 3$ in QCD. The generators of the $SU(N)$ Lie algebra in the fundamental representation are denoted by T^a and satisfy the normalisation

$$\mathrm{Tr}\left(T^a T^b\right) = T_R \delta^{ab}, \quad T_R = \frac{1}{2}. \tag{2.45}$$

When an amplitude is squared, the colour factors of the amplitude and its complex conjugate must be contracted and summed over the corresponding colour indices. As a simple example, consider a colour structure containing a single generator T^a. Since the generators in the fundamental representation are Hermitian, $(T^a)^\dagger = T^a$, the corresponding colour sum can be written as

$$\sum_{i_1, i_2, a} (T^a)_{i_1 i_2} (T^a)_{i_2 i_1} = \sum_a \mathrm{Tr}\,(T^a T^a) = \sum_a T_R \delta^{aa}. \tag{2.46}$$

Since the adjoint representation of $SU(N)$ has dimension $N^2 - 1$, summing over the adjoint index gives

$$\sum_a T_R \delta^{aa} = \frac{1}{2}\left(N^2 - 1\right). \tag{2.47}$$

For processes containing two quark-antiquark colour lines, such as the four-quark final states relevant to the B_4^0 and C_4^0 antennae, the colour algebra involves four fundamental colour indices. A useful relation in this case is the $SU(N)$ completeness relation, also known as the *colour Fierz identity* [2, 4],

$$\sum_a (T^a)_{i_1 i_2} (T^a)_{i_3 i_4} = \frac{1}{2}\left(\delta_{i_1 i_4}\delta_{i_2 i_3} - \frac{1}{N}\delta_{i_1 i_2}\delta_{i_3 i_4}\right). \tag{2.48}$$

We also introduce the quadratic Casimir invariants of the fundamental and adjoint representations, denoted by C_F and C_A, respectively. They are defined through

$$\sum_a T^a T^a = C_F \mathbf{1}, \qquad f^{acd} f^{bcd} = C_A \delta^{ab}. \tag{2.49}$$

For $SU(N)$, with the normalisation $T_R = 1/2$ adopted above, these relations give

$$C_F = \frac{N^2 - 1}{2N}$$
$$C_A = N.$$

The combination $2C_F$ follows directly from the fundamental Casimir invariant above,

$$2C_F = N - \frac{1}{N}, \tag{2.50}$$

and will be used as a normalisation throughout the present work.

After performing the colour contractions and sums, the complete matrix-element contributions can be organised according to their distinct colour structures. For the $q\bar{q}$ antennae considered in this work, these give rise to leading- and subleading-colour contributions and, at loop level, contributions associated with closed quark loops. The corresponding colour-stripped antenna components are conventionally distinguished by their notation: the leading-colour component carries no additional accent, the subleading-colour component is denoted by a tilde, and the closed-quark-loop component by a hat. The explicit identification of these components from the relevant matrix elements is presented in the following subsection.

2.4.4 Antenna Construction

Each antenna set considered in Table 2.3 is now built explicitly in this subsection. As mentioned, every antenna is normalised by the Born-level squared amplitude given as,

$$|\mathcal{M}_2^0|^2 = 4(1-\epsilon)q^2. \tag{2.51}$$

The LO antenna then becomes,

$$A_2^0(1_q, 2_{\bar{q}}) = \frac{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2} = 1. \tag{2.52}$$

Advancing to the NLO tree-level antenna, the relevant amplitude $\mathcal{M}_3^0(1_q, 3_g, 2_{\bar{q}})$ is the sum of the two tree-level diagrams of Fig. 2.1, with the gluon emitted from either the quark or the antiquark line, as given in Eq. (2.10); both diagrams are therefore already contained within the same squared amplitude.

The antenna becomes,

$$A_3^0(1_q, 3_g, 2_{\bar{q}}) = \frac{|\mathcal{M}_3^0(1_q, 3_g, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \tag{2.53}$$

The NLO one-loop antenna is obtained from the interference between the tree-level and one-loop two-parton amplitudes, normalised to the corresponding Born-level squared amplitude,

$$A_2^1(1_q, 2_{\bar{q}}) = \frac{2\,\mathrm{Re}\Big(\mathcal{M}_2^{0\,\dagger}(1_q, 2_{\bar{q}})\mathcal{M}_2^1(1_q, 2_{\bar{q}})\Big)}{|\mathcal{M}_2^0(1_q, 2_{\bar{q}})|^2}. \tag{2.54}$$

The NNLO tree-level A-type antenna set is the first case considered here to contain more than one colour component. For the $qgg\bar{q}$ final state, the two gluons can appear in two different colour orderings along the $q\bar{q}$ colour line. The complete squared matrix element therefore contains contributions from both colour orderings, together with their interference. After performing the colour algebra and extracting the common colour normalisation introduced above, these contributions can be organised into leading- and subleading-colour structures. The resulting decomposition is

$$\begin{aligned} |M_{qgg\bar{q}}^0|^2 = \frac{1}{|\mathcal{M}_2^0|^2}\Bigg(& N\left(|\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}})|^2 + |\mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})|^2\right) \\ & - \frac{1}{N}\left|\mathcal{M}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) + \mathcal{M}_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})\right|^2\Bigg) \\ = N\Big(A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) & + A_4^0(1_q, 4_g, 3_g, 2_{\bar{q}})\Big) - \frac{1}{N}\tilde{A}_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \end{aligned} \tag{2.55}$$

This decomposition identifies the two components of the A_4^0 antenna set: the leading-colour antenna A_4^0 and the subleading-colour antenna $\tilde{A}_4^0$. For all antenna functions considered in this work, details of their explicit calculation using *AntCalc* are presented in Chapter 4, while the complete analytic results are collected in Appendix A.

The NNLO one-loop antenna set combines the features encountered separately in the NLO one-loop and NNLO tree-level antennae. It is obtained from the tree–one-loop interference of the three-parton matrix element and, before ultraviolet renormalisation, decomposes into three colour components,

$$\begin{aligned} |M_{qg\bar{q}}^1|^2 &= \frac{2\,\mathrm{Re}\Big(\mathcal{M}_3^{0\,\dagger}(1_q, 3_g, 2_{\bar{q}})\mathcal{M}_3^1(1_q, 3_g, 2_{\bar{q}})\Big)}{|\mathcal{M}_2^0|^2} \\ &= NA_{3,\mathrm{bare}}^1(1_q, 3_g, 2_{\bar{q}}) - \frac{1}{N}\tilde{A}_{3,\mathrm{bare}}^1(1_q, 3_g, 2_{\bar{q}}) + N_f\hat{A}_{3,\mathrm{bare}}^1(1_q, 3_g, 2_{\bar{q}}). \end{aligned} \tag{2.56}$$

The only diagram contributing to $\hat{A}_{3,\mathrm{bare}}^1$ is a massless-quark self-energy insertion on the external, on-shell gluon leg. Since this bubble carries the on-shell gluon's momentum p_3, and $p_3^2 = 0$, it is a scaleless integral and therefore vanishes identically in dimensional regularisation, such that $\hat{A}_{3,\mathrm{bare}}^1 \equiv 0$. The quark-loop ($N_f$) term that appears below originates entirely from the coupling-renormalisation counterterm.

The same interference can be written with the UV-renormalised antennae (Eq. (2.22)); the β_0 counterterm, which affects the leading-colour and N_f components, appears explicitly,

$$|M_{qg\bar{q}}^1|^2 = N\left(A_{3,\mathrm{ren}}^1 + \frac{11}{6\epsilon}A_3^0\right) - \frac{1}{N}\tilde{A}_{3,\mathrm{ren}}^1 + N_f\left(\hat{A}_{3,\mathrm{ren}}^1 - \frac{1}{3\epsilon}A_3^0\right). \tag{2.57}$$

The renormalised contribution still contains the one-loop correction of the underlying $q\bar{q}$ configuration;

subtracting it from the leading- and subleading-colour components gives

$$|M^1_{qg\bar{q}}|^2 = N\left(A^1_3 + A^0_3\left(\frac{11}{6\epsilon} + A^1_2\right)\right) - \frac{1}{N}\left(\tilde{A}^1_3 + A^1_2 A^0_3\right) + N_f\left(\hat{A}^1_3 - \frac{1}{3\epsilon}A^0_3\right) \tag{2.58}$$

The N_f coefficient vanishes, since $\hat{A}^1_{3,\text{bare}} \equiv 0$, while $\hat{A}^1_3 = A^0_3/(3\epsilon)$ is non-zero. This defines the final NNLO one-loop antenna set, consisting of A^1_3, $\tilde{A}^1_3$, and $\hat{A}^1_3$.

The NNLO two-loop antenna-set contains two distinct terms: the interference between the tree-level and two-loop amplitudes, and the squared one-loop amplitude. These need to be UV renormalised but no subtraction associated with a lower-multiplicity antenna is required in this case. We obtain,

$$\begin{aligned}
|M^2_{q\bar{q}}|^2 &= \frac{2\,\text{Re}\Big(\mathcal{M}^{0\,\dagger}_2(1_q,2_{\bar{q}})\mathcal{M}^2_2(1_q,2_{\bar{q}})\Big) + \mathcal{M}^{1\,\dagger}_2(1_q,2_{\bar{q}})\mathcal{M}^1_2(1_q,2_{\bar{q}})}{|\mathcal{M}^0_2(1_q,2_{\bar{q}})|^2} \\
&= NA^2_{2,\text{bare}}(1_q,2_{\bar{q}}) + \frac{1}{N}\tilde{A}^2_{2,\text{bare}}(1_q,2_{\bar{q}}) + N_f\hat{A}^2_{2,\text{bare}}(1_q,2_{\bar{q}}) + \left(N - \frac{1}{N}\right)\breve{A}^2_{2,\text{bare}}(1_q,2_{\bar{q}}) \\
&= N\left(A^2_2 + \frac{11}{6\epsilon}A^1_2\right) + \frac{1}{N}\tilde{A}^2_2 + N_f\left(\hat{A}^2_2 - \frac{1}{3\epsilon}A^1_2\right) + \left(N - \frac{1}{N}\right)\breve{A}^2_2
\end{aligned} \tag{2.59}$$

defining the four components that make up the set $\{A^2_2, \tilde{A}^2_2, \hat{A}^2_2, \breve{A}^2_2\}$. The $\breve{A}^2_2$ component arises from the self-interference of the one-loop amplitude and carries the single, undivided colour weight $\left(N - \frac{1}{N}\right)$: unlike the tree–two-loop interference, it is not further decomposed into separate leading- and subleading-colour pieces.

The remaining NNLO tree-level antenna functions considered in this work arise from four-quark final states in the decay of a colour-singlet current. For non-identical quark flavours, the corresponding matrix element gives rise to the B^0_4 antenna, whereas for identical flavours additional interference contributions are present, defining the C^0_4 antenna. A further contribution, conventionally denoted by $\hat{B}^0_4$, arises from configurations in which the electroweak current couples to the secondary quark line. This contribution is not required for the antenna functions considered in this thesis and will therefore not be discussed further. The complete four-quark matrix-element contribution can be decomposed as

$$\begin{aligned}
|M^0_{4q}|^2 &= \frac{1}{|\mathcal{M}^0_2|^2}\left(\sum_{q'}|\mathcal{M}^0_4(1_q,3_{q'},4_{\bar{q}'},2_{\bar{q}})|^2 + |\mathcal{M}^0_4(1_q,3_q,4_{\bar{q}},2_{\bar{q}})|^2\right) \\
&= \Bigg[N_f B^0_4(1_q,3_{q'},4_{\bar{q}'},2_{\bar{q}}) + N_{f,\gamma}\hat{B}^0_4(1_q,3_{q'},4_{\bar{q}'},2_{\bar{q}}) \\
&\qquad - \frac{1}{N}\Big(C^0_4(1_q,3_q,4_{\bar{q}},2_{\bar{q}}) + C^0_4(2_{\bar{q}},4_{\bar{q}},3_q,1_q)\Big)\Bigg]
\end{aligned} \tag{2.60}$$

where $N_{f,\gamma}$ is the charge weighted sum of the quark flavours, given as,

$$N_{f,\gamma} = \frac{\left(\sum_q e_q\right)^2}{\sum_q e^2_q}. \tag{2.61}$$

To isolate the B^0_4 and C^0_4 antennae from the complete four-quark matrix element, it is useful to first decompose the corresponding amplitude into its colour structures. Using the $SU(N)$ Fierz identity, the

tree-level amplitude for $\gamma^* \to q\bar{q}q'\bar{q}'$ can be written as,

$$\begin{aligned}\mathcal{M}_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) = ie_q g^2\, \delta_{q_1 q_2}\delta_{q_3 q_4}\left(\delta_{i_1 i_4}\delta_{i_3 i_2} - \frac{1}{N}\delta_{i_1 i_2}\delta_{i_3 i_4}\right)\mathcal{M}_{q\bar{q}\text{-}\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) \\ +(1 \leftrightarrow 3,\ 2 \leftrightarrow 4).\end{aligned} \tag{2.62}$$

Here, $\mathcal{M}_{q\bar{q}\text{-}\gamma}$ denotes the colour-stripped contribution in which the $q_1\bar{q}_2$ pair couples to the electroweak current. The B_4^0 antenna is obtained from the squared colour-stripped amplitude, whereas the C_4^0 antenna is defined by the interference between the two identical-quark assignments:

$$\begin{aligned}B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) &= \frac{|\mathcal{M}_{q\bar{q}\text{-}\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})|^2}{|\mathcal{M}_2^0|^2} \\ C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) &= -\frac{\mathrm{Re}\Big(\mathcal{M}_{q\bar{q}\text{-}\gamma}(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}})\mathcal{M}^{\dagger}_{q\bar{q}\text{-}\gamma}(1_q, 4_{\bar{q}}, 3_q, 2_{\bar{q}})\Big)}{|\mathcal{M}_2^0|^2}.\end{aligned} \tag{2.63}$$

2.4.5 Phase-Space Factorisation

The previous subsections focused on the construction, colour decomposition, and normalisation of the unintegrated antenna functions. In the subtraction formalism, these functions enter local counterterms that must subsequently be integrated analytically over the corresponding antenna phase space. At NLO, as introduced in Eq. (2.13), this gives the integrated subtraction term

$$\int_1 d\sigma^S \tag{2.64}$$

where $\int_1$ denotes the integration associated with one unresolved parton in the antenna phase space. This notation should not be interpreted as an integration over the momentum of the unresolved parton alone. For a three-parton antenna X^l_{ijk}, the antenna phase space contains the momenta of the unresolved parton j and of the two hard radiators i and k. The integration is made possible by the exact factorisation of the full $(m+1)$-particle phase space into the phase space of the reduced m-particle configuration and the antenna phase space, as already given in Eq. (2.31). The antenna contribution can therefore be integrated independently of the reduced m-parton hard process. Schematically, the antenna phase-space integration acts on the antenna function, while the reduced Born contribution remains factorised,

$$\int_1 d\sigma^S \sim \sum_{\text{antennae}} \left[\int d\Phi_{X_{ijk}}\, X^0_{ijk}\right] d\sigma^B. \tag{2.65}$$

At NNLO, the same construction applies to four-parton antennae X^0_{ijkl}, whose antenna phase space contains two unresolved partons.

For the final-final antennae considered here, the antenna phase-space integrals can be related directly to the standard three- and four-particle phase spaces, by setting $m = 2$ in Eqs. (2.31) and (2.38). Specialising the phase-space factorisation to a two-particle reduced configuration gives

$$\begin{aligned}d\Phi_3 &= d\Phi_2\, d\Phi_{X_{ijk}} \\ d\Phi_4 &= d\Phi_2\, d\Phi_{X_{ijkl}}.\end{aligned} \tag{2.66}$$

Since the antenna functions X^l_n depend only on invariants constructed from the antenna momenta $p_i, p_j, p_k, \ldots$, they are independent of the reduced two-particle phase space $d\Phi_2$. The latter can therefore

be integrated separately, giving the constant

$$\Phi_2 = \int d\Phi_2. \tag{2.67}$$

The antenna phase-space integral can therefore be expressed as [5],

$$\int d\Phi_{X_{ijkl\ldots}}\, X_n^l = \frac{1}{\Phi_2} \int d\Phi_n\, X_n^l. \tag{2.68}$$

Including the overall normalisation factor $C(\epsilon, k)$ introduced in Section 2.3, the integrated antenna $\mathcal{X}_n^l$ is then defined, generally, as

$$\mathcal{X}_n^l = C(\epsilon, k) \int d\Phi_{X_{ijkl\ldots}}\, X_n^l. \tag{2.69}$$

For the two-parton antennae A_2^1 and A_2^2, no unresolved real radiation is present and therefore no additional antenna phase space needs to be isolated. Their integrated counterparts are obtained directly from the corresponding two-particle phase-space integration, together with the loop integrations entering the virtual contributions [12]. Thus, while these antennae are also integrated over phase space, the phase-space factorisation of Eq. (2.68), relevant for antennae containing unresolved radiation, is not required. For A_2^2 in particular, the antenna depends only on q^2, so the normalised two-particle phase-space integration is trivial and Eq. (2.69) gives $\mathcal{X}_2^2 = C(\epsilon, 2)\, X_2^2$. This integration is nevertheless the same operation that relates the unintegrated and integrated antennae in every other case.

2.5 Phase-Space Integration and IBP Reduction

In the previous sections, antenna functions were introduced together with their role in the local subtraction of infrared singularities. Eq. (2.69) defined their integration over the unresolved phase space, leading from an unintegrated antenna X_n^l to its integrated counterpart $\mathcal{X}_n^l$. For loop-level antennae ($l \geq 1$), the loop momenta must additionally be integrated.

Both the unintegrated and integrated forms are required in a subtraction calculation, but they play different roles. The unintegrated antenna functions provide local counterterms for real-radiation matrix elements and are therefore required in numerical Monte Carlo (MC) phase-space integration. They must retain the dependence on the local kinematics of each phase-space point, which can be expressed in terms of Mandelstam invariants s_{ij}. This allows the subtraction term to reproduce the singular behaviour of the real-emission matrix element point by point in the unresolved limits. For loop-level antennae, the term *unintegrated* refers here to the unresolved phase space: the loop-momentum integration is first performed analytically, leaving an antenna function that depends on the external kinematic invariants and can be evaluated locally within the phase-space integration. The subsequent analytic integration over the unresolved phase space then produces the corresponding integrated antenna function, whose explicit infrared poles enter the cancellation against virtual contributions.

From Eq. (2.68), we have seen that through phase-space factorisation, the antennae are integrated over the antenna phase space Φ_X. If required, they are also integrated over their loop momenta ℓ_r, $r = 1, \ldots, l$. The propagator or Mandelstam-invariant denominators entering these integrals may generally be defined as,

$$D_j = \left(\sum_{r=1}^{l} c_{jr} \ell_r + r_j \right)^2 - m_j^2 + i0 \tag{2.70}$$

where r_j is built from momenta held fixed during the loop integration, including the mapped hard partons

and unresolved antenna momenta. For the massless, on-shell external partons considered throughout this work, r_j reduces to a combination of Mandelstam invariants s_{ij}.

An algebraic expansion of the integrand of an antenna function can generate multiple integrals with the same kinematic structure but different numerator and denominator powers, since any numerator built from scalar products can itself be absorbed into the same propagator basis $\{D_j\}$. These integrals may then be organised into integral families, which fix the denominators and integration variables, and may then be systematically reduced. These families are denoted as the complete collection [13],

$$\mathcal{F}[\{D_j\};\{\ell_r\},d\Phi_X] = \{I(a_1,...,a_N)\}, \quad a_j \in \mathbb{Z} \tag{2.71}$$

with,

$$I(\vec{a}) = \int \left[\prod_{r=1}^{l} d^d\ell_r\right] d\Phi_X \, \frac{1}{D_1^{a_1}...D_N^{a_N}}, \quad \vec{a} = (a_1,...,a_N). \tag{2.72}$$

The integrals within each family are not independent. Through integral reduction, they can be expressed in terms of a finite and substantially smaller set of master integrals (MI),

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a},\epsilon,\{s_{ij}\})M_k \tag{2.73}$$

where M_k are the master integrals and c_k are coefficients that depend on the integral indices, the dimensional regulator ϵ, and the relevant kinematic invariants. The original integration problem is therefore separated into two tasks: determining the reduction coefficients c_k and evaluating the much smaller set of master integrals M_k. Once these are known, every required integral in the family can be reconstructed from the same master-integral basis. In $\vec{a}$, positive a_j indices denote propagator powers, while non-positive indices account for absent denominators or irreducible scalar products in the numerator.

As stated previously, MC integrators require the unintegrated antenna functions in terms of the Mandelstam invariants s_{ij}, since they must evaluate the subtractions at a sampled physical external phase-space point p. The loop-level antenna required by an MC is ultimately evaluated as a function of the external invariants. Its calculation, however, involves tensor loop integrals with loop-momentum dependence, which must be reduced before the antenna can be evaluated at a sampled phase-space point. This reduction is performed through the Passarino–Veltman reduction mechanism, described in detail in Section B.2, which reduces the tensor loop integrals to linear combinations of scalar one-loop functions, usually denoted by A_0, B_0, ..., while keeping the phase-space variables and integration untouched, as in Ref. [14],

$$\int d^d\ell \, \frac{\ell^{\mu_1}...\ell^{\mu_R}}{\prod_j D_j^{a_j}} \xrightarrow{\text{Passarino–Veltman}} \sum_k c_k\big(\{s_{ij}\}\big)\,\{A_0,B_0,C_0,D_0,...\}_k. \tag{2.74}$$

Since the Passarino–Veltman procedure is a one-loop method, the two-loop integrals of the A_2^2 set are instead reduced by IBP identities (Subsection 2.5.1) to two-loop master integrals when the antenna is built.

To finally reach the integrated antenna functions $\mathcal{X}_n^l$, the remaining phase-space integration is performed through Integration By Parts (IBP) reduction, combined with the method of reverse unitarity to bring the on-shell phase-space constraints into the same algebraic framework as the loop integrals. Both methods, together with the resulting Laurent expansion of the master integrals, are described in full in Subsection 2.5.1.

2.5.1 IBP Identities and the Laporta Algorithm

2.5.1.1 IBP Identities

The integral family has already been defined in Eq. (2.72). This definition allows for the factorisation of the integral into a loop and a phase-space integral term. That factorisation results in,

$$I(\vec{a}) = \int d\Phi_X \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}), \quad \mathcal{I}(\vec{a}; \{D_1, ..., D_N\}) = \int \prod_{r=1}^{l} d^d \ell_r \frac{1}{D_1^{a_1} ... D_N^{a_N}}. \tag{2.75}$$

The overall normalisation factors are irrelevant to the identities and will, therefore, be suppressed throughout the following explanation. The denominators D_j were already defined generally in Eq. (2.70). Take one loop momentum ℓ_s, for $s = 1, ..., l$, and one vector v^μ from the available loop and external momenta. Dimensional regularisation allows us to set a total derivative to zero inside $\mathcal{I}$, as [13, 15],

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \frac{\partial}{\partial \ell_s^\mu} \left[v^\mu \frac{1}{D_1^{a_1} ... D_N^{a_N}} \right] \tag{2.76}$$

In dimensional regularisation, analytic continuation justifies the vanishing of the surface term at infinity, leaving an exact relation among regulated integrals[1].

Expanding the derivative gives,

$$0 = \int \prod_{r=1}^{l} d^d \ell_r \left[\frac{\partial_{\ell_s} \cdot v}{\prod_j D_j^{a_j}} - \sum_{j=1}^{N} a_j \frac{v^\mu \partial_{\ell_s^\mu} D_j}{D_1^{a_1} ... D_j^{a_j+1} ... D_N^{a_N}} \right]. \tag{2.77}$$

For these quadratic denominators D_j, differentiation with respect to ℓ_s^μ is linear in momenta,

$$\partial_{\ell_s^\mu} D_j = 2 c_{js} \left(\sum_r c_{jr} \ell_r + r_j \right)_\mu \tag{2.78}$$

and therefore, $v^\mu \partial_{\ell_s^\mu} D_j$ is a scalar product. Scalar products may be expressed in terms of chosen denominators and irreducible scalar products. The latter are represented by negative family indices a_j, while zero indices denote absent denominators. We are then able to write schematically,

$$\sum_{\vec{b}} \rho_{\vec{b}}(\vec{a}, \epsilon, \{s_{ij}\}) \mathcal{I}(\vec{b}) = 0 \tag{2.79}$$

where each vector $\vec{b}$ labels a family member obtained from $\vec{a}$ through index shifts in one or more denominator powers. $\rho_{\vec{b}}$ is the corresponding coefficient in the IBP relation.

An example of the application of this method is written in Section B.3.

2.5.1.2 Reverse Unitarity and Cut Integrals

In the previous subsubsection, we have derived the IBP identities relative to $\mathcal{I}$. As described in Eq. (2.75), the entire integrated antenna is not merely $\mathcal{I}$, but rather,

$$I(\vec{a}) = \int d\Phi_X \mathcal{I}(\vec{a}). \tag{2.80}$$

The unresolved phase-space measure is not originally an ordinary loop integral, however. Instead, it has on-shell constraints, which have to be considered and transformed before $I(\vec{a})$ is reducible via the

[1] This method is discussed in more detail in Subsection 4.1.1.

present method.

As described in Subsection 2.4.5, phase-space factorisation isolates the antenna measure through $d\Phi_{m+r} = d\Phi_m d\Phi_X$, or in the two-hard-radiator final-final case we are considering throughout the present work, $d\Phi_X = d\Phi_n/d\Phi_2$, for n final-state particles. That n-particle phase space is given generally as,

$$d\Phi_n(q : p_1, ..., p_n) = (2\pi)^d \delta^{(d)}\left(q - \sum_{i=1}^{n} p_i\right) \prod_{i=1}^{n} \frac{d^d p_i}{(2\pi)^{d-1}} \delta^+\left(p_i^2 - m_i^2\right) \tag{2.81}$$

where the $\delta^{(d)}$-function enforces overall momentum conservation, and each δ^+-function imposes the mass-shell condition for the i-th final-state particle, with the superscript + selecting its positive-energy solution. This may be written as,

$$\delta^+\left(p_i^2 - m_i^2\right) = \theta\left(p_i^0\right)\, \delta\left(p_i^2 - m_i^2\right) \tag{2.82}$$

This δ^+-function may be reexpressed as a discontinuity as follows [13, 16],

$$2\pi i\, \delta^+\left(p_i^2 - m_i^2\right) = \left[\frac{1}{p_i^2 - m_i^2 - i0} - \frac{1}{p_i^2 - m_i^2 + i0}\right]_+ \tag{2.83}$$

where the $\pm i0$ terms specify the propagator prescriptions. The subscript + represents the positive-energy cut selection.

These objects are called *cut propagators*, because they come about from cutting a propagator line, thus putting the represented particle on-shell, as a real final-state particle. Algebraically, it also allows phase-space constraints to be written in the same denominator language as virtual propagators. A cut propagator associated with a required final-state particle must be retained during the reduction and sectors in which such a cut is absent do not represent the physical phase-space integral and thus, vanish [13]. Using the discontinuity relation to refactor the δ-functions, we are able to rewrite the integrated antenna expression schematically as,

$$\mathcal{X}_n^l \sim \int \left[\prod_{r=1}^{l} d^d \ell_r\right] \left[\prod_u d^d p_u\right] \frac{N_s(\ell_r, p_u; \tilde{p}_{\text{hard}})}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}} \tag{2.84}$$

where $D_{v,\alpha}$ are ordinary virtual propagators, $D_{c,\beta}$ are cut propagators, $\tilde{p}_{\text{hard}}$ is the hard-radiator combined momenta and p_u are integrated unresolved momenta.

After rewriting the integral in this way, the total derivative introduced in the previous Subsubsection 2.5.1.1 may now be taken with respect to any loop momentum ℓ_r^μ or integrated unresolved momentum p_u^μ. This process produces relations involving both virtual and cut denominators. This then allows us to reduce loop and phase-space integrals simultaneously using IBP reduction [13, 16].

An example of the method used to rewrite $d\Phi_3$ in terms of cut propagators is available in Section B.4.

2.5.1.3 Laporta Algorithm

At NLO and NNLO these reduction systems become very complex and their solution ever more time-intensive. The Laporta algorithm is a systematic elimination procedure that uses IBP relations to express more complicated integrals in terms of a smaller set of unreduced ones (master integrals) [17].

The algorithm first classifies the family members into sectors. These sectors are defined as,

$$\theta_j \equiv \theta(a_j) = \begin{cases} 1, & a_j > 0, \\ 0, & a_j \leq 0, \end{cases} \qquad \vec{\theta} = (\theta_1, ..., \theta_N). \tag{2.85}$$

After fixing an integral family and identifying its admissible sectors, the Laporta algorithm assigns an ordering to the resulting integrals according to their complexity. This hierarchy is based on the number of propagators present t, the excess denominator power r, and the numerator complexity encoded through negative indices s[2]. These quantities are defined using each integral's a-index, taken from the family's $\vec{a}$ vector, as previously defined. They are, generally,

$$t = \sum_{j=1}^{N} \theta(a_j), \quad r = \sum_{j=1}^{N} \max(a_j - 1, 0), \quad s = \sum_{j=1}^{N} \max(-a_j, 0). \tag{2.86}$$

Here, $\theta(a_j)$ records whether denominator D_j is present. The hierarchy is defined by comparing t first, then r and finally s. The algorithm uses this chosen hierarchy to select, within each IBP relation, the most complex integral to eliminate first. An example of the application of the Laporta algorithm is given in Section B.5.

After determining the integral families and sectors present, we apply the IBP identities to the seed integrals. The seeds form a finite set of integrals chosen from each relevant sector, whose IBP identities must be sufficiently numerous to reduce every target integral and every non-zero subtopology reached along the way.

After generating IBP identities for the selected seed integrals, any relation containing these representatives is first solved for its highest-ranked member, $I(1, 1, 1)$ in the example of Section B.5, and the resulting expression is substituted recursively into the remaining system [17].

IBP reduction may produce lower sectors, but may also generate new integrals from the initially relevant sectors. Lastly, some sectors may be cancelled due to symmetry, if one denominator is shifted to another under loop-momentum shift or a relabelling of external momenta. If two sectors become equivalent, one is discarded from the independent reduction and the other is retained as its representative. In short, a sector is deemed relevant if it is an original target sector, or a lower sector reached from another via IBP reduction. The relevant sector must also be non-scaleless in a chosen kinematics, not redundant under a family symmetry and, for a cut family, it must retain all required cut denominators throughout.

It is important to note that cut families add a physical restriction: if an IBP reduction step cancels a cut propagator, the sector it produces is missing one of the required on-shell final-state particles and it no longer represents the intended phase-space. If that happens, the integral is set to zero [13].

The reduction is complete when the remaining non-zero family members cannot be eliminated by the available identities. These survivors form a master-integral basis.

2.5.2 Master Integrals

The IBP reduction method reduces the integral to a linear combination of master integrals as,

$$I(\vec{a}) = \sum_{k=1}^{N_{\text{MI}}} c_k(\vec{a}, \epsilon, \{s_{ij}\}) M_k \tag{2.87}$$

where c_k are the reduction coefficients and represent what comes from the reduction algebra and M_k are the MI. Our remaining task is to evaluate the finite set M_k [17].

Master integrals are not uniquely defined: an integral is a master only relative to the chosen family, kinematics, IBP system, and basis. This means that a different basis with different masters may be used to reduce the same integrals and span the exact same integral space [17].

[2]The present t, r and s metrics are not to be taken as unique Laporta orderings, as these vary by implementation. Instead, they are illustrative objects.

Master integrals are scalar integrals with a fixed denominator/cut structure. They are generically represented as,

$$M_k = \int \left[\prod_r d^d \ell_r\right] \left[\prod_u d^d p_u\right] \frac{1}{\prod_\alpha D_{v,\alpha}^{a_\alpha} \prod_\beta D_{c,\beta}^{a_\beta}}. \tag{2.88}$$

For integrated antennae, the cut denominators in these master integrals encode the real unresolved phase space. Hence, a master may be a pure phase-space volume, a phase-space integral with invariant denominators, or a loop-plus-phase-space cut integral. The MI are also not necessarily finite; in fact, in dimensional regularisation they commonly contain $1/\epsilon$ poles [13].

For a massless fully inclusive antenna system, q^2 is often the only independent kinematic scale once the dimensional-regularisation normalisation factors have been separated. Through dimensional analysis, the MI are strongly constrained to take the form,

$$M_k(q^2, \epsilon) = \left(q^2\right)^{\lambda_k + \kappa_k \epsilon} f_k(\epsilon), \quad f_k(\epsilon) \sim \frac{\Gamma^m(1 - a\epsilon)}{\Gamma^n(1 - b\epsilon)} \tag{2.89}$$

where λ_k is fixed by mass dimension and f_k is dimensionless, depending only on ϵ [13]. Here, Γ denotes the Euler Gamma function, the analytic continuation of the factorial to complex arguments, defined for $\mathrm{Re}(z) > 0$ by

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt, \tag{2.90}$$

and extended to the rest of the complex plane, away from its poles at $z = 0, -1, -2, \ldots$, by analytic continuation.

When evaluating master integrals, different types often call for different approaches. Simple one-scale phase-space masters often integrate directly after the phase-space is parametrised. Angular and energy-fraction integrations typically reduce to Beta functions. When a master depends on a non-trivial kinematic variable, such as $x = m^2/q^2$, the most common approach is to differentiate the master with respect to that variable. The derivative produces integrals in the same family, which IBP is then able to reduce to the master basis, as in Ref. [18],

$$\frac{d}{dx}\vec{M}(x, \epsilon) = A(x, \epsilon)\vec{M}(x, \epsilon) \tag{2.91}$$

where $\vec{M} = (M_1, ..., M_{N_{\mathrm{MI}}})^{\mathrm{T}}$ is the vector of masters and A is a matrix of coefficient functions. Boundary conditions can then be used to determine the solution. The MI used throughout this work and the methods used to compute them are discussed in further detail in Appendix C.

Once the MI are evaluated, we expand them around $\epsilon = 0$, as a Laurent series [13],

$$M_k = \sum_{n=n_{\min}}^{\infty} m_{k,n} \epsilon^n \tag{2.92}$$

to expose their pole and finite structure. The lowest power $n_{\min}$ can be, and often is, negative, resulting in an expansion similar to,

$$M_k = \frac{m_{k,-2}}{\epsilon^2} + \frac{m_{k,-1}}{\epsilon} + m_{k,0} + O(\epsilon). \tag{2.93}$$

The required expansion depth is determined by the reduction coefficient. If $c_k \sim \epsilon^{-p}$, the M_k must be expanded through to ϵ^p to obtain the finite contribution to $c_k M_k$.

The reduction techniques introduced in this chapter provide the tools required for the analytic integration of the antenna functions considered here. Implemented in the *AntCalc* workflow (Chapter 3), they are applied in Chapter 4 to obtain the integrated antennae as series in ϵ.

Chapter 3

Package Framework

A central contribution of this thesis is the development of *AntCalc*[1], a *Wolfram Language* package created from scratch as part of this thesis work to automate the symbolic construction and integration of QCD antenna functions. The package provides direct access to both unintegrated and integrated antenna functions and implements the computational framework described in the previous chapter. The results presented in this thesis were produced with *AntCalc*, version 0.3.2β6, corresponding to commit `5ae114d`.

AntCalc makes use of *FeynCalc* [19, 20, 21, 22] (with *FeynArts* [3] for diagram generation) for symbolic amplitude manipulation and interfaces with *LiteRed2* [23, 24] for IBP reduction. Both are external dependencies that must be installed by the user prior to loading *AntCalc*; the package's own installation procedure only installs the *AntCalc* paclet itself. Its internal architecture is organised as a profile-driven route system: route-specific information is centralised in profiles, which guide the selection and configuration of the appropriate build, reduction, and integration workflows. This establishes a common contract between the package components, reducing the amount of route-specific logic that must be duplicated when existing functionality is extended or new routes are introduced.

The package supports massless and massive antenna routes. All routes were validated against the established literature (the trivial Born-normalisation route A_2^0 excepted), with all massless routes being validated against [5] and [25], the latter being the direct reference for the two-loop antenna set and also providing channel-level results used as cross-checks of assembled sums. The single massive route results were validated against [26]. All routes are public, and each produces stable and reproducible results notwithstanding the current beta version. The release-acceptance procedure that ensures this is described in Section 3.7.

AntCalc's inner working structure can be described using the diagram in Fig. 3.1.

3.1 Design Philosophy and Scope

AntCalc was built to combine the symbolic amplitude capabilities of the *FeynCalc* environment with the IBP-reduction capabilities of *LiteRed2*, adapting and orchestrating the parts of those systems required for the construction and integration of QCD antenna functions.

The central design objective of the package is to return antenna functions in the conventional forms required for higher-order QCD calculations using the antenna subtraction method. Though it supports a wide variety of options to inspect and configure the backend operations, the default outputs of its

[1] Source code available at `https://github.com/HenFar/AntCalc/tree/release/0.3.x`.

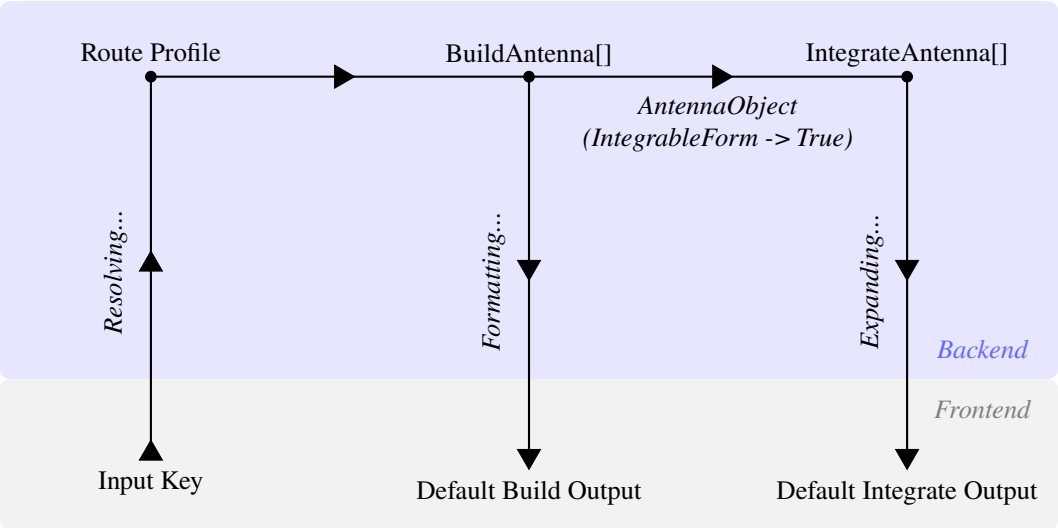


Figure 3.1: Profile-driven architecture of *AntCalc*. Route profiles configure the build and integration workflows, while the build-side route context carries the evolving calculation data. The `AntennaObject` provides the integration-ready handoff between the two stages.

two central functions are designed to reflect this objective. For example, the unintegrated one-loop three-parton antennae $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$ returned by `BuildAntenna[...]` are expressed after Passarino–Veltman reduction, in terms of scalar one-loop functions and the external kinematic invariants, following the conventional representation used in the literature [5]. By default, `IntegrateAntenna[...]` returns the integrated antennae as Laurent series in ϵ, with the truncation order chosen to match the conventions used in the antenna literature: the NLO antennae $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$ are expanded through $O(\epsilon^2)$, while the NNLO antennae $\mathcal{A}_4^0$, $\mathcal{A}_3^1$, $\mathcal{A}_2^2$, $\mathcal{B}_4^0$, and $\mathcal{C}_4^0$ are returned through $O(\epsilon^0)$ [5, 25]. Furthermore, the massive NLO tree-level antenna $\mathcal{A}_3^{0,(m_q)}$ is given up to $O(\epsilon)$ [26].

Its structure was also designed to be modular and extensible. Each of the two central functions (`BuildAntenna` and `IntegrateAntenna`) is implemented as an encapsulated public module that communicates with the rest of the codebase using the profile system: the route profile selects and configures the appropriate workflows and the route context carries the evolving results and diagnostics between stages. Using route profiles and route contexts, as described in Section 3.2, facilitates the addition of further modules while reducing duplicated logic and direct dependencies. New modules do not need function-specific couplings to previously implemented functions: they interoperate through the shared profile and route-context interface, being selected and configured by route profiles, accepting inputs from and delivering outputs to the route contexts.

Currently, *AntCalc* is organised by mass scale, into massless and massive routes. All massless quark-antiquark final-final antenna functions through NNLO have been made public. Alongside these, a single massive route, A_3^0, is likewise public; its construction is validated in the soft, collinear, and massless limits (Appendix B.8). This is the first massive antenna of what is expected to be a complete expansion of *AntCalc*'s capabilities into the massive regime for quark-antiquark final-final antennae. The antenna functions currently implemented are summarised in Table 3.1.

3.2 Profile-Driven Architecture

As described in Section 3.1, *AntCalc* relies on a profile-driven routing system. This system maps the route key to its route-specific profile and workflow.

Table 3.1: Overview of the antenna functions currently implemented in *AntCalc*, by perturbative order and validation source.

Perturbative Order	Antenna	Validated Against
LO	A_2^0	Born normalisation
NLO	A_2^1, A_3^0	[5, 25]
NNLO	A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$, A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0	[5, 25]
NNLO	A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$	[25] ($\check{A}_2^2$ also [27])
NLO, massive	$A_3^{0,(m_q)}$	[26]

A public antenna route key is a three-element list which describes a canonical route whose execution may be refined by options. The route key takes the form,

$$\{\text{Antenna Type}, \text{Final-State Multiplicity}, \text{Loop Order}\}. \tag{3.1}$$

As an example, the route key for the one-loop three-parton antenna set associated with the process-level contribution $|M^1_{qg\bar{q}}|^2$ is $\{A, 3, 1\}$. This key represents the complete set of colour contributions, comprising the leading-colour antenna A_3^1, the subleading-colour antenna $\tilde{A}_3^1$, and the quark-loop contribution $\hat{A}_3^1$. The default public output is therefore the ordered list $\{A_3^1, \tilde{A}_3^1, \hat{A}_3^1\}$.

The package links this key to the profile it identifies and follows that profile's directions on how the route should be constructed and/or integrated. The data produced while the route executes is also stored in the route context association. The profile system consists of a main profile and two further profiles: one build-side, handling reduction, and one integration-side. The principal build-side profile is `AntennaProfile[key]`, which specifies information such as antenna type, multiplicity and loop order, colour normalisation, extraction rules, components, etc. `AntennaReductionProfile[key]` is separate, but still build-side. It specifies the Passarino–Veltman-reduction preferences for the appropriate routes. In this case, up to NNLO, these are the $\{A, 2, 1\}$ and $\{A, 3, 1\}$ loop routes. `AntennaIntegrationProfile[k` is the integration-side profile, which specifies its side's backend, IBP basis families, expansion depth, etc. The specifics of each stage's profiles and options will be discussed in their respective Sections 3.3 and 3.5.

While profiles select and configure computation paths, the workflows themselves perform the calculations. The route dispatcher and workflow code consult the profiles to select and configure each stage. As a representative example, which will be further explained and explored in the two upcoming sections, let us consider the call `BuildAntenna[A,2,1]`. Its full route profile is,

```
AntennaProfile[{A, 2, 1}] :=
    <|
        "Key" -> {A, 2, 1},
        "Name" -> "A21",
        "AntennaType" -> A,
        "NumFinalParticles" -> 2,
        "LoopOrder" -> 1,
        "TreeAmplitude" -> AntennaAmplitude[{A, 2, 0}],
        "BornInterference" -> BornInterference[],
        "Extraction" -> "LoopScalar",
        "ColourNorm" -> colourNorm,
        "ExcludedParticles" ->
            {S[_], V[1], V[2], V[3], U[1], U[2], U[3], U[4]},
```

```
        "DropSumOver" -> True,
        "ReductionProfile" -> AntennaReductionProfile[{A, 2, 1}],
        "ConventionProfile" -> BuildAntennaConventionProfile[{A, 2, 1}]
    |>
```

In this profile, the key, name, antenna type, number of final particles (multiplicity), and loop order have been discussed before. The settings used in this route are explained in Table 3.2.

Table 3.2: Overview of the settings for the A_2^1 route used as a representative example of the route system used in *AntCalc*.

Item	Use
`TreeAmplitude`	Selects the tree-level amplitude. In this case, it is $\mathcal{M}_2^0$.
`BornInterference`	Generates the general normalisation as $\left\|\mathcal{M}_2^0\right\|^2$.
`Extraction`	Selects how the components are extracted.
`ColourNorm`	Colour normalisation: $N - 1/N$.
`ExcludedParticles`	Restricts *FeynArts* internal particle insertions.
`DropSumOver`	Replaces every `SumOver[]` statement with 1.
`ReductionProfile`	Points to the reduction profile.
`ConventionProfile`	Points to the normalisation-convention ruleset profile.

3.3 The Build Stage

The build stage constructs the unintegrated antenna functions from their corresponding matrix elements. Its main public interface is `BuildAntenna[type, multiplicity, loop order]`. The workflow begins with a route key specified through a public call, for example `BuildAntenna[A, 3, 1]`. From this key, *AntCalc* identifies the appropriate construction workflow and the ingredients required for the corresponding antenna set. The relevant amplitudes are then constructed and combined according to the route definition, after which the resulting expressions are processed into the required unintegrated antenna functions. Internally, the intermediate results are stored in an association containing the different components of the calculation. The final step reformats this internal representation into the antenna expression, or ordered set of antenna components, returned to the user. The build workflow can therefore be summarised as follows,

```
BuildAntenna[type, multiplicity, loop order]
    -> route key;
    -> loop-order dispatcher;
    -> route-specific build workflow;
```

```
    -> build-data association / route context;
    -> public-output boundary;
    -> output: expression, component list, AntennaObject, diagnostics,
       or run record.
```

The loop-order dispatcher is a central element of this workflow. Using the route key, it directs each antenna to the appropriate computational path, since different loop orders require different computational treatments, most notably the Passarino–Veltman tensor reduction required for the one-loop antennae and the IBP reduction required for the two-loop antennae.

Functionally, the route context for this stage holds the profile, the generated amplitude, the computed and normalised interference, the extracted components, and the diagnostics; some routes retain additional route-specific intermediates.

A diagram in the spirit of the one in Fig. 3.1, but only relative to the build stage is shown below, in Fig. 3.2.

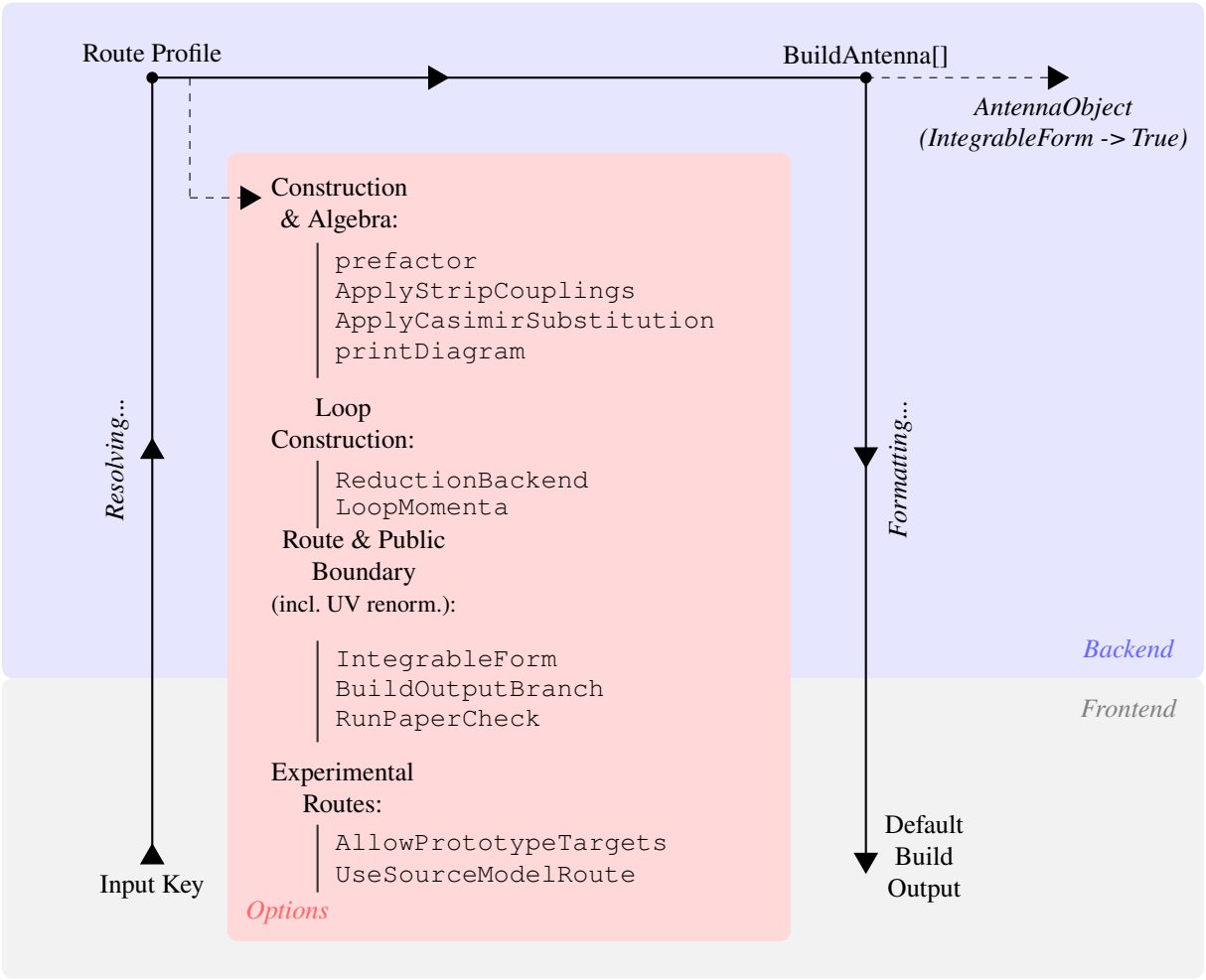


Figure 3.2: Profile-driven architecture of *AntCalc*'s Build stage. The principal build-specific options to the `BuildAntenna` function are displayed on the red card.

For loop routes, this public-output boundary is also where ultraviolet renormalisation is applied. Renormalisation is not a discrete internal stage of the build workflow; instead, it is applied to the unintegrated one- and two-loop antennae precisely at the point where they cross from the internal (backend) representation into the public antenna expression returned to the user, using the coupling

rescaling of Eq. (2.22). This is why no separate renormalisation step appears explicitly in Fig. 3.2: it coincides with the "Route & Public Boundary" crossing already shown there.

3.3.1 Tree-Level Routes

As mentioned, the workflow handles distinct loop-order routes differently, though the differences only appear after the dispatcher is called. Tree-level routes, such as those for the A_3^0 antenna, start by retrieving the profile and computing the relevant process amplitude. It uses the profile's `Production` field to select the physical construction path, and the `Extraction` and `ColourNorm` fields to select the extraction mode and the colour normalisation used for each antenna. The colour normalisation used by default has been described previously in Eq. (2.50), and the `Extraction` mode also introduces the Born normalisation. The routes selected in the `Production` and `Extraction` fields depend on the methodology required to compute the antenna functions and retrieve the components, if required. The public tree-level antenna functions are constructed using different production and extraction modes, as described in Table 3.3.

Table 3.3: Overview of the tree-level production and extraction modes.

Production Mode	Extraction Mode	Current Route
`SelfInterference`	`BornScalar`	A_2^0
`SelfInterference`	`TreeColourCoefficients`	A_3^0
`ColourOrderedAntenna`	`TreeColourCoefficients`	A_4^0, $\tilde{A}_4^0$
`SectorSelfInterference`	`ColourNormScalar`	B_4^0
`SectorSymmetrisedInterference`	`MinusTwoColourNormOverSUNN`	C_4^0
`SelfInterference`	`TreeColourCoefficients`	$A_3^{0,(m_q)}$

These production paths all lead to the extraction of the required antenna functions, but the way in which they do it changes with the antenna's multiplicity. Each production and extraction mode is *AntCalc*'s operational implementation of the construction derived, in mathematical terms, for the corresponding *Current Route* entry in Subsection 2.4.4. On the extraction side, A_2^0 is set to 1; the A_4^0-set antennae are extracted from the usual $NX_n^l - \frac{1}{N}\tilde{X}_n^l$ colour structure, while A_3^0 has a single colour structure; and the B_4^0 and C_4^0 antennae are extracted from the more involved inner structure of the $|M_{4q}^0|^2$ process-level contribution, requiring a correspondingly more complex extraction process.

3.3.2 One-Loop Routes

One-loop routes are dispatched to `BuildLoopAntennaData` where the primary backend reduction method is selected through the nested `ReductionProfile`, which by default is Passarino–Veltman reduction, as described in Section B.2.

The production method consists of generating the tree-level and one-loop amplitudes, using `TreeAmplitude` and `MAmpOneLoop`, respectively, and then forming the tree-loop interference. The interference is colour- and one-loop-normalised, as described in Eqs. (2.20) and (2.50). This normalised interference is then reduced and its components extracted. The production and extraction modes implemented for the supported one-loop routes are summarised in Table 3.4, below.

The two one-loop routes make use of a hybrid production mode that combines the `TreeAmplitude` and `MAmpOneLoop` objects. As shown in Table 3.2, the former is $\mathcal{M}_n^0$, while the latter is its one-loop

Table 3.4: Overview of the one-loop production and extraction modes.

Production Mode	Extraction Mode	Current Route
`TreeAmplitude + MAmpOneLoop`	`LoopScalar`	A_2^1
`TreeAmplitude + MAmpOneLoop`	`LoopColourCoefficients`	$A_3^1, \tilde{A}_3^1, \hat{A}_3^1$

counterparts $\mathcal{M}_n^1$. These two amplitudes are used as described in Subsection 2.4.4. Their extraction modes are the one-loop analogous to the methods used for A_2^0 and A_3^0, as discussed in Subsection 3.3.1.

3.3.3 Two-Loop Routes

Loop-order-two routes are dispatched to `BuildTwoLoopAntennaData`. Within the massless $q\bar{q}$ final-final antennae considered up to NNLO, the only two-loop antenna set is A_2^2. Unlike the one-loop routes, the two-loop source is not reduced with Passarino–Veltman reduction. The two loop momenta ℓ_1 and ℓ_2 are removed by IBP reduction with *LiteRed2*: in a direct `BuildAntenna` call this is part of the build, while in the modular `BuildAndIntegrateAntenna` path used in Chapter 4 it is performed by `IntegrateAntenna`. In either case the known analytic expressions of the resulting master integrals are substituted, so that the public build result is real and expressed in terms of ϵ alone, with the scale set to $\mu^2 = q^2$. Passarino–Veltman reduction enters this route only through the unintegrated A_2^1 antenna, which is attached to the leading-colour and quark-loop components as their ultraviolet counterterm at the public build boundary described in Section 3.3. The four-component set is also not generated like the A_4^0 and A_3^1 sets were; rather, its four components come from two distinct source contributions, as discussed in Subsection 2.4.4. Their production and extraction modes are similar and handled as a bundle inside *AntCalc*, but, as shown in Table 3.5, their sources differ,

Table 3.5: Overview of the two-loop contributions and extraction modes.

Production Mode	Extraction Mode	Current Route
`TwoLoopInterf`	`TwoLoopTTermComponents`	$A_2^2, \tilde{A}_2^2, \hat{A}_2^2$
`OneLoopSelf`	`TwoLoopTTermComponents`	$\check{A}_2^2$

One of the ways in which this route is particular inside *AntCalc*'s architecture is illustrated by their different production modes but the same extraction mode. This route may be schematically represented as shown in Fig. 3.3, below.

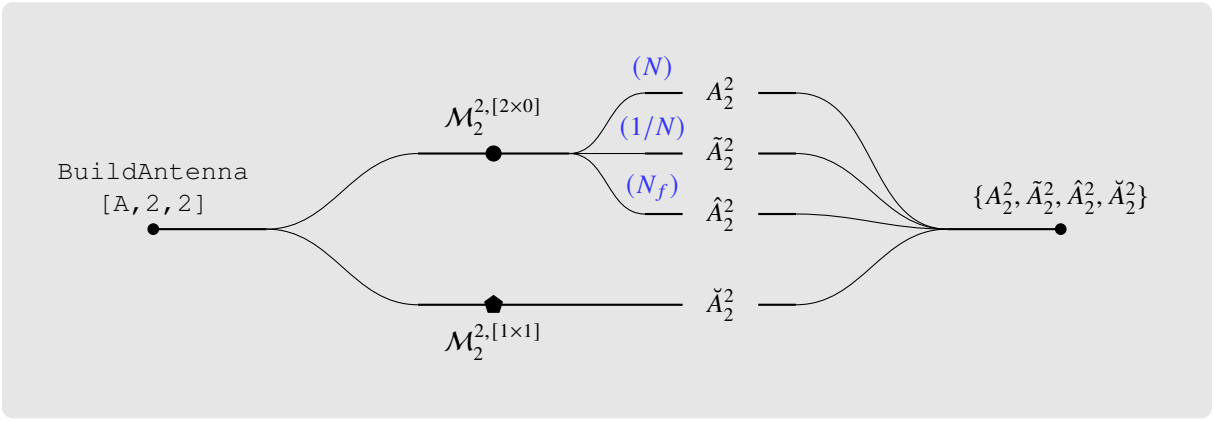


Figure 3.3: Diagram showing the build-side workflow leading to the generation of the A_2^2 antenna set. The two subsets are separated by the interferences used in their generation.

The package reads the profile and generates the two necessary amplitudes, the tree–two-loop interference and the one-loop self-interference, $\mathcal{M}_2^{2,[2\times0]}$ and $\mathcal{M}_2^{2,[1\times1]}$. The former amplitude generates three antenna-set components, while the latter produces only one, but all four are extracted using the same usual method used for tree-level and one-loop antenna functions, applied to two loops.

3.4 From the Build Stage to Integration

The build stage described above produces the unintegrated antenna functions in terms of the external kinematic invariants s_{ij}. This is the natural public representation of the antenna and is suitable for its use as a local subtraction term in a numerical phase-space integration, as discussed in Chapter 2. A standard `BuildAntenna[...]` call therefore returns either a single antenna expression or, where several antenna-set components are present, an ordered list made up of those components.

For the analytic calculation of the integrated antennae, however, the final expression in terms of Mandelstam invariants is not by itself sufficient. The integration stage requires additional information from the construction of the antenna, including the route definition and the integral structures needed for the subsequent reduction and phase-space integration. *AntCalc* therefore provides the option `IntegrableForm -> True`. With this option, `BuildAntenna[...]` returns an `AntennaObject` that retains both the constructed antenna and the additional information required by `IntegrateAntenna[...]`. This object provides the interface between the build and integration stages.

At this point, *AntCalc* can therefore provide the unintegrated antennae in the kinematic form required for local subtraction. The next stage addresses the complementary part of the antenna subtraction procedure: their analytic integration over the unresolved phase space in dimensional regularisation. These integrated antennae are required to combine the subtraction terms with the virtual contributions and achieve the cancellation of explicit infrared poles. The implementation of this integration stage is described in the following section.

3.5 The Integration Stage

The integration stage performs the analytic integration of the antenna functions constructed in the build stage over the corresponding unresolved phase space in $d = 4 - 2\epsilon$ dimensions, as in Eq. (2.69). The integrated antennae are denoted with calligraphic letters $\mathcal{X}_n^l$.

Within *AntCalc*, this stage is initiated by passing the `AntennaObject` produced by `BuildAntenna[..., IntegrableForm -> True]` to `IntegrateAntenna[...]`. A diagram representing this architecture and how it splits from an unchanged `BuildAntenna[...]` call is shown in Figure 3.4.

The route key stored in this object determines the appropriate integration workflow, including the required reduction procedure, master integrals, and normalisation conventions. After these steps have been applied, `IntegrateAntenna[...]` returns the integrated antenna, or the corresponding set of antenna components, expanded as a Laurent series in ϵ. The integration workflow is structured as follows,

```
IntegrateAntenna[AntennaObject]
    -> recover route key and resolve profile;
    -> choose integration backend;
    -> complete IBP/Passarino-Veltman reduction;
```

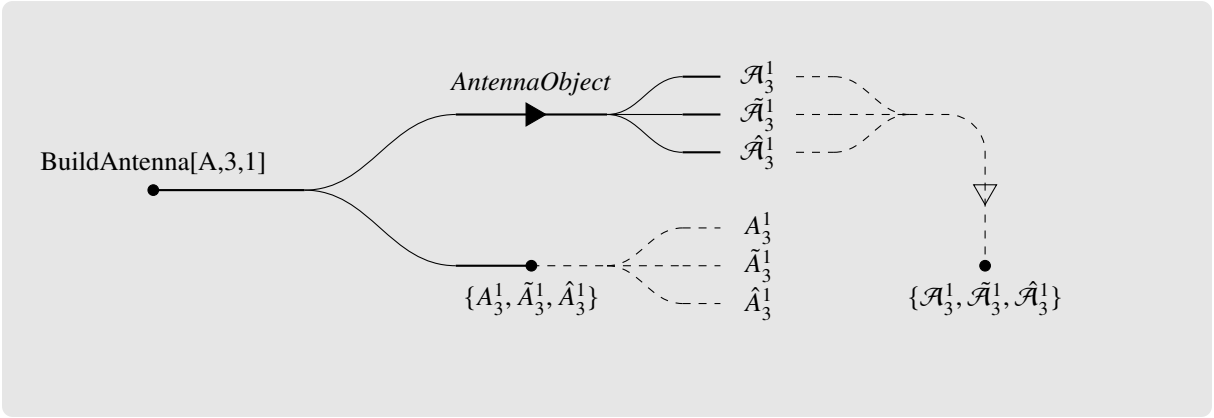


Figure 3.4: Diagram of the build-only and build-and-integrate routes: the combined route is shown above, the build-only route below. The dashed lines mark the options to separate the result into single components (build-only route) or merge them into one ordered list (combined route). The A_3^1 antenna set is used as an example.

```
    -> substitute master integrals when applicable;
    -> extract public integrated antennae;
    -> output: selected component, list, diagnostics, run record, or
       master-combination view.
```

As with `BuildAntenna`, `IntegrateAntenna` requires a profile, since it also encodes the relevant routing information for the reduction and integration process.

Unlike the build stage, whose intermediate data are collected in a single route context, the integration stage does not hold its route state in one context object; it is spread across several objects, accessed as required: the key and profile, the input (unintegrated) antenna and selected component, the IBP bases and master integrals, the series result, and the diagnostics.

An integration-stage-specific diagram in the spirit of the one in Fig. 3.1 can also be drawn in further detail. That diagram is shown in Fig. 3.5.

As for the build stage, ultraviolet renormalisation at the integration stage is applied independently, at the analogous public-output boundary: once the integrated one- and two-loop antennae are extracted, immediately before they are returned by `IntegrateAntenna[...]`. This mirrors, rather than reuses, the renormalisation applied during the build stage, since the two calls can be made independently of one another. No separate renormalisation step therefore appears in Fig. 3.5 either: it coincides with the "Public Integration Views" crossing shown there.

3.5.1 Integration Profiles and Selection

The integration-specific profile for each route selects the kind of backend process (and external package workflow used), and the basis family required for the process. The antennae are divided by topology, that is, number of final-state particles and number of loops, and all of those that share the same topology in the current implementation, share the same basis-family set. Since IBP reduction acts on the kinematic integral structure and is independent of the colour coefficients, the components of a set are reduced separately. *AntCalc* can therefore integrate the complete set or a single selected component, avoiding unnecessary reductions. The exception to this rule is the A_2^2 set, in which the basis-family set depends on which of the two source contributions is being reduced. A diagram of this route's multi-configuration structure is shown in Figure 3.6; comparing it with the upper path of Figure 3.4 shows how the A_2^2 route differs from the single-family routes used for A_3^1 and most others.

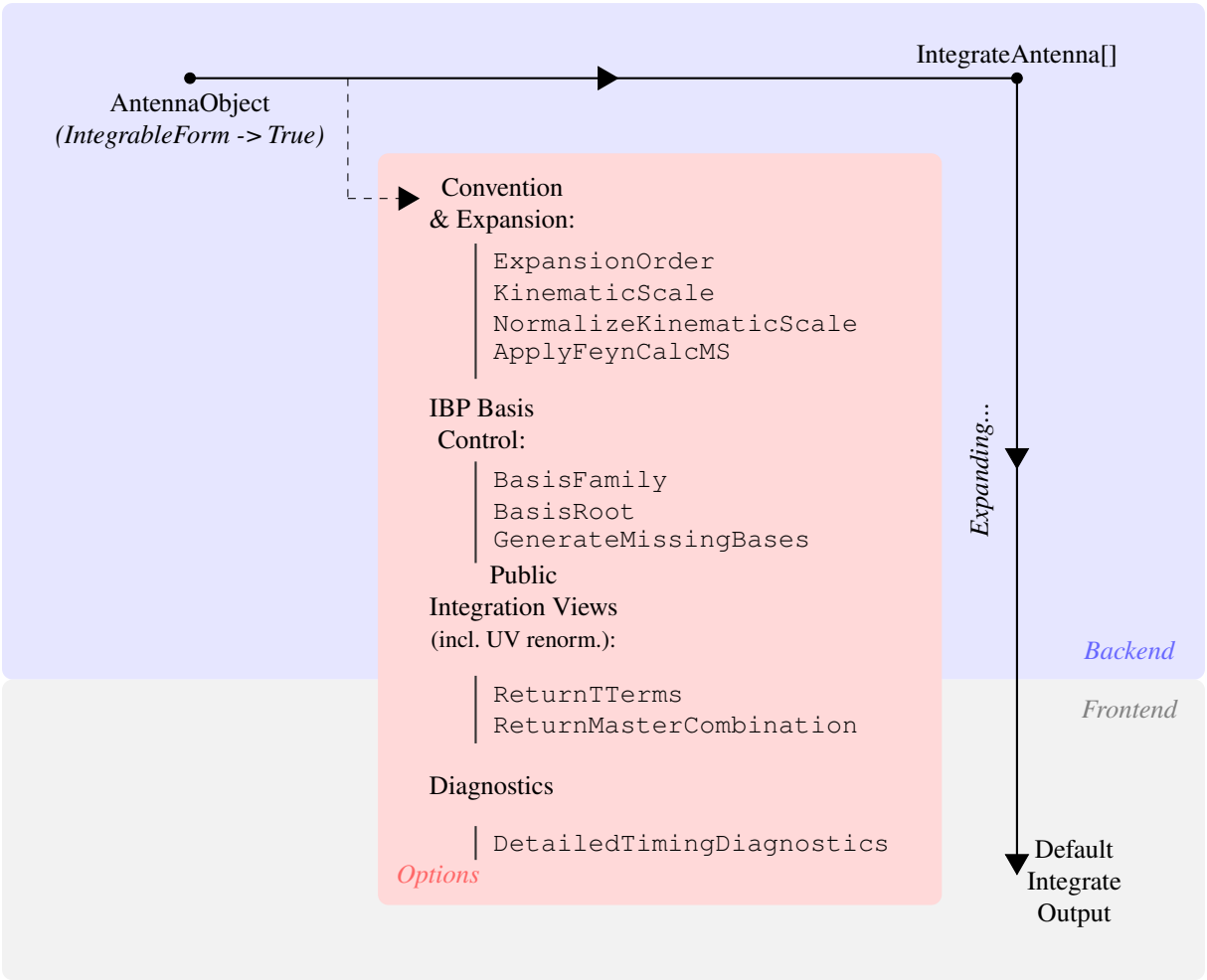


Figure 3.5: Profile-driven architecture of *AntCalc*'s Integration stage. The principal integration-only options are shown inside the red card.

The A_2^1 antenna is also an exception to the use of integral families: its loop integrals are reduced directly to scalar one-loop integrals using Passarino–Veltman reduction, whose known analytic expressions are then substituted and expanded as Laurent series in ϵ. The integration profiles are summarised in Table 3.6.

Table 3.6: Overview of the integration-side route-profiles.

Default Backend	Basis-Family Set	Current Route
Passarino–Veltman / *FeynHelpers*	massless two-parton vertex	$\mathcal{A}_2^1$
IBP / *LiteRed2*	`X30`	$\mathcal{A}_3^0$
IBP / *LiteRed2*	`X40`	$\mathcal{A}_4^0, \tilde{\mathcal{A}}_4^0, \mathcal{B}_4^0, \mathcal{C}_4^0$
IBP / *LiteRed2*	`A31`	$\mathcal{A}_3^1, \tilde{\mathcal{A}}_3^1, \hat{\mathcal{A}}_3^1$
IBP / *LiteRed2*	`A22TwoLoopTree`	$\mathcal{A}_2^2, \tilde{\mathcal{A}}_2^2, \hat{\mathcal{A}}_2^2$
IBP / *LiteRed2*	`A22OneLoopSelf`	$\check{\mathcal{A}}_2^2$
IBP / *LiteRed2*	`MX30`	$\mathcal{A}_3^{0,(m_q)}$

As on the build side, the integration profile does not itself perform any calculation; rather, it specifies the information required to execute the corresponding integration workflow. Consider, for example, the integration of the B_4^0 antenna. From the route key stored in the `AntennaObject`, *AntCalc* identifies the appropriate integration profile and the integral family associated with the X_4^0 phase-space topology. The unintegrated antenna is then mapped onto this integral family and reduced by IBP, using *LiteRed2*, to

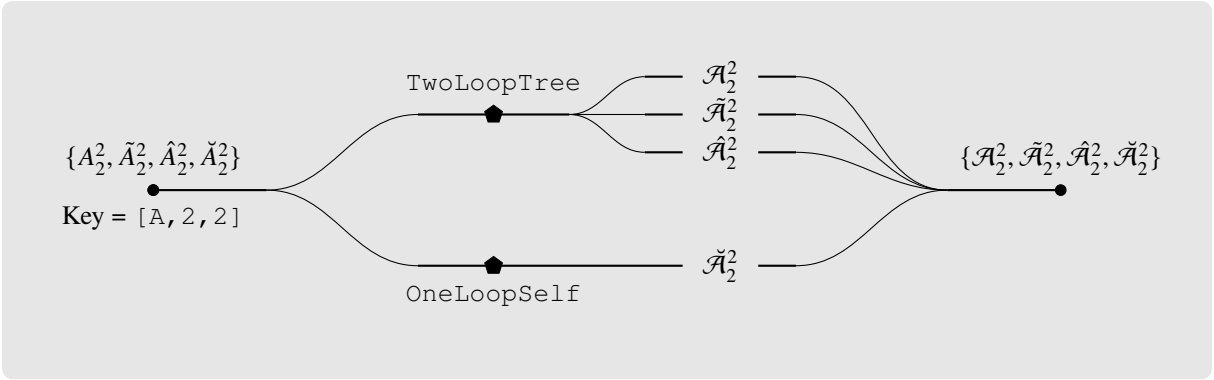


Figure 3.6: Diagram showing the component extraction steps and integration-basis families used in the integration-side workflow of the A_2^2 antenna set.

the corresponding set of master integrals. Their known analytic expressions are subsequently substituted to obtain the integrated antenna.

The master integrals used in these reductions are defined and evaluated in Appendix C, while their role in the analytic integration of antenna functions is illustrated through representative examples in Chapter 4.

3.5.2 Integration Routes

3.5.2.1 Passarino–Veltman Route

The A_2^1 antenna constitutes a special low-multiplicity virtual case and is the only integration route implemented without IBP reduction. As a two-parton one-loop form-factor contribution, its loop integrals can be treated directly using Passarino–Veltman reduction, which reduces the tensor loop integrals to scalar one-loop functions. *AntCalc* evaluates these scalar functions analytically using *Package-X* [28] through *FeynHelpers* [29]. The required branch and scale conventions are then applied, followed by the appropriate kinematic normalisation and Laurent expansion in ϵ, yielding the integrated $\mathcal{A}_2^1$ antenna. The workflow is summarised schematically in Fig. 3.7.

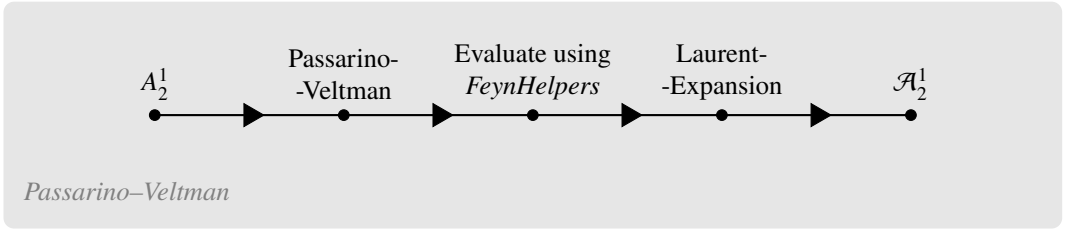


Figure 3.7: Passarino–Veltman reduction for the A_2^1 antenna integration.

3.5.2.2 IBP Route

All remaining antenna functions are integrated using IBP reduction, implemented through *LiteRed2* [23], following the formalism described in Section 2.5. The antenna integrand is first expressed in terms of the momenta, invariants, and denominator structures defining the relevant integral families. The appropriate phase-space cut propagators are then introduced, allowing the phase-space integrals to be treated within the same integral-family framework as loop integrals. After expanding the antenna into individual integral terms, each term is assigned to the corresponding integral family and reduced by

LiteRed2 to a linear combination of master integrals. The reduced terms are subsequently recombined to obtain the complete integrated antenna in terms of the corresponding master-integral basis.

For the public routes, the master integrals (MIs) identified by the IBP reduction are matched to their corresponding analytic expressions known from the literature and evaluated explicitly in this work. Their expressions are then substituted by *AntCalc*. Finally, the required normalisation and dimensional-regularisation conventions are applied, and the result is expanded as a Laurent series in ϵ, yielding the integrated antenna function. The master integrals and their analytic evaluation are presented in Appendix C, and representative applications of this procedure to the integration of antenna functions in Chapter 4. The complete IBP-based integration workflow is summarised schematically in Fig. 3.8.

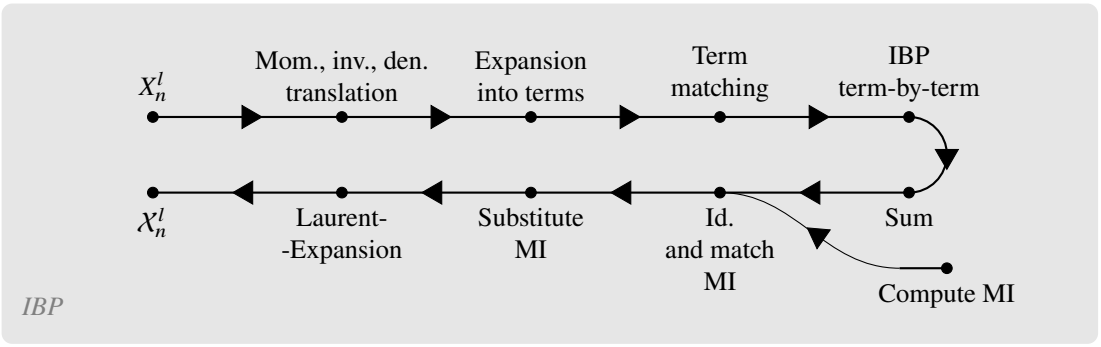


Figure 3.8: IBP reduction workflow for the remaining antennae integrations. The MI computation is external to *AntCalc*'s internal operation, and hence is represented by an external incoming arrow.

3.6 Higher-Level Functions

Besides the two canonical functions, `BuildAntenna` and `IntegrateAntenna`, *AntCalc* also supports higher-level functions. This layer does not introduce new antenna-construction or integration methodology; instead it orchestrates the two canonical functions to compute observables such as the R-ratio, to assemble the topology-grouped combinations used in it, and to streamline repeated build or integrate evaluation. The five higher-level functions are:

- `BuildAndIntegrateAntenna` — chains `BuildAntenna[..., IntegrableForm -> True]` into `IntegrateAntenna` in a single call, forwarding the component-selection and Laurent-expansion-order controls to their respective stages.
- `BuildAllAntennae` — builds every public antenna function in one call.
- `BuildAndIntegrateAllAntennae` — builds and integrates every public antenna function in one call. Both bulk wrappers can be limited to a chosen perturbative order.
- `TObject` — assembles integrated antennae into the topology-grouped combinations used in the R-ratio, defined in Section 5.2 of Chapter 5.
- `BuildRRatio` — assembles the massless hadronic R-ratio from the integrated antenna ingredients required at the selected perturbative order.

The R-ratio produced by `BuildRRatio` is the observable used for *AntCalc*'s global validation. Called as `BuildRRatio[SMQCD, maxOrder -> ...]`[2], it generates the R-ratio through NNLO

[2]The label `SMQCD` is not standard in the literature. It is used here to denote the massless QCD sector of the Standard Model.

and, by default, returns the finite coefficient after applying the fixed weights of Eq. (5.16) to the integrated antennae ($\frac{1}{2}$ for $\tilde{\mathcal{A}}_4^0$ and 2 for C_4^0). This result has been validated against a literature reference target, as discussed in Chapter 5.

3.7 Validation and Development Diagnostics

As outlined in the introduction, the implementation is validated at two levels. Each integrated antenna function is compared against its published Laurent expansion [5, 25]. The unintegrated antennae are, where applicable, also tested locally against the expected eikonal factors and Altarelli–Parisi splitting functions in the relevant unresolved limits (Chapter 4; the one-loop A_3^1 set is the exception, see Table 4.1). Globally, the integrated antennae are assembled into the inclusive R-ratio and compared with the known finite result (Chapter 5).

As discussed in Section 3.2, the route system can retain selected diagnostic information from an evaluation. That makes it easy to locate any inconsistency during a route calculation, which can then be inspected using the built-in diagnostics tool — for example, by retrieving the antenna ingredients of an R-ratio order and checking how their ϵ-poles cancel.

The `Ingredients` association is only one part of the diagnostic information. For a complete retained history of a calculation, *AntCalc* can instead return an `AntennaRunRecord`, which records the relevant route stages and metadata. This information was used to trace the source of discrepancies during development; the resulting validation comparisons are presented in Chapter 5.

3.7.1 Amplitude-Level and Release-Acceptance Validation

The checks described above validate an antenna function against an external reference: a published Laurent series, or a known finite observable. *AntCalc* also implements a complementary validation family that requires no such reference. The function `VerifyWardIdentity[type, numFinalParticles, loopOrder]` acts directly on the unsquared amplitude produced by the build stage, before spin summation, colour decomposition, or phase-space integration is performed. For a chosen external gluon leg, it replaces that gluon's polarisation vector by its own momentum, $\varepsilon^\mu(k) \to k^\mu$, while keeping every other external gluon polarisation explicitly transverse, and checks whether the resulting expression vanishes. This is the standard amplitude-level Ward identity, applied leg by leg to the raw generated amplitude rather than inferred from a property of the extracted or integrated antenna. It is currently applied to every massless public route with an external gluon leg, that is, to A_3^0, A_4^0 and the A_3^1 set, and returns a structured pass, fail, or time-out report for each checked leg. Because it depends only on the internal gauge structure of the generated amplitude, and not on comparison with any external result, it tests a property the literature-comparison checks above cannot: the gauge invariance of the generated amplitude, which would expose errors in the relative signs and couplings of its diagrams before extraction even begins. In principle, this makes it applicable to a route for which no literature result exists at all.

Separately from this route-level testing, *AntCalc*'s development also relies on a release-acceptance practice. Before a version is prepared, every public route is re-evaluated in a fresh kernel, with any stored or cached intermediate result disabled, and its build, integration, and one-shot outputs are compared against each other and against the literature target discussed above. Each run is recorded, together with its timing and the specific checks it satisfied, as dated evidence. This practice has already caught, and allowed the correction of, release-preparation issues that were not visible from the final Laurent-series outputs alone.

3.7.2 Runtime and Memory Characteristics

To characterise the computational cost and performance of the implemented public routes, each antenna was evaluated in a fresh *Wolfram Mathematica* kernel. Package loading was excluded from the measurements. Both the build and integration stages were timed using the `AbsoluteTiming` function. Memory usage was measured using `MaxMemoryUsed[]`, relative to the initial kernel baseline. These measurements were taken using a MacBook Pro (Mac16.8), with an Apple M4 Pro 12 core (8 performance and 4 efficiency) CPU, and 24 GB unified RAM, running macOS 26.6.2 (build 25G83), *Wolfram Mathematica* 15.0.0 for Mac OS X ARM (64-bit), and *AntCalc* 0.3.2β6.

The lower-order routes, up to NLO, complete both stages in approximately one second, with the cost increasing substantially for the NNLO routes, in particular A_2^2, which surpasses 31 minutes. The A_4^0 route has the largest memory requirement, reaching 567 MiB above the kernel benchmark. The complete results are shown in Table 3.7.

Table 3.7: Representative resource measurements for the implemented public antenna routes. The build time measures `BuildAntenna[..., IntegrableForm -> True]`, followed by `IntegrateAntenna[...]` for the integration time. Each route was evaluated in a fresh Wolfram kernel. Peak memory is the maximum kernel-memory increase relative to the initial baseline. Each route is timed on its own, so the entries are not additive to the total `BuildRRatio` runtime.

Route	Build time	Integration time	End-to-end time	Peak memory above baseline
A_2^0	0.09 s	0.49 s	0.58 s	22.7 MiB
A_3^0	0.30 s	0.85 s	1.15 s	24.8 MiB
A_2^1	0.57 s	0.40 s	0.98 s	19.3 MiB
A_3^1	74.67 s	15 min 27 s	16 min 42 s	195.3 MiB
A_2^2	21.79 s	31 min 06 s	31 min 27 s	227.7 MiB
A_4^0	3 min 03 s	13 min 51 s	16 min 53 s	567.3 MiB
B_4^0	8.65 s	17.27 s	25.92 s	143.2 MiB
C_4^0	5.56 s	5 min 26 s	5 min 32 s	174.5 MiB
$A_3^{0,(m_q)}$	2.81 s	0.25 s	3.09 s	12.07 MiB

Chapter 4

Calculation and Integration of QCD Antenna Functions

The antenna functions required for the subtraction of infrared singularities through NNLO were introduced in Chapter 2, while the techniques required for their automated construction and integration within *AntCalc* were described in Chapter 3. This chapter applies those methods to the calculation of the massless final-final $q\bar{q}$ antenna functions through NNLO and presents their unintegrated and analytically integrated forms. The calculation starts from the corresponding QCD amplitudes and proceeds, where required, through integral reduction and master-integral substitution to obtain the integrated antennae as Laurent series in ϵ.

The tree-level three-parton antenna A_3^0 is used first as a detailed example of the complete calculation. It is derived analytically and subsequently reproduced with *AntCalc*, allowing the intermediate and final results to be compared directly. Its infrared behaviour is then validated locally by taking the relevant soft and collinear limits and recovering the corresponding eikonal factor and Altarelli–Parisi splitting functions. This example therefore provides both a transparent derivation of an antenna function and a detailed illustration of the calculation implemented in *AntCalc*.

Having established the procedure with A_3^0, the remaining massless antenna functions are presented more concisely. For each antenna set, the discussion focuses on its perturbative and colour structure, the integral reduction required for its analytic integration, the relevant master integrals, and the resulting Laurent expansion in ϵ. The massless results are compared with the corresponding analytic results in the literature and are subsequently subjected to a global validation in Chapter 5 through the calculation of the massless hadronic R-ratio through NNLO. Section 4.6 summarises the complete set of antenna functions considered in this work and their validation.

The master integrals entering these calculations depend on the antenna and on the corresponding integral family. Different colour components belonging to the same antenna set generally reduce to a common set of master integrals. Table C.1 of Appendix C summarises the master integrals required for the antenna functions considered in this chapter. These master integrals are defined and evaluated there, organised by antenna family; in this chapter, only R_3, which is needed for the worked example of Subsection 4.1.1, and the massive master integrals of Section 4.5 are treated in the main text. Evaluating the master integrals analytically was a crucial part of this work: they were all derived explicitly, and not merely taken from a source, and their results reproduce the analytic expressions known from the literature [5, 13, 26]. For R_6 and $R_{8,b}$ the derivation uses the optical theorem, with the three-loop forward-scattering integrals I_6 and I_8 taken from Ref. [13]. Appendix C presents their definitions and

results, and the derivation of $R_{8,a}$, the most involved of them, is given step by step as an example of the method.

Finally, Section 4.5 explores an extension of the calculation to the massive A_3^0 antenna. This example is treated separately from the massless antenna set and is used to illustrate the modifications introduced by a non-zero quark mass.

As described in Chapter 3, *AntCalc* employs two distinct reduction techniques at different stages of the calculation: Passarino–Veltman reduction and integration-by-parts (IBP) reduction. Passarino–Veltman reduction is used for the relevant one-loop amplitudes to reduce tensor loop integrals to scalar one-loop integrals, allowing the unintegrated antenna functions to be expressed in terms of the external kinematic invariants. IBP identities are instead used in the analytic integration of the antenna functions to reduce the resulting integral families to a basis of master integrals. The two-loop A_2^2 set is the exception: its loop integrals are reduced by IBP identities when the antenna is built, and the subsequent two-particle phase-space integration is trivial (Section 4.2).

Since Passarino–Veltman reduction is required only for the one-loop tensor reduction and does not constitute the main focus of the integrations presented here, its implementation and the corresponding scalar one-loop integrals are discussed separately in Appendix B.2. The remainder of this chapter therefore focuses on the antenna functions and on the master integrals arising from the IBP reduction.

Following the IBP reduction, *LiteRed2* represents the master integrals using its internal $j[\ldots]$ notation, in which the arguments specify the corresponding integral within a given integral family. *AntCalc* retains this notation in its direct reduction output. For the presentation of the results in this chapter, these internal labels are replaced by the conventional symbols used for the corresponding master integrals, such as A_3, A_4, and R_3. The mapping between the *LiteRed2* representation and the master-integral notation adopted here is given in Appendix B.7.

4.1 NLO Antenna Functions

The tree-level three-parton antenna A_3^0 provides the simplest non-trivial example of the antenna calculations considered in this work. Its relative simplicity allows the complete calculation to be carried out analytically, from the underlying matrix element to the unintegrated antenna and its subsequent integration over the antenna phase space. This calculation is therefore used as a benchmark for the corresponding automated procedure implemented in *AntCalc*. In the following, the analytic calculation is presented first and is then reproduced with *AntCalc*, allowing the intermediate and final results to be compared directly.

4.1.1 Tree-Level Three-Parton Antenna A_3^0

Analytic Construction of the Unintegrated A_3^0 Antenna

The A_3^0 antenna is the tree-level final-final quark–antiquark antenna describing the emission of a gluon between the hard $q\bar{q}$ radiators. It is derived from the tree-level process $\gamma^*(p) \to q(k_1)g(k_3)\bar{q}(k_2)$. At this order, the amplitude receives contributions from the two Feynman diagrams shown in Fig. 4.1, corresponding to gluon emission from the quark and antiquark lines. The complete tree-level amplitude is obtained by summing these two contributions, and A_3^0 is constructed from the resulting squared matrix element after normalisation to the corresponding two-parton Born contribution, as defined in Chapter 2.

Applying the QCD Feynman rules to the two diagrams in Fig. 4.1 and summing their contributions gives the tree-level amplitude,



(a) Gluon emitted from the quark line.

(b) Gluon emitted from the antiquark line.

Figure 4.1: Tree-level Feynman diagrams contributing to $\gamma^* \to q(k_1)g(k_3)\bar{q}(k_2)$.

$$\mathcal{M}_3^0 = e_q g_s (T^{a_3})_{i_1 i_2} \left[\bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) - \bar{u}(k_1) \not{\epsilon}(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}^*(k_3) v(k_2) \right] \tag{4.1}$$

where i_1, i_2, and a_3 denote the colour indices of the quark, antiquark, and gluon, respectively, and T^{a_3} is the fundamental $SU(N)$ generator introduced in Subsection 2.4.3.

The amplitude is now squared as follows,

$$\begin{aligned} |\mathcal{M}_3^0|^2 &= e_q^2 g_s^2 (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_2 i_1} \left(M_A M_A^\dagger + M_B M_B^\dagger - M_A M_B^\dagger - M_B M_A^\dagger \right) \\ &\text{with } M_A = \bar{u}(k_1) \not{\epsilon}^*(k_3) \frac{\not{k}_1 + \not{k}_3}{s_{13}} \not{\epsilon}(p) v(k_2) = \bar{u}(k_1) \Gamma_A v(k_2), \\ &\quad M_B = \bar{u}(k_1) \not{\epsilon}(p) \frac{\not{k}_2 + \not{k}_3}{s_{23}} \not{\epsilon}^*(k_3) v(k_2) = \bar{u}(k_1) \Gamma_B v(k_2). \end{aligned} \tag{4.2}$$

Here $\bar{\Gamma} \equiv \gamma^0 \Gamma^\dagger \gamma^0$ denotes the Dirac conjugate of Γ, which reverses the order of the gamma matrices and complex-conjugates the polarisation vectors, such that $M_{A,B}^\dagger = \bar{v}(k_2) \bar{\Gamma}_{A,B} u(k_1)$. In total, the squared amplitude then becomes (the spin sums are understood as traces over Dirac matrices, $\sum_s u\bar{u} = \not{k}$ for massless quarks)

$$\begin{aligned} &= \sum_{a_3, i_1, i_2} (T^{a_3})_{i_1 i_2} (T^{a_3})_{i_2 i_1} \\ &\qquad \sum_{s_1, s_2, \lambda_g, \lambda_\gamma} u(k_1, s_1) \bar{u}(k_1, s_1) \left[\Gamma_A \bar{\Gamma}_A + \Gamma_B \bar{\Gamma}_B - \Gamma_A \bar{\Gamma}_B - \Gamma_B \bar{\Gamma}_A \right] v(k_2, s_2) \bar{v}(k_2, s_2) \end{aligned} \tag{4.3}$$

Each of the Γ terms becomes, using Dirac Algebra,

$$\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} u(k_1,s_1)\bar{u}(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) =$$

$$\begin{aligned}
&= \frac{1}{s_{13}^2}\sum_{\lambda_g,\lambda_\gamma}\varepsilon_\mu^*\varepsilon_\nu\varepsilon_\rho\varepsilon_\sigma^*\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right)\\
&= \frac{1}{s_{13}^2}g_{\mu\nu}g_{\rho\sigma}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma^\sigma(\not{k}_1+\not{k}_3)\gamma^\nu\right)\\
&= \frac{1}{s_{13}^2}\,\mathrm{Tr}\left(\not{k}_1\gamma^\mu(\not{k}_1+\not{k}_3)\gamma^\rho\not{k}_2\gamma_\rho(\not{k}_1+\not{k}_3)\gamma_\mu\right)\\
&= \left(\frac{2-d}{s_{13}}\right)^2\mathrm{Tr}(\not{k}_1\not{P}\not{k}_2\not{P}),\quad \not{P}=(\not{k}_1+\not{k}_3)\\
&= 4\left(\frac{2-d}{s_{13}}\right)^2\left(2(k_1\cdot P)(k_2\cdot P)-(k_1\cdot k_2)P^2\right)\\
&= 2(d-2)^2\frac{s_{23}}{s_{13}}
\end{aligned} \tag{4.4}$$

$$\begin{aligned}
&\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} u(k_1,s_1)\bar{u}(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) =\\
&\qquad = 2(d-2)^2\frac{s_{13}}{s_{23}}
\end{aligned} \tag{4.5}$$

$$\begin{aligned}
&\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} u(k_1,s_1)\bar{u}(k_1,s_1)\left(\Gamma_A\bar{\Gamma}_B\right)v(k_2,s_2)\bar{v}(k_2,s_2) =\\
&\qquad = 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.6}$$

$$\begin{aligned}
&\sum_{s_1,s_2,\lambda_g,\lambda_\gamma} u(k_1,s_1)\bar{u}(k_1,s_1)\left(\Gamma_B\bar{\Gamma}_A\right)v(k_2,s_2)\bar{v}(k_2,s_2) =\\
&\qquad = 2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}
\end{aligned} \tag{4.7}$$

Summing the squared matrix element over the final-state colour indices gives,

$$\sum_{a_3,i_1,i_2}(T^{a_3})_{i_1i_2}(T^{a_3})_{i_2i_1} = \frac{1}{2}(N^2-1) \tag{4.8}$$

and the tree-level squared matrix element is obtained in d dimensions (with $d = 4-2\epsilon$) as

$$\begin{aligned}
|\mathcal{M}_3^0|^2 &= e_q^2g_s^2(N^2-1)\left((d-2)^2\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-2(d-2)\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= (d-2)e_q^2g_s^2(N^2-1)\left((d-2)\left(\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-2\frac{(d-4)s_{13}s_{23}-2s_{12}^2-2s_{12}(s_{13}+s_{23})}{s_{13}s_{23}}\right)\\
&= \frac{4e_q^2g_s^2\left(N^2-1\right)}{s_{13}s_{23}}\left(2q^2s_{12}+s_{13}^2+s_{23}^2-2\epsilon\left(q^2s_{12}+s_{13}^2+s_{23}^2+s_{13}s_{23}\right)+\epsilon^2\left(s_{13}+s_{23}\right)^2\right)
\end{aligned} \tag{4.9}$$

To construct A_3^0, the three-parton squared matrix element above must be normalised to the corresponding two-parton Born contribution, as defined in Chapter 2. The relevant Born process is $\gamma^*(p) \to q(k_1)\bar{q}(k_2)$, with tree-level amplitude

$$\mathcal{M}_2^0 = e_q \delta_{i_1 i_2} \bar{u}(k_1) \not{\epsilon}(p) v(k_2). \tag{4.10}$$

Applying the same spin, polarisation and colour sums as above, and considering that, for a two-particle final state $s_{12} = q^2$, we get

$$\begin{aligned} |\mathcal{M}_2^0|^2 = 2e_q^2(d-2)Ns_{12} &= 2e_q^2(d-2)Nq^2 \\ &= 4e_q^2(1-\epsilon)Nq^2 \end{aligned} \tag{4.11}$$

The antenna is now ready to be built. That is accomplished using the following formula,

$$A_3^0 = \frac{1}{2g_s^2 C_F} \frac{|\mathcal{M}_3^0|^2}{|\mathcal{M}_2^0|^2}. \tag{4.12}$$

By taking $C_F = (N^2-1)/(2N)$, the formula yields, using Eqs. (4.9) and (4.11),

$$\begin{aligned} A_3^0 &= \frac{2q^2 s_{12} + s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}} - \epsilon \frac{(s_{13}+s_{23})^2}{q^2 s_{13} s_{23}} \\ &= \frac{1}{q^2}\left[\left(\frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} + 2\frac{s_{12}q^2}{s_{13}s_{23}} \right) - \epsilon \left(2 + \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}} \right) \right] \end{aligned} \tag{4.13}$$

The electric charge cancels between the three- and two-parton squared matrix elements, while the remaining coupling and colour factors are removed by the antenna normalisation. Equation (4.13) is therefore purely kinematic. In Chapter 2, these factors were already absorbed into the Born-normalised matrix-element notation; here they have been retained explicitly to illustrate the normalisation.

Phase-Space Integration through R_3

As discussed in Chapter 2, the antenna must be integrated over the unresolved phase space to obtain the integrated antenna $\mathcal{A}_3^0$, whose explicit infrared poles in ϵ cancel the corresponding poles of the virtual contribution. We therefore integrate A_3^0 over the three-particle antenna phase space in $d = 4-2\epsilon$ dimensions. Expressing the phase-space measure in terms of the Mandelstam invariants s_{ij} and the total invariant mass q^2, the integral takes the form,

$$\begin{aligned} \mathcal{A}_3^0 &= C(\epsilon,1) \int d\Phi_{X_{ijk}}\, A_3^0 \\ &= \mathcal{N}(\epsilon,1)\,(4\pi)^{3-2d} \left(q^2\right)^{1-\frac{d}{2}} \int ds_{12} ds_{13} ds_{23} d\Omega\, (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} \delta\left(q^2 - s_{12} - s_{13} - s_{23}\right) A_3^0. \end{aligned} \tag{4.14}$$

with $d\Phi_{X_{ijk}} = d\Phi_3/\Phi_2$. $C(\epsilon,1)$ is the integral normalisation which is here given as

$$C(\epsilon,1) = 8\pi^2 (4\pi)^{-\epsilon} e^{\epsilon\gamma_E}. \tag{4.15}$$

For A_3^0, the phase-space integration can be performed directly. However, in order to illustrate the

general integration strategy employed for the more involved antenna functions considered later, we instead apply the IBP reduction procedure introduced in Subsection 2.5.1. The phase-space integrals are thereby reduced to a linear combination of master integrals, whose analytic expressions can then be substituted to obtain the integrated antenna.

After IBP reduction, the integrated antenna can be expressed in terms of a single master integral, R_3, as

$$\mathcal{A}_3^0 = \frac{C(\epsilon,1)}{\Phi_2}\frac{2(2-6\epsilon+5\epsilon^2-2\epsilon^3)}{q^2\epsilon^2}R_3 \tag{4.16}$$

The IBP reduction therefore involves a single master integral, R_3, which corresponds to the volume of the massless three-particle phase space,

$$R_3 = \int d\Phi_3. \tag{4.17}$$

Here, in d dimensions, the massless two- and three-particle phase-space measures are [13, 16],

$$\begin{aligned} d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta\Big(q^2 - s_{12}\Big) \\ d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} \Big(q^2\Big)^{\frac{2-d}{2}} (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta\Big(q^2 - s_{123}\Big), \end{aligned} \tag{4.18}$$

where the shorthands $d\mathcal{S}_n$ and s_{123} and the solid-angle factors Ω_n are defined, together with the four-particle measure, in Appendix C.1. With this parametrisation, the integral can be written as [30],

$$\begin{aligned} R_3 &= \int d\Phi_3 \\ &= \frac{1}{4}\frac{\Omega_{d-1}\Omega_{d-2}}{(2\pi)^{2d-3}} \iiint dE_1 dE_2 d\theta_{12} \, (E_1 E_2 \sin\theta_{12})^{d-3} \delta\Big(q^2 - s_{123}\Big) \\ &= (2\pi)^{3-2d} 2^{-1-d} \left(q^2\right)^{\frac{2-d}{2}} \Omega_{d-1}\Omega_{d-2} \int d\mathcal{S}_3 \, (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} \delta\Big(q^2 - s_{123}\Big). \end{aligned} \tag{4.19}$$

Its analytic evaluation in $d = 4 - 2\epsilon$ dimensions gives [13],

$$R_3 = 2^{-7+4\epsilon}\pi^{-3+2\epsilon}\frac{\Gamma^3(1-\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}\Big(q^2\Big)^{1-2\epsilon}. \tag{4.20}$$

Substitution of this result into Eq. (4.16) then yields the integrated antenna $\mathcal{A}_3^0$.

IBP Reduction of a Representative Term

To illustrate explicitly how the IBP reduction maps the terms of A_3^0 onto the master integral R_3, we consider the eikonal term,

$$\mathcal{S}_{12}^{(3)} = \frac{2s_{12}}{s_{13}s_{23}}. \tag{4.21}$$

We define,

$$\begin{aligned} y_{13} = \frac{s_{13}}{q^2}, \quad y_{23} = \frac{s_{23}}{q^2}, \quad y_{12} = \frac{s_{12}}{q^2} = 1 - y_{13} - y_{23} \\ J(a,b,c) = \int_\Delta dy_{13} dy_{23} \, y_{13}^{a-1} y_{23}^{b-1} y_{12}^{c-1}, \quad \Delta : y_{13}, y_{23}, y_{12} \geq 0, \;\; y_{13} + y_{23} + y_{12} = 1 \end{aligned} \tag{4.22}$$

and with these,

$$\begin{aligned}\Phi_3 &\equiv \int d\Phi_3 = R_3 \\ &= \mathcal{K}\Big(q^2,\epsilon\Big)\int_\Delta (y_{13}y_{23}y_{12})^{-\epsilon}\,dy_{13}dy_{23} \\ &= \mathcal{K}\Big(q^2,\epsilon\Big)\,J(1-\epsilon,1-\epsilon,1-\epsilon), \quad \mathcal{K}\Big(q^2,\epsilon\Big) = \frac{2^{-7+4\epsilon}\pi^{-3+2\epsilon}}{\Gamma(2-2\epsilon)}\Big(q^2\Big)^{1-2\epsilon}.\end{aligned} \tag{4.23}$$

In terms of this notation, the phase-space integral of the eikonal term becomes

$$\int d\Phi_3\, \mathcal{S}_{12}^{(3)} = \frac{2\mathcal{K}}{q^2} J(-\epsilon,-\epsilon,2-\epsilon). \tag{4.24}$$

Applying an IBP identity with respect to y_{13} and recalling that $y_{12} = 1 - y_{13} - y_{23}$ gives,

$$\begin{aligned}0 &= \int_\Delta dy_{13}dy_{23}\,\frac{\partial}{\partial y_{13}}\Big(y_{13}^a y_{23}^{b-1} y_{12}^{c-1}\Big) \\ &= aJ(a,b,c) - (c-1)J(a+1,b,c-1)\end{aligned} \tag{4.25}$$

such that, for y_{13},

$$J(-\epsilon,-\epsilon,2-\epsilon) = -\frac{1-\epsilon}{\epsilon}J(1-\epsilon,-\epsilon,1-\epsilon) \tag{4.26}$$

and for y_{23},

$$J(1-\epsilon,-\epsilon,2-\epsilon) = -\frac{1-\epsilon}{\epsilon}J(1-\epsilon,1-\epsilon,1-\epsilon). \tag{4.27}$$

By the symmetry $y_{13} \leftrightarrow y_{12}$,

$$J(2-\epsilon,-\epsilon,1-\epsilon) = J(1-\epsilon,-\epsilon,2-\epsilon). \tag{4.28}$$

The remaining integral in the first relation, $J(1-\epsilon,-\epsilon,1-\epsilon)$, is related to the previous ones by multiplying the integrand of $J(a,b,c)$ by $1 = y_{13} + y_{23} + y_{12}$, which gives

$$J(a,b,c) = J(a+1,b,c) + J(a,b+1,c) + J(a,b,c+1). \tag{4.29}$$

At $(a,b,c) = (1-\epsilon,-\epsilon,1-\epsilon)$, and using the symmetry above together with the relation for y_{23},

$$J(1-\epsilon,-\epsilon,1-\epsilon) = 2J(1-\epsilon,-\epsilon,2-\epsilon) + J(1-\epsilon,1-\epsilon,1-\epsilon) = \frac{3\epsilon-2}{\epsilon}J(1-\epsilon,1-\epsilon,1-\epsilon). \tag{4.30}$$

Substituting this in the relation for y_{13} gives, hence,

$$J(-\epsilon,-\epsilon,2-\epsilon) = \frac{(1-\epsilon)(2-3\epsilon)}{\epsilon^2}J(1-\epsilon,1-\epsilon,1-\epsilon) \tag{4.31}$$

which finally makes,

$$\int d\Phi_3\,\frac{2s_{12}}{s_{13}s_{23}} = \frac{2(1-\epsilon)(2-3\epsilon)}{\epsilon^2 q^2}R_3. \tag{4.32}$$

Master Integral Substitution and Laurent Expansion

Substituting Eq. (4.20) into Eq. (4.16) gives

$$\begin{aligned}\mathcal{A}_3^0 &= \frac{C(\epsilon,1)}{\Phi_2}\Big(q^2\Big)^{-2\epsilon}(4\pi)^{-3+2\epsilon}(2-6\epsilon+5\epsilon^2-2\epsilon^3)\frac{\Gamma^3(1-\epsilon)}{\epsilon^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)} \\ &= \Big(q^2\Big)^{-\epsilon} e^{\epsilon\gamma_E}(1-2\epsilon)\big(2-2\epsilon+\epsilon^2\big)\frac{\Gamma^2(-\epsilon)}{\Gamma(3-3\epsilon)}\end{aligned} \tag{4.33}$$

and, after expanding the integrated expression as a Laurent series in ϵ, we arrive at the Laurent expansion,

$$\mathcal{A}_3^0 = \left(q^2\right)^{-\epsilon}\left[\frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{57-7\pi^2}{12}+\epsilon\left(\frac{84-21\pi^2-8\psi^{(2)}(1)+108\psi^{(2)}(3)}{24}\right)\right.$$
$$\left.+\epsilon^2\left(\frac{35640-3990\pi^2-71\pi^4-720\psi^{(2)}(1)+9720\psi^{(2)}(3)}{1440}\right)\right]$$
$$=\left(q^2\right)^{-\epsilon}\left[\frac{1}{\epsilon^2}+\frac{3}{2\epsilon}+\frac{19}{4}-\frac{7\pi^2}{12}+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)\right] \quad (4.34)$$

where the polygamma identities, $\psi^{(2)}(1) = -2\zeta_3$ and $\psi^{(2)}(3) = -2\zeta_3 + \frac{9}{4}$, were employed.

Implementation in *AntCalc*

Using *AntCalc*, building the antenna is very simple, since the function `BuildAntenna[type, numFinalParticles, numLoops]` already encodes the entire logical chain and is able to return all intermediate steps using the appropriate options.

First, by running the function naively we get the final result in Eq. (4.13),

```
In[2]:= BuildAntenna[A,3,0]//Collect[#,Epsilon]&
Out[2]:=
```

$$\frac{2s_{12}}{q^2s_{13}}+\frac{2s_{12}}{q^2s_{23}}+\frac{2s_{12}^2}{q^2s_{13}s_{23}}+\frac{s_{13}}{q^2s_{23}}+\frac{s_{23}}{q^2s_{13}}+\epsilon\left(\frac{-2}{q^2}-\frac{s_{13}}{q^2s_{23}}-\frac{s_{23}}{q^2s_{13}}\right)$$

We can also use the `ReturnRecord` option to access all intermediate steps, and check that *AntCalc* goes through the same steps as the by-hand derivation in Subsection 4.1.1. This is done with,

```
In[3]:=
BuildAntenna[A,3,0,ReturnRecord->True]["IntermediateSteps"]
Out[3]:=
   Amplitude:
```

$$T_{i_1i_2}^{a_3}\left[\frac{\bar{u}(k_1)\,\not{\epsilon}^*(k_3)\,(\not{k}_1+\not{k}_3)\,\not{\epsilon}(p)\,v(k_2)}{(k_1+k_3)^2}-\frac{\bar{u}(k_1)\,\not{\epsilon}(p)\,(\not{k}_2+\not{k}_3)\,\not{\epsilon}^*(k_3)\,v(k_2)}{(k_2+k_3)^2}\right]$$

```
   Interference:
```

$$\frac{4(\epsilon-1)(N^2-1)\Big(-2s_{12}^2-2s_{12}(s_{13}+s_{23})+(\epsilon-1)s_{13}^2+2\epsilon\,s_{13}s_{23}+(\epsilon-1)s_{23}^2\Big)}{s_{13}s_{23}}$$

```
   Result:
```

$$-\frac{2\epsilon}{q^2}+\frac{2s_{12}^2}{q^2s_{13}s_{23}}+\frac{2s_{12}}{q^2s_{13}}+\frac{2s_{12}}{q^2s_{23}}+\frac{s_{13}}{q^2s_{23}}-\frac{\epsilon\,s_{13}}{q^2s_{23}}+\frac{s_{23}}{q^2s_{13}}-\frac{\epsilon\,s_{23}}{q^2s_{13}}$$

where `Amplitude`, `Interference` and `Result` respectively refer to the expressions in Eqs. (4.1), (4.9) and (4.13).

Integrating antennae using *AntCalc* is also simple. Every step of the analytic integration above is handled by a single function: `IntegrateAntenna[antenna]`.

By default, `BuildAntenna` returns the unintegrated antenna as an expression in Mandelstam invariants, providing a compact representation suitable for direct inspection and applications requiring the local antenna. For analytic integration, however, additional information must be retained in order to map the antenna onto the integral families used for IBP reduction with *LiteRed2*. As described in Chapter 3, this is achieved by setting `IntegrableForm -> True`, as in `BuildAntenna[A,3,0,IntegrableForm -> True]`. In this mode, `BuildAntenna` returns an `AntennaObject` containing the information required to construct the corresponding *LiteRed2* integral representation and perform the subsequent IBP reduction through `IntegrateAntenna`.

We may now proceed to the integration. The whole process goes as follows,

```
In[4]:= intA30 = BuildAntenna[A,3,0,IntegrableForm -> True];
In[5]:= IntegrateAntenna[intA30,ExpansionOrder->2]
Out[5]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

As with `BuildAntenna`, `IntegrateAntenna` allows intermediate stages of the calculation to be inspected. In particular, the result can be returned either in its master-integral representation, containing R_3, or after substituting the analytic expression for R_3. These outputs correspond respectively to the intermediate results shown in Eqs. (4.16) and (4.33), prior to the final normalisation. The corresponding commands and outputs are,

```
In[6]:= resA30 = BuildAndIntegrateAntenna[A,3,0,
ReturnMasterCombination->True];
In[7]:= resA30//Simplify//Collect[#,R3]&
Out[7]:=
```

$$\frac{-4(d-3)\big(4+d(\epsilon-2)-4\epsilon\big)}{(d-4)^2q^2}R_3$$

```
In[8]:= resA30["MasterSubstitutedExpression"]//Simplify
Out[8]:=
```

$$-\left(q^2\right)^{-2\epsilon}\,\frac{2^{-5+4\epsilon}(d-3)\big(4+d(\epsilon-2)-4\epsilon\big)\pi^{-3+2\epsilon}\Gamma^3(1-\epsilon)}{(d-4)^2\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}$$

At this intermediate stage, the result retains both d and ϵ explicitly. This reflects the internal sequence of operations in `IntegrateAntenna`: the master-integral substitution is performed before the replacement $d = 4 - 2\epsilon$ is applied to the complete expression. After this substitution, the result depends only on ϵ and can be expanded as a Laurent series.

The complete construction and integration procedure can also be performed in a single step using `BuildAndIntegrateAntenna[type,numFinalParticles,numLoops]`. This function combines the build and integration stages described above and returns the integrated antenna directly as a Laurent series in ϵ. For A_3^0, retaining terms through $O(\epsilon^2)$,

```
In[9]:= BuildAndIntegrateAntenna[A,3,0,ExpansionOrder->2]
Out[9]:=
```

$$\frac{19}{4}+\epsilon^{-2}+\frac{3}{2\epsilon}-\frac{7\pi^2}{12}+\epsilon^2\left(\frac{639}{16}-\frac{133\pi^2}{48}-\frac{71\pi^4}{1440}-\frac{25\zeta_3}{2}\right)+\epsilon\left(\frac{109}{8}-\frac{7\pi^2}{8}-\frac{25\zeta_3}{3}\right)$$

The agreement between this result and the explicit analytic derivation above (the latter with $\mu^2 = q^2$ understood) provides a direct validation of the automated construction and integration of A_3^0 in *AntCalc*. The local infrared behaviour of the unintegrated antenna is tested independently in the following subsection through its soft and collinear limits.

4.1.2 Local Infrared Validation of A_3^0

To be usable as a subtraction term, an antenna function's unintegrated form must also reproduce the singular behaviour of the corresponding matrix element locally in phase space. In particular, A_3^0 must reproduce the universal eikonal factor when the gluon becomes soft and the Altarelli–Parisi (AP) splitting function in the relevant collinear limits. These two quantities provide local validation of the results obtained analytically and using *AntCalc*.

Soft Limit and the Eikonal Factor

When a gluon becomes soft, gauge theory squared matrix elements factorise universally into the squared matrix element without the soft gluon times a universal eikonal factor as [8],

$$|\mathcal{M}_{n+1}^0(...,i,j,k,...)|^2 \xrightarrow{j\to 0} -g_s^2 \sum_{i,k} T_i \cdot T_k \mathcal{S}_{ik}^{(j)} |\mathcal{M}_n^0(...,i,k,...)|^2 \tag{4.35}$$

where T_i are colour-charge operators and,

$$\mathcal{S}_{ik}^{(j)} = \frac{2s_{ik}}{s_{ij}s_{jk}} \tag{4.36}$$

is the eikonal factor.

Since this property is universal, we can use it to validate our results. For the A_3^0 antenna, the eikonal factor may be computed as,

$$\mathcal{S}_{12}^{(3)} = \lim_{k_3\to 0} A_3^0(1_q, 3_g, 2_{\bar{q}}). \tag{4.37}$$

After building the antenna using *AntCalc*, the limit may be computed using *Wolfram Mathematica* as follows,

```
In[2]:= A30 = BuildAntenna[A,3,0];

softSubs = {s13 -> lambda * s13, s23 -> lambda * s23};
A30Soft = A30 /.  softSubs;

A30SoftSeries = Series[A30Soft, {lambda, 0, 0}];

eikonalFactor = SeriesCoefficient[A30SoftSeries, -2] /.  q2 -> s12
// Simplify;
Out[2]:
```

$$\frac{2s_{12}}{s_{13}s_{23}}$$

This result agrees with the expected eikonal factor.

Collinear Limit and the Altarelli–Parisi Splitting Function

In the same manner, when two partons i and j become collinear, the squared matrix element factorises to [8, 31],

$$|\mathcal{M}^0_{n+1}(...,i,j,...)|^2 \xrightarrow{i\|j} \frac{8\pi\alpha_s}{s_{ij}} P_{IJ\to K}(z)|\mathcal{M}^0_n(...,K,...)|^2 \tag{4.38}$$

where K is the parent parton in which i and j recombine, z is the momentum fraction carried by one of the daughter partons, and $P_{IJ\to K}$ is the splitting function. This function is universal but channel-dependent and, since it carries the spin and helicity structure of the collinear splitting, it depends on the partons involved. For A^0_3, where a gluon may become collinear with a quark, the function appears as,

$$P_{qg\to q}(z) = C_F \frac{1+z^2}{1-z}. \tag{4.39}$$

Noting that the building process already normalises the fundamental Casimir C_F, using *AntCalc* in *Wolfram Mathematica*, the AP splitting function may be computed as,

```
In[3]:= A30 = BuildAntenna[A,3,0];

collinearSubs = {s13 -> lambda, s12 -> z q2, s23 -> (1 - z) q2,
Epsilon -> 0};
A30Collinear = A30 /.  collinearSubs;

A30CollinearSeries = Series[A30Collinear, {lambda, 0, 0}];

apSplittingFunc = SeriesCoefficient[A30CollinearSeries, -1]
// Simplify;
Out[3]:=
```

$$\frac{1+z^2}{1-z}$$

This result agrees with the expected colour-stripped $qg \to q$ AP kernel.

4.1.3 Virtual Two-Parton Antenna A^1_2

The A^1_2 antenna is obtained from the one-loop virtual correction to the two-parton process $\gamma^* \to q\bar{q}$, whose contributing Feynman diagram is shown in Fig. 4.2. Unlike the tree-level A^0_3 antenna considered above, its construction involves one-loop tensor integrals. As described in Chapter 3, *AntCalc* applies Passarino–Veltman reduction to express these tensor integrals in terms of scalar one-loop functions. For A^1_2, the resulting expression involves the massless two-point (bubble) function B_0 and three-point (triangle) function C_0, whose definitions and analytic expressions are given in Appendix B.2. The unintegrated A^1_2 returned by *AntCalc* is,

```
In[2]:= BuildAntenna[A,2,1]
Out[2]:=
```

$$-\frac{1}{2}\Big[(3+2\epsilon)B_0\Big(q^2;0,0\Big)\Big] - q^2 C_0\Big(0,0,q^2;0,0,0\Big)$$

The corresponding integrated antenna is obtained by evaluating these scalar one-loop functions and integrating the resulting expression over the two-particle phase space. Unlike the real-radiation A^0_3

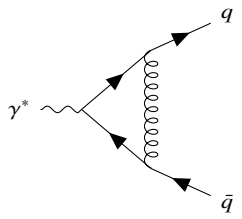


Figure 4.2: The one-loop Feynman diagram contributing to $\gamma^* \to q\bar{q}$ (vertex correction), from which A_2^1 arises. The self-energy corrections on the external quark and antiquark legs are scaleless and vanish for massless on-shell partons in dimensional regularisation.

antenna, no phase-space IBP reduction is required for this two-parton contribution. Using *AntCalc*, the result through $O(\epsilon^2)$ is [5],

```
In[3]:= BuildAndIntegrateAntenna[A,2,1,ExpansionOrder->2]
Out[3]:=
```

$$-\frac{1}{\epsilon^2}-\frac{3}{2\epsilon}-4+\frac{7\pi^2}{12}+\epsilon\left(-8+\frac{7\pi^2}{8}+\frac{7\zeta_3}{3}\right)+\epsilon^2\left(-16+\frac{7\pi^2}{3}-\frac{73\pi^4}{1440}+\frac{7\zeta_3}{2}\right)$$

The Laurent expansion displays the infrared singularity structure of the one-loop virtual correction, with leading $1/\epsilon^2$ and $1/\epsilon$ poles. These poles are equal in magnitude and opposite in sign to those of the integrated tree-level real-radiation antenna $\mathcal{A}_3^0$,

$$\mathcal{A}_2^1\big|_{\text{poles}} = -\frac{1}{\epsilon^2}-\frac{3}{2\epsilon}, \qquad \mathcal{A}_3^0\big|_{\text{poles}} = \frac{1}{\epsilon^2}+\frac{3}{2\epsilon}.$$

Their sum is therefore free of infrared poles, providing an explicit illustration of the cancellation between virtual and integrated real-radiation singularities at NLO discussed in Chapter 2. Terms through $O(\epsilon^2)$ are retained because higher-order coefficients of lower-order antennae can contribute when these quantities multiply singular Laurent series at higher perturbative orders.

4.2 NNLO Two-Parton Antennae: The A_2^2 Set

At NNLO, the two-parton virtual contribution to $\gamma^* \to q\bar{q}$ gives rise to the four components A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$ and $\check{A}_2^2$ defined in Chapter 2: the first three are the colour structures of the tree–two-loop interference, while $\check{A}_2^2$ originates from the square of the one-loop amplitude.

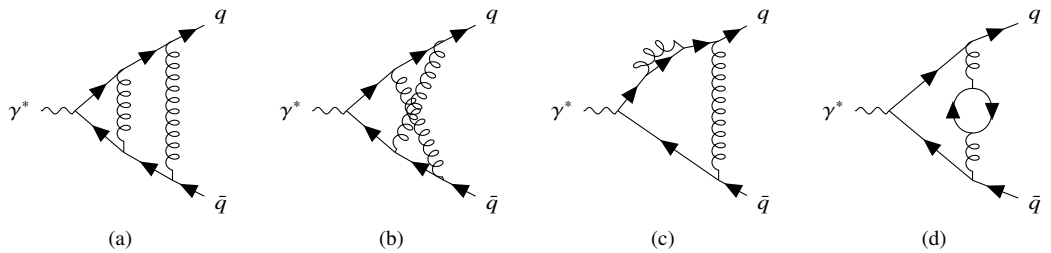


Figure 4.3: Representative two-loop Feynman diagrams for $\gamma^* \to q\bar{q}$: (a) planar and (b) crossed ladder, (c) a quark self-energy insertion, and (d) a closed quark loop on the gluon propagator, which gives rise to the N_f-dependent component $\hat{A}_2^2$. Panel (c) is a self-energy insertion on an internal propagator, which contributes. Self-energy corrections on the external legs and tadpoles are not generated: they are scaleless for massless on-shell partons and vanish in dimensional regularisation.

4.2.1 A_2^2

We begin with the leading-colour component A_2^2. This contribution is obtained from the leading-colour part of the interference between the two-loop and tree-level amplitudes for $\gamma^* \to q\bar{q}$. Representative two-loop Feynman diagrams contributing to this amplitude are shown in Fig. 4.3.

Using *AntCalc*, the unintegrated leading-colour contribution is obtained in the master-integral basis with

```
In[2]:= A22Lead = BuildAndIntegrateAntenna[A,2,2,
Component->Leading,ReturnMasterCombination->True];
In[3]:= (A22Lead/.A22MISubs)//
Collect[#,{A22LO,A3,A4,B0,C0},Simplify]&
Out[3]:=
```

$$-\frac{\left(-2\epsilon^2+\epsilon-2\right)^2}{8\pi^4\epsilon^2}A_{22,\mathrm{LO}}+\frac{24\epsilon^6-40\epsilon^5+76\epsilon^4-88\epsilon^3+68\epsilon^2-33\epsilon+6}{4\pi^4\epsilon^3(2\epsilon-1)}A_3$$
$$+\frac{12\epsilon^6+20\epsilon^5-101\epsilon^4+103\epsilon^3-88\epsilon^2+47\epsilon-6}{4\pi^4\epsilon^2\left(6\epsilon^2-13\epsilon+6\right)}A_4$$
$$+\frac{11\cos(\pi\epsilon)}{6\epsilon}C_0^\dagger+\frac{11(3+2\epsilon)\cos(\pi\epsilon)}{12\epsilon}B_0^\dagger$$

The result above is the unintegrated leading-colour antenna A_2^2, expressed in the integral basis obtained after IBP reduction. *AntCalc* exposes this combination through the `ReturnMasterCombination` option of the integration call because, in its modular build-and-integrate path, the IBP reduction of the loop integrals is carried out inside `IntegrateAntenna`; no phase-space integration has been performed at this stage. The genuine two-loop contribution is written in terms of the master integrals $A_{22,\mathrm{LO}}$, A_3, and A_4, whose definitions and analytic expressions are given in Appendix C.2. The additional one-loop functions $B_0^\dagger$ and $C_0^\dagger$ arise from the coupling-renormalisation contribution proportional to the one-loop antenna A_2^1. Thus, although A_2^2 is a two-loop antenna, its renormalised expression contains both genuine two-loop master integrals and lower-order integrals inherited from ultraviolet renormalisation.

Two further steps are required when substituting the analytic expressions for these integrals. First, the master integrals are expressed using the S_Γ normalisation of Eq. (C.5), while the antenna functions are defined in terms of the S_ϵ normalisation introduced in Chapter 2. The appropriate conversion, $e^{2\gamma_E\epsilon}/\Gamma^2(1-\epsilon)$ together with the constant loop-measure factors that cancel the $1/\pi^4$ above, is therefore applied. Second, the loop-integral expressions contain powers of $-q^2$, which are real in the spacelike region $q^2<0$, whereas the antenna describes the physical timelike decay $\gamma^* \to q\bar{q}$, for which $q^2>0$. The corresponding analytic continuation is performed using the Feynman prescription, $-q^2 \to -q^2 - i0$. In particular, the two-loop dependence gives $(-q^2-i0)^{-2\epsilon} = (q^2)^{-2\epsilon}e^{2i\pi\epsilon}$, which enters as $\cos(2\pi\epsilon)$, since the antenna is defined through the real part of the interference, Eq. (2.40). The counterterm functions already carry their one-loop continuation, $\cos(\pi\epsilon)$, explicitly in `Out[3]`. The two steps are common to the whole set, except that $\check{A}_2^2$ involves no such timelike phase factor: it is the modulus squared of the one-loop amplitude, Eq. (2.59), so the phase $e^{i\pi\epsilon}$ carried by $(-q^2-i0)^{-\epsilon}$ in $\mathcal{M}_2^1$ cancels against the phase $e^{-i\pi\epsilon}$ of its complex conjugate. With these steps applied, `BuildAntenna` returns the unintegrated antenna with the master integrals substituted,

```
In[4]:= BuildAntenna[A,2,2,Component->Leading]
```

```
Out[4]:=
```

$$4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\left[\frac{1}{4\epsilon^4}+\frac{17}{8\epsilon^3}+\cdots\right]$$

where only the leading terms of the bracket are shown. The output is real, with the renormalisation scale set to $\mu^2 = q^2$. The master combination in `Out[3]` already carries the normalisation $C(\epsilon, 2)$ of Eq. (2.69), which `BuildAntenna` removes; this is the origin of the prefactor $1/C(\epsilon, 2)$.

The integrated antenna $\mathcal{A}_2^2$ is subsequently obtained from the definition introduced in Chapter 2, Eqs. (2.68) and (2.69),

$$\mathcal{A}_2^2 = C(\epsilon, 2)\,\frac{1}{\Phi_2}\int d\Phi_2\, A_2^2 = C(\epsilon, 2)\, A_2^2.$$

In the present case, the antenna contains only the two hard radiators, so that the remaining integration is over the $1 \to 2$ two-particle phase space. This phase-space integration is trivial, but it is conceptually the same operation that relates the unintegrated and integrated antenna functions throughout this work. In *AntCalc*, the complete construction and phase-space integration can be performed with `BuildAndIntegrateAntenna`, yielding the integrated antenna through $O(\epsilon^0)$,

```
In[5]:= BuildAndIntegrateAntenna[A,2,2,Component->Leading,
ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{4\epsilon^4}+\frac{17}{8\epsilon^3}+\frac{433-72\pi^2}{144\epsilon^2}+\frac{121350-44820\pi^2+15120\zeta_3}{25920\epsilon}+\frac{-45415-64590\pi^2+4734\pi^4+37440\zeta_3}{25920}$$

The Laurent expansion exhibits the infrared pole structure expected for a two-loop virtual contribution. In particular, the leading $1/\epsilon^4$ singularity represents an increase by two powers of $1/\epsilon$ relative to the one-loop A_2^1 antenna, whose leading singularity is $1/\epsilon^2$. The poles through $1/\epsilon$ encode the overlapping infrared singularities of the two-loop contribution and enter the cancellation of infrared divergences among the double-virtual, real–virtual, and double-real contributions at NNLO. The finite coefficient contains, in addition to rational terms, transcendental constants such as π^2, ζ_3, and π^4, arising from the ϵ-expansion of the corresponding loop and phase-space integrals.

4.2.2 $\tilde{A}_2^2$

The subleading-colour component $\tilde{A}_2^2$ is obtained by selecting the corresponding colour structure of the two-parton NNLO contribution. Using *AntCalc*, its unintegrated form is obtained in the master-integral basis with

```
In[2]:= A22Sub = BuildAndIntegrateAntenna[A,2,2,
Component->Subleading,ReturnMasterCombination->True];
In[3]:= (A22Sub/.A22MISubs)//Collect[#,{A22LO,A3,A4,A6},Simplify]&
```

```
Out[3]:=
```

$$\frac{\left(-2\epsilon^2+\epsilon-2\right)^2}{8\pi^4\epsilon^2}A_{22,\mathrm{LO}}+\frac{108\epsilon^7+780\epsilon^6-2641\epsilon^5+3305\epsilon^4-2476\epsilon^3+1210\epsilon^2-330\epsilon+36}{8\pi^4\epsilon^3\left(8\epsilon^2-6\epsilon+1\right)}A_3$$
$$+\frac{-42\epsilon^6-285\epsilon^5+659\epsilon^4-618\epsilon^3+372\epsilon^2-138\epsilon+20}{8\pi^4\epsilon^2(3\epsilon-2)(4\epsilon-1)}A_4-\frac{\epsilon^3+4\epsilon^2-2\epsilon+2}{8\pi^4(4\epsilon-1)}A_6$$

As for the leading-colour component, the IBP reduction expresses the loop integrals in terms of the master integrals $A_{22,\mathrm{LO}}$, A_3, A_4, and A_6, whose definitions and analytic expressions are presented in Appendix C.2. After the two steps described in Subsection 4.2.1, `BuildAntenna` returns

```
In[4]:= BuildAntenna[A,2,2,Component->Subleading]
Out[4]:=
```

$$4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\left[-\frac{1}{4\epsilon^4}-\frac{3}{4\epsilon^3}+\cdots\right]$$

and the trivial two-particle phase-space integration gives the integrated subleading-colour antenna through $O(\epsilon^0)$,

```
In[5]:= BuildAndIntegrateAntenna[A,2,2,Component->Subleading,
ExpansionOrder->0]
Out[5]:=
```

$$-\frac{1}{4\epsilon^4}-\frac{3}{4\epsilon^3}+\frac{-1476+312\pi^2}{576\epsilon^2}+\frac{-3978+864\pi^2+1536\zeta_3}{576\epsilon}$$
$$+\frac{-10359+2850\pi^2-118\pi^4+4176\zeta_3}{576}$$

4.2.3 $\hat{A}_2^2$

The $\hat{A}_2^2$ antenna denotes the contribution proportional to the number of light-quark flavours N_f. It originates from two-loop diagrams containing a closed quark loop. Representative examples of these diagrams are shown in Fig. 4.3(d). This contribution therefore isolates a distinct flavour and colour structure of the two-loop $q\bar{q}$ form factor. Using *AntCalc*, its unintegrated form is obtained in the master-integral basis with

```
In[2]:= A22Nf = BuildAndIntegrateAntenna[A,2,2,
Component->Nf,ReturnMasterCombination->True];
In[3]:= (A22Nf/.A22MISubs)//Collect[#,{A4,B0,C0},Simplify]&
Out[3]:=
```

$$-\frac{(\epsilon-1)\left(6\epsilon^3-5\epsilon^2+3\epsilon-2\right)}{2\pi^4\epsilon(2\epsilon-3)(3\epsilon-2)}A_4$$
$$-\frac{\cos(\pi\epsilon)}{3\epsilon}C_0^\dagger-\frac{(3+2\epsilon)\cos(\pi\epsilon)}{6\epsilon}B_0^\dagger$$

After IBP reduction, the genuine two-loop part of the N_f contribution reduces to the single master integral A_4, in contrast to the leading- and subleading-colour components considered above. In addition, the counterterm functions $B_0^\dagger$ and $C_0^\dagger$ appear. The definition and analytic evaluation of A_4 are presented in Appendix C.2. After the two steps described in Subsection 4.2.1, `BuildAntenna` returns

```
In[4]:= BuildAntenna[A,2,2,Component->Nf]
Out[4]:=
```

$$4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\left[-\frac{1}{4\epsilon^3}-\frac{1}{9\epsilon^2}+\cdots\right]$$

and the trivial two-particle phase-space integration gives the integrated quark-loop antenna through $O(\epsilon^0)$,

```
In[5]:= BuildAndIntegrateAntenna[A,2,2,Component->Nf,
ExpansionOrder->0]
Out[5]:=
```

$$-\frac{1}{4\epsilon^3}-\frac{1}{9\epsilon^2}+\frac{65+9\pi^2}{216\epsilon}+\frac{4085-546\pi^2+72\zeta_3}{1296}$$

Unlike the leading- and subleading-colour components, the Laurent expansion of $\hat{A}_2^2$ begins at $1/\epsilon^3$, so the N_f-dependent component does not contribute to the leading quartic pole.

4.2.4 $\breve{A}_2^2$

The $\breve{A}_2^2$ antenna originates from the one-loop-squared contribution to the NNLO two-parton matrix element,

$$\mathcal{M}_2^{1\dagger}\mathcal{M}_2^1,$$

in contrast to the A_2^2, $\tilde{A}_2^2$, and $\hat{A}_2^2$ components discussed above, which arise from the interference of the two-loop amplitude with the tree-level amplitude. Using *AntCalc*, the unintegrated antenna is obtained in the master-integral basis with

```
In[2]:= A22Breve = BuildAndIntegrateAntenna[A,2,2,
Component->Breve,ReturnMasterCombination->True];
In[3]:= (A22Breve/.A22MISubs)//Collect[#,A22LO,Simplify]&
Out[3]:=
```

$$\frac{\left(-2\epsilon^2+\epsilon-2\right)^2}{16\pi^4\epsilon^2}A_{22,\mathrm{LO}}$$

which is expressed through the single master integral $A_{22,\mathrm{LO}}$, whose definition and analytic evaluation are presented in Appendix C.2. `Out[3]` is quoted before *AntCalc*'s conversion factor between the raw and the public master normalisations, which is exactly $-1/\cos(2\pi\epsilon)$ and is included when the master integral is substituted. After the two steps described in Subsection 4.2.1, `BuildAntenna` returns

```
In[4]:= BuildAntenna[A,2,2,Component->Breve]
Out[4]:=
```

$$4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\left[\frac{1}{4\epsilon^4}+\frac{3}{4\epsilon^3}+\cdots\right]$$

where only part of the output is displayed; the full expression is collected in Appendix A. The trivial two-particle phase-space integration then gives the integrated antenna through $O(\epsilon^0)$,

```
In[5]:= BuildAndIntegrateAntenna[A,2,2,Component->Breve,
ExpansionOrder->0]
```

```
Out[5]:=
```

$$\frac{1}{4\epsilon^4}+\frac{3}{4\epsilon^3}+\frac{1}{\epsilon^2}\left(\frac{41}{16}-\frac{\pi^2}{24}\right)+\frac{1}{\epsilon}\left(7-\frac{\pi^2}{8}-\frac{7\zeta_3}{6}\right)$$
$$+18-\frac{41\pi^2}{96}-\frac{7\pi^4}{480}-\frac{7\zeta_3}{2}$$

The leading $1/\epsilon^4$ pole reflects the one-loop-squared origin of $\check{A}_2^2$: the one-loop $q\bar{q}$ amplitude has infrared poles beginning at $1/\epsilon^2$, so its square begins at $1/\epsilon^4$.

The integrated Laurent series agree with the established $\mathcal{A}_2^2$ set results [25] apart from the exception discussed below.

The integrated $\check{A}_2^2$ obtained directly with *AntCalc* was found to differ from the expression given in Eq. (4.10) of Ref. [25] in the sign of the $7\zeta_3/(6\epsilon)$ term. Since *AntCalc* constructs this contribution independently from the one-loop-squared matrix element, this discrepancy prompted an additional check of the published expression.

An independent indication of the same sign discrepancy arises in the NNLO massless hadronic R-ratio calculation presented in Chapter 5. When the expression printed in Ref. [25] is used, the infrared poles do not cancel completely. Decomposing the remaining pole according to its colour structure traces the mismatch to the $\check{A}_2^2$ contribution, whereas the result obtained with *AntCalc* yields the required cancellation.

The origin of the discrepancy appears to be a typographical error in the published expression of Ref. [25], specifically in the sign of the $7\zeta_3/(6\epsilon)$ term. The corrected sign is also documented independently by Jakubčík, Marcoli, and Stagnitto in Eq. (B.7) of Ref. [27].

4.3 NNLO Three-Parton Antennae: The A_3^1 Set

At NNLO, the three-parton contribution to the quark–antiquark antenna family is provided by the one-loop A_3^1 antenna set. These antennae are derived from the interference between the one-loop and tree-level amplitudes for

$$\gamma^*(q) \to q(k_1)\, g(k_3)\, \bar{q}(k_2),$$

and therefore constitute the real–virtual counterpart of the tree-level A_3^0 antenna discussed in Subsection 4.1.1.

The colour decomposition of the complete tree–one-loop interference gives rise to three antenna components: A_3^1 is the leading-colour component, $\tilde{A}_3^1$ the subleading-colour component, and $\hat{A}_3^1$ the N_f-dependent UV-renormalisation contribution. Unlike the A_2^2 antenna set, there is no separate one-loop-squared contribution at this perturbative order: the A_3^1 set is entirely obtained from the tree–one-loop interference.

Unlike the purely virtual two-parton antennae of the previous section, their integrated forms require the one-loop three-parton contribution to be integrated over the $1 \to 3$ antenna phase space. The three components are considered in turn below.

4.3.1 A_3^1

We begin with the leading-colour component A_3^1. Representative one-loop Feynman diagrams entering its construction are shown in Fig. 4.4. Before defining the antenna function, the one-loop contribution

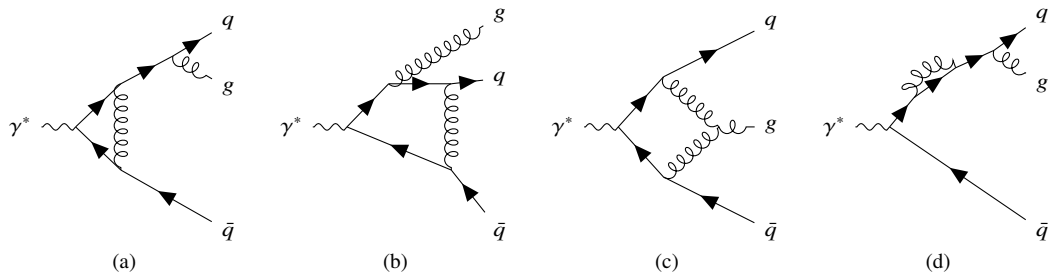


Figure 4.4: Representative one-loop Feynman diagrams for $\gamma^* \to qg\bar{q}$: (a) a vertex correction, (b) a box, in which the loop gluon spans the emission of the real gluon, (c) a diagram with the three-gluon vertex, and (d) a self-energy insertion on the internal quark propagator, which contributes. Self-energy corrections on the external legs and tadpoles are not generated: they are scaleless for massless on-shell partons and vanish in dimensional regularisation.

must be ultraviolet renormalised. In dimensional regularisation, the unrenormalised one-loop matrix element contains both ultraviolet and infrared poles in ϵ. Following the renormalisation procedure described by Eq. (2.22), the ultraviolet poles are removed through renormalisation of the strong coupling in the $\overline{\text{MS}}$ scheme. The resulting one-loop antenna is therefore UV finite, although it retains the infrared poles associated with the one-loop amplitude. The corresponding UV-renormalised unintegrated antenna is obtained with

```
In[2]:= BuildAntenna[A,3,1,Component->Leading]
Out[2]:=
```

$$\frac{22s_{12}^6 - 22\epsilon s_{12}^6 + 66s_{12}^5 s_{13} - 66\epsilon s_{12}^5 s_{13} + \cdots}{6(\epsilon-1)\epsilon\, q^2 s_{13}s_{23}(s_{12}+s_{13})^2(s_{12}+s_{23})^2}$$

For readability, only part of the complete unintegrated expression is displayed here; the full expression is retained by *AntCalc* for the subsequent integration.

As a one-loop matrix element, the unintegrated A_3^1 antenna contains explicit infrared poles in ϵ originating from the loop integration, with singularities extending up to $1/\epsilon^2$. To obtain the integrated antenna $\mathcal{A}_3^1$, this expression must additionally be integrated over the corresponding $1 \to 3$ antenna phase space. The resulting integrals are reduced by IBP identities to the master-integral basis required for this antenna. Retaining the master-integral representation, this calculation is performed in *AntCalc* using

```
In[3]:= A31Lead = BuildAndIntegrateAntenna[A,3,1,
Component->Leading,ReturnMasterCombination->True];
In[4]:= (A31Lead/.A31MISubs)//Collect[#,{R3,V5a,V5b,V8},Simplify]&
Out[4]:=
```

$$\frac{11\cdot 4^{\epsilon-2}e^{-\gamma_E\epsilon}\left(2\epsilon^3-5\epsilon^2+6\epsilon-2\right)\pi^{\epsilon-4}}{3\epsilon^3}R_3$$
$$-\frac{i\left(24\epsilon^6-40\epsilon^5+76\epsilon^4-88\epsilon^3+68\epsilon^2-33\epsilon+6\right)}{2\pi^2\epsilon^3(2\epsilon-1)}V_{5,a}$$
$$+\frac{i\left(9\epsilon^5+25\epsilon^4-34\epsilon^3+41\epsilon^2-20\epsilon+4\right)}{2\pi^2\epsilon^3}V_{5,b}$$

which expresses the integrated antenna in terms of the relevant master integrals. The first term originates from ultraviolet renormalisation of the one-loop antenna. More specifically, it is the leading-colour contribution from the renormalisation of the strong coupling, proportional to the N-dependent part of the first QCD beta-function coefficient,

$$\beta_0 = \frac{11N - 2N_f}{6}.$$

After separating the leading-colour factor N, the corresponding counterterm contributes $-11/(6\epsilon)$ times the tree-level antenna A_3^0. Since the latter has already been integrated over the three-particle phase space, this contribution appears in the master-integral representation through R_3, the phase-space master integral introduced in Subsection 4.1.1. The remaining terms originate from the genuine one-loop contribution to the real–virtual antenna and are reduced to the master integrals $V_{5,a}$ and $V_{5,b}$. For the leading-colour component, the coefficient of V_8 vanishes, so that V_8 does not contribute to the result. The definitions and analytic expressions of these master integrals are presented in Appendix C.3. The masters are evaluated at $q^2 = 1$, and the combination is shown up to an overall normalisation common to the three components.

Substituting the analytic expressions for the master integrals, applying the appropriate normalisation, and expanding in ϵ, the integrated leading-colour antenna through $O(\epsilon^0)$ is obtained with

```
In[5]:= BuildAndIntegrateAntenna[A,3,1,Component->Leading,
ExpansionOrder->0]
Out[5]:=
```

$$-\frac{1}{4\epsilon^4}-\frac{31}{12\epsilon^3}+\frac{-9540+660\pi^2}{1440\epsilon^2}+\frac{-38820+3520\pi^2+11040\zeta_3}{1440\epsilon}$$
$$+\frac{-156930+12240\pi^2-123\pi^4+55120\zeta_3}{1440}+O(\epsilon)$$

The increase in the maximal pole order relative to the unintegrated antenna can be understood from the additional integration over the $1 \to 3$ antenna phase space. In addition to the explicit infrared poles generated by the one-loop integration, the phase-space integration includes unresolved regions in which the final-state gluon becomes soft or collinear with either of the hard quark radiators. Integration over these singular regions generates additional poles in ϵ, resulting in poles up to $1/\epsilon^4$ in the integrated antenna. The pole structure therefore reflects the combined loop and unresolved phase-space singularities of the NNLO real–virtual contribution and enters the cancellation of infrared singularities among the NNLO subtraction terms.

4.3.2 $\tilde{A}_3^1$

The subleading-colour component of the one-loop three-parton antenna set is denoted by $\widetilde{A}_3^1$. Following the colour decomposition described above, its unintegrated form is obtained with *AntCalc* using

```
In[2]:= BuildAntenna[A,3,1,Component->Subleading]
Out[2]:=
```

$$-\frac{4s_{12}^6}{(\epsilon-1)q^2(s_{12}+s_{13})^2(s_{12}+s_{23})^2(s_{13}+s_{23})^2}B_0\left(q^2;0,0\right)+\cdots$$

The subsequent integration and IBP reduction express the subleading-colour contribution in terms of the master integrals $V_{5,a}$, $V_{5,b}$, and V_8. The corresponding master-integral representation is obtained with

```
In[3]:= A31Sub = BuildAndIntegrateAntenna[A,3,1,
Component->Subleading,ReturnMasterCombination->True];
In[4]:= (A31Sub/.A31MISubs)//Collect[#,{V5a,V5b,V8},Simplify]&
```

```
Out[4]:=
```

$$\frac{i\left(\epsilon^3+4\epsilon^2-2\epsilon+2\right)}{2\pi^2(4\epsilon-1)}V_8$$
$$-\frac{i\left(108\epsilon^7+780\epsilon^6-2641\epsilon^5+3305\epsilon^4-2476\epsilon^3+1210\epsilon^2-330\epsilon+36\right)}{4\pi^2\epsilon^3\left(8\epsilon^2-6\epsilon+1\right)}V_{5,a}$$
$$+\frac{i\left(49\epsilon^5+125\epsilon^4-242\epsilon^3+185\epsilon^2-94\epsilon+20\right)}{2\pi^2\epsilon^3}V_{5,b}$$

Substituting the master integrals, applying the appropriate normalisation, and expanding through $O(\epsilon^0)$ gives

```
In[5]:= BuildAndIntegrateAntenna[A,3,1,Component->Subleading,
ExpansionOrder->0]
Out[5]:=
```

$$\frac{-15+4\pi^2}{24\epsilon^2}+\frac{-19+\pi^2+28\zeta_3}{4\epsilon}+\frac{1215\pi^2+56\pi^4+540(-35+18\zeta_3)}{720}$$

In contrast to the leading-colour result, the integrated subleading-colour antenna contains no $1/\epsilon^4$ or $1/\epsilon^3$ poles. This reduced pole degree originates from cancellations in the infrared structure of $\tilde{A}_3^1$, including those associated with the $A_2^1 A_3^0$ subtraction required in the definition of the one-loop antenna. After this subtraction, the $1/\epsilon^2$ loop poles cancel in the subleading-colour component, leaving $1/\epsilon$ terms proportional to logarithms of ratios of invariants. In the soft-gluon limit, these logarithmic factors suppress the leading soft behaviour, while the quark–gluon collinear singularities remain. Their phase-space integration therefore generates one additional $1/\epsilon$ pole, such that the integrated $\tilde{\mathcal{A}}_3^1$ begins at $1/\epsilon^2$ [5].

4.3.3 $\hat{A}_3^1$

The N_f-dependent component $\hat{A}_3^1$ has a different origin from the leading- and subleading-colour components considered above. The unrenormalised tree–one-loop interference contains no N_f contribution. The non-zero $\hat{A}_3^1$ antenna therefore arises entirely from the N_f-dependent part of the UV coupling-renormalisation term, which is proportional to the tree-level antenna A_3^0. *AntCalc* incorporates this renormalisation contribution directly when the N_f component is selected,

```
In[2]:= BuildAntenna[A,3,1,Component->Nf]
Out[2]:=
```

$$\frac{1}{3\epsilon q^2}\left(\frac{2s_{12}^2}{s_{13}s_{23}}+\frac{2s_{12}}{s_{13}}+\frac{2s_{12}}{s_{23}}+\frac{s_{13}}{s_{23}}+\frac{s_{23}}{s_{13}}\right)-\frac{(s_{13}+s_{23})^2}{3q^2s_{13}s_{23}}$$

yielding the corresponding $\hat{A}_3^1$ antenna, which is exactly $A_3^0/(3\epsilon)$. Since the unrenormalised one-loop interference contains no N_f component, there is no N_f-dependent one-loop contribution to reduce to the $V_{5,a}$, $V_{5,b}$, and V_8 master integrals appearing in the leading- and subleading-colour components. Instead, because the renormalisation contribution is proportional to A_3^0, the integration of $\hat{A}_3^1$ follows directly from the tree-level calculation presented in Subsection 4.1.1. In particular, its phase-space integration inherits the master integral R_3 appearing in the integrated tree-level antenna $\mathcal{A}_3^0$.

The integrated N_f component is obtained directly with

```
In[5]:= BuildAndIntegrateAntenna[A,3,1,Component->Nf,
ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{3\epsilon^3}+\frac{1}{2\epsilon^2}+\frac{114-14\pi^2}{72\epsilon}+\frac{327-21\pi^2-200\zeta_3}{72}+O(\epsilon)$$

The pole structure is consistent with this renormalisation origin. The UV coupling-renormalisation term contains a pole in ϵ and multiplies the tree-level antenna A_3^0. After integration over the $1 \to 3$ antenna phase space, the infrared poles of $\mathcal{A}_3^0$ therefore generate poles of higher order in $\hat{\mathcal{A}}_3^1$, with the result above beginning at $1/\epsilon^3$.

4.4 NNLO Four-Parton Antennae

At NNLO, double-real radiation contributions involve tree-level final states containing two additional partons relative to the underlying two-parton $q\bar{q}$ configuration. For the massless final-final quark–antiquark antennae considered in this work, the corresponding four-parton antenna functions are obtained from the processes

$$\gamma^* \to q\bar{q}gg, \qquad \gamma^* \to q\bar{q}q'\bar{q}',$$

with the appropriate colour and flavour decompositions. These antennae describe the double-real contribution and contain the single- and double-unresolved limits associated with the four-parton final states.

The $\gamma^* \to q\bar{q}gg$ process gives rise to the leading- and subleading-colour antennae A_4^0 and $\widetilde{A}_4^0$. Four-quark final states give rise to the B_4^0 and C_4^0 antennae, with the latter associated with the interference between the different fermion assignments present for identical quarks, as discussed in Chapter 2. Together, these constitute the four-parton antennae calculated and integrated in this section.

In contrast to the real–virtual A_3^1 antennae of Section 4.3, the four-parton antennae are tree-level quantities and therefore contain no explicit $1/\epsilon$ poles before phase-space integration; these are generated when the integration over the $1 \to 4$ antenna phase space reaches the regions where one or two partons become unresolved. For each antenna, *AntCalc* maps the terms of the unintegrated expression onto the integral families of Appendix C.4 for IBP reduction. The full expressions are collected in Appendix A, and their local infrared validation is given in Appendix B.8.

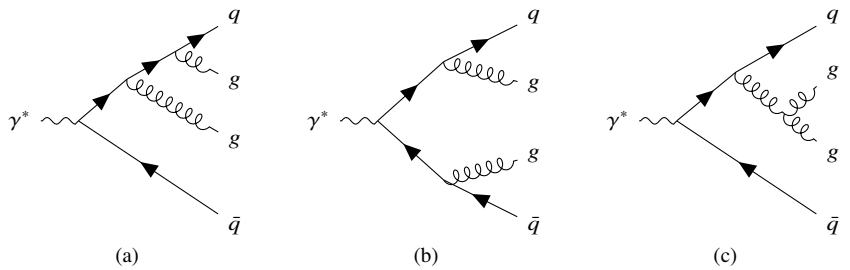


Figure 4.5: Representative tree-level Feynman diagrams for $\gamma^* \to qgg\bar{q}$: (a) both gluons emitted from the quark line, (b) one gluon emitted from each of the quark and antiquark lines, and (c) a diagram with the three-gluon vertex.

4.4.1 A_4^0

We begin with the leading-colour four-parton antenna A_4^0, obtained from the tree-level process

$$\gamma^*(q) \to q(k_1)\, g(k_3)\, g(k_4)\, \bar{q}(k_2).$$

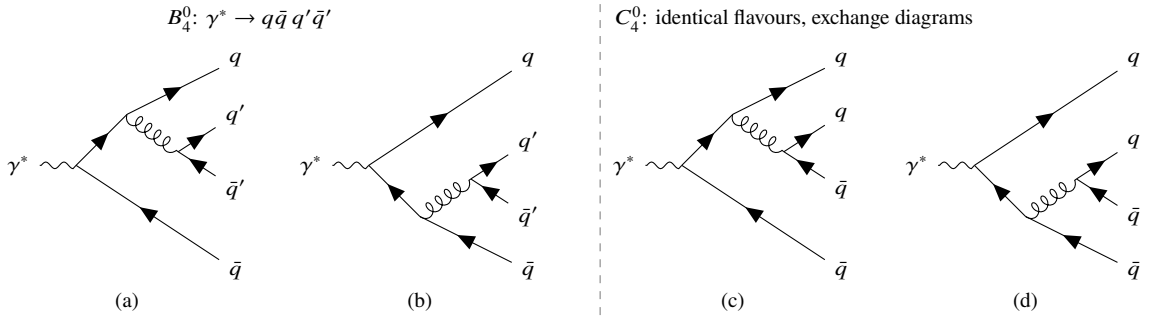


Figure 4.6: Representative tree-level Feynman diagrams for $\gamma^* \to q\bar{q}\,q'\bar{q}'$ (B_4^0), in which the secondary pair is produced by gluon splitting from (a) the quark line or (b) the antiquark line. For identical flavours ($q' = q$), the exchange diagrams (c) and (d), with the final-state quarks or antiquarks interchanged, also contribute, and C_4^0 arises from their interference with the direct diagrams.

Representative tree-level Feynman diagrams contributing to this process are shown in Fig. 4.5. As discussed in Chapter 2, the complete squared matrix element contains different colour structures. The leading-colour contribution, after normalisation to the corresponding two-parton Born matrix element, defines the A_4^0 antenna. The corresponding unintegrated leading-colour antenna is constructed in *AntCalc* using,

```
In[2]:= BuildAntenna[A,4,0,Component->Leading]
Out[2]:=
```

$$-\frac{3s_{12}}{2(-1+\epsilon)q^2s_{134}^2}+\frac{5\epsilon s_{12}}{2(-1+\epsilon)q^2s_{134}^2}-\frac{2\epsilon^2 s_{12}}{(-1+\epsilon)q^2s_{134}^2}+\frac{\epsilon^3 s_{12}}{(-1+\epsilon)q^2s_{134}^2}$$
$$+\frac{1}{(-1+\epsilon)q^2s_{134}}+\frac{2\epsilon}{(-1+\epsilon)q^2s_{134}}-\frac{11\epsilon^2}{2(-1+\epsilon)q^2s_{134}}+\frac{\epsilon^3}{(-1+\epsilon)q^2s_{134}}+\cdots$$

The unintegrated A_4^0 antenna contains the single- and double-unresolved limits of the $qgg\bar{q}$ final state. Its infrared poles arise only after integration over the $1 \to 4$ antenna phase space, where the integration probes the soft and collinear regions of the four-parton final state. To perform this integration analytically, the terms in A_4^0 are mapped onto the corresponding four-particle phase-space integral family and reduced by IBP identities. The resulting master-integral representation is obtained with

```
In[3]:= A40Lead = BuildAndIntegrateAntenna[A,4,0,
Component->Leading,ReturnMasterCombination->True];
In[4]:= (A40Lead/.X40MISubs)//Collect[#,{R4,R6,R8a,R8b},Simplify]&
Out[4]:=
```

$$\left[\frac{1632\epsilon^{10}-13528\epsilon^9+55268\epsilon^8-136894\epsilon^7+220017\epsilon^6}{2(1-2\epsilon)^2\epsilon^4\left(6\epsilon^2-13\epsilon+6\right)\left(q^2\right)^2}\right.$$
$$\left.+\frac{-238829\epsilon^5+177763\epsilon^4-89380\epsilon^3+28844\epsilon^2-5352\epsilon+432}{2(1-2\epsilon)^2\epsilon^4\left(6\epsilon^2-13\epsilon+6\right)\left(q^2\right)^2}\right]R_4$$
$$+\frac{-12\epsilon^6-20\epsilon^5+101\epsilon^4-103\epsilon^3+88\epsilon^2-47\epsilon+6}{\epsilon^2\left(6\epsilon^2-13\epsilon+6\right)}R_6$$

The integrated antenna is thereby reduced to the master integrals R_4 and R_6, whose definitions and analytic evaluation are discussed in Appendix C.4. Substituting the analytic expressions for these master

integrals, applying the normalisation of Eq. (2.21), and expanding around $\epsilon = 0$, the integrated antenna through $O(\epsilon^0)$ is obtained using

```
In[5]:= BuildAndIntegrateAntenna[A,4,0,Component->Leading,
ExpansionOrder->0]
Out[5]:=
```

$$\frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{217}{18} - \frac{13\pi^2}{12}\right) + \frac{1}{\epsilon}\left(\frac{43223}{864} - \frac{589\pi^2}{144} - \frac{71\zeta_3}{4}\right) + \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon)$$

The leading $1/\epsilon^4$ pole is consistent with the double-unresolved singular structure of A_4^0. In particular, the $qgg\bar{q}$ final state contains configurations in which the two gluons become simultaneously soft, in addition to the corresponding double- and triple-collinear limits. These singular regions contribute to the pole structure of the integrated antenna required in the NNLO double-real subtraction term.

4.4.2 $\tilde{A}_4^0$

The subleading-colour contribution to the $q\bar{q}gg$ antenna set is described by $\widetilde{A}_4^0$. It is obtained from the same tree-level process considered in Subsection 4.4.1, but corresponds to the subleading term in the colour decomposition of the squared matrix element. The unintegrated antenna is obtained in *AntCalc* by selecting the subleading-colour component,

```
In[2]:= BuildAntenna[A,4,0,Component->Subleading]
Out[2]:=
```

$$-\frac{\epsilon}{(-1+\epsilon)q^2 s_{13}} + \frac{\epsilon^2}{(-1+\epsilon)q^2 s_{13}} - \frac{8 s_{12}}{(-1+\epsilon)q^2 s_{134}^2} + \frac{10\epsilon s_{12}}{(-1+\epsilon)q^2 s_{134}^2} - \frac{2\epsilon^2 s_{12}}{(-1+\epsilon)q^2 s_{134}^2} + \frac{2}{(-1+\epsilon)q^2 s_{134}} - \frac{2\epsilon^2}{(-1+\epsilon)q^2 s_{134}} + \frac{s_{12}}{(-1+\epsilon)q^2 s_{13} s_{134}} + \cdots$$

Only part of the complete expression is displayed; the full result is given in Appendix A. As for the leading-colour component, integration over the four-particle antenna phase space is performed by mapping the unintegrated expression onto the corresponding integral family and applying IBP reduction. The master-integral representation is obtained with,

```
In[3]:= A40Sub = BuildAndIntegrateAntenna[A,4,0,
Component->Subleading,ReturnMasterCombination->True];
In[4]:= (A40Sub/.X40MISubs)//Collect[#,{R4,R6,R8a,R8b},Simplify]&
```

```
Out[4]:=
```

$$\left[\frac{26016\epsilon^{10}-21800\epsilon^9-124084\epsilon^8+326554\epsilon^7-392893\epsilon^6}{2(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)\left(q^2\right)^2}\right.$$
$$\left.+\frac{297255\epsilon^5-155324\epsilon^4+56744\epsilon^3-13782\epsilon^2+1960\epsilon-120}{2(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)\left(q^2\right)^2}\right]R_4$$
$$+\frac{\left(\epsilon^3+4\epsilon^2-2\epsilon+2\right)\left(q^2\right)^2}{2-8\epsilon}R_{8,a}+\frac{2\left(\epsilon^3+4\epsilon^2-2\epsilon+2\right)\left(q^2\right)^2}{2-8\epsilon}R_{8,b}$$
$$+\frac{-138\epsilon^6-341\epsilon^5+687\epsilon^4-620\epsilon^3+420\epsilon^2-150\epsilon+20}{24\epsilon^4-22\epsilon^3+4\epsilon^2}R_6$$

In contrast to the leading-colour antenna, whose reduction involves R_4 and R_6, the subleading-colour contribution additionally requires the master integrals $R_{8,a}$ and $R_{8,b}$, discussed in Appendix C.4. Substituting their analytic expressions, applying the appropriate normalisation, and expanding through $O(\epsilon^0)$ gives

```
In[5]:= BuildAndIntegrateAntenna[A,4,0,Component->Subleading,
ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{\epsilon^4}+\frac{3}{\epsilon^3}+\frac{1}{\epsilon^2}\left(13-\frac{3\pi^2}{2}\right)+\frac{1}{\epsilon}\left(\frac{845}{16}-\frac{9\pi^2}{2}-\frac{80\zeta_3}{3}\right)$$
$$+\frac{6921}{32}-\frac{473\pi^2}{24}+\frac{17\pi^4}{72}-80\zeta_3+O(\epsilon)$$

4.4.3 B_4^0

We now turn to the four-quark contribution to the NNLO double-real antennae. The B_4^0 antenna is obtained from the tree-level process

$$\gamma^* \to q\bar{q}q'\bar{q}',$$

where q' denotes a quark flavour different from that of the primary $q\bar{q}$ pair. Representative tree-level Feynman diagrams contributing to this process are shown in Fig. 4.6. In these contributions, the secondary $q'\bar{q}'$ pair is produced through gluon splitting, $g^* \to q'\bar{q}'$. After the colour and flavour decomposition described in Chapter 2 and normalisation to the corresponding two-parton Born matrix element, the resulting contribution defines the B_4^0 antenna. Its unintegrated form is constructed in *AntCalc* using,

```
In[2]:= BuildAntenna[B,4,0]
Out[2]:=
```

$$-\frac{\epsilon s_{12}}{(-1+\epsilon)q^2s_{134}^2}+\frac{\epsilon^2 s_{12}}{(-1+\epsilon)q^2s_{134}^2}-\frac{\epsilon s_{12}}{(-1+\epsilon)q^2s_{234}^2}+\frac{\epsilon^2 s_{12}}{(-1+\epsilon)q^2s_{234}^2}$$
$$-\frac{2s_{12}}{(-1+\epsilon)q^2s_{134}s_{234}}+\frac{2\epsilon s_{12}}{(-1+\epsilon)q^2s_{134}s_{234}}+\frac{2\epsilon^2 s_{12}}{(-1+\epsilon)q^2s_{134}s_{234}}-\frac{2s_{12}s_{13}s_{14}}{(-1+\epsilon)q^2s_{134}^2s_{34}^2}+\cdots$$

The integrated antenna is obtained by integrating B_4^0 over the four-particle antenna phase space and reducing the resulting integrals by IBP identities. The master-integral representation is obtained with

```
In[3]:= B40 = BuildAndIntegrateAntenna[B,4,0,
ReturnMasterCombination->True];
In[4]:= (B40/.X40MISubs)//Collect[#,{R4,R6,R8a,R8b},Simplify]&
Out[4]:=
```

$$\frac{2\left(6\epsilon^4-11\epsilon^3+8\epsilon^2-5\epsilon+2\right)}{\epsilon\left(6\epsilon^2-13\epsilon+6\right)}R_6$$
$$-\frac{2\left(48\epsilon^6-184\epsilon^5+335\epsilon^4-372\epsilon^3+249\epsilon^2-88\epsilon+12\right)}{\epsilon^3\left(6\epsilon^2-13\epsilon+6\right)\left(q^2\right)^2}R_4$$

For B_4^0, the IBP reduction involves the master integrals R_4 and R_6. Substituting these expressions, applying the appropriate normalisation, and expanding through $O(\epsilon^0)$ we obtain,

```
In[5]:= BuildAndIntegrateAntenna[B,4,0,ExpansionOrder->0]
Out[5]:=
```

$$-\frac{1}{12\epsilon^3}-\frac{7}{18\epsilon^2}+\frac{1}{\epsilon}\left(-\frac{407}{216}+\frac{11\pi^2}{72}\right)-\frac{11753}{1296}+\frac{77\pi^2}{108}+\frac{67\zeta_3}{18}+O(\epsilon)$$

4.4.4 C_4^0

The C_4^0 antenna describes the identical-flavour four-quark contribution. In this case, the identical quarks allow different fermion assignments in the tree-level amplitude, and C_4^0 arises from the interference between these assignments. This distinguishes it from the B_4^0 antenna considered in the previous subsection, which describes the non-identical-flavour contribution. The unintegrated antenna is constructed in *AntCalc* using,

```
In[2]:= BuildAntenna[C,4,0]
Out[2]:=
```

$$-\frac{\epsilon s_{12}}{2(-1+\epsilon)q^2 s_{123}s_{134}}+\frac{\epsilon^2 s_{12}}{(-1+\epsilon)q^2 s_{123}s_{134}}+\frac{\epsilon^3 s_{12}}{2(-1+\epsilon)q^2 s_{123}s_{134}}+\frac{s_{13}}{(-1+\epsilon)q^2 s_{123}s_{134}}$$
$$-\frac{2\epsilon s_{13}}{(-1+\epsilon)q^2 s_{123}s_{134}}+\frac{\epsilon^3 s_{13}}{(-1+\epsilon)q^2 s_{123}s_{134}}-\frac{\epsilon s_{14}}{2(-1+\epsilon)q^2 s_{123}s_{134}}+\frac{\epsilon^2 s_{14}}{(-1+\epsilon)q^2 s_{123}s_{134}}+\cdots$$

The integration over the four-particle antenna phase space is then performed using the same IBP-reduction framework as for the preceding four-parton antennae. The corresponding master-integral representation is obtained from,

```
In[3]:= C40 = BuildAndIntegrateAntenna[C,4,0,
ReturnMasterCombination->True];
In[4]:= (C40/.X40MISubs)//Collect[#,{R4,R6,R8a,R8b},Simplify]&
```

```
Out[4]:=
```

$$\left[\frac{4128\epsilon^{10}+88\epsilon^9-44412\epsilon^8+115994\epsilon^7-154689\epsilon^6}{8(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)\left(q^2\right)^2}+\frac{132651\epsilon^5-78114\epsilon^4+31324\epsilon^3-8058\epsilon^2+1176\epsilon-72}{8(1-2\epsilon)^2\epsilon^4(3\epsilon-2)(4\epsilon-1)\left(q^2\right)^2}\right]R_4+\frac{\left(\epsilon^3+4\epsilon^2-2\epsilon+2\right)\left(q^2\right)^2}{4-16\epsilon}R_{8,b}+\frac{54\epsilon^6-229\epsilon^5+631\epsilon^4-616\epsilon^3+324\epsilon^2-126\epsilon+20}{96\epsilon^4-88\epsilon^3+16\epsilon^2}R_6$$

After substituting the analytic master-integral expressions and expanding through $O(\epsilon^0)$ we obtain,

```
In[5]:= BuildAndIntegrateAntenna[C,4,0,ExpansionOrder->0]
Out[5]:=
```

$$\frac{1}{\epsilon}\left(-\frac{13}{32}+\frac{\pi^2}{16}-\frac{\zeta_3}{4}\right)-\frac{339}{64}+\frac{17\pi^2}{48}-\frac{\pi^4}{45}+\frac{21\zeta_3}{8}+O(\epsilon)$$

The single $1/\epsilon$ pole has a direct interpretation in terms of the infrared structure of C_4^0. As an interference antenna, its only non-vanishing unresolved limit is the triple-collinear limit, while all other single- and double-unresolved limits vanish. Accordingly, no higher-order infrared poles are generated upon integration.

4.5 Exploratory Massive Extension

The massive A_3^0 antenna provides an exploratory extension of *AntCalc* beyond the massless antenna functions considered in the preceding sections. It is currently the only massive antenna route implemented in the package. The overall construction follows the same strategy as for the massless A_3^0 antenna, but the presence of a non-zero quark mass modifies both the kinematics and the integral families required for the phase-space integration; the general consequences of this extension for the antenna subtraction and reduction framework are discussed in Appendix B.6.

For the process

$$\gamma^*(q) \to Q(k_1)\,\bar{Q}(k_2)\,g(k_3),$$

the final-state momenta satisfy

$$k_1^2 = k_2^2 = m_q^2, \qquad k_3^2 = 0.$$

In contrast to the massless case, the kinematics therefore depends on two dimensionful scales, q^2 and m_q^2. Their ratio can conveniently be expressed through the dimensionless threshold variable

$$r = 1 - \frac{4m_q^2}{q^2}. \tag{4.40}$$

Thus $r = 0$ corresponds to the heavy-quark production threshold, while $r \to 1$ approaches the massless limit.

The non-zero quark mass also modifies the cut propagators used to represent the phase-space constraints in the IBP reduction. With the conventions adopted in *AntCalc*, the two massive cut denominators and the massless-gluon cut denominator are

$$D_1 = m_q^2 - k_1^2, \qquad D_2 = m_q^2 - k_2^2, \qquad D_3 = k_3^2. \tag{4.41}$$

We retain the convention $s_{ij} = 2k_i \cdot k_j$. For massive external momenta this may equivalently be written as,

$$s_{ij} = (q - k_k)^2 - m_i^2 - m_j^2, \quad \{i, j, k\} = \{1, 2, 3\}. \tag{4.42}$$

The massive on-shell conditions define an integral family different from the massless A_3^0 family. The corresponding family is therefore implemented separately in *LiteRed2* and reduced through IBP identities. The reduction yields two master integrals. However, the master basis selected by *LiteRed2* is not identical to the basis convention used for the analytic results in the literature. A change of master-integral basis is therefore required before the known analytic expressions can be substituted.

Denoting the two masters selected by the IBP reduction by $J_1^{(m_q,0,m_q)}$ and $J_2^{(m_q,0,m_q)}$, the corresponding *LiteRed2* representation is given below. The complete basis, named `MX30Basis123`, uses the denominators $\{m_q^2 - k_1^2, m_q^2 - k_2^2, k_3^2, -s_{13}, -s_{23}\}$ in that order:

$$\begin{aligned} J_1^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 1, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1 D_2 D_3}\right], \\ J_2^{(m_q,0,m_q)} &= j[\text{MX30Basis123}, 2, 1, 1, 0, 0] = \mathcal{N}_{\text{LiteRed2}} \int \text{Cut}\left[\frac{1}{D_1^2 D_2 D_3}\right]. \end{aligned} \tag{4.43}$$

Here, $\mathcal{N}_{\text{LiteRed2}}$ represents the normalisation implicit in the cut-integral convention used for the family in *LiteRed2*. The *LiteRed2* masters J_1 and J_2 must subsequently be converted to the master-integral basis I_1 and I_2 used for the available analytic expressions. The change of basis follows from the explicit reduction of the numerator master $I_2^{(m_q,0,m_q)}$ in the *LiteRed2* family. The unintegrated massive antenna includes the explicit numerator factor of 4 of Eq. (3.1) of Ref. [26]. Comparing the coefficients of $J_1^{(m_q,0,m_q)}$ and $J_2^{(m_q,0,m_q)}$ obtained from the reduction with those of the literature result then fixes an effective map between the two master bases. This map is verified independently in Section C.5 of Appendix C, where, with the overall normalisation fixed by $J_1 = I_1$, J_2 is obtained directly as a mass derivative of the phase-space volume. The resulting transformation is,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= J_1^{(m_q,0,m_q)}, \\ I_2^{(m_q,0,m_q)} &= a J_1^{(m_q,0,m_q)} + b J_2^{(m_q,0,m_q)}. \end{aligned} \tag{4.44}$$

with,

$$\begin{aligned} a &= \frac{(1-2\epsilon)m_q^2 + (1-\epsilon)q^2}{3(1-\epsilon)}, \\ b &= \frac{m_q^2\left(4m_q^2 - q^2\right)}{3(1-\epsilon)}. \end{aligned} \tag{4.45}$$

4.5.1 Massive A_3^0: Master Integrals

The integration of the massive A_3^0 antenna requires two master integrals, denoted by $I_1^{(m_q,0,m_q)}$ and $I_2^{(m_q,0,m_q)}$. These form the literature basis onto which the master integrals generated by the IBP reduction in Section 4.5 are mapped. They are defined by [26],

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)} \\ I_2^{(m_q,0,m_q)} &= \int d\Phi_X^{(m_q,0,m_q)}\, s_{13}. \end{aligned} \tag{4.46}$$

Here $d\Phi_X^{(m_q,0,m_q)}$ denotes the massive antenna phase-space measure associated with

$$\gamma^*(q) \to Q(k_1)\, g(k_3)\, \bar{Q}(k_2),$$

and is defined through the factorisation of the three-particle phase space relative to the corresponding reduced two-particle configuration,

$$d\Phi_X^{(m_q,0,m_q)} = \frac{d\Phi_3^{(m_q,0,m_q)}(q; k_1, k_3, k_2)}{d\Phi_2^{(m_q,m_q)}(q; \tilde{k}_I, \tilde{k}_K)}. \tag{4.47}$$

The external momenta satisfy $k_1^2 = k_2^2 = m_q^2$, $k_3^2 = 0$, with $q^2 > 4m_q^2$. Thus, I_1 corresponds to the massive antenna phase-space volume, while I_2 is its first moment weighted by the invariant s_{13}.

The two master integrals are known analytically [26] and can be expressed in terms of Gauss hypergeometric functions. They were re-derived in this work, as shown in Section C.5 of Appendix C. Using the threshold variable of Eq. (4.40) they take the form,

$$\begin{aligned} I_1^{(m_q,0,m_q)} &= \left(q^2\right)^{1-\epsilon} r^{2-2\epsilon} 2^{-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; r\right) \\ I_2^{(m_q,0,m_q)} &= \left(q^2\right)^{2-\epsilon} r^{3-2\epsilon} 2^{1-2\epsilon} \pi^{-2+\epsilon} \frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)} {}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; r\right). \end{aligned} \tag{4.48}$$

4.5.2 Construction and Integration of the Massive A_3^0

The massive A_3^0 antenna can be constructed in *AntCalc* using the same interface as in the massless case discussed in Subsection 4.1.1, with the quark mass specified through the option `quarkMass -> mQ`, where `mQ` corresponds to m_q. The unintegrated antenna is obtained as,

```
In[2]:= BuildAntenna[A,3,0,quarkMass->mQ]
Out[2]:=
```

$$\begin{aligned} &\frac{1}{s_{13}^2 s_{23}^2 \left[2(-2+\epsilon)m_q^2 + (-1+\epsilon)(s_{12}+s_{13}+s_{23})\right]} \Big\{ -4(-2+\epsilon)m_q^4 \left(s_{13}^2 + s_{23}^2\right) \\ &+(-1+\epsilon)s_{13}s_{23}\left[2s_{12}^2 - (-1+\epsilon)s_{13}^2 - 2\epsilon s_{13}s_{23} - (-1+\epsilon)s_{23}^2 + 2s_{12}(s_{13}+s_{23})\right] \\ &-2m_q^2\Big[s_{12}\left((-1+\epsilon)s_{13}^2 - 2(-2+\epsilon)s_{13}s_{23} + (-1+\epsilon)s_{23}^2\right) \\ &+(-1+\epsilon)\left(s_{13}^3 + s_{13}^2 s_{23} + s_{13}s_{23}^2 + s_{23}^3\right)\Big]\Big\} \end{aligned}$$

which is the complete unintegrated antenna, exact in ϵ. In the massless limit $m_q \to 0$ it reduces exactly to the massless A_3^0 of Subsection 4.1.1, and at $\epsilon = 0$ it coincides with Eq. (3.1) of Ref. [26]. The corresponding integrated antenna is obtained with,

```
In[3]:= BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ,
ExpansionOrder->0]
```

Out[3]:=

$$\frac{1}{2100m_q^2\left(1+2m_q^2\right)^2\pi^2}\left[\frac{1}{\epsilon}\left(70m_q^2\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+...\right)\right.$$
$$\left.+\left(10\left(-1+2m_q^2+8m_q^4\right){}_2F_1\left(\frac{1}{2},2;\frac{7}{2};1-4m_q^2\right)+...\right)\right]+O(\epsilon)$$

This result is computed entirely within *AntCalc*: the unintegrated antenna is reduced with *LiteRed2* to the masters $J_1^{(m_q,0,m_q)}$ and $J_2^{(m_q,0,m_q)}$, these are converted to the literature masters through Eq. (4.44), and the analytic expressions of $I_1^{(m_q,0,m_q)}$ and $I_2^{(m_q,0,m_q)}$ given above are then substituted. Before the expansion in ϵ, the resulting integrated antenna coincides with the analytic expression of Ref. [26] to all orders in ϵ. As for the other integrated antennae presented in this chapter, *AntCalc* displays the result by default in units of the total invariant mass, $q^2 = 1$. Consequently, the threshold variable of Eq. (4.40) appears in the output as $r \to 1 - 4m_q^2$. Only representative terms of the integrated antenna are displayed here; the complete expressions are collected in Appendix A. This is the first massive route implemented in *AntCalc*: it agrees with the literature, including in its local soft and quasi-collinear limits (Appendix B.8.5), but no massive global observable has been computed yet.

An important feature of the massive result is that its Laurent expansion contains only a single $1/\epsilon$ infrared pole, in contrast to the $1/\epsilon^2$ leading behaviour of the corresponding massless A_3^0 antenna. The non-zero quark mass regulates the quark–gluon collinear singularities, while the soft-gluon singularity remains. Consequently, the integrated massive antenna exhibits only a single infrared pole.

4.6 Summary and Validation of the Calculated Antennae

This chapter has presented the construction and analytic integration of the massless quark–antiquark antenna functions implemented in *AntCalc* through NNLO, together with an exploratory extension to the massive A_3^0 antenna. The purpose of this final section is to summarise the resulting antenna set, the integration methods required for each contribution, and the checks used to validate the results.

The implemented routes span the different perturbative contributions discussed throughout this chapter: the NLO tree-level and virtual antennae A_3^0 and A_2^1; the NNLO double-virtual A_2^2 set; the real–virtual A_3^1 set; and the double-real four-parton antennae A_4^0, $\tilde{A}_4^0$, B_4^0, and C_4^0. The exploratory massive A_3^0 route discussed in Section 4.5 is included separately.

Table 4.1 summarises, for each antenna, the construction and integration method together with the corresponding validation evidence. It provides a compact overview of the calculations presented in this chapter and distinguishes the massless antenna set from the exploratory massive extension.

Table 4.1: Construction, integration and validation overview of every antenna route implemented in *AntCalc*. The API call of each route is `BuildAndIntegrateAntenna[type, multiplicity, loop order]`; only the arguments are shown. "Not applicable" marks routes with no real unresolved parton to take a soft/collinear limit of (A_2^0, A_2^1, A_2^2); "Not performed" marks the one remaining route, the real–virtual A_3^1 set, whose real-radiation limits would require one-loop soft-current and splitting-amplitude factorisation rather than the tree-level method used throughout this table — its infrared structure is instead validated via the pole cancellation shown in Chapter 5. The massive A_3^0 route is the first massive route in *AntCalc*: it agrees with the literature, including its local soft and quasi-collinear limits, but no massive global observable has been computed yet.

Route	**Construction and integration**	**Literature (order)**	**Validation**	**Status**
A_2^0: $\gamma^* \to q\bar{q}$, LO; `[A,2,0]`	Self-interference, Born-normalised reference. Trivial (reference, $A_2^0 \equiv 1$).	— (reference normalisation)	Local: not applicable. Global: LO normalisation of the R-ratio.	By construction
A_3^0: $\gamma^* \to qg\bar{q}$, NLO real; `[A,3,0]`	Self-interference; colour-coefficient extraction; matched to by-hand derivation. IBP / *LiteRed2*, `X30` basis; MI: R_3.	[5] (NLO)	Local: eikonal factor and AP splitting function reproduced. Global: feeds r_{NLO}.	Pass
A_2^1: $\gamma^* \to q\bar{q}$, NLO virtual; `[A,2,1]`	Tree–loop interference; Passarino–Veltman-reduced scalar extraction. Passarino–Veltman / *FeynHelpers*; MI: B_0, C_0.	[5] (NLO)	Local: not applicable. Global: feeds r_{NLO}.	Pass
A_3^1 set (A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$): $\gamma^* \to qg\bar{q}$, NNLO, one loop; `[A,3,1]`	Tree–loop interference; colour-coefficient extraction (leading/subleading/quark-loop). IBP / *LiteRed2*, `A31` basis; build-side MI are Passarino–Veltman scalar functions (B_0, C_0, D_0, Table C.1); integration MI: $V_{5,a}$, $V_{5,b}$, V_8.	[5, 25] (NNLO)	Local: not performed. Global: feeds r_{NNLO}.	Pass
A_2^2 set (A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$, $\check{A}_2^2$): $\gamma^* \to q\bar{q}$, NNLO, two loops; `[A,2,2]`	Two-loop tree interference and one-loop self-interference (two source contributions). IBP / *LiteRed2*, `A22TwoLoopTree`/ `A22OneLoopSelf` bases; MI: $A_{22,\mathrm{LO}}$, A_3, A_4 (also A_6 for $\tilde{A}_2^2$).	[25] (NNLO; A_2^2, $\tilde{A}_2^2$, $\hat{A}_2^2$), [27] (NNLO; $\check{A}_2^2$)	Local: not applicable. Global: feeds r_{NNLO}.	Pass
A_4^0, $\tilde{A}_4^0$: $\gamma^* \to qgg\bar{q}$, NNLO, double-real; `[A,4,0]`	Colour-ordered amplitude; colour-coefficient extraction. IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6 (A_4^0); R_4, R_6, $R_{8,a}$, $R_{8,b}$ ($\tilde{A}_4^0$).	[5] (NNLO)	Local: both colour components validated: eikonal and AP/spin-correlated splitting functions for the leading term; dipole eikonal and absent gg collinear singularity for the subleading term (Appendix B.8). Global: feeds r_{NNLO}.	Pass

Table 4.1 continued.

Route	Construction and integration	Literature (order)	Validation	Status
B_4^0: $\gamma^* \to q\bar{q}q'\bar{q}'$, NNLO; `[B,4,0]`	Sector self-interference (direct amplitude only). IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6.	[5] (NNLO)	Local: single-collinear ($q'\bar{q}' \to g$), double-soft, and triple-collinear limits reproduced (Appendix B.8.3). Global: feeds r_{NNLO}.	Pass
C_4^0: $\gamma^* \to q\bar{q}q\bar{q}$, NNLO; `[C,4,0]`	Sector symmetrised interference (direct and exchange). IBP / *LiteRed2*, `X40` basis; MI: R_4, R_6, $R_{8,b}$.	[5] (NNLO)	Local: single-collinear region confirmed regular; identical-flavour triple-collinear splitting function reproduced (Appendix B.8.4). Global: feeds r_{NNLO}.	Pass
Massive A_3^0: $\gamma^* \to q(m_q)g\bar{q}(m_q)$, exploratory; `[A,3,0, quarkMass-> mQ]`	Same route as massless A_3^0, with the `quarkMass` option. IBP / *LiteRed2*, massive family (default call performs the reduction and substitutes the literature master values); MI mapped from *LiteRed2*'s J_1, J_2 to the literature $I_1^{(m_q,0,m_q)}$, $I_2^{(m_q,0,m_q)}$.	[26]	Local: soft-gluon eikonal and quasi-collinear splitting function reproduced (Appendix B.8.5). Global: not applicable (does not feed the massless R-ratio).	Literature agreement, incl. local soft/quasi-collinear limits; no global observable validation yet

The complete massless antenna set is globally validated in Chapter 5 through the cancellation of all infrared poles and the reproduction of the known NNLO hadronic R-ratio result. Further work may complement this global validation with a systematic study of all the double-unresolved limits of the four-parton antennae, the one-loop single-unresolved limits of the real–virtual antennae, and direct comparisons of the loop pole structure with universal infrared factorisation formulae [32].

Chapter 5

Global Validation: The Massless Hadronic R-Ratio

The antenna functions constructed and integrated in Chapter 4 are now subjected to a global validation through the calculation of the massless hadronic R-ratio up to NNLO. This fully inclusive observable provides a particularly stringent test of the antenna set, since the different real and virtual contributions enter at each perturbative order and their infrared poles must cancel in the inclusive sum, as discussed in Chapter 2. The resulting finite NLO and NNLO coefficients can then be compared with the known R-ratio results obtained independently using methods other than antenna subtraction [33].

5.1 The Hadronic R-Ratio

The hadronic R-ratio is defined as the ratio of the total cross section for electron–positron annihilation into hadrons to the point-like cross section for muon-pair production,

$$\begin{aligned} R_{\text{had}} &= \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} \\ &= \frac{3q^2}{4\pi\alpha_{\text{em}}^2}\,\sigma(e^+e^- \to \text{hadrons}). \end{aligned} \tag{5.1}$$

For massless quarks, the hadronic cross section can be related, through the optical theorem, to the imaginary part of the electromagnetic current correlator [13, 34]. Equivalently, its perturbative evaluation can be organised as an inclusive sum over the partonic final states produced by the virtual photon. This makes the R-ratio particularly well suited to the global validation performed in this chapter. At leading order, the hadronic R-ratio is [2],

$$R^{(0)} = N \sum_q e_q^2, \tag{5.2}$$

where the sum runs over the active quark flavours and e_q denotes the quark electric charge in units of the positron charge. It is convenient to factor out this Born-level contribution and define

$$R_{\text{had}} = R^{(0)} R(\alpha_s), \tag{5.3}$$

where the normalised ratio admits the perturbative expansion,

$$R(\alpha_s) = 1 + \frac{\alpha_s}{2\pi}\, r_{\text{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\text{NNLO}} + O\left(\alpha_s^3\right). \tag{5.4}$$

At each perturbative order, the coefficients r_{NLO} and r_{NNLO} receive contributions from partonic final states of different multiplicities and loop orders. In the antenna formulation used in this work, these contributions are expressed in terms of the corresponding integrated antenna functions derived in Chapter 4. At NLO, the real and virtual contributions involve $\mathcal{A}_3^0$ and $\mathcal{A}_2^1$, respectively, while at NNLO the complete set of two-, three-, and four-parton contributions is required.

The normalised quantity $R(\alpha_s)$ will be reconstructed from these integrated antenna functions in the following sections and compared with the known result obtained independently using methods other than antenna subtraction [33].

5.2 Assembly from Integrated Antenna Functions

The antenna functions constructed in Chapter 4 are colour-stripped kinematic quantities. To reconstruct the contributions to the inclusive hadronic rate, the antennae associated with each partonic final state must therefore be combined with the appropriate colour factors and normalisation. In this work, these colour-restored, final-state-specific combinations are denoted by T-objects.

The notation for a T-object identifies both the corresponding partonic final state and its perturbative order. The subscript specifies the final-state particles, while the superscript is the total power of couplings (e and g_s) in the corresponding squared-amplitude contribution, with 2, 4, and 6 corresponding to LO, NLO, and NNLO, respectively. The LO two-parton contribution, associated with the process $\gamma^* \to q\bar{q}$, provides the common normalisation and is given by

$$T^2_{q\bar{q}} = N\left|\mathcal{M}^0_2\right|^2 = 4N(1-\epsilon)q^2. \tag{5.5}$$

For convenience, we define

$$P = 2C_F T^2_{q\bar{q}}. \tag{5.6}$$

At NLO, the inclusive rate receives contributions from the virtual two-parton process $\gamma^* \to q\bar{q}$ and the real-emission three-parton process $\gamma^* \to q\bar{q}g$. In terms of the corresponding antenna functions, their T-objects are

$$T^4_{q\bar{q}} = P\,A^1_2, \qquad T^4_{q\bar{q}g} = P\,A^0_3. \tag{5.7}$$

At NNLO, contributions arise from two-, three-, and four-parton final states. The two-parton contribution contains the different colour components of the two-loop antenna set, together with the one-loop-squared contribution,

$$T^6_{q\bar{q}} = P\left(NA^2_2 + \frac{1}{N}\tilde{A}^2_2 + N_f\hat{A}^2_2 + \left(N - \frac{1}{N}\right)\breve{A}^2_2\right). \tag{5.8}$$

The three-parton real–virtual contribution is

$$T^6_{q\bar{q}g} = P\left[N\left(A^1_3 + A^1_2A^0_3\right) - \frac{1}{N}\left(\tilde{A}^1_3 + A^1_2A^0_3\right) + N_f\hat{A}^1_3\right]. \tag{5.9}$$

Finally, the four-parton double-real contributions are separated according to their partonic final states,

$$T^6_{q\bar{q}q'\bar{q}'} = P\,(N_f - 1)B^0_4, \tag{5.10}$$

$$T^6_{q\bar{q}q\bar{q}} = P\left(B^0_4 - \frac{2}{N}C^0_4\right), \tag{5.11}$$

$$T^6_{q\bar{q}gg} = P\left(NA^0_4 - \frac{1}{2N}\tilde{A}^0_4\right). \tag{5.12}$$

Upon integration over the corresponding final-state phase spaces, the antenna functions appearing in these expressions are replaced by their integrated counterparts, $\mathcal{A}$, $\mathcal{B}$, and C, obtained in Chapter 4. The factors $\frac{1}{2}$ and 2 multiplying $\tilde{\mathcal{A}}_4^0$ and C_4^0 are included explicitly in Eq. (5.16), following Ref. [25]; their origin is explained in Section 5.4. We denote the resulting integrated combinations by $\mathcal{T}$-objects. These quantities provide the individual two-, three-, and four-parton contributions entering the inclusive R-ratio.

5.3 NLO Validation

At NLO, the inclusive hadronic R-ratio receives contributions from the virtual correction to the two-parton final state, $\gamma^* \to q\bar{q}$, and from the real-emission process, $\gamma^* \to q\bar{q}g$. In terms of the integrated antenna functions obtained in Chapter 4, these contributions are described by $\mathcal{A}_2^1$ and $\mathcal{A}_3^0$, respectively. Using the T-object normalisation introduced in the previous section, the NLO coefficient can therefore be written as

$$\begin{aligned} r_{\mathrm{NLO}} &= \frac{\mathcal{T}_{q\bar{q}}^4 + \mathcal{T}_{q\bar{q}g}^4}{T_{q\bar{q}}^2} = 2C_F \left(\mathcal{A}_2^1 + \mathcal{A}_3^0\right) \\ &= 2C_F \left[\left(-\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \ldots\right) + \left(\frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \ldots\right)\right] = \frac{3}{2}C_F. \end{aligned} \tag{5.13}$$

Thus, the $1/\epsilon^2$ and $1/\epsilon$ infrared poles of the virtual contribution cancel exactly against those generated by the integrated real-emission contribution, as required for the inclusive observable. This cancellation provides a first global check of the relative normalisation and combination of the integrated antenna functions. For $N = 3$, where $C_F = 4/3$, the NLO contribution becomes,

$$\frac{\alpha_s}{2\pi}\frac{3}{2}C_F = \frac{\alpha_s}{\pi}, \tag{5.14}$$

reproducing the standard NLO correction to the massless hadronic R-ratio.

5.4 NNLO Validation and the Hadronic R-Ratio

At NNLO, the global validation requires the simultaneous combination of the double-virtual two-parton, real–virtual three-parton, and double-real four-parton contributions. In contrast to the NLO case discussed above, several colour structures and partonic final states contribute. Using the integrated $\mathcal{T}$-objects defined in Section 5.2, the normalised hadronic R-ratio can be assembled order by order as,

$$\begin{aligned} R(\alpha_s) &= 1 + \frac{\alpha_s}{2\pi}\frac{\mathcal{T}_{q\bar{q}}^4 + \mathcal{T}_{q\bar{q}g}^4}{T_{q\bar{q}}^2} \\ &\quad + \left(\frac{\alpha_s}{2\pi}\right)^2 \frac{\mathcal{T}_{q\bar{q}}^6 + \mathcal{T}_{q\bar{q}g}^6 + \mathcal{T}_{q\bar{q}q\bar{q}}^6 + \mathcal{T}_{q\bar{q}q'\bar{q}'}^6 + \mathcal{T}_{q\bar{q}gg}^6}{T_{q\bar{q}}^2} + O\left(\alpha_s^3\right) \\ &= 1 + \frac{\alpha_s}{2\pi} r_{\mathrm{NLO}} + \left(\frac{\alpha_s}{2\pi}\right)^2 r_{\mathrm{NNLO}} + O\left(\alpha_s^3\right). \end{aligned} \tag{5.15}$$

The NNLO coefficient therefore tests the complete massless antenna set calculated in Chapter 4. Substituting the definitions of the T-objects, it can be expressed entirely in terms of the corresponding integrated

antenna functions,

$$r_{\mathrm{NNLO}} = 2C_F \Bigg[N\Big(\mathcal{A}_2^1\mathcal{A}_3^0 + \mathcal{A}_3^1 + \mathcal{A}_4^0 + \mathcal{A}_2^2\Big) - \frac{1}{N}\Big(\mathcal{A}_2^1\mathcal{A}_3^0 + \tilde{\mathcal{A}}_3^1 + \tfrac{1}{2}\tilde{\mathcal{A}}_4^0 + 2\,C_4^0 - \tilde{\mathcal{A}}_2^2\Big) + N_f\Big(\mathcal{B}_4^0 + \hat{\mathcal{A}}_3^1 + \hat{\mathcal{A}}_2^2\Big) + 2C_F\check{\mathcal{A}}_2^2 \Bigg]. \tag{5.16}$$

The relative factors multiplying the four-parton antennae follow directly from the matrix-element decompositions given in Eqs. (2.55) and (2.60). In the $q\bar{q}gg$ channel, an identical-gluon symmetry factor $1/2!$ cancels the two equivalent leading-colour orderings after integration, whereas the subleading-colour $\tilde{A}_4^0$ contribution occurs only once, leaving the factor $1/2$ multiplying $\tilde{\mathcal{A}}_4^0$. For the identical-flavour four-quark contribution, Eq. (2.60) contains two C_4^0 interference terms which give equal contributions after integration over the full four-particle phase space, resulting in the factor $2\,C_4^0$.

Each contribution in Eq. (5.16) is individually infrared divergent. After inserting the integrated antenna functions obtained in Chapter 4, however, the poles cancel in the sum. In particular, the coefficients of

$$\frac{1}{\epsilon^4}, \qquad \frac{1}{\epsilon^3}, \qquad \frac{1}{\epsilon^2}, \qquad \frac{1}{\epsilon}$$

vanish, leaving a finite NNLO coefficient. This cancellation simultaneously tests the relative normalisation, colour decomposition, and assembly of the two-, three-, and four-parton antenna contributions.

Combining the finite NLO result of Section 5.3 with the NNLO contribution gives

$$R(\alpha_s) = 1 + \frac{\alpha_s}{2\pi}\frac{3}{2}C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right] + O\left(\alpha_s^3\right). \tag{5.17}$$

The result is finite and agrees with the known massless hadronic R-ratio through NNLO [33], and with its earlier antenna-based assembly in Ref. [25]. Importantly, the first comparison is with a result obtained independently of the antenna-subtraction construction used here. The agreement therefore provides a global validation of the complete massless antenna set calculated in Chapter 4: not only do the infrared poles cancel when the different final-state contributions are combined, but the remaining finite coefficient reproduces the independently known inclusive result.

5.5 Implementation in *AntCalc*

The assembly of the hadronic R-ratio described in the previous sections is automated in *AntCalc* through the function `BuildRRatio`. For the massless QCD antenna set considered in this work, the calculation is performed with,

```
In[2]:= BuildRRatio[SMQCD]
Out[2]:=
```

$$1 + \left(\frac{\alpha_s}{2\pi}\right)\frac{3}{2}C_F + \left(\frac{\alpha_s}{2\pi}\right)^2 C_F\left[N\left(\frac{243}{16} - 11\zeta_3\right) + \frac{3}{16N} + N_f\left(-\frac{11}{4} + 2\zeta_3\right)\right]$$

The function performs the assembly described explicitly in Sections 5.2–5.4, combining the integrated antenna contributions with their corresponding colour factors and normalisations at each perturbative order. The cancellation of the infrared poles occurs within this assembly, so that the returned result is finite through NNLO. The output of `BuildRRatio[SMQCD]` therefore reproduces the analytic result obtained in Eq. (5.17), which in turn agrees with the independently known massless hadronic R-ratio

through NNLO [33]. This provides an automated global check of both the integrated massless antenna set and its implementation in *AntCalc*.

5.6 Comparison with Experimental Data

As an additional illustration of the perturbative result obtained above, the *AntCalc* prediction for the hadronic R-ratio can be compared with the experimental data compiled by the Particle Data Group (PDG) [35]. Figure 5.1 shows this comparison over a broad range of centre-of-mass energies. The coupling is $\alpha_s(M_Z) = 0.1180$, evolved at two loops with $N_f = 3, 4, 5$ matched at $m_c = 1.27$ and $m_b = 4.18$ GeV, with $\mu = Q = \sqrt{s}$; the coefficients use $N_f = 3, 4, 5$ below 3.73 GeV, up to 10.56 GeV, and above. The purpose of this comparison is not to describe the full experimental spectrum. The result derived in this work is a massless fixed-order perturbative QCD prediction and therefore does not include several effects that are important in specific energy regions. At low energies, the data exhibit pronounced hadronic resonance and non-perturbative structure. Heavy-quark thresholds and quarkonium resonances also produce features that are not described by the massless approximation, while at energies around the Z pole electroweak contributions dominate the observed behaviour. Consequently, agreement with the data should only be expected away from such regions, where a fixed-order perturbative description is appropriate.

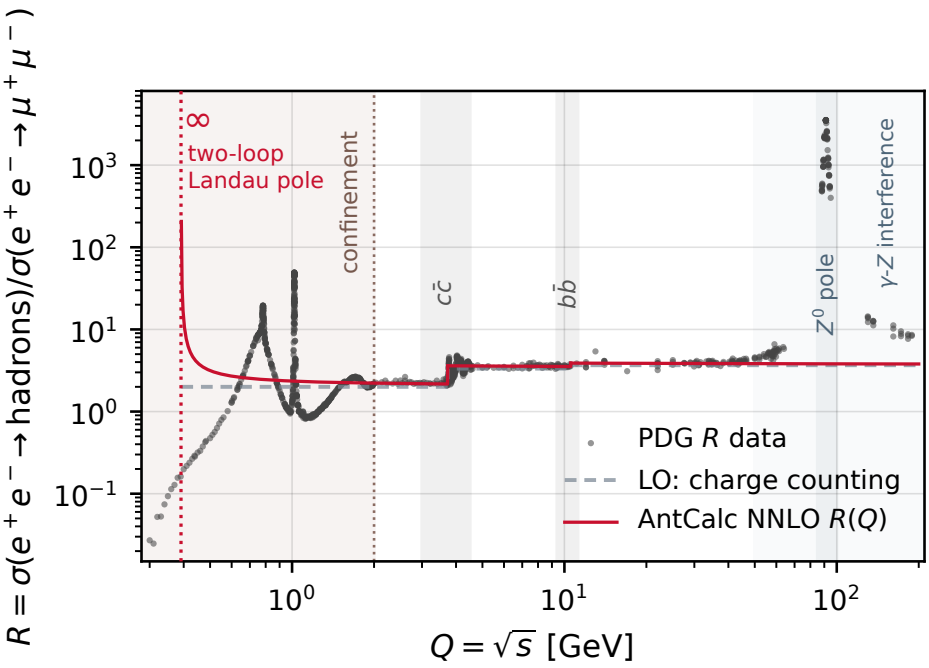


Figure 5.1: Comparison of the massless NNLO R-ratio obtained with *AntCalc* with the PDG data over the full energy range considered. The LO charge-counting result is shown for reference. The indicated regions highlight the low-energy non-perturbative regime, heavy-quark thresholds, and the electroweak Z^0-pole region, illustrating the limited domain in which the massless fixed-order QCD prediction can be meaningfully compared with the data.

Figure 5.2 focuses on the energy range $2 \lesssim \sqrt{s} \lesssim 30$ GeV, where the comparison with the massless perturbative QCD prediction is more informative. The charm- and bottom-quark threshold regions, where the massless approximation is not expected to provide an adequate description, are indicated by the grey-shaded bands. Away from these regions, the inclusion of higher-order QCD corrections improves the overall description of the experimental data relative to the LO prediction. This is particularly visible in the lower panel, where the results are normalised to the LO prediction and the NLO and NNLO curves

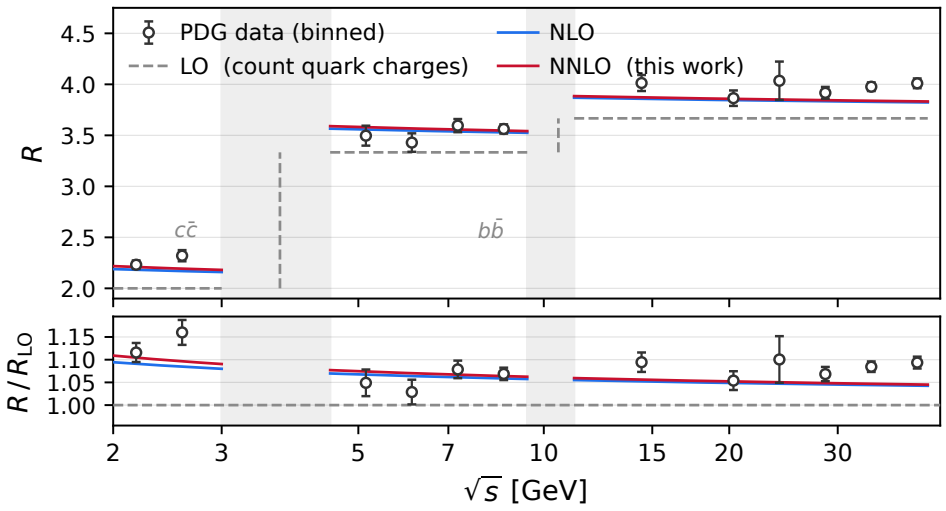


Figure 5.2: Comparison of the LO, NLO, and NNLO massless perturbative predictions for the hadronic R-ratio with the PDG data in the range $2 \lesssim \sqrt{s} \lesssim 30\,\mathrm{GeV}$. The grey-shaded bands indicate the charm- and bottom-quark threshold regions. The lower panel shows the results normalised to the LO prediction, highlighting the improved overall description obtained after including the higher-order QCD corrections.

are generally shifted towards the experimental measurements. Moreover, the relatively small change from NLO to NNLO compared with that from LO to NLO is indicative of good convergence of the perturbative expansion in this energy range.

5.7 Summary

The massless antenna functions calculated in Chapter 4 have been subjected to a global validation through the hadronic R-ratio. At both NLO and NNLO, the infrared poles of the individual real and virtual contributions cancel in the inclusive sum, providing a stringent check of the relative normalisation, colour structure, and assembly of the integrated antenna functions. The resulting finite coefficients reproduce the known massless hadronic R-ratio through NNLO, previously obtained using methods independent of antenna subtraction [33]. This agreement therefore confirms not only the internal consistency of the antenna set, but also its consistency with independently established QCD results.

The comparison with the PDG data provides an additional phenomenological illustration of the calculation. In the energy region where the massless fixed-order approximation is applicable, and away from the heavy-quark threshold regions, the inclusion of successive perturbative orders generally improves the agreement between the theoretical prediction and the experimental measurements. Moreover, the comparatively small shift from NLO to NNLO relative to that from LO to NLO is indicative of good convergence of the perturbative expansion in this region. Conversely, the deviations observed in regions dominated by non-perturbative, heavy-quark threshold, resonance, or electroweak effects illustrate the expected limitations of the massless fixed-order approximation.

Taken together, the cancellation of the infrared poles, the agreement with the independently known perturbative R-ratio, and the comparison with experimental data provide complementary checks of the results obtained in this work. In particular, the first two constitute the principal global validation of the complete massless antenna set implemented in *AntCalc* through NNLO. This validation matters because these antenna functions are intended as building blocks for NNLO subtraction terms.

Chapter 6

Conclusions and Future Work

This thesis has developed *AntCalc*, a computational framework for the construction and analytic integration of QCD antenna functions. The principal objective was to automate a technically demanding component of antenna subtraction: the derivation of the antenna functions from the corresponding matrix elements, their reduction to suitable integral representations, and their analytic integration over the unresolved phase space. *AntCalc* was developed from the ground up in this work and provides a common framework in which these different stages can be carried out consistently.

The main result is the implementation of the complete set of massless final–final quark–antiquark antenna functions considered in this work through NNLO. This includes the tree-level, one-loop and two-loop two-parton contributions, the three-parton tree-level and real–virtual contributions, and the four-parton double-real antennae with both gluonic and fermionic final states. The antenna functions are available both in their unintegrated form, required for the construction of local subtraction terms, and in analytically integrated form, required for the cancellation of the explicit infrared poles of virtual contributions. Their integration required the reduction of the corresponding phase-space and loop integrals to master integrals, using IBP reduction where appropriate, followed by the substitution of their analytic expressions and expansion in the dimensional-regularisation parameter ϵ.

The results were subjected to several complementary validation procedures. The integrated antenna functions were compared with established results in the literature, the four-parton antennae were also compared with explicit unintegrated expressions, and the infrared behaviour of the tree-level antennae was tested in the relevant unresolved limits. In particular, the massless A_3^0 antenna was used to demonstrate explicitly the recovery of the soft eikonal factor and the relevant Altarelli–Parisi splitting functions. At the integrated level, the complete massless antenna set was subjected to the global validation presented in Chapter 5. When assembled into the massless hadronic R-ratio, the infrared poles of the separate real and virtual contributions cancel at NLO and NNLO, and the remaining finite coefficients reproduce the independently known result. This provides a particularly stringent test of the relative normalisation, colour decomposition, and integrated infrared structure of the antenna set.

This validation is important in view of the intended application of these functions. Antenna functions are not isolated observables, but universal building blocks used to construct subtraction terms for higher-order QCD calculations. A consistent NNLO subtraction calculation requires the appropriate unintegrated antennae to reproduce the local unresolved behaviour of real-radiation matrix elements, while their integrated counterparts must provide the infrared poles required to cancel those of the corresponding virtual contributions. The local and global checks performed in this thesis therefore test precisely the properties required for the antenna functions to be used within the antenna-subtraction framework.

Beyond the complete massless quark–antiquark set, this work has also explored the extension of

AntCalc to massive antenna functions through the implementation of the massive A_3^0 antenna. This required a separate massive integral family and a transformation between the master-integral basis produced by the IBP reduction and that used for the known analytic results. Although this route has not yet undergone the same level of global validation as the massless antenna set, it demonstrates that the architecture of *AntCalc* is not intrinsically restricted to massless kinematics and provides a starting point for a systematic extension to massive antenna functions.

Future Work

The results presented in this thesis provide several natural directions for the further development of *AntCalc*. From the physics perspective, the most immediate extension is the completion of the massive quark–antiquark final–final antenna set. The massive A_3^0 antenna implemented in this work provides a first demonstration that the framework developed for the massless case can be extended to massive kinematics. Completing this sector would require the corresponding massive four-parton antennae, A_4^0, B_4^0, and C_4^0, followed by the massive real–virtual and double-virtual antennae A_3^1 and A_2^2. Importantly, this extension would go beyond reproducing antenna functions that are already analytically known: some of the required massive antenna functions remain to be calculated. The framework developed and validated in this thesis therefore provides a possible starting point for obtaining genuinely new ingredients for massive NNLO antenna subtraction.

A broader extension would be to include antenna functions associated with other pairs of hard radiators. The present implementation focuses on quark–antiquark antennae, while a general antenna-subtraction calculation also requires quark–gluon and gluon–gluon antenna functions, conventionally organised into the D- and E-type and F-, G-, and H-type families, respectively. Extending *AntCalc* to these antenna families would broaden the range of partonic processes for which the required subtraction ingredients could be constructed within a common framework. Similarly, extending the present final–final implementation to initial–final and initial–initial kinematics would be necessary for applications involving coloured partons in the initial state. In the longer term, the same framework could provide a basis for investigating the antenna functions required at perturbative orders beyond NNLO, in particular at $\mathrm{N^3LO}$. Further work could also complement the global R-ratio validation with additional local checks, including the full double-unresolved limits of the four-parton antennae, the one-loop single-unresolved limits of the real–virtual antennae, and comparisons of the loop pole structure with universal infrared factorisation formulae [32].

These extensions would substantially increase the algebraic complexity and computational cost of both the construction and integration stages. They therefore also motivate the further development of the computational architecture of *AntCalc*. In its current form, the package is implemented entirely within *Wolfram Mathematica*. This environment has provided a flexible setting in which to develop, test, and validate the symbolic workflow presented in this thesis, but the more involved calculations already reveal limitations in execution time and memory usage. For example, the complete `BuildRRatio` calculation currently requires approximately 53 minutes, with its execution becoming sensitive to the available memory. While this does not affect the results obtained in the present work, it indicates that a more scalable implementation would be desirable before substantially larger antenna families and integral reductions are attempted.

One possible architecture for this development is illustrated in Fig. 6.1. Rather than replacing the high-level workflow established in this thesis, the objective would be to make its implementation more modular and to delegate computationally intensive stages to specialised tools. A *Python*-based

layer could coordinate the workflow, with the process-dependent profiles stored in a portable, machine-readable format such as `JSON`. *FeynArts* and *FeynCalc* could continue to provide diagram generation and amplitude manipulation, respectively, while large symbolic manipulations could be delegated to *FORM* [36]. Similarly, the IBP-reduction stage could be transferred from *LiteRed2* to *Kira* [37], while *Mathematica* could be retained for symbolic operations for which it remains advantageous. The purpose of this restructuring would therefore not simply be to improve the performance of calculations that *AntCalc* can already perform, but to make the framework sufficiently scalable for the more demanding physics applications outlined above.

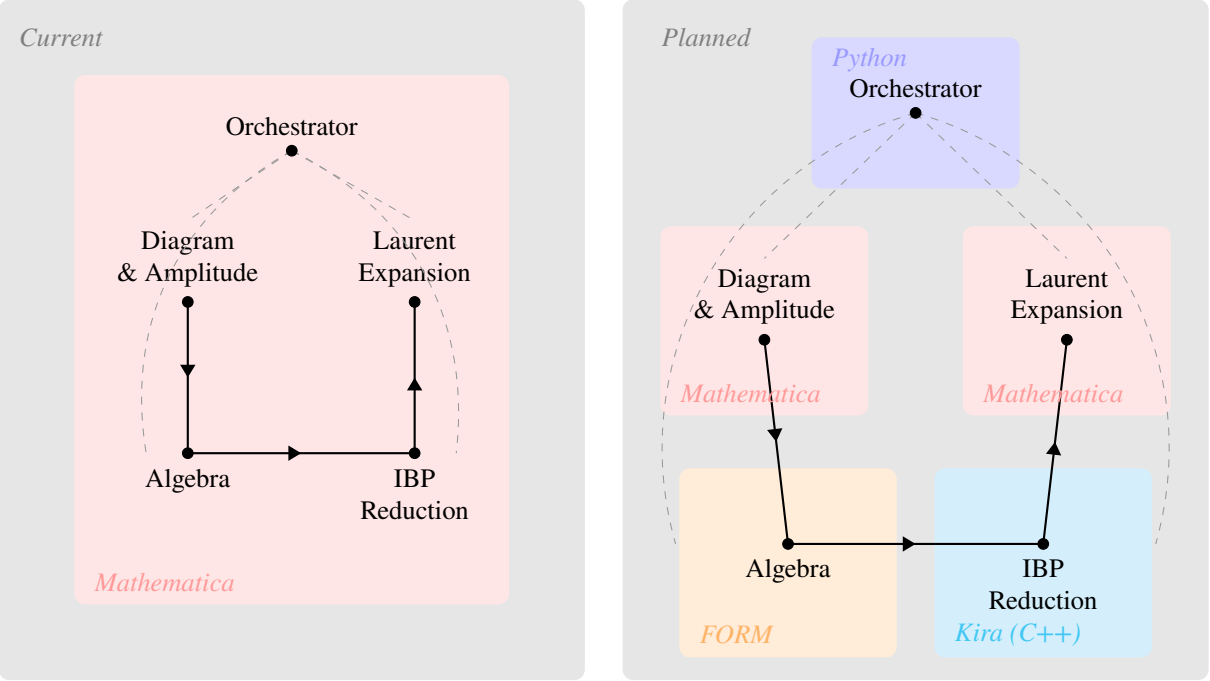


Figure 6.1: Diagram showing the current and planned workflow steps relative to their programming languages.

The present work thus provides both a validated set of massless quark–antiquark antenna functions through NNLO and a framework from which the calculation of further antenna functions can be pursued. The extension to the massive A_3^0 antenna demonstrates the first step beyond the massless sector, while the remaining massive antennae provide a particularly well-defined direction in which the methods developed here can be applied to calculations that are not yet complete in the literature. Extending the framework to additional radiator configurations and kinematics would further broaden its applicability to antenna subtraction. In this sense, the developments proposed above are not separate from the work presented in this thesis, but represent a continuation of its central objective: to make the construction and analytic integration of the antenna functions required for higher-order QCD calculations more systematic, reproducible, and computationally accessible.

Appendix A

Complete Antenna Expressions

This appendix includes the complete expressions of the antennae computed using *AntCalc*. It shows the unintegrated and integrated antennae, along with the commands used to compute them.

A note on conventions: the unintegrated antenna functions below are given exactly as *AntCalc* returns them, in terms of s_{ij} and q^2. The integrated antennae, however, are quoted at $q^2 = 1$, matching `IntegrateAntenna`'s default convention. This does not break generality as established in Subsection 2.5.2 and Eq. (2.89), a quantity of this kind scales with q^2 as a single power fixed by its mass dimension and ϵ-scaling, so the complete q^2-dependent result is recovered by multiplying the expression given here by $(q^2)^{\lambda+\kappa\epsilon}$ for the appropriate λ and κ. The explicit case of $\mathcal{A}_3^0$, which scales as $(q^2)^{-\epsilon}$, is worked out in full in Eq. (4.33).

A.1 LO Antenna

A.1.1 A_2^0

Running `BuildAntenna[A,2,0]`, we get the built result,

$$A_2^0 = 1. \tag{A.1}$$

A.2 NLO Antennae

A.2.1 A_2^1

Running `BuildAntenna[A,2,1]`, we get the built result,

$$A_2^1 = -\frac{1}{2}(3+2\epsilon)B_0\Big(q^2;0,0\Big) - q^2 C_0\Big(0,0,q^2;0,0,0\Big). \tag{A.2}$$

By running `BuildAndIntegrateAntenna[A,2,1]` we get the integrated result,

$$\mathcal{A}_2^1 = -\frac{1}{\epsilon^2} - \frac{3}{2\epsilon} - 4 + \frac{7\pi^2}{12} + \epsilon\left(\frac{7\zeta_3}{3} - 8 + \frac{7\pi^2}{8}\right) + \epsilon^2\left(\frac{7\zeta_3}{2} - 16 + \frac{7\pi^2}{3} - \frac{73\pi^4}{1440}\right) + O\Big(\epsilon^3\Big). \tag{A.3}$$

A.2.2 A_3^0

Running `BuildAntenna[A,3,0]`, we get the built result,

$$A_3^0 = \frac{2s_{12}^2 + 2s_{12}(s_{13}+s_{23})}{q^2 s_{13} s_{23}} - \epsilon\frac{2}{q^2} + (1-\epsilon)\frac{s_{13}^2 + s_{23}^2}{q^2 s_{13} s_{23}}. \tag{A.4}$$

By running `BuildAndIntegrateAntenna[A,3,0]` we get the integrated result,

$$\begin{aligned}\mathcal{A}_3^0 = \frac{1}{\epsilon^2} + \frac{3}{2\epsilon} + \frac{19}{4} - \frac{7\pi^2}{12} + \epsilon\left(-\frac{25\zeta_3}{3} + \frac{109}{8} - \frac{7\pi^2}{8}\right) \\ +\epsilon^2\left(-\frac{25\zeta_3}{2} + \frac{639}{16} - \frac{133\pi^2}{48} - \frac{71\pi^4}{1440}\right) + O\left(\epsilon^3\right).\end{aligned} \tag{A.5}$$

A.3 NNLO Antennae

A.3.1 A_2^2

Running `BuildAntenna[A,2,2,Component->Leading]`, we get the built result,

$$\begin{aligned}A_2^2 = 4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\Bigg[\frac{1}{4\epsilon^4} + \frac{17}{8\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{433}{144} - \frac{\pi^2}{2}\right) + \frac{1}{\epsilon}\left(\frac{4045}{864} - \frac{83\pi^2}{48} + \frac{7\zeta_3}{12}\right) \\ -\frac{9083}{5184} - \frac{2153\pi^2}{864} + \frac{263\pi^4}{1440} + \frac{13\zeta_3}{9} + O(\epsilon)\Bigg],\end{aligned} \tag{A.6}$$

where the prefactor equals $1/C(\epsilon,2)$, with $C(\epsilon,2) = G_2/S_\epsilon^2$.

By running `BuildAndIntegrateAntenna[A,2,2,Component->Leading]`, the trivial two-particle phase-space integration of Eq. (2.69) gives the integrated result,

$$\begin{aligned}\mathcal{A}_2^2 = C(\epsilon,2)\,A_2^2 = \frac{1}{4\epsilon^4} + \frac{17}{8\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{433}{144} - \frac{\pi^2}{2}\right) + \frac{1}{\epsilon}\left(\frac{4045}{864} - \frac{83\pi^2}{48} + \frac{7\zeta_3}{12}\right) \\ -\frac{9083}{5184} - \frac{2153\pi^2}{864} + \frac{263\pi^4}{1440} + \frac{13\zeta_3}{9} + O(\epsilon).\end{aligned} \tag{A.7}$$

A.3.2 $\tilde{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Subleading]`, we get the built result,

$$\begin{aligned}\tilde{A}_2^2 = 4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\Bigg[-\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} - \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{13\pi^2}{24}\right) + \frac{1}{\epsilon}\left(-\frac{221}{32} + \frac{3\pi^2}{2} + \frac{8\zeta_3}{3}\right) \\ -\frac{1151}{64} + \frac{475\pi^2}{96} - \frac{59\pi^4}{288} + \frac{29\zeta_3}{4} + O(\epsilon)\Bigg],\end{aligned} \tag{A.8}$$

where the prefactor equals $1/C(\epsilon,2)$, with $C(\epsilon,2) = G_2/S_\epsilon^2$.

By running `BuildAndIntegrateAntenna[A,2,2,Component->Subleading]`, the trivial two-particle phase-space integration of Eq. (2.69) gives the integrated result,

$$\begin{aligned}\tilde{\mathcal{A}}_2^2 = C(\epsilon,2)\,\tilde{A}_2^2 = -\frac{1}{4\epsilon^4} - \frac{3}{4\epsilon^3} - \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{13\pi^2}{24}\right) + \frac{1}{\epsilon}\left(-\frac{221}{32} + \frac{3\pi^2}{2} + \frac{8\zeta_3}{3}\right) \\ -\frac{1151}{64} + \frac{475\pi^2}{96} - \frac{59\pi^4}{288} + \frac{29\zeta_3}{4} + O(\epsilon).\end{aligned} \tag{A.9}$$

A.3.3 $\hat{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Nf]`, we get the built result,

$$\hat{A}_2^2 = 4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\left[-\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{1}{\epsilon}\left(\frac{65}{216} + \frac{\pi^2}{24}\right) + \frac{4085}{1296} - \frac{91\pi^2}{216} + \frac{\zeta_3}{18} + O(\epsilon)\right], \tag{A.10}$$

where the prefactor equals $1/C(\epsilon,2)$, with $C(\epsilon,2) = G_2/S_\epsilon^2$.

By running `BuildAndIntegrateAntenna[A,2,2,Component->Nf]`, the trivial two-particle phase-space integration of Eq. (2.69) gives the integrated result,

$$\hat{\mathcal{A}}_2^2 = C(\epsilon,2)\,\hat{A}_2^2 = -\frac{1}{4\epsilon^3} - \frac{1}{9\epsilon^2} + \frac{1}{\epsilon}\left(\frac{65}{216} + \frac{\pi^2}{24}\right) + \frac{4085}{1296} - \frac{91\pi^2}{216} + \frac{\zeta_3}{18} + O(\epsilon). \tag{A.11}$$

A.3.4 $\breve{A}_2^2$

Running `BuildAntenna[A,2,2,Component->Breve]`, we get the built result,

$$\begin{aligned}\breve{A}_2^2 = 4^{2\epsilon-3}\pi^{2\epsilon-4}e^{-2\gamma_E\epsilon}\Bigg[&\frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{\pi^2}{24}\right) + \frac{1}{\epsilon}\left(7 - \frac{\pi^2}{8} - \frac{7\zeta_3}{6}\right)\\ &+18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} + O(\epsilon)\Bigg],\end{aligned} \tag{A.12}$$

where the prefactor equals $1/C(\epsilon,2)$, with $C(\epsilon,2) = G_2/S_\epsilon^2$.

By running `BuildAndIntegrateAntenna[A,2,2,Component->Breve]`, the trivial two-particle phase-space integration of Eq. (2.69) gives the integrated result,

$$\begin{aligned}\breve{\mathcal{A}}_2^2 = C(\epsilon,2)\,\breve{A}_2^2 = &\frac{1}{4\epsilon^4} + \frac{3}{4\epsilon^3} + \frac{1}{\epsilon^2}\left(\frac{41}{16} - \frac{\pi^2}{24}\right) + \frac{1}{\epsilon}\left(7 - \frac{\pi^2}{8} - \frac{7\zeta_3}{6}\right)\\ &+18 - \frac{41\pi^2}{96} - \frac{7\pi^4}{480} - \frac{7\zeta_3}{2} + O(\epsilon).\end{aligned} \tag{A.13}$$

A.3.5 A_3^1

Running `BuildAntenna[A,3,1,Component->Leading]`, *AntCalc* returns a single rational expression in the invariants, q^2 and ϵ, together with the Passarino–Veltman functions B_0, C_0 and D_0 of Appendix B.2; its leading terms are shown in Chapter 4. Collecting the expression by these functions and factorising the coefficients, and writing $u = s_{13} + s_{23}$ and $v = s_{13}s_{23}$, the built result is

$$\begin{aligned}A_3^1 = \frac{1}{6\epsilon q^2 s_{13}s_{23}\,(\epsilon-1)\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^2}\Big\{&\\ -11\,(\epsilon-1)\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^2\,\Big[&\\ &\big(2s_{12}^2 + 2s_{12}u + u^2 - 2v\big)\\ &+\epsilon\big(-u^2\big)\\ \Big]&\\ -\,3\epsilon\,(s_{12}+s_{13}+s_{23})\,\Big[&\\ &\big(-6s_{12}^5 - 12s_{12}^4u - 9s_{12}^3u^2 + 2s_{12}^3v - 3s_{12}^2u^3\\ &+2s_{12}^2uv - 2s_{12}u^2v + 4s_{12}v^2\big)\\ &+\epsilon\big(2s_{12}^5 + 4s_{12}^4u + 6s_{12}^3u^2 - 10s_{12}^3v + 4s_{12}^2u^3\\ &-8s_{12}^2uv + 5s_{12}u^2v - 8s_{12}v^2 + 2uv^2\big)\\ &+\epsilon^2\big(4s_{12}^5 + 8s_{12}^4u + 5s_{12}^3u^2 + s_{12}^2u^3 - 2s_{12}^2uv\\ &-s_{12}u^2v - 4s_{12}v^2 - 2uv^2\big)\\ &+\epsilon^3\big(-2s_{12}^3u^2 - 2s_{12}^2u^3 - 4s_{12}^2uv - 6s_{12}u^2v - 4uv^2\big)\\ \Big]\,B_0(q^2;0,0)&\end{aligned}$$

$$
\begin{aligned}
&+\ 3\epsilon s_{23}\,(s_{12}+s_{13})^2\ \Big[\\
&\qquad \big(4s_{12}^2 s_{13}+4s_{12}s_{13}^2+4s_{12}s_{13}s_{23}+3s_{13}^2 s_{23}\big)\\
&\qquad +\epsilon\big(-s_{12}^3-7s_{12}^2 s_{13}-3s_{12}^2 s_{23}-3s_{12}s_{13}^2-8s_{12}s_{13}s_{23}\\
&\qquad -3s_{12}s_{23}^2-2s_{13}^2 s_{23}-s_{13}s_{23}^2-s_{23}^3\big)\\
&\qquad +\epsilon^2\big(s_{12}^3+4s_{12}^2 s_{23}-3s_{12}s_{13}^2+5s_{12}s_{23}^2-3s_{13}^2 s_{23}\\
&\qquad +2s_{23}^3\big)\\
&\qquad +\epsilon^3\big(-s_{12}^2 s_{13}-s_{12}^2 s_{23}-2s_{12}s_{13}^2-4s_{12}s_{13}s_{23}\\
&\qquad -2s_{12}s_{23}^2-2s_{13}^2 s_{23}-3s_{13}s_{23}^2-s_{23}^3\big)\\
&\quad \Big]\,B_0(s_{13};0,0)\ +\ (s_{13}\leftrightarrow s_{23})\\
&-\ 3\epsilon s_{13}\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^2\ \Big[\\
&\qquad \big(-2s_{12}^2-2s_{12}u-u^2+2v\big)\\
&\qquad +\epsilon\big(2s_{12}^2+2s_{12}u+2u^2+v\big)\\
&\qquad +\epsilon^2\big(-u^2+v\big)\\
&\quad \Big]\,C_0(0,0,s_{13};0,0,0)\ +\ (s_{13}\leftrightarrow s_{23})\\
&-\ 3\epsilon\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^3\ \Big[\\
&\qquad \big(-2s_{12}^2-2s_{12}u-u^2+2v\big)\\
&\qquad +\epsilon\big(2s_{12}^2+2s_{12}u+2u^2-5v\big)\\
&\qquad +\epsilon^2\big(-u^2-v\big)\\
&\quad \Big]\,C_0(0,s_{13},q^2;0,0,0)\ +\ (s_{13}\leftrightarrow s_{23})\\
&-\ 3\epsilon s_{13}s_{23}\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^2\ \Big[\\
&\qquad \big(-2s_{12}^2-2s_{12}u-u^2+2v\big)\\
&\qquad +\epsilon\big(2s_{12}^2+2s_{12}u+2u^2-5v\big)\\
&\qquad +\epsilon^2\big(-u^2-v\big)\\
&\quad \Big]\,D_0(0,0,0;q^2,s_{13},s_{23};0,0,0,0)\\
&\Big\}.
\end{aligned}
\tag{A.14}
$$

where $(s_{13}\leftrightarrow s_{23})$ repeats the preceding term with the two invariants exchanged, including the Passarino–Veltman function.

By running `BuildAndIntegrateAntenna[A,3,1,Component->Leading]` we get the integrated result,

$$
\begin{aligned}
\mathcal{A}_3^1 = &-\frac{1}{4\epsilon^4}-\frac{31}{12\epsilon^3}-\frac{1}{\epsilon^2}\left(\frac{53}{8}-\frac{11\pi^2}{24}\right)-\frac{1}{\epsilon}\left(\frac{647}{24}-\frac{22\pi^2}{9}-\frac{23\zeta_3}{3}\right)\\
&-\frac{5231}{48}+\frac{17\pi^2}{2}-\frac{41\pi^4}{480}+\frac{689\zeta_3}{18}+O(\epsilon).
\end{aligned}
\tag{A.15}
$$

A.3.6 $\tilde{A}_3^1$

Running `BuildAntenna[A,3,1,Component->Subleading]`, the built result is returned in the same form. Collected by the Passarino–Veltman functions, with u and v as above,

$$
\begin{aligned}
\tilde{A}_3^1 = \frac{1}{2q^2 s_{13} s_{23} (\epsilon-1) (s_{12}+s_{13})^2 (s_{12}+s_{23})^2 (s_{13}+s_{23})^2} \Big\{ \\
&+ (s_{12}+s_{13}+s_{23}) \Big[\\
&\quad \big(-6s_{12}^5 u^2 + 4s_{12}^5 v - 12 s_{12}^4 u^3 + 12 s_{12}^4 uv - 9 s_{12}^3 u^4 \\
&\quad + 10 s_{12}^3 u^2 v + 8 s_{12}^3 v^2 - 3 s_{12}^2 u^5 + 16 s_{12}^2 u v^2 \\
&\quad - 4 s_{12} u^4 v + 8 s_{12} u^2 v^2 + 4 s_{12} v^3 - 2 u^3 v^2 + 4 u v^3\big) \\
&\quad + \epsilon\big(2 s_{12}^5 u^2 + 4 s_{12}^4 u^3 - 4 s_{12}^4 uv + 6 s_{12}^3 u^4 \\
&\quad - 10 s_{12}^3 u^2 v + 4 s_{12}^2 u^5 - 8 s_{12}^2 u v^2 + 9 s_{12} u^4 v \\
&\quad - 8 s_{12} u^2 v^2 + 6 u^3 v^2 - 4 u v^3\big) \\
&\quad + \epsilon^2\big(4 s_{12}^5 u^2 + 8 s_{12}^4 u^3 + 5 s_{12}^3 u^4 - 4 s_{12}^3 u^2 v + s_{12}^2 u^5 \\
&\quad - 8 s_{12}^2 u^3 v - 3 s_{12} u^4 v - 8 s_{12} u^2 v^2 - 4 u^3 v^2\big) \\
&\quad + \epsilon^3\big(-2 s_{12}^3 u^4 - 2 s_{12}^2 u^5 - 4 s_{12}^2 u^3 v - 6 s_{12} u^4 v - 4 u^3 v^2\big) \\
&\Big] B_0(q^2;0,0) \\
&- 4 s_{12} s_{13} s_{23} (s_{12}+s_{13})^2 (s_{12}+s_{23})^2 (-2\epsilon s_{13} - 2\epsilon s_{23} + s_{12} + 2 s_{13} + 2 s_{23})\, B_0(s_{12};0,0) \\
&+ s_{23} (s_{12}+s_{13})^2 (s_{13}+s_{23})^2 \Big[\\
&\quad \big(-4 s_{12}^2 s_{13} - 2 s_{12} s_{13}^2 - 4 s_{12} s_{13} s_{23} - s_{13}^2 s_{23}\big) \\
&\quad + \epsilon\big(-s_{12}^3 + 5 s_{12}^2 s_{13} - 3 s_{12}^2 s_{23} - s_{12} s_{13}^2 + 4 s_{12} s_{13} s_{23} \\
&\quad - 3 s_{12} s_{23}^2 - 2 s_{13}^2 s_{23} - s_{13} s_{23}^2 - s_{23}^3\big) \\
&\quad + \epsilon^2\big(s_{12}^3 + 4 s_{12}^2 s_{13} + 4 s_{12}^2 s_{23} + 5 s_{12} s_{13}^2 + 8 s_{12} s_{13} s_{23} \\
&\quad + 5 s_{12} s_{23}^2 + 5 s_{13}^2 s_{23} + 4 s_{13} s_{23}^2 + 2 s_{23}^3\big) \\
&\quad + \epsilon^3\big(-s_{12}^2 s_{13} - s_{12}^2 s_{23} + 2 s_{12} s_{13}^2 - 2 s_{12} s_{23}^2 \\
&\quad + 2 s_{13}^2 s_{23} + s_{13} s_{23}^2 - s_{23}^3\big) \\
&\Big] B_0(s_{13};0,0) \;+\; (s_{13} \leftrightarrow s_{23}) \\
&- s_{12} (s_{12}+s_{13})^2 (s_{12}+s_{23})^2 (s_{13}+s_{23})^2 \Big[\\
&\quad (2 s_{12} u + u^2 - 2v) \\
&\quad + \epsilon^2\big(-2u^2 + 2v\big) \\
&\quad + \epsilon^3\big(-u^2\big) \\
&\Big] C_0(0,0,s_{12};0,0,0) \\
&- s_{13} (s_{12}+s_{13})^2 (s_{12}+s_{23})^2 (s_{13}+s_{23})^2 \Big[\\
&\quad \big(-2 s_{12}^2 - 2 s_{12} s_{23} - s_{23}^2\big) \\
&\quad + \epsilon\big(2 s_{12}^2 + 2 s_{12} s_{13} + 2 s_{12} s_{23} + 2 s_{13}^2 + 2 s_{13} s_{23} + 2 s_{23}^2\big) \\
&\quad + \epsilon^2\big(-3 s_{13}^2 - 3 s_{13} s_{23} - s_{23}^2\big)
\end{aligned}
$$

$$
\begin{aligned}
&\quad + \epsilon^3\left(-s_{13}^2 - s_{13}s_{23}\right) \\
&\Big] C_0(0,0,s_{13};0,0,0) \;+\; (s_{13} \leftrightarrow s_{23}) \\
&+\; (s_{12}+s_{13})^2\,(s_{12}+s_{23})^2\,(s_{13}+s_{23})^3 \Big[\\
&\quad \left(-4s_{12}^2 - 2s_{12}u - u^2 + 2v\right) \\
&\quad + \epsilon\left(4s_{12}^2 + 4s_{12}u + 4u^2 - 4v\right) \\
&\quad + \epsilon^2\left(-4u^2 + 2v\right) \\
&\quad + \epsilon^3\left(-u^2\right) \\
&\Big] C_0(0,s_{12},q^2;0,0,0) \\
&+\; (s_{12}+s_{13})^2\,(s_{13}+s_{23})^2\,(s_{12}+s_{23})^3 \Big[\\
&\quad \left(-2s_{12}^2 - 2s_{12}s_{23} - s_{23}^2\right) \\
&\quad + \epsilon\left(2s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{23} + 2s_{13}^2 + 2s_{13}s_{23} + 2s_{23}^2\right) \\
&\quad + \epsilon^2\left(-3s_{13}^2 - 3s_{13}s_{23} - s_{23}^2\right) \\
&\quad + \epsilon^3\left(-s_{13}^2 - s_{13}s_{23}\right) \\
&\Big] C_0(0,s_{13},q^2;0,0,0) \;+\; (s_{13} \leftrightarrow s_{23}) \\
&+\; s_{12}s_{13}\,(s_{12}+s_{13})^2\,(s_{12}+s_{23})^2\,(s_{13}+s_{23})^2 \Big[\\
&\quad \left(-2s_{12}^2 - 2s_{12}s_{23} - s_{23}^2\right) \\
&\quad + \epsilon\left(2s_{12}^2 + 2s_{12}s_{13} + 2s_{12}s_{23} + 2s_{13}^2 + 2s_{13}s_{23} + 2s_{23}^2\right) \\
&\quad + \epsilon^2\left(-3s_{13}^2 - 3s_{13}s_{23} - s_{23}^2\right) \\
&\quad + \epsilon^3\left(-s_{13}^2 - s_{13}s_{23}\right) \\
&\Big] D_0(0,0,0;q^2,s_{12},s_{13};0,0,0,0) \;+\; (s_{13} \leftrightarrow s_{23}) \\
&\Big\}.
\end{aligned}
\tag{A.16}
$$

where $(s_{13} \leftrightarrow s_{23})$ has the same meaning as above.

By running `BuildAndIntegrateAntenna[A,3,1,Component->Subleading]` we get the integrated result,

$$
\tilde{\mathcal{A}}_3^1 = -\frac{1}{\epsilon^2}\left(\frac{5}{8} - \frac{\pi^2}{6}\right) - \frac{1}{\epsilon}\left(\frac{19}{4} - \frac{\pi^2}{4} - 7\zeta_3\right) - \frac{105}{4} + \frac{27\pi^2}{16} + \frac{7\pi^4}{90} + \frac{27\zeta_3}{2} + O(\epsilon). \tag{A.17}
$$

A.3.7 $\hat{A}_3^1$

Running `BuildAntenna[A,3,1,Component->Nf]`, we get the built result,

$$
\hat{A}_3^1 = \frac{1}{3\epsilon q^2}\left(\frac{2s_{12}^2}{s_{13}s_{23}} + \frac{2s_{12}}{s_{13}} + \frac{2s_{12}}{s_{23}} + \frac{s_{13}}{s_{23}} + \frac{s_{23}}{s_{13}}\right) - \frac{(s_{13}+s_{23})^2}{3q^2 s_{13}s_{23}} = \frac{1}{3\epsilon}A_3^0. \tag{A.18}
$$

By running `BuildAndIntegrateAntenna[A,3,1,Component->Nf]` we get the integrated result,

$$
\hat{\mathcal{A}}_3^1 = \frac{1}{3\epsilon^3} + \frac{1}{2\epsilon^2} + \frac{1}{\epsilon}\left(\frac{19}{12} - \frac{7\pi^2}{36}\right) + \frac{109}{24} - \frac{7\pi^2}{24} - \frac{25\zeta_3}{9} + O(\epsilon). \tag{A.19}
$$

A.3.8 A_4^0

Running `BuildAntenna[A,4,0,Component->Leading]`, we get the built result,

$$A_4^0 = \frac{1}{2(\epsilon-1)q^2}\left(A_4^{0,(0)} + \epsilon A_4^{0,(1)} + \epsilon^2 A_4^{0,(2)} + \epsilon^3 A_4^{0,(3)}\right) \tag{A.20}$$

where,

$$\begin{aligned}
A_4^{0,(0)} = {} & \frac{2s_{12}s_{14}}{s_{13}s_{134}^2} + \frac{s_{14}s_{23}}{s_{13}s_{134}^2} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}^2} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{s_{24}}{s_{134}s_{234}} - \frac{5s_{24}}{s_{134}^2} \\
& - \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} - \frac{2s_{34}}{s_{13}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{4s_{12}^2 s_{34}}{s_{13}s_{134}s_{234}s_{24}} \\
& - \frac{s_{12}s_{34}}{s_{234}^2 s_{24}} - \frac{2s_{13}s_{34}}{s_{234}^2 s_{24}} - \frac{s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{2s_{12}}{s_{13}s_{134}} + \frac{2}{s_{134}} - \frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}} + \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} \\
& - \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{s_{13}}{s_{134}s_{234}} - \frac{2s_{12}^2}{s_{13}s_{134}s_{234}} + \frac{2}{s_{234}} - \frac{4s_{14}}{s_{13}s_{24}} - \frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\
& + \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{3s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{13}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{134}s_{24}} - \frac{2s_{12}}{s_{234}s_{24}} - \frac{3s_{12}s_{14}}{s_{13}s_{234}s_{24}} \\
& - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2 s_{14}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} + \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} + \frac{6s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\
& - \frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{12}^2 s_{23}}{s_{13}s_{134}s_{234}s_{24}} - \frac{4s_{12}^2}{s_{13}s_{234}s_{24}} - \frac{2s_{12}^2}{s_{134}s_{234}s_{24}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}} \\
& - \frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}} - \frac{2s_{12}s_{23}}{s_{234}^2 s_{24}} - \frac{4s_{13}s_{23}}{s_{234}^2 s_{24}} + \frac{s_{14}s_{23}}{s_{234}^2 s_{24}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2 s_{34}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}} - \frac{s_{14}}{s_{134}s_{34}} - \frac{6s_{12}s_{14}}{s_{134}^2 s_{34}} + \frac{2s_{14}^2 s_{23}}{s_{13}s_{134}^2 s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}} - \frac{3s_{23}}{s_{134}s_{34}} \\
& - \frac{4s_{14}^2 s_{24}}{s_{13}s_{134}^2 s_{34}} - \frac{6s_{14}s_{24}}{s_{134}^2 s_{34}} + \frac{2s_{24}}{s_{134}s_{34}} - \frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} \\
& - \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{4s_{13}s_{24}}{s_{134}^2 s_{34}} - \frac{3s_{12}s_{24}}{s_{234}^2 s_{34}} - \frac{4s_{13}s_{24}}{s_{234}^2 s_{34}} \\
& + \frac{4s_{12}}{s_{134}s_{34}} + \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} + \frac{4s_{12}}{s_{234}s_{34}} + \frac{2s_{13}}{s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} \\
& - \frac{3s_{14}}{s_{234}s_{34}} + \frac{s_{14}^2 s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{6s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
& - \frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{23}}{s_{234}s_{34}} - \frac{2s_{12}^2}{s_{13}s_{234}s_{34}} + \frac{8s_{12}^2}{s_{134}s_{234}s_{34}} - \frac{s_{13}^2}{s_{134}s_{234}s_{34}} + \frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}} \\
& - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}} + \frac{3s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}} \\
& - \frac{6s_{12}^2 s_{14}}{s_{13}s_{134}s_{24}s_{34}} + \frac{5s_{14}^2 s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}} + \frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}} - \frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}} \\
& - \frac{2s_{12}^2}{s_{134}s_{24}s_{34}} - \frac{2s_{14}^2 s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} \\
& - \frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{3s_{14}^2 s_{23}}{s_{13}s_{234}s_{24}s_{34}} + \frac{4s_{12}s_{14}^2 s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}} + \frac{s_{14}^2 s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}}
\end{aligned}$$

$$
\begin{aligned}
&+\frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}}+\frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}+\frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}+\frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}}-\frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}}\\
&-\frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}-\frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}}-\frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}}-\frac{4s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}}+\frac{2s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}}-\frac{3s_{12}s_{13}}{s_{134}^2s_{34}}\\
&-\frac{6s_{12}s_{23}}{s_{234}^2s_{34}}-\frac{6s_{13}s_{23}}{s_{234}^2s_{34}}+\frac{2}{s_{34}}-\frac{3s_{12}}{s_{134}^2}-\frac{3s_{12}}{s_{234}^2}-\frac{5s_{13}}{s_{234}^2}-\frac{3s_{14}^2}{s_{134}s_{34}^2}\\
&-\frac{5s_{12}s_{14}^2}{s_{134}^2s_{34}^2}-\frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{3s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{s_{12}s_{24}^2}{s_{234}^2s_{34}^2}-\frac{s_{13}s_{24}^2}{s_{234}^2s_{34}^2}+\frac{6s_{12}s_{14}}{s_{134}s_{34}^2}\\
&-\frac{s_{13}s_{14}}{s_{134}s_{34}^2}-\frac{2s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2}+\frac{4s_{14}}{s_{34}^2}+\frac{3s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2}-\frac{4s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}-\frac{3s_{14}s_{23}}{s_{134}s_{34}^2}+\frac{s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}\\
&-\frac{s_{13}s_{23}}{s_{134}s_{34}^2}+\frac{4s_{23}}{s_{34}^2}-\frac{3s_{14}^2s_{24}}{s_{134}^2s_{34}^2}+\frac{4s_{14}s_{24}}{s_{134}s_{34}^2}+\frac{2s_{12}s_{24}}{s_{234}s_{34}^2}-\frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}\\
&-\frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{14}s_{24}}{s_{234}s_{34}^2}+\frac{s_{14}^2s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{23}s_{24}}{s_{234}s_{34}^2}\\
&-\frac{3s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{6s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{s_{13}^2s_{24}}{s_{134}^2s_{34}^2}-\frac{2s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{2s_{12}s_{13}}{s_{134}s_{34}^2}\\
&+\frac{s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2}+\frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2}+\frac{3s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}-\frac{3s_{23}^2}{s_{234}s_{34}^2}+\frac{s_{14}^2s_{23}}{s_{13}s_{234}s_{34}^2}+\frac{3s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}\\
&+\frac{6s_{12}s_{23}}{s_{234}s_{34}^2}+\frac{4s_{13}s_{23}}{s_{234}s_{34}^2}+\frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2}-\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{3s_{14}s_{23}}{s_{234}s_{34}^2}\\
&-\frac{s_{13}^2s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{3s_{14}^2s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2}-\frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2}+\frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2}+\frac{3s_{14}^3s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}\\
&-\frac{4s_{14}^2s_{23}}{s_{13}s_{24}s_{34}^2}+\frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}+\frac{s_{14}^2s_{23}}{s_{134}s_{24}s_{34}^2}-\frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2}+\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2}\\
&-\frac{2s_{14}^2s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{2s_{14}^3s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{3s_{14}^2s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}\\
&+\frac{s_{14}^2s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}+\frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}+\frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2}-\frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2}+\frac{3s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2}\\
&-\frac{s_{12}s_{13}^2}{s_{134}^2s_{34}^2}-\frac{5s_{12}s_{23}^2}{s_{234}^2s_{34}^2}-\frac{3s_{13}s_{23}^2}{s_{234}^2s_{34}^2}
\end{aligned}
\tag{A.21}
$$

$$
\begin{aligned}
A_4^{0,(1)} = &\frac{4s_{12}^3}{s_{13}s_{134}s_{234}s_{24}}+\frac{2s_{12}^2}{s_{13}s_{134}s_{234}}+\frac{4s_{12}^2}{s_{13}s_{134}s_{24}}+\frac{2s_{12}^2s_{14}}{s_{13}s_{134}s_{234}s_{24}}+\frac{2s_{12}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}}\\
&+\frac{4s_{12}^2}{s_{13}s_{234}s_{24}}+\frac{2s_{12}^2}{s_{134}s_{234}s_{24}}+\frac{2s_{12}^2}{s_{13}s_{234}s_{34}}-\frac{8s_{12}^2}{s_{134}s_{234}s_{34}}+\frac{6s_{12}^2s_{14}}{s_{13}s_{134}s_{24}s_{34}}+\frac{2s_{12}^2}{s_{134}s_{24}s_{34}}\\
&-\frac{8s_{12}^2s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}}+\frac{6s_{12}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}-\frac{s_{12}s_{14}}{s_{13}s_{134}^2}+\frac{2s_{12}s_{24}}{s_{13}s_{134}s_{234}}+\frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}}\\
&-\frac{2s_{12}s_{14}s_{34}}{s_{13}s_{134}s_{234}s_{24}}-\frac{2s_{12}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}}+\frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}}+\frac{2s_{12}s_{34}}{s_{234}^2s_{24}}+\frac{2s_{12}s_{34}}{s_{13}s_{134}^2}+\frac{2s_{12}}{s_{13}s_{134}}\\
&+\frac{s_{12}s_{14}}{s_{13}s_{134}s_{234}}+\frac{6s_{12}}{s_{134}s_{234}}+\frac{5s_{12}s_{14}}{s_{13}s_{134}s_{24}}+\frac{5s_{12}s_{23}}{s_{13}s_{134}s_{24}}+\frac{2s_{12}}{s_{13}s_{24}}+\frac{5s_{12}s_{14}}{s_{13}s_{234}s_{24}}
\end{aligned}
$$

$$+\frac{5s_{12}s_{23}}{s_{13}s_{234}s_{24}}+\frac{s_{12}s_{23}}{s_{134}s_{234}s_{24}}+\frac{2s_{12}s_{13}}{s_{134}s_{234}s_{24}}+\frac{2s_{12}}{s_{234}s_{24}}-\frac{s_{12}s_{23}}{s_{234}^2s_{24}}+\frac{3s_{12}s_{14}^2}{s_{13}s_{134}^2s_{34}}+\frac{4s_{12}s_{14}}{s_{13}s_{134}s_{34}}$$
$$+\frac{4s_{12}s_{14}}{s_{134}^2s_{34}}+\frac{s_{12}s_{14}s_{24}}{s_{13}s_{134}s_{234}s_{34}}+\frac{s_{12}s_{24}}{s_{134}s_{234}s_{34}}+\frac{5s_{12}s_{24}}{s_{234}^2s_{34}}-\frac{4s_{12}}{s_{134}s_{34}}+\frac{s_{12}s_{14}}{s_{13}s_{234}s_{34}}$$
$$-\frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{234}s_{34}}+\frac{s_{12}s_{23}}{s_{13}s_{234}s_{34}}-\frac{s_{12}s_{23}}{s_{134}s_{234}s_{34}}+\frac{s_{12}s_{13}}{s_{134}s_{234}s_{34}}-\frac{4s_{12}}{s_{234}s_{34}}$$
$$+\frac{3s_{12}s_{14}^2}{s_{13}s_{134}s_{24}s_{34}}+\frac{4s_{12}s_{14}}{s_{13}s_{24}s_{34}}+\frac{s_{12}s_{14}}{s_{134}s_{24}s_{34}}-\frac{s_{12}s_{14}s_{23}}{s_{13}s_{134}s_{24}s_{34}}+\frac{4s_{12}s_{23}}{s_{13}s_{24}s_{34}}+\frac{s_{12}s_{23}}{s_{134}s_{24}s_{34}}$$
$$-\frac{4s_{12}s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}}+\frac{3s_{12}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}}-\frac{4s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}s_{34}}-\frac{s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{24}s_{34}}-\frac{s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}$$
$$+\frac{s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{24}s_{34}}+\frac{4s_{12}s_{23}}{s_{234}s_{24}s_{34}}+\frac{3s_{12}s_{23}^2}{s_{234}^2s_{24}s_{34}}+\frac{5s_{12}s_{13}}{s_{134}^2s_{34}}+\frac{4s_{12}s_{23}}{s_{234}^2s_{34}}+\frac{5s_{12}}{s_{134}^2}$$
$$+\frac{5s_{12}}{s_{234}^2}+\frac{6s_{12}s_{14}^2}{s_{134}^2s_{34}^2}+\frac{2s_{12}s_{24}^2}{s_{234}^2s_{34}^2}-\frac{6s_{12}s_{14}}{s_{134}s_{34}^2}+\frac{8s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{12}s_{24}}{s_{234}s_{34}^2}$$
$$-\frac{2s_{12}s_{13}}{s_{134}s_{34}^2}-\frac{2s_{12}s_{14}s_{23}}{s_{13}s_{234}s_{34}^2}+\frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{6s_{12}s_{23}}{s_{234}s_{34}^2}-\frac{6s_{12}s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}^2}+\frac{8s_{12}s_{14}s_{23}}{s_{13}s_{24}s_{34}^2}$$
$$-\frac{2s_{12}s_{14}s_{23}}{s_{134}s_{24}s_{34}^2}+\frac{4s_{12}s_{14}^2s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}-\frac{6s_{12}s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2}+\frac{2s_{12}s_{13}^2}{s_{134}^2s_{34}^2}+\frac{6s_{12}s_{23}^2}{s_{234}^2s_{34}^2}$$
$$+\frac{8s_{14}}{s_{13}s_{134}}-\frac{2s_{14}s_{23}}{s_{13}s_{134}^2}+\frac{8s_{23}}{s_{13}s_{134}}+\frac{8s_{14}s_{24}}{s_{13}s_{134}^2}+\frac{4s_{24}}{s_{134}s_{234}}+\frac{14s_{24}}{s_{134}^2}$$
$$+\frac{6s_{24}s_{34}}{s_{13}s_{134}^2}+\frac{4s_{34}}{s_{13}s_{134}}+\frac{6s_{34}}{s_{13}s_{24}}+\frac{4s_{34}}{s_{234}s_{24}}+\frac{6s_{13}s_{34}}{s_{234}^2s_{24}}+\frac{4}{s_{134}}+\frac{2s_{14}}{s_{134}s_{234}}$$
$$+\frac{2s_{23}}{s_{134}s_{234}}+\frac{4s_{13}}{s_{134}s_{234}}+\frac{4}{s_{234}}+\frac{8s_{14}}{s_{13}s_{24}}-\frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}}+\frac{8s_{23}}{s_{13}s_{24}}+\frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{24}}$$
$$+\frac{8s_{14}}{s_{234}s_{24}}+\frac{2s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{24}}-\frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}}+\frac{8s_{23}}{s_{234}s_{24}}+\frac{8s_{13}s_{23}}{s_{234}^2s_{24}}-\frac{2s_{14}s_{23}}{s_{234}^2s_{24}}$$
$$+\frac{8s_{14}^2}{s_{13}s_{134}s_{34}}+\frac{2s_{24}^2}{s_{134}s_{234}s_{34}}+\frac{7s_{14}}{s_{134}s_{34}}+\frac{s_{14}^2s_{23}}{s_{13}s_{134}^2s_{34}}+\frac{6s_{14}s_{23}}{s_{13}s_{134}s_{34}}+\frac{s_{14}s_{23}}{s_{134}^2s_{34}}$$
$$+\frac{5s_{23}}{s_{134}s_{34}}+\frac{8s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}}+\frac{11s_{14}s_{24}}{s_{134}^2s_{34}}-\frac{2s_{24}}{s_{134}s_{34}}+\frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}}-\frac{2s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}}$$
$$+\frac{6s_{13}s_{24}}{s_{134}s_{234}s_{34}}+\frac{2s_{24}}{s_{234}s_{34}}+\frac{11s_{13}s_{24}}{s_{134}^2s_{34}}+\frac{11s_{13}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{13}}{s_{134}s_{34}}-\frac{2s_{13}}{s_{234}s_{34}}$$
$$+\frac{5s_{14}}{s_{234}s_{34}}-\frac{s_{14}^2s_{23}}{s_{13}s_{134}s_{234}s_{34}}+\frac{2s_{14}s_{23}}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{23}}{s_{134}s_{234}s_{34}}+\frac{7s_{23}}{s_{234}s_{34}}+\frac{2s_{13}^2}{s_{134}s_{234}s_{34}}$$
$$+\frac{8s_{14}^2}{s_{13}s_{24}s_{34}}-\frac{5s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}}+\frac{8s_{23}^2}{s_{13}s_{24}s_{34}}-\frac{5s_{14}^2s_{23}}{s_{13}s_{134}s_{24}s_{34}}+\frac{6s_{14}s_{23}}{s_{13}s_{24}s_{34}}$$
$$-\frac{5s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}}-\frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}}+\frac{8s_{23}^2}{s_{234}s_{24}s_{34}}-\frac{5s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}}-\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}}$$
$$+\frac{6s_{14}s_{23}}{s_{234}s_{24}s_{34}}+\frac{8s_{13}s_{23}^2}{s_{234}^2s_{24}s_{34}}+\frac{s_{14}s_{23}^2}{s_{234}^2s_{24}s_{34}}+\frac{11s_{13}s_{23}}{s_{234}^2s_{34}}+\frac{s_{14}s_{23}}{s_{234}^2s_{34}}-\frac{2}{s_{34}}$$
$$+\frac{14s_{13}}{s_{234}^2}+\frac{3s_{14}^2}{s_{134}s_{34}^2}+\frac{s_{14}s_{24}^2}{s_{134}s_{234}s_{34}^2}+\frac{5s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}+\frac{3s_{13}s_{24}^2}{s_{234}^2s_{34}^2}+\frac{s_{13}s_{14}}{s_{134}s_{34}^2}$$

$$
\begin{aligned}
&- \frac{4s_{14}}{s_{34}^2} - \frac{3s_{14}^3 s_{23}}{s_{13}s_{134}^2 s_{34}^2} + \frac{4s_{14}^2 s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{2s_{14}^2 s_{23}}{s_{134}^2 s_{34}^2} + \frac{3s_{14}s_{23}}{s_{134}s_{34}^2} - \frac{3s_{13}s_{14}s_{23}}{s_{134}^2 s_{34}^2} \\
&+ \frac{s_{13}s_{23}}{s_{134}s_{34}^2} - \frac{4s_{23}}{s_{34}^2} + \frac{3s_{14}^2 s_{24}}{s_{134}^2 s_{34}^2} - \frac{4s_{14}s_{24}}{s_{134}s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2 s_{34}^2} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{s_{14}s_{24}}{s_{234}s_{34}^2} - \frac{s_{14}^2 s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{4s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{s_{23}s_{24}}{s_{234}s_{34}^2} + \frac{5s_{13}^2 s_{24}}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{3s_{13}^2 s_{24}}{s_{134}^2 s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2 s_{34}^2} - \frac{3s_{14}s_{23}s_{24}}{s_{234}^2 s_{34}^2} - \frac{s_{14}^2 s_{23}^2}{s_{13}s_{134}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{3s_{23}^2}{s_{234}s_{34}^2} - \frac{s_{14}^2 s_{23}}{s_{13}s_{234}s_{34}^2} - \frac{s_{14}^2 s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{4s_{13}s_{23}}{s_{234}s_{34}^2} - \frac{4s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{3s_{14}s_{23}}{s_{234}s_{34}^2} \\
&+ \frac{s_{13}^2 s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{3s_{14}^2 s_{23}^2}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{13}s_{24}s_{34}^2} - \frac{s_{14}s_{23}^2}{s_{134}s_{24}s_{34}^2} - \frac{3s_{14}^3 s_{23}}{s_{13}s_{134}s_{24}s_{34}^2} + \frac{4s_{14}^2 s_{23}}{s_{13}s_{24}s_{34}^2} \\
&- \frac{s_{14}^2 s_{23}}{s_{134}s_{24}s_{34}^2} + \frac{2s_{14}^2 s_{23}^3}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{13}s_{234}s_{24}s_{34}^2} + \frac{2s_{14}^3 s_{23}^2}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} - \frac{3s_{14}^2 s_{23}^2}{s_{13}s_{234}s_{24}s_{34}^2} \\
&- \frac{s_{14}^2 s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} - \frac{s_{13}s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}^2} + \frac{4s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} - \frac{3s_{14}s_{23}^3}{s_{234}^2 s_{24}s_{34}^2} + \frac{3s_{13}s_{23}^2}{s_{234}^2 s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2 s_{34}^2}
\end{aligned}
\tag{A.22}
$$

$$
\begin{aligned}
A_4^{0,(2)} = &- \frac{s_{13}^2}{s_{134}s_{234}s_{34}} - \frac{2s_{13}^2 s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}^2 s_{24}}{s_{134}^2 s_{34}^2} - \frac{s_{12}s_{13}^2}{s_{134}^2 s_{34}^2} - \frac{6s_{13}s_{34}}{s_{234}^2 s_{24}} - \frac{7s_{13}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{134}s_{234}s_{24}} \\
&- \frac{4s_{13}s_{23}}{s_{234}^2 s_{24}} - \frac{12s_{13}s_{24}}{s_{134}s_{234}s_{34}} - \frac{7s_{13}s_{24}}{s_{134}^2 s_{34}} - \frac{7s_{13}s_{24}}{s_{234}^2 s_{34}} - \frac{s_{13}}{s_{134}s_{34}} + \frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{13}s_{23}}{s_{134}s_{234}s_{34}} \\
&- \frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{4s_{13}s_{23}^2}{s_{234}^2 s_{24}s_{34}} - \frac{2s_{12}s_{13}}{s_{134}^2 s_{34}} - \frac{5s_{13}s_{23}}{s_{234}^2 s_{34}} - \frac{11s_{13}}{s_{234}^2} \\
&- \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{24}^2}{s_{234}^2 s_{34}^2} + \frac{2s_{12}s_{13}s_{14}}{s_{134}^2 s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}^2 s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}^2 s_{34}^2} + \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{23}s_{24}}{s_{234}^2 s_{34}^2} + \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{13}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{7s_{24}}{s_{134}s_{234}} \\
&- \frac{11s_{24}}{s_{134}^2} - \frac{2s_{34}}{s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{134}s_{234}s_{24}} - \frac{8s_{34}}{s_{234}s_{24}} - \frac{3s_{12}s_{34}}{s_{234}^2 s_{24}} - \frac{11}{s_{134}} - \frac{4s_{14}}{s_{134}s_{234}} \\
&- \frac{4s_{23}}{s_{134}s_{234}} + \frac{2s_{12}}{s_{134}s_{234}} - \frac{11}{s_{234}} - \frac{s_{12}}{s_{134}s_{24}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{24}} - \frac{4s_{14}}{s_{234}s_{24}} - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{24}} \\
&- \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{24}} - \frac{8s_{23}}{s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{234}^2 s_{24}} + \frac{3s_{14}s_{23}}{s_{234}^2 s_{24}} - \frac{s_{24}^2}{s_{134}s_{234}s_{34}} - \frac{5s_{14}}{s_{134}s_{34}} + \frac{2s_{12}s_{14}}{s_{134}^2 s_{34}} \\
&- \frac{s_{14}s_{23}}{s_{134}^2 s_{34}} - \frac{2s_{23}}{s_{134}s_{34}} - \frac{5s_{14}s_{24}}{s_{134}^2 s_{34}} - \frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{23}s_{24}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}} \\
&- \frac{s_{24}}{s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2 s_{34}} + \frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{14}}{s_{234}s_{34}} + \frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}} + \frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}} \\
&- \frac{5s_{23}}{s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{24}s_{34}} - \frac{8s_{23}^2}{s_{234}s_{24}s_{34}} - \frac{s_{14}^2 s_{23}}{s_{134}s_{234}s_{24}s_{34}} - \frac{3s_{14}s_{23}^2}{s_{234}^2 s_{24}s_{34}} \\
&+ \frac{2s_{12}s_{23}}{s_{234}^2 s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2 s_{34}} - \frac{4s_{12}}{s_{134}^2} - \frac{4s_{12}}{s_{234}^2} - \frac{s_{12}s_{14}^2}{s_{134}^2 s_{34}^2} - \frac{s_{12}s_{24}^2}{s_{234}^2 s_{34}^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{2s_{14}^2 s_{23}}{s_{134}^2 s_{34}^2} + \frac{2s_{12}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{12}s_{23}s_{24}}{s_{234}^2 s_{34}^2} + \frac{2s_{14}s_{23}s_{24}}{s_{234}^2 s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} \\
&- \frac{2s_{14}^2 s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{s_{12}s_{23}^2}{s_{234}^2 s_{34}^2} - \frac{2s_{14}s_{23}^2}{s_{234}^2 s_{34}^2} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{234}s_{24}} - \frac{8s_{14}}{s_{13}s_{134}} \\
&+ \frac{3s_{12}s_{14}}{s_{13}s_{134}^2} + \frac{3s_{14}s_{23}}{s_{13}s_{134}^2} - \frac{4s_{23}}{s_{13}s_{134}} - \frac{4s_{14}s_{24}}{s_{13}s_{134}^2} - \frac{s_{12}s_{24}}{s_{13}s_{134}s_{234}} - \frac{6s_{24}s_{34}}{s_{13}s_{134}^2} \\
&- \frac{8s_{34}}{s_{13}s_{134}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{14}s_{23}s_{34}}{s_{13}s_{134}s_{234}s_{24}} - \frac{6s_{34}}{s_{13}s_{24}} \\
&- \frac{3s_{12}s_{34}}{s_{13}s_{134}^2} - \frac{s_{12}}{s_{13}s_{234}} - \frac{2s_{12}s_{14}}{s_{13}s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{s_{12}s_{23}}{s_{13}s_{134}s_{234}} + \frac{3s_{12}s_{14}}{s_{13}s_{134}s_{24}} \\
&- \frac{4s_{14}}{s_{13}s_{24}} - \frac{s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{12}s_{23}}{s_{13}s_{134}s_{24}} - \frac{4s_{23}}{s_{13}s_{24}} - \frac{2s_{12}s_{14}}{s_{13}s_{234}s_{24}} + \frac{3s_{12}s_{23}}{s_{13}s_{234}s_{24}} \\
&- \frac{s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{8s_{14}^2}{s_{13}s_{134}s_{34}} - \frac{3s_{14}^2 s_{23}}{s_{13}s_{134}^2 s_{34}} - \frac{4s_{14}^2 s_{24}}{s_{13}s_{134}^2 s_{34}} + \frac{s_{14}s_{23}s_{24}}{s_{13}s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}^2}{s_{13}s_{134}s_{234}s_{34}} \\
&+ \frac{2s_{14}^2 s_{23}}{s_{13}s_{134}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{13}s_{234}s_{34}} - \frac{4s_{14}^2}{s_{13}s_{24}s_{34}} + \frac{2s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{4s_{23}^2}{s_{13}s_{24}s_{34}} - \frac{3s_{14}^2 s_{23}}{s_{13}s_{134}s_{24}s_{34}} \\
&- \frac{3s_{14}s_{23}^2}{s_{13}s_{234}s_{24}s_{34}} + \frac{2s_{14}^2 s_{23}}{s_{13}s_{234}s_{24}s_{34}}
\end{aligned}
\tag{A.23}
$$

$$
\begin{aligned}
A_4^{0,(3)} = &\frac{2s_{12}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{134}s_{234}s_{24}} + \frac{2s_{12}s_{34}}{s_{234}^2 s_{24}} + \frac{2s_{12}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{12}}{s_{134}s_{234}} \\
&+ \frac{2s_{12}}{s_{134}^2} + \frac{2s_{12}}{s_{234}^2} - \frac{2s_{14}s_{23}}{s_{13}s_{134}^2} + \frac{2s_{24}}{s_{134}s_{234}} + \frac{2s_{24}}{s_{134}^2} + \frac{2s_{24}s_{34}}{s_{13}s_{134}^2} + \frac{4s_{34}}{s_{13}s_{134}} \\
&+ \frac{4s_{34}}{s_{134}s_{234}} + \frac{2s_{34}}{s_{13}s_{24}} + \frac{4s_{34}}{s_{234}s_{24}} + \frac{2s_{13}s_{34}}{s_{234}^2 s_{24}} + \frac{2}{s_{134}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} \\
&+ \frac{2}{s_{234}} - \frac{2s_{14}s_{23}}{s_{13}s_{134}s_{24}} - \frac{2s_{14}s_{23}}{s_{13}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{134}s_{234}s_{24}} - \frac{2s_{14}s_{23}}{s_{234}^2 s_{24}} + \frac{2s_{13}}{s_{234}^2}.
\end{aligned}
\tag{A.24}
$$

By running `BuildAndIntegrateAntenna[A,4,0,Component->Leading]` we get the integrated result,

$$
\begin{aligned}
\mathcal{A}_4^0 = &\frac{3}{4\epsilon^4} + \frac{65}{24\epsilon^3} + \frac{434 - 39\pi^2}{36\epsilon^2} + \frac{1}{\epsilon}\left(\frac{43223}{864} - \frac{589\pi^2}{144} - \frac{71\zeta_3}{4}\right) \\
&+ \frac{1076717}{5184} - \frac{7955\pi^2}{432} + \frac{373\pi^4}{1440} - \frac{1327\zeta_3}{18} + O(\epsilon)\,.
\end{aligned}
\tag{A.25}
$$

A.3.9 $\tilde{A}_4^0$

Running `BuildAntenna[A,4,0,Component->Subleading]`, we get the built result,

$$
\tilde{A}_4^0 = \frac{1}{2(\epsilon-1)q^2}\left(\tilde{A}_4^{0,(0)} + \epsilon\tilde{A}_4^{0,(1)} + \epsilon^2\tilde{A}_4^{0,(2)} + \epsilon^3\tilde{A}_4^{0,(3)}\right) \tag{A.26}
$$

where,

$$
\tilde{A}_4^{0,(0)} = -\frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}}\left[-\frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2 s_{14}} + \frac{4s_{14}^3 s_{23}}{s_{13}s_{134}^2 s_{34}^2}\right.
$$

$$
-\frac{2s_{14}^2\left(2s_{12}-3s_{23}+2s_{24}\right)}{s_{13}s_{134}^2s_{34}}-\frac{2\left(3s_{12}s_{14}+3s_{12}s_{34}-3s_{14}s_{23}+2s_{14}s_{24}-s_{23}s_{34}+s_{24}s_{34}\right)}{s_{13}s_{134}^2}
$$

$$
-\frac{2s_{12}s_{34}(s_{12}+s_{24})}{s_{13}s_{134}s_{14}s_{23}}-\frac{2s_{12}s_{34}(s_{12}+s_{23})}{s_{13}s_{134}s_{14}s_{24}}
$$

$$
-\frac{2s_{12}^2\left(2s_{12}+s_{14}+s_{24}+2s_{34}\right)}{s_{13}s_{134}s_{23}s_{234}}-\frac{2s_{12}\left(2s_{12}+2s_{24}+s_{34}\right)}{s_{13}s_{134}s_{23}}
$$

$$
-\frac{2s_{14}^2s_{23}^2\left(2s_{12}+s_{14}+s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}s_{34}^2}+\frac{2s_{14}s_{23}\left(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}-s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}s_{34}}
$$

$$
-\frac{2s_{12}\left(2s_{12}^2+s_{12}s_{14}+s_{12}s_{23}+2s_{12}s_{34}-3s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}s_{24}}-\frac{2s_{14}^2s_{23}\left(2s_{12}+s_{14}-s_{24}\right)}{s_{13}s_{134}s_{234}s_{34}^2}
$$

$$
+\frac{2s_{14}\left(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+s_{23}^2+s_{23}s_{24}\right)}{s_{13}s_{134}s_{234}s_{34}}
$$

$$
-\frac{2\left(2s_{12}^2-2s_{12}s_{14}+s_{12}s_{23}+s_{12}s_{24}+s_{12}s_{34}-s_{14}s_{23}\right)}{s_{13}s_{134}s_{234}}
$$

$$
+\frac{4s_{14}^2s_{23}\left(2s_{12}+s_{14}+s_{23}\right)}{s_{13}s_{134}s_{24}s_{34}^2}-\frac{2s_{14}\left(4s_{12}^2+2s_{12}s_{14}-2s_{12}s_{23}-3s_{14}s_{23}-3s_{23}^2\right)}{s_{13}s_{134}s_{24}s_{34}}
$$

$$
-\frac{2\left(4s_{12}^2+3s_{12}s_{14}+2s_{12}s_{23}+s_{12}s_{34}-s_{14}s_{23}-s_{23}^2\right)}{s_{13}s_{134}s_{24}}-\frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2}
$$

$$
+\frac{4s_{14}(s_{12}-2s_{23}+s_{24})}{s_{13}s_{134}s_{34}}+\frac{2(s_{12}+s_{14}-2s_{23}+s_{24})}{s_{13}s_{134}}
$$

$$
-\frac{4s_{12}^3}{s_{13}s_{14}s_{23}s_{24}}-\frac{2s_{12}^2}{s_{13}s_{14}s_{23}}-\frac{2s_{12}^2}{s_{13}s_{14}s_{24}}-\frac{2s_{12}}{s_{13}s_{14}}
$$

$$
-\frac{2s_{12}s_{34}(s_{12}+s_{14})}{s_{13}s_{23}s_{234}s_{24}}-\frac{2s_{12}(2s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}}-\frac{2s_{12}^2}{s_{13}s_{23}s_{24}}
$$

$$
+\frac{4s_{14}s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{24}s_{34}^2}-\frac{2s_{23}\left(4s_{12}^2-2s_{12}s_{14}+2s_{12}s_{23}-3s_{14}^2-3s_{14}s_{23}\right)}{s_{13}s_{234}s_{24}s_{34}}
$$

$$
-\frac{2\left(4s_{12}^2+2s_{12}s_{14}+3s_{12}s_{23}+s_{12}s_{34}-s_{14}^2-s_{14}s_{23}\right)}{s_{13}s_{234}s_{24}}+\frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{34}^2}
$$

$$
-\frac{4\left(2s_{12}^2+s_{12}s_{23}-s_{14}^2-s_{14}s_{23}\right)}{s_{13}s_{234}s_{34}}-\frac{2(2s_{12}-s_{14})}{s_{13}s_{234}}
$$

$$
-\frac{6s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{24}s_{34}^2}+\frac{2\left(4s_{12}^2-2s_{14}^2-3s_{14}s_{23}-2s_{23}^2\right)}{s_{13}s_{24}s_{34}}
$$

$$
-\frac{2(s_{12}+2s_{14}+2s_{23}+s_{34})}{s_{13}s_{24}}+\frac{2s_{14}s_{23}}{s_{13}s_{34}^2}+\frac{2s_{23}}{s_{13}s_{34}}
$$

$$
+\frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2}-\frac{2s_{13}^2(2s_{12}+2s_{23}-3s_{24})}{s_{134}^2s_{14}s_{34}}
$$

$$
-\frac{2\left(3s_{12}s_{13}+3s_{12}s_{34}+2s_{13}s_{23}-3s_{13}s_{24}+s_{23}s_{34}-s_{24}s_{34}\right)}{s_{134}^2s_{14}}
$$

$$
-\frac{\begin{array}{l}4\big(2s_{12}s_{13}^2+4s_{12}s_{13}s_{14}+2s_{12}s_{14}^2+s_{13}^2s_{23}\\ \quad -s_{13}^2s_{24}+s_{13}s_{14}s_{23}+s_{13}s_{14}s_{24}-s_{14}^2s_{23}+s_{14}^2s_{24}\big)\end{array}}{s_{134}^2s_{34}^2}
$$

$$
-\frac{10\left(2s_{12}s_{13}+2s_{12}s_{14}+s_{13}s_{23}+s_{14}s_{24}\right)}{s_{134}^2s_{34}}-\frac{2(8s_{12}+s_{23}+s_{24})}{s_{134}^2}
$$

$$
\begin{aligned}
&- \frac{2s_{13}^2 s_{24}^2 (2s_{12} + s_{13} + s_{24})}{s_{134} s_{14} s_{23} s_{234} s_{34}^2} + \frac{2s_{13} s_{24} \left(4s_{12}^2 + 2s_{12} s_{13} + 2s_{12} s_{24} - s_{13} s_{24}\right)}{s_{134} s_{14} s_{23} s_{234} s_{34}} \\
&- \frac{2s_{12} \left(2s_{12}^2 + s_{12} s_{13} + s_{12} s_{24} + 2s_{12} s_{34} - 3s_{13} s_{24}\right)}{s_{134} s_{14} s_{23} s_{234}} + \frac{4s_{13}^2 s_{24} (2s_{12} + s_{13} + s_{24})}{s_{134} s_{14} s_{23} s_{34}^2} \\
&- \frac{2s_{13} \left(4s_{12}^2 + 2s_{12} s_{13} - 2s_{12} s_{24} - 3s_{13} s_{24} - 3s_{24}^2\right)}{s_{134} s_{14} s_{23} s_{34}} \\
&- \frac{2 \left(4s_{12}^2 + 3s_{12} s_{13} + 2s_{12} s_{24} + s_{12} s_{34} - s_{13} s_{24} - s_{24}^2\right)}{s_{134} s_{14} s_{23}} \\
&- \frac{2s_{12}^2 (2s_{12} + s_{13} + s_{23} + 2s_{34})}{s_{134} s_{14} s_{234} s_{24}} - \frac{2s_{13}^2 s_{24} (2s_{12} + s_{13} - s_{23})}{s_{134} s_{14} s_{234} s_{34}^2} \\
&+ \frac{2s_{13} \left(4s_{12}^2 + 2s_{12} s_{13} + 2s_{12} s_{24} + s_{23} s_{24} + s_{24}^2\right)}{s_{134} s_{14} s_{234} s_{34}} \\
&- \frac{2 \left(2s_{12}^2 - 2s_{12} s_{13} + s_{12} s_{23} + s_{12} s_{24} + s_{12} s_{34} - s_{13} s_{24}\right)}{s_{134} s_{14} s_{234}} \\
&- \frac{2s_{12} (2s_{12} + 2s_{23} + s_{34})}{s_{134} s_{14} s_{24}} - \frac{8s_{13}^2 s_{24}}{s_{134} s_{14} s_{34}^2} \\
&+ \frac{4s_{13} (s_{12} + s_{23} - 2s_{24})}{s_{134} s_{14} s_{34}} + \frac{2(s_{12} + s_{13} + s_{23} - 2s_{24})}{s_{134} s_{14}} \\
&- \frac{2s_{13} s_{24}^2 (2s_{12} - s_{14} + s_{24})}{s_{134} s_{23} s_{234} s_{34}^2} + \frac{2s_{24} \left(4s_{12}^2 + 2s_{12} s_{13} + 2s_{12} s_{24} + s_{13}^2 + s_{13} s_{14}\right)}{s_{134} s_{23} s_{234} s_{34}} \\
&- \frac{2 \left(2s_{12}^2 + s_{12} s_{13} + s_{12} s_{14} - 2s_{12} s_{24} + s_{12} s_{34} - s_{13} s_{24}\right)}{s_{134} s_{23} s_{234}} \\
&+ \frac{4s_{13} s_{24} (2s_{12} + s_{13} + s_{24})}{s_{134} s_{23} s_{34}^2} - \frac{4 \left(2s_{12}^2 + s_{12} s_{13} - s_{13} s_{24} - s_{24}^2\right)}{s_{134} s_{23} s_{34}} - \frac{2(2s_{12} - s_{24})}{s_{134} s_{23}} \\
&- \frac{2s_{14} s_{23}^2 (2s_{12} - s_{13} + s_{23})}{s_{134} s_{234} s_{24} s_{34}^2} + \frac{2s_{23} \left(4s_{12}^2 + 2s_{12} s_{14} + 2s_{12} s_{23} + s_{13} s_{14} + s_{14}^2\right)}{s_{134} s_{234} s_{24} s_{34}} \\
&- \frac{2 \left(2s_{12}^2 + s_{12} s_{13} + s_{12} s_{14} - 2s_{12} s_{23} + s_{12} s_{34} - s_{14} s_{23}\right)}{s_{134} s_{234} s_{24}} \\
&- \frac{\begin{aligned}2\big(&4s_{12} s_{13} s_{23} + 6s_{12} s_{13} s_{24} + 6s_{12} s_{14} s_{23} + 4s_{12} s_{14} s_{24} \\ &- s_{13} s_{14} s_{23} - s_{13} s_{14} s_{24} - s_{13} s_{23} s_{24} - s_{14} s_{23} s_{24}\big)\end{aligned}}{s_{134} s_{234} s_{34}^2} \\
&+ \frac{2}{s_{134} s_{234} s_{34}} \Big(8s_{12}^2 - 2s_{12} s_{13} - 2s_{12} s_{14} - 2s_{12} s_{23} - 2s_{12} s_{24} - s_{13}^2 - 2s_{13} s_{14} + s_{13} s_{24} \\
&\qquad - s_{14}^2 + s_{14} s_{23} - s_{23}^2 - 2s_{23} s_{24} - s_{24}^2\Big) \\
&- \frac{2(2s_{12} + s_{13} + s_{14} + s_{23} + s_{24})}{s_{134} s_{234}} \\
&+ \frac{4s_{14} s_{23} (2s_{12} + s_{14} + s_{23})}{s_{134} s_{24} s_{34}^2} - \frac{4 \left(2s_{12}^2 + s_{12} s_{14} - s_{14} s_{23} - s_{23}^2\right)}{s_{134} s_{24} s_{34}} - \frac{2(2s_{12} - s_{23})}{s_{134} s_{24}} \\
&+ \frac{4 \left(4s_{12} s_{13} + 4s_{12} s_{14} - s_{13}^2 - 2s_{13} s_{14} + s_{13} s_{23} - s_{13} s_{24} - s_{14}^2 - s_{14} s_{23} + s_{14} s_{24}\right)}{s_{134} s_{34}^2} \\
&+ \frac{2(8s_{12} - s_{13} - s_{14} + s_{23} + s_{24})}{s_{134} s_{34}} + \frac{4}{s_{134}}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{2s_{12}s_{34}(s_{12}+s_{13})}{s_{14}s_{23}s_{234}s_{24}} + \frac{4s_{13}s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{234}s_{34}^2} \\
&- \frac{2s_{24}\left(4s_{12}^2-2s_{12}s_{13}+2s_{12}s_{24}-3s_{13}^2-3s_{13}s_{24}\right)}{s_{14}s_{23}s_{234}s_{34}} \\
&- \frac{2\left(4s_{12}^2+2s_{12}s_{13}+3s_{12}s_{24}+s_{12}s_{34}-s_{13}^2-s_{13}s_{24}\right)}{s_{14}s_{23}s_{234}} - \frac{2s_{12}^2}{s_{14}s_{23}s_{24}} \\
&- \frac{6s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{34}^2} + \frac{2\left(4s_{12}^2-2s_{13}^2-3s_{13}s_{24}-2s_{24}^2\right)}{s_{14}s_{23}s_{34}} \\
&- \frac{2(s_{12}+2s_{13}+2s_{24}+s_{34})}{s_{14}s_{23}} - \frac{2s_{12}(2s_{12}+2s_{13}+s_{34})}{s_{14}s_{234}s_{24}} \\
&+ \frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{234}s_{34}^2} - \frac{4\left(2s_{12}^2+s_{12}s_{24}-s_{13}^2-s_{13}s_{24}\right)}{s_{14}s_{234}s_{34}} \\
&- \frac{2(2s_{12}-s_{13})}{s_{14}s_{234}} + \frac{2s_{13}s_{24}}{s_{14}s_{34}^2} + \frac{2s_{24}}{s_{14}s_{34}} \\
&- \frac{2s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}} + \frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2} \\
&- \frac{2s_{24}^2(2s_{12}-3s_{13}+2s_{14})}{s_{23}s_{234}^2s_{34}} - \frac{2\left(3s_{12}s_{24}+3s_{12}s_{34}-3s_{13}s_{24}-s_{13}s_{34}+2s_{14}s_{24}+s_{14}s_{34}\right)}{s_{23}s_{234}^2} \\
&- \frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2} + \frac{4s_{24}(s_{12}-2s_{13}+s_{14})}{s_{23}s_{234}s_{34}} \\
&+ \frac{2(s_{12}-2s_{13}+s_{14}+s_{24})}{s_{23}s_{234}} - \frac{2s_{12}}{s_{23}s_{24}} + \frac{2s_{13}s_{24}}{s_{23}s_{34}^2} + \frac{2s_{13}}{s_{23}s_{34}} \\
&+ \frac{4s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2} - \frac{2s_{23}^2(2s_{12}+2s_{13}-3s_{14})}{s_{234}^2s_{24}s_{34}} \\
&- \frac{2\left(3s_{12}s_{23}+3s_{12}s_{34}+2s_{13}s_{23}+s_{13}s_{34}-3s_{14}s_{23}-s_{14}s_{34}\right)}{s_{234}^2s_{24}} \\
&- \frac{\begin{aligned}4\big(&2s_{12}s_{23}^2+4s_{12}s_{23}s_{24}+2s_{12}s_{24}^2+s_{13}s_{23}^2 \\ &+s_{13}s_{23}s_{24}-s_{13}s_{24}^2-s_{14}s_{23}^2+s_{14}s_{23}s_{24}+s_{14}s_{24}^2\big)\end{aligned}}{s_{234}^2s_{34}^2} \\
&- \frac{10\left(2s_{12}s_{23}+2s_{12}s_{24}+s_{13}s_{23}+s_{14}s_{24}\right)}{s_{234}^2s_{34}} - \frac{2(8s_{12}+s_{13}+s_{14})}{s_{234}^2} \\
&- \frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2} + \frac{4s_{23}(s_{12}+s_{13}-2s_{14})}{s_{234}s_{24}s_{34}} \\
&+ \frac{2(s_{12}+s_{13}-2s_{14}+s_{23})}{s_{234}s_{24}} \\
&+ \frac{4\left(4s_{12}s_{23}+4s_{12}s_{24}+s_{13}s_{23}-s_{13}s_{24}-s_{14}s_{23}+s_{14}s_{24}-s_{23}^2-2s_{23}s_{24}-s_{24}^2\right)}{s_{234}s_{34}^2} \\
&+ \frac{2(8s_{12}+s_{13}+s_{14}-s_{23}-s_{24})}{s_{234}s_{34}} + \frac{4}{s_{234}} \\
&+ \frac{2s_{14}s_{23}}{s_{24}s_{34}^2} + \frac{2s_{14}}{s_{24}s_{34}} - \frac{4(2s_{12}-s_{13}-s_{14}-s_{23}-s_{24})}{s_{34}^2} - \frac{4}{s_{34}}\Bigg]
\end{aligned}
\tag{A.27}
$$

$$
\begin{aligned}
\tilde{A}_4^{0,(1)} = \frac{2}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[& \frac{4s_{12}s_{34}^2}{s_{13}s_{134}^2s_{14}} - \frac{4s_{14}^3s_{23}}{s_{13}s_{134}^2s_{34}^2} + \frac{2s_{14}^2(2s_{12}-3s_{23}+4s_{24})}{s_{13}s_{134}^2s_{34}} \\
& + \frac{2(3s_{12}s_{14}+6s_{12}s_{34}-6s_{14}s_{23}+4s_{14}s_{24}-2s_{23}s_{34}+3s_{24}s_{34})}{s_{13}s_{134}^2} \\
& + \frac{2s_{12}s_{34}(s_{12}+2s_{24}+s_{34})}{s_{13}s_{134}s_{14}s_{23}} + \frac{2s_{12}s_{34}(s_{12}+2s_{23}+s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
& + \frac{2s_{12}(2s_{12}^2+s_{12}s_{14}+s_{12}s_{24}+2s_{14}s_{24}+s_{14}s_{34}+s_{24}s_{34}+2s_{34}^2)}{s_{13}s_{134}s_{23}s_{234}} \\
& + \frac{4s_{12}(s_{12}+2s_{24}+s_{34})}{s_{13}s_{134}s_{23}} + \frac{2s_{14}^2s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}^2} \\
& - \frac{4s_{12}s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{234}s_{24}s_{34}} \\
& + \frac{2(2s_{12}^3+s_{12}^2s_{14}+s_{12}^2s_{23}-s_{12}s_{14}s_{34}-s_{12}s_{23}s_{34}+s_{14}^2s_{23}+s_{14}s_{23}^2)}{s_{13}s_{134}s_{234}s_{24}} \\
& + \frac{2s_{14}^2s_{23}(2s_{12}+s_{14}-s_{24})}{s_{13}s_{134}s_{234}s_{34}^2} \\
& - \frac{2s_{14}(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+3s_{14}s_{23}+2s_{14}s_{24}+s_{23}^2+s_{23}s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
& + \frac{2(2s_{12}^2+3s_{12}s_{14}+s_{12}s_{23}+s_{12}s_{24}+2s_{12}s_{34}-s_{14}^2-2s_{14}s_{23}-s_{14}s_{24}-2s_{14}s_{34})}{s_{13}s_{134}s_{234}} \\
& - \frac{4s_{14}^2s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{134}s_{24}s_{34}^2} \\
& + \frac{2s_{14}(4s_{12}^2+2s_{12}s_{14}-2s_{12}s_{23}-3s_{14}s_{23}-5s_{23}^2)}{s_{13}s_{134}s_{24}s_{34}} \\
& + \frac{2(4s_{12}^2+3s_{12}s_{14}+4s_{12}s_{23}+2s_{12}s_{34}-2s_{14}s_{23}-2s_{23}^2-s_{23}s_{34})}{s_{13}s_{134}s_{24}} \\
& + \frac{8s_{14}^2s_{23}}{s_{13}s_{134}s_{34}^2} + \frac{4s_{14}(-s_{12}+2s_{14}+2s_{23}-2s_{24})}{s_{13}s_{134}s_{34}} \\
& + \frac{2(-s_{12}+2s_{14}+4s_{23}-2s_{24}+s_{34})}{s_{13}s_{134}} + \frac{4s_{12}(s_{12}^2+s_{12}s_{34}+s_{34}^2)}{s_{13}s_{14}s_{23}s_{24}} \\
& + \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{14}s_{23}} + \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{14}s_{24}} + \frac{2s_{12}}{s_{13}s_{14}} \\
& + \frac{2s_{12}s_{34}(s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}s_{24}} + \frac{4s_{12}(s_{12}+2s_{14}+s_{34})}{s_{13}s_{23}s_{234}} \\
& + \frac{2s_{12}(s_{12}+s_{34})}{s_{13}s_{23}s_{24}} + \frac{4s_{12}}{s_{13}s_{23}} \\
& - \frac{4s_{14}s_{23}^2(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{24}s_{34}^2} \\
& + \frac{2s_{23}(4s_{12}^2-2s_{12}s_{14}+2s_{12}s_{23}-5s_{14}^2-3s_{14}s_{23})}{s_{13}s_{234}s_{24}s_{34}} \\
& + \frac{2(4s_{12}^2+4s_{12}s_{14}+3s_{12}s_{23}+2s_{12}s_{34}-2s_{14}^2-2s_{14}s_{23}-s_{14}s_{34})}{s_{13}s_{234}s_{24}}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{234}s_{34}^2} + \frac{4(2s_{12}^2+s_{12}s_{23}-2s_{14}^2-s_{14}s_{23})}{s_{13}s_{234}s_{34}} \\
&+ \frac{4(s_{12}-s_{14})}{s_{13}s_{234}} + \frac{6s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{13}s_{24}s_{34}^2} \\
&+ \frac{8(-s_{12}^2+s_{14}^2+s_{14}s_{23}+s_{23}^2)}{s_{13}s_{24}s_{34}} + \frac{2(-2s_{12}+3s_{14}+3s_{23}+s_{34})}{s_{13}s_{24}} \\
&- \frac{2s_{14}s_{23}}{s_{13}s_{34}^2} - \frac{2(2s_{14}+s_{23})}{s_{13}s_{34}} - \frac{2}{s_{13}} \\
&- \frac{4s_{13}^3s_{24}}{s_{134}^2s_{14}s_{34}^2} + \frac{2s_{13}^2(2s_{12}+4s_{23}-3s_{24})}{s_{134}^2s_{14}s_{34}} \\
&+ \frac{2(3s_{12}s_{13}+6s_{12}s_{34}+4s_{13}s_{23}-6s_{13}s_{24}+3s_{23}s_{34}-2s_{24}s_{34})}{s_{134}^2s_{14}} \\
&+ \frac{\begin{array}{l}4(2s_{12}s_{13}^2+4s_{12}s_{13}s_{14}+2s_{12}s_{14}^2+s_{13}^2s_{23} \\ \quad -s_{13}^2s_{24}+s_{13}s_{14}s_{23}+s_{13}s_{14}s_{24}-s_{14}^2s_{23}+s_{14}^2s_{24})\end{array}}{s_{134}^2s_{34}^2} \\
&+ \frac{2(10s_{12}s_{13}+10s_{12}s_{14}+9s_{13}s_{23}+2s_{13}s_{24}+2s_{14}s_{23}+9s_{14}s_{24})}{s_{134}^2s_{34}} \\
&+ \frac{4(5s_{12}+s_{23}+s_{24})}{s_{134}^2} \\
&+ \frac{2s_{13}^2s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}^2} \\
&- \frac{4s_{12}s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{234}s_{34}} \\
&+ \frac{2(2s_{12}^3+s_{12}^2s_{13}+s_{12}^2s_{24}-s_{12}s_{13}s_{34}-s_{12}s_{24}s_{34}+s_{13}^2s_{24}+s_{13}s_{24}^2)}{s_{134}s_{14}s_{23}s_{234}} \\
&- \frac{4s_{13}^2s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{14}s_{23}s_{34}^2} \\
&+ \frac{2s_{13}(4s_{12}^2+2s_{12}s_{13}-2s_{12}s_{24}-3s_{13}s_{24}-5s_{24}^2)}{s_{134}s_{14}s_{23}s_{34}} \\
&+ \frac{2(4s_{12}^2+3s_{12}s_{13}+4s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-2s_{24}^2-s_{24}s_{34})}{s_{134}s_{14}s_{23}} \\
&+ \frac{2s_{12}(2s_{12}^2+s_{12}s_{13}+s_{12}s_{23}+2s_{13}s_{23}+s_{13}s_{34}+s_{23}s_{34}+2s_{34}^2)}{s_{134}s_{14}s_{234}s_{24}} \\
&+ \frac{2s_{13}^2s_{24}(2s_{12}+s_{13}-s_{23})}{s_{134}s_{14}s_{234}s_{34}^2} \\
&- \frac{2s_{13}(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+2s_{13}s_{23}+3s_{13}s_{24}+s_{23}s_{24}+s_{24}^2)}{s_{134}s_{14}s_{234}s_{34}} \\
&+ \frac{2(2s_{12}^2+3s_{12}s_{13}+s_{12}s_{23}+s_{12}s_{24}+2s_{12}s_{34}-s_{13}^2-s_{13}s_{23}-2s_{13}s_{24}-2s_{13}s_{34})}{s_{134}s_{14}s_{234}} \\
&+ \frac{4s_{12}(s_{12}+2s_{23}+s_{34})}{s_{134}s_{14}s_{24}} + \frac{8s_{13}^2s_{24}}{s_{134}s_{14}s_{34}^2} \\
&+ \frac{4s_{13}(-s_{12}+2s_{13}-2s_{23}+2s_{24})}{s_{134}s_{14}s_{34}} + \frac{2(-s_{12}+2s_{13}-2s_{23}+4s_{24}+s_{34})}{s_{134}s_{14}}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{2s_{13}s_{24}^2(2s_{12}-s_{14}+s_{24})}{s_{134}s_{23}s_{234}s_{34}^2}\\
&-\frac{2s_{24}(4s_{12}^2+2s_{12}s_{13}+2s_{12}s_{24}+s_{13}^2+s_{13}s_{14}+3s_{13}s_{24}+2s_{14}s_{24})}{s_{134}s_{23}s_{234}s_{34}}\\
&+\frac{2(2s_{12}^2+s_{12}s_{13}+s_{12}s_{14}+3s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-s_{14}s_{24}-s_{24}^2-2s_{24}s_{34})}{s_{134}s_{23}s_{234}}\\
&-\frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{134}s_{23}s_{34}^2}+\frac{4(2s_{12}^2+s_{12}s_{13}-s_{13}s_{24}-2s_{24}^2)}{s_{134}s_{23}s_{34}}\\
&+\frac{4(s_{12}-s_{24})}{s_{134}s_{23}}+\frac{2s_{14}s_{23}^2(2s_{12}-s_{13}+s_{23})}{s_{134}s_{234}s_{24}s_{34}^2}\\
&-\frac{2s_{23}(4s_{12}^2+2s_{12}s_{14}+2s_{12}s_{23}+s_{13}s_{14}+2s_{13}s_{23}+s_{14}^2+3s_{14}s_{23})}{s_{134}s_{234}s_{24}s_{34}}\\
&+\frac{2(2s_{12}^2+s_{12}s_{13}+s_{12}s_{14}+3s_{12}s_{23}+2s_{12}s_{34}-s_{13}s_{23}-2s_{14}s_{23}-s_{23}^2-2s_{23}s_{34})}{s_{134}s_{234}s_{24}}\\
&+\frac{\begin{array}{l}2(4s_{12}s_{13}s_{23}+6s_{12}s_{13}s_{24}+6s_{12}s_{14}s_{23}+4s_{12}s_{14}s_{24}\\ \quad -s_{13}s_{14}s_{23}-s_{13}s_{14}s_{24}-s_{13}s_{23}s_{24}-s_{14}s_{23}s_{24})\end{array}}{s_{134}s_{234}s_{34}^2}\\
&+\frac{2}{s_{134}s_{234}s_{34}}\Big((-8s_{12}^2+2s_{12}s_{13}+2s_{12}s_{14}+2s_{12}s_{23}+2s_{12}s_{24}+s_{13}^2+2s_{13}s_{14}+2s_{13}s_{23}+s_{14}^2\\
&\qquad +2s_{14}s_{24}+s_{23}^2+2s_{23}s_{24}+s_{24}^2)\Big)\\
&+\frac{2(16s_{12}+3s_{13}+3s_{14}+3s_{23}+3s_{24})}{s_{134}s_{234}}\\
&-\frac{4s_{14}s_{23}(2s_{12}+s_{14}+s_{23})}{s_{134}s_{24}s_{34}^2}+\frac{4(2s_{12}^2+s_{12}s_{14}-s_{14}s_{23}-2s_{23}^2)}{s_{134}s_{24}s_{34}}\\
&+\frac{4(s_{12}-s_{23})}{s_{134}s_{24}}\\
&+\frac{4(-4s_{12}s_{13}-4s_{12}s_{14}+s_{13}^2+2s_{13}s_{14}-s_{13}s_{23}+s_{13}s_{24}+s_{14}^2+s_{14}s_{23}-s_{14}s_{24})}{s_{134}s_{34}^2}\\
&+\frac{2(-8s_{12}+5s_{13}+5s_{14}-3s_{23}-3s_{24})}{s_{134}s_{34}}\\
&+\frac{2s_{12}s_{34}(s_{12}+2s_{13}+s_{34})}{s_{14}s_{23}s_{234}s_{24}}\\
&-\frac{4s_{13}s_{24}^2(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{234}s_{34}^2}\\
&+\frac{2s_{24}(4s_{12}^2-2s_{12}s_{13}+2s_{12}s_{24}-5s_{13}^2-3s_{13}s_{24})}{s_{14}s_{23}s_{234}s_{34}}\\
&+\frac{2(4s_{12}^2+4s_{12}s_{13}+3s_{12}s_{24}+2s_{12}s_{34}-2s_{13}^2-2s_{13}s_{24}-s_{13}s_{34})}{s_{14}s_{23}s_{234}}\\
&+\frac{2s_{12}(s_{12}+s_{34})}{s_{14}s_{23}s_{24}}+\frac{6s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{23}s_{34}^2}\\
&+\frac{8(-s_{12}^2+s_{13}^2+s_{13}s_{24}+s_{24}^2)}{s_{14}s_{23}s_{34}}+\frac{2(-2s_{12}+3s_{13}+3s_{24}+s_{34})}{s_{14}s_{23}}\\
&+\frac{4s_{12}(s_{12}+2s_{13}+s_{34})}{s_{14}s_{234}s_{24}}-\frac{4s_{13}s_{24}(2s_{12}+s_{13}+s_{24})}{s_{14}s_{234}s_{34}^2}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{4(2s_{12}^2+s_{12}s_{24}-2s_{13}^2-s_{13}s_{24})}{s_{14}s_{234}s_{34}}+\frac{4(s_{12}-s_{13})}{s_{14}s_{234}}\\
&+\frac{4s_{12}}{s_{14}s_{24}}-\frac{2s_{13}s_{24}}{s_{14}s_{34}^2}-\frac{2(2s_{13}+s_{24})}{s_{14}s_{34}}-\frac{2}{s_{14}}\\
&+\frac{4s_{12}s_{34}^2}{s_{23}s_{234}^2s_{24}}-\frac{4s_{13}s_{24}^3}{s_{23}s_{234}^2s_{34}^2}\\
&+\frac{2s_{24}^2(2s_{12}-3s_{13}+4s_{14})}{s_{23}s_{234}^2s_{34}}\\
&+\frac{2(3s_{12}s_{24}+6s_{12}s_{34}-6s_{13}s_{24}-2s_{13}s_{34}+4s_{14}s_{24}+3s_{14}s_{34})}{s_{23}s_{234}^2}\\
&+\frac{8s_{13}s_{24}^2}{s_{23}s_{234}s_{34}^2}+\frac{4s_{24}(-s_{12}+2s_{13}-2s_{14}+2s_{24})}{s_{23}s_{234}s_{34}}\\
&+\frac{2(-s_{12}+4s_{13}-2s_{14}+2s_{24}+s_{34})}{s_{23}s_{234}}+\frac{2s_{12}}{s_{23}s_{24}}\\
&-\frac{2s_{13}s_{24}}{s_{23}s_{34}^2}-\frac{2(s_{13}+2s_{24})}{s_{23}s_{34}}-\frac{2}{s_{23}}\\
&-\frac{4s_{14}s_{23}^3}{s_{234}^2s_{24}s_{34}^2}+\frac{2s_{23}^2(2s_{12}+4s_{13}-3s_{14})}{s_{234}^2s_{24}s_{34}}\\
&+\frac{2(3s_{12}s_{23}+6s_{12}s_{34}+4s_{13}s_{23}+3s_{13}s_{34}-6s_{14}s_{23}-2s_{14}s_{34})}{s_{234}^2s_{24}}\\
&+\frac{\begin{array}{l}4(2s_{12}s_{23}^2+4s_{12}s_{23}s_{24}+2s_{12}s_{24}^2+s_{13}s_{23}^2\\ \quad +s_{13}s_{23}s_{24}-s_{13}s_{24}^2-s_{14}s_{23}^2+s_{14}s_{23}s_{24}+s_{14}s_{24}^2)\end{array}}{s_{234}^2s_{34}^2}\\
&+\frac{2(10s_{12}s_{23}+10s_{12}s_{24}+9s_{13}s_{23}+2s_{13}s_{24}+2s_{14}s_{23}+9s_{14}s_{24})}{s_{234}^2s_{34}}\\
&+\frac{4(5s_{12}+s_{13}+s_{14})}{s_{234}^2}+\frac{8s_{14}s_{23}^2}{s_{234}s_{24}s_{34}^2}\\
&+\frac{4s_{23}(-s_{12}-2s_{13}+2s_{14}+2s_{23})}{s_{234}s_{24}s_{34}}\\
&+\frac{2(-s_{12}-2s_{13}+4s_{14}+2s_{23}+s_{34})}{s_{234}s_{24}}\\
&+\frac{4(-4s_{12}s_{23}-4s_{12}s_{24}-s_{13}s_{23}+s_{13}s_{24}+s_{14}s_{23}-s_{14}s_{24}+s_{23}^2+2s_{23}s_{24}+s_{24}^2)}{s_{234}s_{34}^2}\\
&+\frac{2(-8s_{12}-3s_{13}-3s_{14}+5s_{23}+5s_{24})}{s_{234}s_{34}}\\
&-\frac{2s_{14}s_{23}}{s_{24}s_{34}^2}-\frac{2(s_{14}+2s_{23})}{s_{24}s_{34}}-\frac{2}{s_{24}}\\
&+\frac{4(2s_{12}-s_{13}-s_{14}-s_{23}-s_{24})}{s_{34}^2}-\frac{8}{s_{34}}\Bigg]
\end{aligned}
\tag{A.28}
$$

$$
\tilde{A}_4^{0,(2)}=\frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[-\frac{4s_{14}^2s_{24}}{s_{13}s_{134}^2s_{34}}-\frac{2\left(3s_{12}s_{34}-3s_{14}s_{23}+2s_{14}s_{24}+3s_{24}s_{34}\right)}{s_{13}s_{134}^2}-\frac{2s_{12}s_{34}(s_{24}+2s_{34})}{s_{13}s_{134}s_{14}s_{23}}
$$

$$
\begin{aligned}
&- \frac{2s_{12}s_{34}(s_{23}+2s_{34})}{s_{13}s_{134}s_{14}s_{24}} \\
&- \frac{2s_{12}\left(2s_{14}s_{24}+s_{14}s_{34}+s_{24}s_{34}+3s_{34}^2\right)}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}(2s_{24}+s_{34})}{s_{13}s_{134}s_{23}} \\
&- \frac{2s_{34}(s_{12}s_{34}-s_{14}s_{23})}{s_{13}s_{134}s_{234}s_{24}} + \frac{4s_{14}^2(s_{23}+s_{24})}{s_{13}s_{134}s_{234}s_{34}} \\
&- \frac{2\left(2s_{12}s_{14}+2s_{12}s_{34}-s_{14}^2-2s_{14}s_{23}-s_{14}s_{24}-3s_{14}s_{34}\right)}{s_{13}s_{134}s_{234}} \\
&+ \frac{4s_{14}s_{23}^2}{s_{13}s_{134}s_{24}s_{34}} - \frac{2\left(2s_{12}s_{23}+3s_{12}s_{34}-3s_{14}s_{23}-s_{23}^2-2s_{23}s_{34}\right)}{s_{13}s_{134}s_{24}} \\
&- \frac{4s_{14}(2s_{14}-s_{24})}{s_{13}s_{134}s_{34}} - \frac{2(3s_{14}+2s_{23}-s_{24}+2s_{34})}{s_{13}s_{134}} \\
&- \frac{6s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{23}} - \frac{2s_{12}s_{34}}{s_{13}s_{14}s_{24}} \\
&- \frac{2s_{12}s_{34}(s_{14}+2s_{34})}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}(2s_{14}+s_{34})}{s_{13}s_{23}s_{234}} \\
&- \frac{2s_{12}s_{34}}{s_{13}s_{23}s_{24}} - \frac{4s_{12}}{s_{13}s_{23}} + \frac{4s_{14}^2s_{23}}{s_{13}s_{234}s_{24}s_{34}} \\
&- \frac{2\left(2s_{12}s_{14}+3s_{12}s_{34}-s_{14}^2-3s_{14}s_{23}-2s_{14}s_{34}\right)}{s_{13}s_{234}s_{24}} \\
&+ \frac{4s_{14}^2}{s_{13}s_{234}s_{34}} + \frac{2s_{14}}{s_{13}s_{234}} - \frac{4(s_{14}^2+s_{23}^2)}{s_{13}s_{24}s_{34}} \\
&- \frac{2(s_{14}+s_{23})}{s_{13}s_{24}} + \frac{4s_{14}}{s_{13}s_{34}} + \frac{2}{s_{13}} \\
&- \frac{4s_{13}^2s_{23}}{s_{134}^2s_{14}s_{34}} - \frac{2\left(3s_{12}s_{34}+2s_{13}s_{23}-3s_{13}s_{24}+3s_{23}s_{34}\right)}{s_{134}^2s_{14}} \\
&- \frac{4\left(2s_{13}s_{23}+s_{13}s_{24}+s_{14}s_{23}+2s_{14}s_{24}\right)}{s_{134}^2s_{34}} - \frac{2(2s_{12}+s_{23}+s_{24})}{s_{134}^2} \\
&- \frac{2s_{34}(s_{12}s_{34}-s_{13}s_{24})}{s_{134}s_{14}s_{23}s_{234}} + \frac{4s_{13}s_{24}^2}{s_{134}s_{14}s_{23}s_{34}} \\
&- \frac{2\left(2s_{12}s_{24}+3s_{12}s_{34}-3s_{13}s_{24}-s_{24}^2-2s_{24}s_{34}\right)}{s_{134}s_{14}s_{23}} \\
&- \frac{2s_{12}\left(2s_{13}s_{23}+s_{13}s_{34}+s_{23}s_{34}+3s_{34}^2\right)}{s_{134}s_{14}s_{234}s_{24}} \\
&+ \frac{4s_{13}^2(s_{23}+s_{24})}{s_{134}s_{14}s_{234}s_{34}} - \frac{2\left(2s_{12}s_{13}+2s_{12}s_{34}-s_{13}^2-s_{13}s_{23}-2s_{13}s_{24}-3s_{13}s_{34}\right)}{s_{134}s_{14}s_{234}} \\
&- \frac{2s_{12}(2s_{23}+s_{34})}{s_{134}s_{14}s_{24}} - \frac{4s_{13}(2s_{13}-s_{23})}{s_{134}s_{14}s_{34}} \\
&- \frac{2(3s_{13}-s_{23}+2s_{24}+2s_{34})}{s_{134}s_{14}} + \frac{4s_{24}^2(s_{13}+s_{14})}{s_{134}s_{23}s_{234}s_{34}} \\
&- \frac{2\left(2s_{12}s_{24}+2s_{12}s_{34}-2s_{13}s_{24}-s_{14}s_{24}-s_{24}^2-3s_{24}s_{34}\right)}{s_{134}s_{23}s_{234}} \\
&+ \frac{4s_{24}^2}{s_{134}s_{23}s_{34}} + \frac{2s_{24}}{s_{134}s_{23}} + \frac{4s_{23}^2(s_{13}+s_{14})}{s_{134}s_{234}s_{24}s_{34}}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{2\left(2s_{12}s_{23} + 2s_{12}s_{34} - s_{13}s_{23} - 2s_{14}s_{23} - s_{23}^2 - 3s_{23}s_{34}\right)}{s_{134}s_{234}s_{24}} \\
&+ \frac{8(s_{13}+s_{14})(s_{23}+s_{24})}{s_{134}s_{234}s_{34}} - \frac{4\left(4s_{12} + 2s_{13} + 2s_{14} + 2s_{23} + 2s_{24} + s_{34}\right)}{s_{134}s_{234}} \\
&+ \frac{4s_{23}^2}{s_{134}s_{24}s_{34}} + \frac{2s_{23}}{s_{134}s_{24}} - \frac{4(2s_{13} + 2s_{14} - s_{23} - s_{24})}{s_{134}s_{34}} - \frac{4}{s_{134}} \\
&- \frac{2s_{12}s_{34}(s_{13} + 2s_{34})}{s_{14}s_{23}s_{234}s_{24}} + \frac{4s_{13}^2 s_{24}}{s_{14}s_{23}s_{234}s_{34}} \\
&- \frac{2\left(2s_{12}s_{13} + 3s_{12}s_{34} - s_{13}^2 - 3s_{13}s_{24} - 2s_{13}s_{34}\right)}{s_{14}s_{23}s_{234}} \\
&- \frac{2s_{12}s_{34}}{s_{14}s_{23}s_{24}} - \frac{4(s_{13}^2 + s_{24}^2)}{s_{14}s_{23}s_{34}} - \frac{2(s_{13}+s_{24})}{s_{14}s_{23}} \\
&- \frac{2s_{12}(2s_{13} + s_{34})}{s_{14}s_{234}s_{24}} + \frac{4s_{13}^2}{s_{14}s_{234}s_{34}} + \frac{2s_{13}}{s_{14}s_{234}} \\
&- \frac{4s_{12}}{s_{14}s_{24}} + \frac{4s_{13}}{s_{14}s_{34}} + \frac{2}{s_{14}} \\
&- \frac{4s_{14}s_{24}^2}{s_{23}s_{234}^2 s_{34}} - \frac{2\left(3s_{12}s_{34} - 3s_{13}s_{24} + 2s_{14}s_{24} + 3s_{14}s_{34}\right)}{s_{23}s_{234}^2} \\
&+ \frac{4s_{24}(s_{14} - 2s_{24})}{s_{23}s_{234}s_{34}} - \frac{2(2s_{13} - s_{14} + 3s_{24} + 2s_{34})}{s_{23}s_{234}} \\
&+ \frac{4s_{24}}{s_{23}s_{34}} + \frac{2}{s_{23}} - \frac{4s_{13}s_{23}^2}{s_{234}^2 s_{24}s_{34}} \\
&- \frac{2\left(3s_{12}s_{34} + 2s_{13}s_{23} + 3s_{13}s_{34} - 3s_{14}s_{23}\right)}{s_{234}^2 s_{24}} \\
&- \frac{4\left(2s_{13}s_{23} + s_{13}s_{24} + s_{14}s_{23} + 2s_{14}s_{24}\right)}{s_{234}^2 s_{34}} - \frac{2(2s_{12} + s_{13} + s_{14})}{s_{234}^2} \\
&+ \frac{4s_{23}(s_{13} - 2s_{23})}{s_{234}s_{24}s_{34}} + \frac{2(s_{13} - 2s_{14} - 3s_{23} - 2s_{34})}{s_{234}s_{24}} \\
&+ \frac{4(s_{13} + s_{14} - 2s_{23} - 2s_{24})}{s_{234}s_{34}} - \frac{4}{s_{234}} + \frac{4s_{23}}{s_{24}s_{34}} + \frac{2}{s_{24}}\Bigg]
\end{aligned}
\tag{A.29}
$$

$$
\begin{aligned}
\tilde{A}_4^{0,(3)} = \frac{1}{s_{14}s_{23}\mathcal{D}_{4,gg}}\Bigg[&- \frac{2s_{12}s_{34}^2}{s_{13}s_{134}^2 s_{14}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{23}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{14}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{134}s_{23}s_{234}} - \frac{2s_{12}s_{34}^2}{s_{13}s_{14}s_{23}s_{24}} \\
&- \frac{2s_{12}s_{34}^2}{s_{13}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{134}s_{14}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{14}s_{23}s_{234}s_{24}} - \frac{2s_{12}s_{34}^2}{s_{23}s_{234}^2 s_{24}} \\
&+ \frac{2s_{34}(s_{23}+s_{24})}{s_{13}s_{134}^2} + \frac{2s_{14}s_{34}}{s_{13}s_{134}s_{234}} + \frac{2s_{23}s_{34}}{s_{13}s_{134}s_{24}} \\
&+ \frac{2s_{14}s_{34}}{s_{13}s_{234}s_{24}} + \frac{2s_{34}(s_{23}+s_{24})}{s_{134}^2 s_{14}} + \frac{2s_{24}s_{34}}{s_{134}s_{14}s_{23}} \\
&+ \frac{2s_{13}s_{34}}{s_{134}s_{14}s_{234}} + \frac{2s_{24}s_{34}}{s_{134}s_{23}s_{234}} + \frac{2s_{23}s_{34}}{s_{134}s_{234}s_{24}} \\
&+ \frac{2s_{13}s_{34}}{s_{14}s_{23}s_{234}} + \frac{2s_{34}(s_{13}+s_{14})}{s_{23}s_{234}^2} + \frac{2s_{34}(s_{13}+s_{14})}{s_{234}^2 s_{24}}
\end{aligned}
$$

$$+\frac{6s_{34}}{s_{13}s_{134}}+\frac{4s_{34}}{s_{13}s_{24}}+\frac{6s_{34}}{s_{134}s_{14}}+\frac{8s_{34}}{s_{134}s_{234}}+\frac{4s_{34}}{s_{14}s_{23}}+\frac{6s_{34}}{s_{23}s_{234}}+\frac{6s_{34}}{s_{234}s_{24}}\Bigg] \tag{A.30}$$

using,

$$\mathcal{D}_{4,gg}=s_{13}s_{24}s_{34}^2s_{134}^2s_{234}^2$$

By running `BuildAndIntegrateAntenna[A,4,0,Component->Subleading]` we get the integrated result,

$$\begin{aligned}\tilde{\mathcal{A}}_4^0=&\frac{1}{\epsilon^4}+\frac{3}{\epsilon^3}+\frac{1}{\epsilon^2}\left(13-\frac{3\pi^2}{2}\right)+\frac{1}{\epsilon}\left(\frac{845}{16}-\frac{9\pi^2}{2}-\frac{80\zeta_3}{3}\right)\\&+\frac{6921}{32}-\frac{473\pi^2}{24}+\frac{17\pi^4}{72}-80\zeta_3+\mathcal{O}(\epsilon).\end{aligned} \tag{A.31}$$

A.3.10 B_4^0

Running `BuildAntenna[B,4,0]`, we get the built result,

$$\begin{aligned}B_4^0=&\epsilon^2\Bigg(\frac{4s_{12}}{s_{134}s_{234}}+\frac{2s_{12}}{s_{134}^2}+\frac{2s_{12}}{s_{234}^2}-\frac{2s_{14}s_{23}}{s_{134}^2s_{34}}-\frac{2s_{13}s_{23}}{s_{134}^2s_{34}}-\frac{2s_{14}s_{24}}{s_{134}^2s_{34}}\\&-\frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}}-\frac{4s_{13}s_{24}}{s_{134}s_{234}s_{34}}-\frac{2s_{13}s_{24}}{s_{134}^2s_{34}}-\frac{2s_{13}s_{24}}{s_{234}^2s_{34}}-\frac{2s_{14}s_{24}}{s_{234}^2s_{34}}\\&-\frac{4s_{14}s_{23}}{s_{134}s_{234}s_{34}}-\frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}}-\frac{2s_{13}s_{23}}{s_{234}^2s_{34}}-\frac{2s_{14}s_{23}}{s_{234}^2s_{34}}\Bigg)\\&+\epsilon\Bigg(\frac{4s_{12}^2}{s_{134}s_{234}s_{34}}+\frac{4s_{12}}{s_{134}s_{234}}+\frac{2s_{12}s_{14}}{s_{134}^2s_{34}}+\frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}}+\frac{2s_{12}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}\\&+\frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}}+\frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}}+\frac{2s_{12}s_{13}}{s_{134}^2s_{34}}+\frac{2s_{12}s_{23}}{s_{234}^2s_{34}}-\frac{2s_{12}}{s_{134}^2}-\frac{2s_{12}}{s_{234}^2}\\&+\frac{4s_{12}s_{13}s_{14}}{s_{134}^2s_{34}^2}-\frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{4s_{12}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}}{s_{134}^2s_{34}}+\frac{4s_{13}s_{23}}{s_{134}^2s_{34}}\\&+\frac{4s_{14}s_{24}}{s_{134}^2s_{34}}+\frac{4s_{14}s_{24}}{s_{134}s_{234}s_{34}}+\frac{2s_{13}s_{24}}{s_{134}^2s_{34}}+\frac{2s_{13}s_{24}}{s_{234}^2s_{34}}+\frac{4s_{14}s_{24}}{s_{234}^2s_{34}}+\frac{4s_{13}s_{23}}{s_{134}s_{234}s_{34}}\\&+\frac{4s_{13}s_{23}}{s_{234}^2s_{34}}+\frac{2s_{14}s_{23}}{s_{234}^2s_{34}}-\frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}s_{24}^2}{s_{234}^2s_{34}^2}-\frac{2s_{14}^2s_{23}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{23}}{s_{134}^2s_{34}^2}\\&+\frac{2s_{13}s_{14}s_{24}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}+\frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{13}^2s_{24}}{s_{134}s_{234}s_{34}^2}\\&-\frac{2s_{13}^2s_{24}}{s_{134}^2s_{34}^2}+\frac{2s_{13}s_{23}s_{24}}{s_{234}^2s_{34}^2}+\frac{2s_{14}s_{23}s_{24}}{s_{234}^2s_{34}^2}-\frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2}-\frac{2s_{14}^2s_{23}}{s_{134}s_{234}s_{34}^2}\\&+\frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2}-\frac{2s_{14}s_{23}^2}{s_{234}^2s_{34}^2}\Bigg)\\&-\frac{4s_{12}}{s_{134}s_{234}}-\frac{2s_{12}s_{14}}{s_{134}^2s_{34}}-\frac{2s_{13}s_{23}}{s_{134}^2s_{34}}-\frac{2s_{14}s_{24}}{s_{134}^2s_{34}}-\frac{2s_{12}s_{24}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{24}}{s_{234}^2s_{34}}\\&-\frac{2s_{14}s_{24}}{s_{234}^2s_{34}}-\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{23}}{s_{134}s_{234}s_{34}}-\frac{4s_{12}^2}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{13}}{s_{134}s_{234}s_{34}}-\frac{2s_{12}s_{13}}{s_{134}^2s_{34}}\end{aligned}$$

$$
\begin{aligned}
&- \frac{2s_{12}s_{23}}{s_{234}^2 s_{34}} - \frac{2s_{13}s_{23}}{s_{234}^2 s_{34}} + \frac{2s_{13}s_{24}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}s_{24}^2}{s_{234}^2 s_{34}^2} - \frac{4s_{12}s_{13}s_{14}}{s_{134}^2 s_{34}^2} + \frac{2s_{14}^2 s_{23}}{s_{134}^2 s_{34}^2} \\
&- \frac{2s_{13}s_{14}s_{23}}{s_{134}^2 s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}^2 s_{34}^2} - \frac{2s_{13}s_{14}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{134}s_{234}s_{34}^2} \\
&+ \frac{2s_{13}^2 s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{13}s_{24}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{13}^2 s_{24}}{s_{134}^2 s_{34}^2} - \frac{4s_{12}s_{23}s_{24}}{s_{234}^2 s_{34}^2} - \frac{2s_{13}s_{23}s_{24}}{s_{234}^2 s_{34}^2} - \frac{2s_{14}s_{23}s_{24}}{s_{234}^2 s_{34}^2} \\
&+ \frac{2s_{14}s_{23}^2}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}^2 s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{4s_{12}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} - \frac{2s_{13}s_{14}s_{23}}{s_{134}s_{234}s_{34}^2} + \frac{2s_{14}s_{23}^2}{s_{234}^2 s_{34}^2}.
\end{aligned}
\tag{A.32}
$$

By running `BuildAndIntegrateAntenna[B,4,0]` we get the integrated result,

$$
\mathcal{B}_4^0 = -\frac{1}{12\epsilon^3} - \frac{7}{18\epsilon^2} - \frac{1}{\epsilon}\frac{11(37-3\pi^2)}{216} - \frac{11753}{1296} + \frac{77\pi^2}{108} + \frac{67\zeta_3}{18} + O(\epsilon).
\tag{A.33}
$$

A.3.11 C_4^0

Running `BuildAntenna[C,4,0]`, we get the built result,

$$
\begin{aligned}
C_4^0 = \epsilon^3 \Bigg(& \frac{s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}} + \frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{24}^2}{s_{23}s_{234}^2 s_{34}} + \frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}} \\
& - \frac{s_{12}s_{24}}{s_{123}s_{134}s_{23}} + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}} + \frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}} \\
& - \frac{s_{14}s_{24}}{s_{123}s_{134}s_{34}} + \frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}} + \frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}} - \frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}} - \frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}} \\
& + \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} + \frac{s_{13}s_{24}}{s_{234}^2 s_{34}} - \frac{s_{14}s_{24}}{s_{234}^2 s_{34}} - \frac{s_{12}s_{24}}{s_{23}s_{234}^2} + \frac{s_{13}s_{24}}{s_{23}s_{234}^2} \\
& + \frac{s_{14}}{s_{123}s_{134}} - \frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}} - \frac{s_{12}s_{34}}{s_{23}s_{234}^2} \\
& + \frac{2s_{13}}{s_{123}s_{134}} + \frac{s_{12}}{s_{123}s_{134}} + \frac{2s_{13}}{s_{123}s_{234}} + \frac{s_{14}}{s_{123}s_{234}} + \frac{s_{14}}{s_{134}s_{234}} \\
& + \frac{s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} + \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} \\
& - \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{234}^2 s_{34}} + \frac{s_{12}}{s_{234}^2} + \frac{2s_{13}}{s_{234}^2} + \frac{s_{14}}{s_{234}^2} \Bigg) \\
+ \epsilon^2 \Bigg(& - \frac{s_{12}^2}{s_{123}s_{134}s_{23}} - \frac{s_{12}^2}{s_{134}s_{23}s_{234}} - \frac{3s_{12}s_{24}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{12}s_{24}}{s_{134}s_{23}s_{234}} \\
& - \frac{2s_{12}s_{34}}{s_{123}s_{134}s_{23}} - \frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}} + \frac{2s_{12}s_{34}}{s_{23}s_{234}^2} + \frac{2s_{12}}{s_{123}s_{134}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}} \\
& - \frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}} - \frac{2s_{12}}{s_{123}s_{234}} + \frac{s_{12}}{s_{134}s_{234}} - \frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}} \\
& - \frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}} + \frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}} - \frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} \\
& - \frac{2s_{12}s_{24}}{s_{234}^2 s_{34}} + \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{2s_{12}}{s_{234}^2} + \frac{2s_{14}}{s_{123}s_{134}} + \frac{2s_{13}s_{24}}{s_{123}s_{134}s_{23}} \\
& + \frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}} - \frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}} - \frac{2s_{13}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{14}s_{24}}{s_{23}s_{234}^2} - \frac{2s_{13}}{s_{123}s_{234}} \\
& + \frac{s_{14}}{s_{123}s_{234}} - \frac{2s_{14}}{s_{134}s_{234}} - \frac{2s_{13}}{s_{134}s_{234}} - \frac{s_{14}^2}{s_{123}s_{134}s_{34}} + \frac{3s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}}
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{2s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}}+\frac{2s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}}-\frac{2s_{14}s_{23}}{s_{123}s_{134}s_{34}}-\frac{3s_{14}s_{24}}{s_{123}s_{134}s_{34}}\\
&+\frac{2s_{13}s_{24}}{s_{123}s_{134}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{2s_{13}^2s_{24}}{s_{123}s_{134}s_{23}s_{34}}-\frac{2s_{14}s_{24}}{s_{123}s_{234}s_{34}}-\frac{2s_{14}s_{24}}{s_{134}s_{234}s_{34}}\\
&+\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{24}}{s_{234}^2s_{34}}\\
&-\frac{s_{14}^2}{s_{123}s_{234}s_{34}}-\frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}}-\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}+\frac{2s_{14}s_{23}}{s_{234}^2s_{34}}\\
&-\frac{4s_{13}}{s_{234}^2}-\frac{2s_{14}}{s_{234}^2}\Bigg)\\
&+\epsilon\Bigg(\frac{s_{12}^2}{s_{123}s_{134}s_{23}}-\frac{s_{12}^2}{s_{123}s_{23}s_{234}}+\frac{2s_{12}^2}{s_{134}s_{23}s_{234}}+\frac{2s_{12}s_{24}}{s_{123}s_{134}s_{23}}+\frac{s_{12}s_{24}}{s_{123}s_{23}s_{234}}\\
&+\frac{s_{12}s_{24}}{s_{134}s_{23}s_{234}}+\frac{3s_{12}s_{24}}{s_{23}s_{234}^2}+\frac{s_{12}s_{34}}{s_{123}s_{134}s_{23}}-\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}+\frac{2s_{12}s_{34}}{s_{134}s_{23}s_{234}}\\
&-\frac{s_{12}s_{34}}{s_{23}s_{234}^2}-\frac{s_{12}}{s_{123}s_{134}}-\frac{s_{12}s_{14}}{s_{123}s_{134}s_{23}}-\frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}}+\frac{s_{12}}{s_{134}s_{234}}\\
&+\frac{s_{12}s_{13}}{s_{123}s_{23}s_{234}}-\frac{2s_{12}s_{14}}{s_{123}s_{23}s_{234}}+\frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}}+\frac{s_{12}s_{13}}{s_{134}s_{23}s_{234}}-\frac{s_{12}s_{14}}{s_{123}s_{134}s_{34}}\\
&-\frac{s_{12}s_{13}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{2s_{12}s_{24}}{s_{123}s_{234}s_{34}}+\frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}}-\frac{2s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}}+\frac{4s_{12}s_{24}}{s_{234}^2s_{34}}\\
&-\frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}}+\frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}}-\frac{2s_{12}s_{14}}{s_{134}s_{234}s_{34}}+\frac{s_{12}}{s_{234}^2}-\frac{s_{14}}{s_{123}s_{134}}\\
&-\frac{s_{13}s_{24}}{s_{123}s_{134}s_{23}}+\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}+\frac{2s_{14}s_{24}}{s_{134}s_{23}s_{234}}+\frac{s_{13}s_{24}}{s_{23}s_{234}^2}+\frac{4s_{14}s_{24}}{s_{23}s_{234}^2}\\
&-\frac{4s_{13}}{s_{123}s_{134}}+\frac{s_{14}}{s_{123}s_{234}}+\frac{s_{14}^2}{s_{123}s_{134}s_{34}}-\frac{2s_{13}s_{24}^2}{s_{123}s_{134}s_{23}s_{34}}-\frac{s_{13}s_{24}^2}{s_{123}s_{23}s_{234}s_{34}}\\
&-\frac{s_{13}s_{24}^2}{s_{134}s_{23}s_{234}s_{34}}-\frac{3s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}}-\frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}}+\frac{s_{14}s_{23}}{s_{123}s_{134}s_{34}}+\frac{2s_{14}s_{24}}{s_{123}s_{134}s_{34}}\\
&-\frac{s_{13}s_{24}}{s_{123}s_{134}s_{34}}-\frac{s_{13}s_{14}s_{24}}{s_{123}s_{134}s_{23}s_{34}}+\frac{s_{14}s_{24}}{s_{123}s_{234}s_{34}}+\frac{s_{14}s_{24}}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}}\\
&-\frac{s_{13}^2s_{24}}{s_{123}s_{23}s_{234}s_{34}}-\frac{2s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}}-\frac{s_{13}^2s_{24}}{s_{134}s_{23}s_{234}s_{34}}+\frac{s_{13}s_{24}}{s_{234}^2s_{34}}\\
&+\frac{3s_{14}s_{24}}{s_{234}^2s_{34}}+\frac{2s_{14}^2}{s_{123}s_{234}s_{34}}-\frac{s_{14}^2}{s_{134}s_{234}s_{34}}+\frac{s_{13}s_{14}}{s_{123}s_{234}s_{34}}+\frac{s_{13}s_{14}}{s_{134}s_{234}s_{34}}\\
&+\frac{2s_{14}s_{23}}{s_{123}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}-\frac{s_{14}s_{23}}{s_{234}^2s_{34}}+\frac{2s_{13}}{s_{234}^2}+\frac{s_{14}}{s_{234}^2}\Bigg)\\
&-\frac{s_{13}s_{24}}{s_{123}s_{23}s_{234}}-\frac{s_{13}s_{24}}{s_{134}s_{23}s_{234}}-\frac{2s_{12}s_{24}}{s_{23}s_{234}^2}-\frac{2s_{14}s_{24}}{s_{23}s_{234}^2}+\frac{s_{12}s_{34}}{s_{123}s_{23}s_{234}}\\
&-\frac{s_{12}s_{34}}{s_{134}s_{23}s_{234}}+\frac{2s_{13}}{s_{123}s_{134}}+\frac{2s_{12}s_{13}}{s_{123}s_{134}s_{23}}-\frac{s_{14}}{s_{123}s_{234}}+\frac{s_{14}}{s_{134}s_{234}}\\
&+\frac{s_{12}}{s_{123}s_{234}}-\frac{s_{12}}{s_{134}s_{234}}+\frac{s_{12}s_{14}}{s_{123}s_{23}s_{234}}-\frac{s_{12}s_{14}}{s_{134}s_{23}s_{234}}+\frac{s_{12}^2}{s_{123}s_{23}s_{234}}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{s_{12}^2}{s_{134}s_{23}s_{234}} + \frac{2s_{13}s_{24}^2}{s_{23}s_{234}^2s_{34}} + \frac{2s_{13}s_{14}}{s_{123}s_{134}s_{34}} - \frac{s_{13}s_{24}}{s_{123}s_{234}s_{34}} - \frac{s_{13}s_{24}}{s_{134}s_{234}s_{34}} \\
&- \frac{s_{12}s_{13}s_{24}}{s_{123}s_{23}s_{234}s_{34}} + \frac{s_{13}s_{14}s_{24}}{s_{123}s_{23}s_{234}s_{34}} - \frac{s_{13}s_{14}s_{24}}{s_{134}s_{23}s_{234}s_{34}} + \frac{s_{12}s_{13}s_{24}}{s_{134}s_{23}s_{234}s_{34}} - \frac{2s_{12}s_{24}}{s_{234}^2s_{34}} \\
&- \frac{2s_{14}s_{24}}{s_{234}^2s_{34}} + \frac{2s_{12}s_{13}s_{14}}{s_{123}s_{134}s_{23}s_{34}} - \frac{s_{14}^2}{s_{123}s_{234}s_{34}} + \frac{s_{14}^2}{s_{134}s_{234}s_{34}} - \frac{s_{12}s_{14}}{s_{123}s_{234}s_{34}} \\
&+ \frac{s_{12}s_{14}}{s_{134}s_{234}s_{34}} - \frac{s_{14}s_{23}}{s_{123}s_{234}s_{34}} + \frac{s_{14}s_{23}}{s_{134}s_{234}s_{34}}.
\end{aligned} \tag{A.34}
$$

By running `BuildAndIntegrateAntenna[C,4,0]` we get the integrated result,

$$
C_4^0 = \frac{1}{\epsilon}\frac{(-13 + 2\pi^2 - 8\zeta_3)}{32} - \frac{339}{64} + \frac{17\pi^2}{48} - \frac{\pi^4}{45} + \frac{21\zeta_3}{8} + O(\epsilon). \tag{A.35}
$$

A.4 Exploratory Massive A_3^0

Running `BuildAntenna[A,3,0,quarkMass->mQ]`, we get the built result,

$$
\begin{aligned}
A_3^{0,(m_q)} = \frac{1}{s_{13}^2 s_{23}^2\left[2(-2+\epsilon)m_q^2 + (-1+\epsilon)(s_{12}+s_{13}+s_{23})\right]}\Big\{ &-4(-2+\epsilon)m_q^4\left(s_{13}^2 + s_{23}^2\right) \\
&+(-1+\epsilon)s_{13}s_{23}\left[2s_{12}^2 - (-1+\epsilon)s_{13}^2 - 2\epsilon s_{13}s_{23} - (-1+\epsilon)s_{23}^2 + 2s_{12}(s_{13}+s_{23})\right] \\
-2m_q^2\Big[s_{12}\left((-1+\epsilon)s_{13}^2 - 2(-2+\epsilon)s_{13}s_{23} + (-1+\epsilon)s_{23}^2\right) &+ (-1+\epsilon)\left(s_{13}^3 + s_{13}^2 s_{23} + s_{13}s_{23}^2 + s_{23}^3\right)\Big]\Big\}.
\end{aligned} \tag{A.36}
$$

By running `BuildAndIntegrateAntenna[A,3,0,quarkMass->mQ]` we get the integrated result,

$$
\begin{aligned}
\mathcal{A}_3^{0,(m_q)} = \frac{2^{-2(1+\epsilon)}(1-\epsilon)\pi^{\epsilon-2}\left(1-4m_q^2\right)^{1-2\epsilon}}{\epsilon(2\epsilon-1)m_q^2(\epsilon-2m_q^2-1)}\Bigg[&-\frac{24\Gamma(4-3\epsilon)\Gamma(3-2\epsilon)}{\Gamma(8-6\epsilon)}(\epsilon-1)\left(4m_q^2-1\right) \\
&\cdot\left(\epsilon^2\left(8m_q^2-2\right) + \epsilon\left(3-4m_q^2\right) - 4m_q^2 - 2\right){}_2F_1\left(\frac{1}{2}, 3-2\epsilon; \frac{9}{2}-3\epsilon; 1-4m_q^2\right) \\
&-\frac{4\Gamma(3-3\epsilon)\Gamma(2-2\epsilon)}{\Gamma(6-6\epsilon)}\Big(4m_q^4 + \epsilon^3\left(8m_q^2-2\right) + 6m_q^2 + \epsilon^2\left(16m_q^4 - 12m_q^2 + 5\right) \\
&-\epsilon\left(16m_q^4 + 4m_q^2 + 5\right) + 2\Big){}_2F_1\left(\frac{1}{2}, 2-2\epsilon; \frac{7}{2}-3\epsilon; 1-4m_q^2\right)\Bigg].
\end{aligned} \tag{A.37}
$$

Appendix B

Supporting Derivations

B.1 Feynman Rules

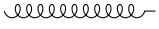
For the massless final-final quark-antiquark antenna functions considered throughout this work, in CDR, with $d = 4 - 2\epsilon$, and using the Feynman gauge $\lambda = 1$, the required Feynman rules are as follows.

An application of these rules is shown in full in Subsection 4.1.1.

B.1.1 Propagators

Each internal line in a Feynman diagram represents a propagator: the way a quantum field carries an excitation from one spacetime point to another. These propagators differ particle-by-particle both in mathematical structure and in the way they are graphically represented in the diagrams. A description of propagators encountered throughout this work is given in Table B.1 [2, 9].

Table B.1: Overview of the propagators found throughout the thesis, alongside their mathematical and graphical representations.

Particle-type	**Mathematical Representation**	**Graphical Representation**
Quark (q)	$\frac{i\delta_{ij}\not{p}}{p^2+i0}$ $\left(m_q \neq 0 : \frac{i(\not{p}+m_q)}{p^2-m_q^2+i0}\right)$	
Gluon (g)	$\frac{i\delta^{ab}}{p^2+i0}\left(-g_{\mu\nu}+(1-\lambda)\frac{p_\mu p_\nu}{p^2}\right)$ $\left(\lambda = 1 : \frac{-i\delta^{ab}g_{\mu\nu}}{p^2+i0}\right)$	
Ghost (c)	$\frac{i\delta^{ab}}{p^2+i0}$	

For external massless quarks and antiquarks,

$$\sum_s u_s(p)\bar{u}_s(p) = \not{p}, \quad \sum_s v_s(p)\bar{v}_s(p) = \not{p}. \tag{B.1}$$

For external gluons,

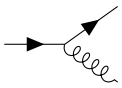
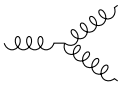
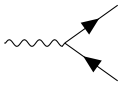
$$\sum_{\lambda} \varepsilon_{\mu}^{(\lambda)}(p)\varepsilon_{\nu}^{(\lambda)\,*}(p) = -g_{\mu\nu} + t_g, \quad t_g = \frac{p_\mu n_\nu + n_\mu p_\nu}{p \cdot n} \tag{B.2}$$

where n is a light-like reference vector and t_g removes the unphysical (timelike and longitudinal) gluon polarisations. With a single external gluon (A_3^0 and the A_3^1 set), t_g drops out by the Ward identity and $-g_{\mu\nu}$ alone is used. With two external gluons (A_4^0 and $\tilde{A}_4^0$) it does not drop out, and *AntCalc* keeps it, taking each gluon's reference vector to be the other gluon's momentum.

B.1.2 Vertices

These lines meet in vertices, which represent the way in which quantum fields interact. The vertices encountered throughout this work are shown in Table B.2 [2, 9].

Table B.2: Overview of the vertices encountered throughout this thesis, alongside their mathematical and graphical representations.

Interaction	Mathematical Representation	Graphical Representation
$q\bar{q}g$	$ig_s\gamma^\mu\,(T^a)_{ij}$	
ggg	$g_s f^{abc}\,(g^{\mu\nu}(p-q)^\rho + g^{\nu\rho}(q-r)^\mu + g^{\rho\mu}(r-p)^\nu)$ $p+q+r=0$	
$gggg$	$-ig_s^2\Big(f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho})$ $+ f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho})$ $+ f^{ade}f^{bce}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma})\Big)$	
$c\bar{c}g$	$-g_s f^{abc}p^\mu$ p is the outgoing ghost momentum	
$\gamma^* q\bar{q}$	$ie_q\gamma^\mu\delta_{ij}$	

B.2 Passarino–Veltman Reduction

At one loop, amplitudes produce tensor integrals similar to those in Eq. (2.74) [14]. With the massless[1] denominators defined as,

$$D_0 = \ell^2 + i0, \quad D_i = (\ell + r_i)^2 + i0 \tag{B.3}$$

[1]Throughout this work the internal masses vanish and the physical external partons satisfy $p_a^2 = 0$. The shifted momenta r_i, introduced in Eq. (B.3), entering propagators are combinations of external momenta and need not be lightlike.

the tensor integral takes the form,

$$I_N^{\mu_1 \ldots \mu_R} = \mu^{4-d} \int \frac{d^d \ell}{i\pi^{d/2}} \frac{\ell^{\mu_1} \ldots \ell^{\mu_R}}{D_0 D_1 \ldots D_{N-1}} \tag{B.4}$$

where R is the tensor rank and r_i are the combinations of external momenta. The reduction is carried out in $d = 4 - 2\epsilon$ dimensions, so that,

$$g_{\mu\nu} g^{\mu\nu} = d. \tag{B.5}$$

Passarino–Veltman reduction removes the Lorentz indices in the numerator $\ell^\mu \ell^\nu \ell^\rho \ldots$ terms and expresses them in terms of external momenta p_i^μ, the metric tensor $g_{\mu\nu}$ and scalar one-loop integrals, through Lorentz covariance [14].

For a rank-one N-point integral, the only available vectors are the independent external momenta, such that,

$$I_N^\mu = \sum_{i=1}^{N-1} p_i^\mu C_i \tag{B.6}$$

where C_i are scalar coefficient functions. For rank-two, the relation becomes,

$$I_N^{\mu\nu} = \sum_{i,j=1}^{N-1} p_i^\mu p_j^\nu C_{ij} + g^{\mu\nu} C_{00} \tag{B.7}$$

and for rank-three,

$$I_N^{\mu\nu\rho} = \sum_{i,j,k=1}^{N-1} p_i^\mu p_j^\nu p_k^\rho C_{ijk} + \sum_{i=1}^{N-1} \left(g^{\mu\nu} p_i^\rho + g^{\mu\rho} p_i^\nu + g^{\nu\rho} p_i^\mu \right) C_{00i} \tag{B.8}$$

and so on for higher ranks.

The reduction mechanism works because scalar products involving the loop momentum can be rewritten using the propagator denominators as,

$$2\ell \cdot r_i = D_i - D_0 - r_i^2 \tag{B.9}$$

which lets us cancel denominator terms, through a method like the following example set at $i = 1$,

$$\frac{\ell \cdot r_1}{D_0 D_1} = \frac{1}{2} \left(\frac{1}{D_0} - \frac{1}{D_1} - \frac{r_1^2}{D_0 D_1} \right) \tag{B.10}$$

thereby rewriting loop-momentum scalar products in terms of denominators and scalar integrals.

While using this reduction scheme, we might also encounter Gram matrices, defined as,

$$G_{ji} = 2 p_j \cdot p_i. \tag{B.11}$$

Contracting a general rank-one integral, as described in Eq. (B.6), with p_j gives,

$$p_j \cdot I_N = \sum_i (p_j \cdot p_i) C_i = \frac{1}{2} \sum_i G_{ji} C_i. \tag{B.12}$$

If we now define $R_j \equiv p_j \cdot I_N$, we are then able to compute the C_i coefficients in the Lorentz decomposition by taking,

$$G_{ji} C_i = 2 R_j \Rightarrow C_i = 2 \left(G^{-1} \right)_{ij} R_j. \tag{B.13}$$

B.2.1 Rank-One Bubble Example

To illustrate the method, let us consider the rank-one bubble,

$$B^{\mu}(p;0,0) = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^{\mu}}{D_0 D_1} \tag{B.14}$$

where the denominators follow Eq. (B.3). Lorentz covariance gives,

$$B^{\mu} = p^{\mu} B_1 \tag{B.15}$$

Contracting with p_{μ} yields,

$$p^2 B_1 = p_{\mu} B^{\mu} = \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell \cdot p}{D_0 D_1}. \tag{B.16}$$

Using the relation in Eq. (B.9), setting $r_1 = p$, we get,

$$p^2 B_1 = \frac{1}{2}\left[A_0(0) - A_0(0) - p^2 B_0\Big(p^2;0,0\Big)\right] = -\frac{1}{2} p^2 B_0\Big(p^2;0,0\Big) \tag{B.17}$$

and therefore[2],

$$B^{\mu} = -\frac{1}{2} p^{\mu} B_0\Big(p^2;0,0\Big)\,. \tag{B.18}$$

Here, A_0 and B_0 are the scalar one-loop integrals mentioned throughout. These two types can be defined as,

$$\begin{aligned} A_0(0) &= \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{\ell^2 + i0} = 0 \\ B_0\Big(p^2;0,0\Big) &= \mu^{4-d} \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{D_0 D_1}. \end{aligned} \tag{B.19}$$

For massless propagators A_0 is scaleless and therefore vanishes in dimensional regularisation.

B.2.2 Passarino–Veltman Scalar Functions

As mentioned, only three of the MI considered are Passarino–Veltman scalar functions. Those are B_0, C_0 and D_0 and are the Passarino–Veltman two-, three- and four-point scalar integrals, respectively. The Passarino–Veltman method also features the A_0 (one-point) scalar function, which does not feature in *AntCalc*. It also accounts for scalar integrals with more than four external legs, as those can be further reduced in four dimensions to combinations of D_0 functions [38, 39].

B.2.2.1 The B_0 Function

The B_0 function represents a scalar one-loop diagram with two external lines and two internal propagators. It is represented in d dimensions as,

$$B_0\Big(k^2;0,0\Big) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p \frac{1}{\big(p^2 + i\varepsilon\big)\Big((p+k)^2 + i\varepsilon\Big)} \tag{B.20}$$

where k is the external momentum and μ is the 't Hooft normalisation scale. This differs from the normalisation used above by an overall factor $(4\pi)^{\epsilon}$.

[2]The result shown in Eq. (B.18) assumes generic kinematics with $p^2 \neq 0$. Exceptional cases are obtained through an appropriate limit or alternative reduction.

It is also noteworthy that this integral, unlike higher-point ones, is UV divergent and generates the pole term Δ_B,

$$\Delta_B = \frac{2}{4-d} - \gamma_E + \log(4\pi) \tag{B.21}$$

where γ_E is Euler's gamma constant. This function may also be written as an integral over the Feynman parameter x. That yields the explicit analytical form [14],

$$\begin{aligned} B_0\Big(k^2;0,0\Big) &= \Delta_B - \int_0^1 dx\, \log\left[\frac{-x(1-x)k^2 - i\varepsilon}{\mu^2}\right] + O(d-4) \\ &= \Delta_B + 2 - \log\left(\frac{-k^2 - i\varepsilon}{\mu^2}\right) + O(d-4). \end{aligned} \tag{B.22}$$

B.2.2.2 The C_0 Function

The C_0 function represents a scalar loop with three external lines and three internal propagators. It is a common feature of one-loop vertex corrections. It is written in d dimensions as,

$$\begin{aligned} C_0\Big(k_1^2, k_2^2, (k_1+k_2)^2; 0,0,0\Big) = &\frac{(2\pi\mu)^{4-d}}{i\pi^2} \\ &\int d^d p\, \frac{1}{\Big(p^2 + i\varepsilon\Big)\Big((p+k_1)^2 + i\varepsilon\Big)\Big((p+k_1+k_2)^2 + i\varepsilon\Big)}. \end{aligned} \tag{B.23}$$

This integral is UV finite (for massless propagators and external legs it does, however, contain infrared singularities) and may be rewritten as a double integral over the Feynman parameters x and y as,

$$C_0 \propto \int_0^1 dx \int_0^{1-x} dy\, \left[ax^2 + by^2 + cxy + dx + ey + f\right]^{-1} \tag{B.24}$$

where the coefficients a, b, c, d, e and f are linear combinations of the external momentum invariants [40].

B.2.2.3 The D_0 Function

The D_0 function describes a scalar loop with four external lines and four internal propagators. It is often called the scalar box integral. It is written as,

$$D_0\Big(k_1^2, k_2^2, k_3^2; k_4^2, (k_1+k_2)^2, (k_2+k_3)^2; 0,0,0,0\Big) = \frac{(2\pi\mu)^{4-d}}{i\pi^2} \int d^d p\, \frac{1}{\mathcal{D}_0\mathcal{D}_1\mathcal{D}_2\mathcal{D}_3} \tag{B.25}$$

with denominator factors,

$$\begin{aligned} \mathcal{D}_0 &= p^2 + i\varepsilon \\ \mathcal{D}_i &= \left(p + \sum_{a=1}^{i} k_a\right)^2 + i\varepsilon, \end{aligned} \tag{B.26}$$

with $k_4 = -(k_1 + k_2 + k_3)$.

Evaluating D_0 is highly non-trivial as it features numerous physical cuts and branch points. The resulting analytical expression is expressed in terms of Spence functions (dilogarithms) [40].

B.3 IBP Reduction Example

The method behind IBP reduction, as discussed in Subsection 2.5.1 may be made easier to visualise with an example. Consider a one-loop bubble, akin to that studied via Passarino–Veltman reduction in Section B.2. Take,

$$\mathcal{I}(a_0, a_1) = \int d^d\ell \, \frac{1}{D_0^{a_0} D_1^{a_1}} \tag{B.27}$$

where,

$$D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell + p)^2 - m_1^2. \tag{B.28}$$

We now choose $v^\mu = \ell^\mu$, and build and expand the relation from Eq. (2.76) as,

$$\begin{aligned} 0 &= \int d^d\ell \, \frac{\partial}{\partial \ell^\mu} \left(\frac{\ell^\mu}{D_0^{a_0} D_1^{a_1}} \right) \\ &= d\mathcal{I}(a_0, a_1) - 2a_0 \int d^d\ell \, \frac{\ell^2}{D_0^{a_0+1} D_1^{a_1}} - 2a_1 \int d^d\ell \, \frac{\ell \cdot (\ell + p)}{D_0^{a_0} D_1^{a_1+1}}. \end{aligned} \tag{B.29}$$

We can now use,

$$\ell^2 = D_0 + m_0^2 \tag{B.30}$$

$$2\ell \cdot (\ell + p) = D_0 + D_1 + m_0^2 + m_1^2 - p^2 \tag{B.31}$$

and finally get the explicit shifted-index identity,

$$\begin{aligned} 0 = &(d - 2a_0 - a_1)\mathcal{I}(a_0, a_1) \\ &- 2a_0 m_0^2 \mathcal{I}(a_0 + 1, a_1) \\ &- a_1 \mathcal{I}(a_0 - 1, a_1 + 1) \\ &- a_1 \left(m_0^2 + m_1^2 - p^2 \right) \mathcal{I}(a_0, a_1 + 1). \end{aligned} \tag{B.32}$$

B.4 Three-Particle Phase-Space to Cut Propagators

Let us now consider the massless three-particle phase-space Φ_3. It may be described as,

$$d\Phi_3(q : p_1, p_2, p_3) \propto d^d p_1 d^d p_2 d^d p_3 \, \delta^+\left(p_1^2\right) \, \delta^+\left(p_2^2\right) \, \delta^+\left(p_3^2\right) \, \delta^{(d)}(q - p_1 - p_2 - p_3). \tag{B.33}$$

We start by using momentum conservation to eliminate p_3,

$$p_3 = q - p_1 - p_2. \tag{B.34}$$

This first step eliminates the rightmost δ-function. Then, using reverse unitarity, each of the remaining δ-functions may be rewritten as cut propagators. Schematically, it gives,

$$d\Phi_3 \sim d^d p_1 d^d p_2 \frac{1}{[p_1^2]_c [p_2^2]_c [(q - p_1 - p_2)^2]_c} \tag{B.35}$$

with

$$\left[\frac{1}{p^2} \right]_c \equiv \frac{1}{2\pi i} \left[\frac{1}{p^2 - i0} - \frac{1}{p^2 + i0} \right]_+ . \tag{B.36}$$

B.5 Laporta Algorithm Implementation Example

Consider an arbitrary family defined as $\mathcal{F}[\{D_1, D_2, D_3\}; d\Phi]$, with the denominators chosen as $D_1 = s_{12}$, $D_2 = s_{13}$ and $D_3 = s_{23}$. We may take the integrals below, all members of this family, and organise them using their respective sector vectors as follows,

$$I_1 = \int d\Phi \, \frac{s_{13}}{s_{12}} \Rightarrow I_1 = I(1, -1, 0), \tag{B.37}$$

$$I_2 = \int d\Phi \, \frac{2}{s_{12} s_{13} s_{23}} \Rightarrow I_2 = 2I(1, 1, 1), \tag{B.38}$$

$$I_3 = \int d\Phi \, \frac{1}{s_{12}^3} \Rightarrow I_3 = I(3, 0, 0) \tag{B.39}$$

where $d\Phi$ is the common phase-space measure. Importantly, two integrals may share a sector if they contain the same denominators, independently of those denominators' powers. Also, only positive indices denote denominator presence, meaning that integrals I_1 and I_3 in the example above are members of the same sector, $I_1, I_3 \in (1, 0, 0)$. In our example, only two sectors appear: $(1, 0, 0)$ and $(1, 1, 1)$. These are our two target sectors. The three integrals are organised in terms of the t, r and s as follows,

	t	r	s
$I(1, -1, 0)$	1	0	1
$I(1, 1, 1)$	3	0	0
$I(3, 0, 0)$	1	2	0

(B.40)

then, by order of complexity, the algorithm orders the integrals as,

$$I(1, 1, 1) \succ I(3, 0, 0) \succ I(1, -1, 0). \tag{B.41}$$

B.6 The Massive Extension

The main body of this thesis focuses on the massless regime. This appendix collects the supporting extension of the discussed methods into the massive regime.

The fundamental change is the massive on-shell condition; the remainder of this section traces its consequences for the antenna construction and integration. Since final-state particles now have mass, their squared momenta are not set to zero; they are set to the square of their mass.

$$p_q^2 = p_{\tilde{q}}^2 = 0 \xrightarrow{\text{Massive Regime}} p_q^2 = p_{\tilde{q}}^2 = m_q^2. \tag{B.42}$$

where p_i represents the momentum and m_i represents the mass of particle i.

The antenna subtraction formalism and the reduction architecture that was developed and described throughout Chapter 2 are retained, however. The intermediate steps and kinematics change: the massive on-shell condition changes the kinematics themselves and the unresolved limits, with soft radiation remaining singular but heavy-quark collinear behaviour becoming quasi-collinear. Massive integration is also different, as the characteristic phase-space scale in the massive regime is not equal to its massless counterpart. A new scale of m_q^2/q^2 is also introduced and leads to distinct cut-integral families and master integrals [26].

B.6.1 Massive Kinematics and Infrared Structure

In Subsection 2.2.1, the Mandelstam invariants were introduced as $s_{ij} = 2p_i \cdot p_j$. This definition is retained in the massive regime, as in Section 4.5, so that the squared invariant mass of a pair of partons now also carries their masses,

$$(p_i + p_j)^2 = s_{ij} + m_i^2 + m_j^2. \tag{B.43}$$

Through momentum conservation, $q = \sum_i p_i$, the total invariant mass likewise becomes,

$$q^2 = \sum_i m_i^2 + \sum_{i<j} s_{ij}. \tag{B.44}$$

Soft radiation remains singular in the massive regime and therefore continues to generate explicit infrared poles after integration. A non-zero mass regulates collinear limits involving a massive parton, whereas collinear limits between two massless partons remain singular. If the two final-state collinear partons are massless, then the limit follows the massless collinear limit from the massless regime. If the two partons are massive, no strict collinear singularity exists. But if one of the collinear partons is massive (particle i, of mass m_i and energy E_i) and the other is massless (particle j, with $m_j = 0$), then,

$$p_i \cdot p_j = E_i E_j (1 - \beta_i \cos\theta), \quad \beta_i = \sqrt{1 - \frac{m_i^2}{E_i^2}} < 1. \tag{B.45}$$

As the angle between the partons θ tends towards zero, the mass prevents the invariant from reaching the collinear singular surface,

$$\lim_{\theta \to 0} p_i \cdot p_j = E_i E_j (1 - \beta_i) \neq 0. \tag{B.46}$$

This regulated collinear region still carries physically important information, particularly in the joint high-energy and small-angle region, $E_i \gg m_i$ and $\theta \ll 1$,

$$p_i \cdot p_j \sim \frac{E_i E_j}{2} \left(\theta^2 + \frac{m_i^2}{E_i^2} \right) \tag{B.47}$$

and, consequently, since

$$\int_0^{\theta_{\max}^2} d\theta^2 \, \frac{1}{p_i \cdot p_j} \sim \int_0^{\theta_{\max}^2} d\theta^2 \left(\theta^2 + \frac{m_i^2}{E_i^2} \right)^{-1} \sim \ln\left(\frac{E_i^2}{m_i^2} \right) \tag{B.48}$$

the massive-massless quasi-collinear region replaces the strict collinear pole by a mass logarithm. The quasi-collinear limit, then, is the limit in which both the mass m_i and the opening angle θ become small together. It is through this limit that the massive antennae recover the massless splitting behaviour, as $m_i \to 0$ [26].

Mass does not regulate UV divergences, as $\ell^2 \gg m_i^2, p_i^2$. It is noteworthy, however, that in a theory with massive quarks, the coupling renormalisation of Eq. (2.22) must be supplemented by the renormalisation of the quark mass and quark field,

$$m_{q,0} = Z_m m_q, \quad \psi_{q,0} = \sqrt{Z_2^q} \psi_q. \tag{B.49}$$

B.6.2 Massive Phase-Space Integration and Reduction Methods

The n-particle phase-space measure of Eq. (2.81) already carries the final-state masses through its on-shell conditions $\delta^+(p_i^2 - m_i^2)$. The massless regime of the main text sets $m_i = 0$, whereas in the

massive regime these conditions impose $p_i^2 = m_i^2$, with $p_i^0 > 0$, for the heavy quarks.

The phase-space factorisations of Eqs. (2.31) and (2.38) also still hold, with the antenna phase space evaluated at $q = \tilde{p}_I + \tilde{p}_K$ (or $q = \tilde{p}_I + \tilde{p}_L$), which expresses momentum conservation across the mapping. The only change is that the momentum mapping must now also preserve the mass-shell conditions of the hard radiators, $\tilde{p}_I^2 = m_i^2$ and $\tilde{p}_K^2 = m_k^2$.

Antenna integration is computed in the same manner as in the massless regime,

$$\mathcal{X}_n^l \sim \int d\Phi_X \left(\prod_{r=1}^{l} d^d \ell_r \right) X_n^l(\{p\}, \{\ell\}) \sim \left(q^2\right)^{\lambda + \kappa\epsilon} f\left(\left\{ \frac{m^2}{q^2} \right\}, \epsilon \right). \tag{B.50}$$

Passarino–Veltman and IBP reduction remain applicable in the massive regime, following the same principles introduced for the massless case. For IBP reduction, however, the massive case is significantly harder analytically: the phase-space is non-trivial and has mass-dependent thresholds; the master integrals are generally multi-scale, and can involve logarithms, polylogarithms and more complicated transcendental functions; and lastly, boundary conditions are less direct, although common choices are the massless limit at $m \to 0$ and the threshold limit at $\beta \to 0$ [26].

B.7 Master Integral Substitution Arrays

As described in Chapter 4, *AntCalc*'s IBP reduction, handled internally by *LiteRed2*, organises each antenna family's master integrals as members of that family's own basis, addressed through *LiteRed2*'s *j*[basis, ...] notation rather than the named masters used throughout Chapter 4. At the master combination stage for each antenna, the raw *j*[...]-form result is substituted through a rule list specific to that family's basis, translating it into the named masters. The rule lists used are collected below, one per antenna family.

B.7.1 The A_2^2 Set

```
In[1]:= A22MISubs = {
j[A22TwoLoopTreeBasis1, 0, 0, 0, 1, 1, 0, 1] -> A3,
j[A22TwoLoopTreeBasis1, 1, 0, 1, 1, 1, 0, 0] -> A4,
j[A22TwoLoopTreeBasis4, 1, 0, 1, 0, 0, 1, 0] -> A3,
j[A22TwoLoopTreeBasis4, 1, 0, 0, 1, 0, 1, 1] -> A4,
j[A22TwoLoopTreeBasis5, 0, 0, 1, 0, 0, 1, 1] -> A3,
j[A22TwoLoopTreeBasis6, 0, 0, 1, 1, 0, 0, 1] -> A3,
j[A22TwoLoopTreeBasis6, 1, 0, 1, 1, 0, 1, 0] -> A4,
j[A22TwoLoopTreeBasis7, 0, 0, 1, 0, 1, 0, 1] -> A3,
j[A22TwoLoopTreeBasis7, 1, 0, 1, 0, 1, 1, 0] -> A4,
j[A22TwoLoopTreeBasis7, 1, 1, 0, 1, 0, 1, 0] -> A22LO,
j[A22TwoLoopTreeBasis8, 0, 1, 0, 0, 0, 1, 1] -> A3,
j[A22TwoLoopTreeBasis8, 1, 0, 1, 1, 1, 0, 0] -> A4,
j[A22TwoLoopTreeBasis8, 1, 1, 1, 1, 1, 1, 0] -> A6,
j[A22OneLoopSelfBasis, 1, 0, 1, 1, 0, 1, 0] -> A22LO
};
```

B.7.2 The A_3^1 Set

```
In[1]:= A31MISubs = {
j[A31Basis10, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis12, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis2, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis6, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis6, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis9, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis9, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b,
j[A31Basis3, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis3, 1, 1, 1, 0, 0, 0, 1, 1, 0] -> V5b,
j[A31Basis4, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5a,
j[A31Basis4, 1, 1, 1, 0, 0, 1, 0, 1, 0] -> V5b,
j[A31Basis4, 1, 1, 1, 0, 1, 1, 1, 1, 1] -> 2 V8,
j[A31Basis7, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis7, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b,
j[A31Basis7, 1, 1, 1, 1, 1, 1, 1, 1, 0] -> V8,
j[A31Basis8, 1, 1, 1, 1, 0, 0, 0, 1, 0] -> V5a,
j[A31Basis8, 1, 1, 1, 0, 1, 0, 0, 1, 0] -> V5b
};
```

B.7.3 The Four-Parton (X40) Set

```
In[1]:= X40MISubs = {
j[chainBasis1342, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1342, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis1432, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1432, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[boxBasis1324, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[boxBasis1324, 1, 1, 1, 1, 1, 1, 1, 1, 0] -> R8a,
j[chainBasis1324, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1423, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1234, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1234, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis1234, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b,
j[hybridBasis1243, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[hybridBasis1243, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis1243, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b,
j[chainBasis1234, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis1234, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis2134, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis2134, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[chainBasis2314, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
j[chainBasis2314, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis2431, 1, 1, 1, 1, 0, 0, 0, 0, 0] -> R4,
```

```
j[hybridBasis2431, 1, 1, 1, 1, 0, 0, 0, 1, 1] -> R6,
j[hybridBasis2431, 1, 1, 1, 1, 1, 0, 1, 1, 1] -> R8b
};
```

B.8 Local Validation of Additional Routes

Much like the A_3^0 massless unintegrated antenna was locally validated by computing its local soft and (single) collinear limits, so have other antenna functions been validated in that way. In this section we present those validation methods and results.

In addition to the unresolved-limit checks described below, the constructed A_4^0, $\widetilde{A}_4^0$, B_4^0, and C_4^0 antennae were compared with the explicit $\epsilon \to 0$ expressions given in Eqs. (5.29), (5.30), (5.38), and (5.43) of Ref. [5] (v3). Numerical evaluations at randomly chosen kinematic points showed exact agreement. The two-parton antennae A_2^1 and A_2^2 contain no unresolved final-state radiation and therefore have no corresponding local unresolved limits.

B.8.1 A_4^0

The leading $A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}})$ massless antenna component has five relevant limits: two soft limits, and three collinear limits, for $1_q \parallel 3_g$, $3_g \parallel 4_g$, and $2_{\bar{q}} \parallel 4_g$. The two qg collinear limits, $1_q \parallel 3_g$ and $2_{\bar{q}} \parallel 4_g$, and the two soft-$g$ limits are symmetric under the reversal symmetry of the colour ordering, as $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$.

B.8.1.1 Soft Limit and the Eikonal Factor

In this limit we set $p_g \to 0$. As mentioned, the limit is symmetric for both gluons. Here we consider the soft-3_g limit.

The limit is thus computed as,

$$\lim_{\lambda\to 0} \lambda^2 A_4^0\Big|_{s_{13},s_{23},s_{34}\to\lambda s_{13},\lambda s_{23},\lambda s_{34}} = \mathcal{S}_{14}^{(3)} A_3^0(1_q, 4_g, 2_{\bar{q}}), \quad \mathcal{S}_{14}^{(3)} = \frac{2s_{14}}{s_{13}s_{34}}. \tag{B.51}$$

This limit is computed using *AntCalc* as follows [5].

```
In[1]:= A40 = BuildAntenna[A,4,0,Component->Leading];

        soft3Rules = {s13 -> lambda s13, s23 -> lambda s23,
            s34 -> lambda s34, s134 -> s14, s234 -> s24, q2 -> s124};
        SeriesCoefficient[A40 /. soft3Rules, {lambda, 0, -2}] //
   Simplify

Out[1]=
```

$$\frac{4s_{12}^2 + 4s_{12}(s_{14}+s_{24}) - 2(\epsilon-1)s_{14}^2 - 4\epsilon s_{14}s_{24} - 2(\epsilon-1)s_{24}^2}{s_{124}s_{13}s_{24}s_{34}}$$

Since the result is larger than a simple factor, unlike for A_3^0, it is compared with the right-hand side of the equation above after the calculation. Using Eq. (4.13) for $A_3^0(1_q, 4_g, 2_{\bar{q}})$, with $s_{13} \to s_{14}$, $s_{23} \to s_{24}$ and $q^2 \to s_{124}$, that right-hand side is

$$\mathcal{S}_{14}^{(3)} A_3^0(1_q, 4_g, 2_{\bar{q}}) = \frac{2s_{14}}{s_{13}s_{34}} \frac{2s_{12}s_{124} + s_{14}^2 + s_{24}^2 - \epsilon(s_{14}+s_{24})^2}{s_{124}s_{14}s_{24}}, \tag{B.52}$$

which coincides with the result above once $s_{124} = s_{12} + s_{14} + s_{24}$ is used.

B.8.1.2 Collinear $1_q \parallel 3_g$ Limit

For the $1_q \parallel 3_g$ limit, the quark and gluon pair recombine into a parent quark as,

$$p_1 = z p_{13}, \quad p_3 = (1-z) p_{13} \tag{B.53}$$

where z is the momentum fraction carried by the original quark. In this limit, $s_{13} \to 0$.

We now introduce the Mandelstam notation $s_{(13)j}$ to represent the invariant computed from the momenta of the parent quark and some other j particle. The remaining invariants become,

$$\begin{aligned} s_{12} &\to z s_{(13)2}, \quad s_{23} \to (1-z) s_{(13)2}, \\ s_{14} &\to z s_{(13)4}, \quad s_{34} \to (1-z) s_{(13)4}, \\ s_{24} &\to s_{24}. \end{aligned} \tag{B.54}$$

We also define the reduced configuration of the three-parton tree-level antenna as,

$$A_3^0\Big((13)_q, 4_g, 2_{\bar{q}}\Big). \tag{B.55}$$

At leading power, the four-parton antenna factorises as,

$$A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \xrightarrow{s_{13}\to 0} \frac{1}{s_{13}} P_{qg\to q}(z,\epsilon) A_3^0\Big((13)_q, 4_g, 2_{\bar{q}}\Big), \quad P_{qg\to q}(z,\epsilon) = \frac{1+z^2}{1-z} - \epsilon(1-z). \tag{B.56}$$

Here $P_{qg\to q}(z,\epsilon)$ is the $O(\epsilon)$-extended version of the splitting function used for A_3^0's collinear limit in Subsection 4.1.2; the $-\epsilon(1-z)$ term is the known d-dimensional extension from the literature, quoted here since A_4^0 is an NNLO antenna where it no longer drops out [5]. The check below is performed at $\epsilon = 0$, and therefore verifies the $(1+z^2)/(1-z)$ piece only.

```
In[1]:= A40 = BuildAntenna[A,4,0,Component->Leading];

        col13Rules = {s13 -> lambda, s12 -> z sP2, s23 -> (1-z) sP2,
            s14 -> z sP4, s34 -> (1-z) sP4, s134 -> sP4,
            s234 -> (1-z) sP2 + s24 + (1-z) sP4, q2 -> sP2 + sP4 + s24,
            Epsilon -> 0};
        SeriesCoefficient[Together[A40 /. col13Rules],
            {lambda, 0, -1}] // FullSimplify

Out[1]:=
```

$$-\frac{(1+z^2)\left(s_{24}^2 + 2s_{24}s_{P2} + 2s_{P2}^2 + 2s_{P2}s_{P4} + s_{P4}^2\right)}{s_{24}s_{P4}(z-1)(s_{24}+s_{P2}+s_{P4})}$$

This is the $\epsilon = 0$ kernel $(1+z^2)/(1-z)$ times $A_3^0\big((13)_q, 4_g, 2_{\bar{q}}\big)$: with $s_{12} \to s_{P2}$, $s_{13} \to s_{P4}$, $s_{23} \to s_{24}$ and $q^2 \to s_{P2} + s_{P4} + s_{24}$, Eq. (4.13) gives

$$A_3^0 = \frac{s_{24}^2 + 2s_{24}s_{P2} + 2s_{P2}^2 + 2s_{P2}s_{P4} + s_{P4}^2}{s_{24}s_{P4}(s_{24}+s_{P2}+s_{P4})}.$$

B.8.1.3 Collinear $3_g \parallel 4_g$ Limit

For the $3_g \parallel 4_g$ limit, the gluon-gluon pair recombine into a parent gluon as,

$$p_3 = zp_{34}, \quad p_4 = (1-z)p_{34}. \tag{B.57}$$

In this limit, $s_{34} \to 0$.

The remaining invariants become,

$$\begin{aligned} s_{13} &\to z s_{1(34)}, \quad s_{14} \to (1-z)s_{1(34)}, \\ s_{23} &\to z s_{(34)2}, \quad s_{24} \to (1-z)s_{(34)2}, \\ s_{12} &\to s_{12}. \end{aligned} \tag{B.58}$$

The reduced configuration of the three-parton tree-level antenna also becomes,

$$A_3^0\Big(1_q, (34)_g, 2_{\bar{q}}\Big). \tag{B.59}$$

The parent parton here is now a gluon, not a quark. Hence, the local factorisation formula involves the spin-correlated $g \to gg$ splitting tensor:

$$A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \xrightarrow{3_g \| 4_g} \frac{1}{s_{34}} \hat{P}^{\mu\nu}_{gg\to G}(z, k_\perp) \big(A_3^0\big)_{\mu\nu}\Big(1_q, (34)_g, 2_{\bar{q}}\Big). \tag{B.60}$$

Here, $\big(A_3^0\big)_{\mu\nu}$ is the reduced three-parton antenna with the parent-gluon polarisation indices left uncontracted. Equivalently, in terms of scalar antenna functions, the unaveraged collinear limit contains an angular contribution,

$$A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \xrightarrow{3_g \| 4_g} \frac{1}{s_{34}} P_{gg\to G}(z) A_3^0\Big(1_q, (34)_g, 2_{\bar{q}}\Big) + \text{ang.}, \tag{B.61}$$

where ang. denotes a spin-correlation contribution. This term vanishes after integration over the unresolved azimuthal angle. Consequently,

$$\lim_{s_{34}\to 0} s_{34} \left\langle A_4^0(1_q, 3_g, 4_g, 2_{\bar{q}}) \right\rangle_\phi = P_{gg\to G}(z) A_3^0\Big(1_q, (34)_g, 2_{\bar{q}}\Big), \tag{B.62}$$

where

$$P_{gg\to G}(z) = 2\left[\frac{z}{1-z} + \frac{1-z}{z} + z(1-z)\right]. \tag{B.63}$$

The direct invariant-space limit produces a non-vanishing residual after subtracting the spin-averaged contribution, consistently with the angular term in the local gluon–gluon collinear factorisation [5].

```
In[1]:= A40 = BuildAntenna[A,4,0,Component->Leading];

        col34Rules = {s34 -> lambda, s13 -> z s1P, s14 -> (1-z) s1P,
            s23 -> z sP2, s24 -> (1-z) sP2, s134 -> s1P + lambda,
            s234 -> sP2 + lambda, q2 -> s12 + s1P + sP2 + lambda,
            Epsilon -> 0};
        SeriesCoefficient[Together[A40 /. col34Rules],
            {lambda, 0, -1}] // FullSimplify
```

```
Out[1]:=
```

$$\frac{-4\,(1+2z(z-1))\,s_{12}\,(s_{12}+s_{1P}+s_{P2})-2\left(s_{1P}^2+s_{P2}^2\right)(1+z(z-1))^2}{s_{1P}s_{P2}\,z(z-1)\,(s_{12}+s_{1P}+s_{P2})}$$

Subtracting $P_{gg\to G}(z)A_3^0\left(1_q,(34)_g,2_{\bar{q}}\right)$, the spin-averaged term of the equation above, from this result leaves exactly the angular term.

Note that the ang. quantity was found to be,

$$\text{ang.} = \frac{4s_{12}z(z-1)}{s_{1(34)}s_{(34)2}} \tag{B.64}$$

but this value is not to be understood as a general result.

B.8.2 $\tilde{A}_4^0$

The subleading $\tilde{A}_4^0(1_q,3_g,4_g,2_{\bar{q}})$ massless antenna component has the same five kinematic regions as the leading term, but the singular structure of each region differs. As seen in Subsection 2.4.4, $\tilde{A}_4^0$ is obtained from the squared sum of both colour orderings, $\left|\mathcal{M}_4^0(1_q,3_g,4_g,2_{\bar{q}})+\mathcal{M}_4^0(1_q,4_g,3_g,2_{\bar{q}})\right|^2$, so its gluons are not colour ordered and it must be symmetric under the exchange of the two gluon labels, $3\leftrightarrow 4$. The check below confirms this: once the physical relations for s_{134} and s_{234} are imposed, the difference between $\tilde{A}_4^0$ and its gluon-exchanged copy vanishes, so the antenna is symmetric in the gluon indices.

```
In[1]:= tA40 = BuildAntenna[A,4,0,Component->Subleading];

        physicalRelations = {s134 -> s13 + s14 + s34,
            s234 -> s23 + s24 + s34};

        tA40Physical = Together[tA40 /. physicalRelations];

        gluonExchangeRules = {s13 -> s14, s14 -> s13, s23 -> s24,
            s24 -> s23};

        FullSimplify[tA40Physical - (tA40Physical /. gluonExchangeRules)
   ]
Out[1]:=
                                    0
```

All other limits are checked using the same method as for A_4^0.

B.8.2.1 Soft Limit and the Eikonal Factor

This limit is also symmetric for the 4_g gluon. It also generates the eikonal factor [5].

```
In[1]:= tA40 = BuildAntenna[A,4,0,Component->Subleading];

        soft3Rules = {s13 -> lambda s13, s23 -> lambda s23,
            s34 -> lambda s34, s134 -> s14, s234 -> s24, q2 -> s124};
        SeriesCoefficient[tA40 /. soft3Rules, {lambda, 0, -2}] //
```

```
    Simplify

Out[1]:=
```

$$\frac{2s_{12}\left[2s_{12}(s_{12}+s_{14}+s_{24})+s_{14}^2+s_{24}^2-\epsilon(s_{14}+s_{24})^2\right]}{s_{124}s_{13}s_{14}s_{23}s_{24}}$$

The reduced antenna keeps its ϵ dependence. With Eq. (4.13) for $A_3^0(1_q,4_g,2_{\bar{q}})$ as above, the result equals, for general ϵ,

$$\frac{2s_{12}}{s_{13}s_{23}}A_3^0(1_q,4_g,2_{\bar{q}})=\frac{2s_{12}}{s_{13}s_{23}}\,\frac{2s_{12}s_{124}+s_{14}^2+s_{24}^2-\epsilon(s_{14}+s_{24})^2}{s_{124}s_{14}s_{24}}, \tag{B.65}$$

that is, the eikonal factor $\mathcal{S}_{12}^{(3)}$ between the quark and the antiquark times the reduced antenna.

B.8.2.2 Collinear $1_q \parallel 3_g$ Limit

The $1_q \parallel 3_g$ limit also behaves in the same way as for the leading term, again checked here at $\epsilon = 0$ against the $(1+z^2)/(1-z)$ kernel [5].

```
In[1]:= tA40 = BuildAntenna[A,4,0,Component->Subleading];

        col13Rules = {s13 -> lambda, s12 -> z sP2, s23 -> (1-z) sP2,
            s14 -> z sP4, s34 -> (1-z) sP4, s134 -> sP4,
            s234 -> (1-z) sP2 + s24 + (1-z) sP4, q2 -> sP2 + sP4 + s24,
            Epsilon -> 0};
        SeriesCoefficient[Together[tA40 /. col13Rules],
            {lambda, 0, -1}] // FullSimplify

Out[1]:=
```

$$-\frac{(1+z^2)\left(s_{24}^2+2s_{24}s_{P2}+2s_{P2}^2+2s_{P2}s_{P4}+s_{P4}^2\right)}{s_{24}s_{P4}(z-1)(s_{24}+s_{P2}+s_{P4})}$$

The result is identical to that of the leading-colour component above.

B.8.2.3 Collinear $3_g \parallel 4_g$ Limit

The $3_g \parallel 4_g$ limit is the one that differs significantly from its analogue for A_4^0. Using the same parametrisation as for A_4^0, the expansion of $\tilde{A}_4^0$ in s_{34} starts at order s_{34}^0: the coefficient of $1/s_{34}$, computed below for general ϵ, vanishes identically, so $\tilde{A}_4^0$ has no $3_g \parallel 4_g$ collinear singularity. The gluon-exchange symmetry shown above is not, on its own, the reason for this, since the sum of the two colour-ordered A_4^0 terms is equally symmetric and is singular. The reason is that the only diagram carrying the $1/s_{34}$ propagator is the one with the three-gluon vertex, and, as the colour-ordered three-gluon vertex is antisymmetric in its gluon legs, this diagram enters $\mathcal{M}_4^0(1_q,3_g,4_g,2_{\bar{q}})$ and $\mathcal{M}_4^0(1_q,4_g,3_g,2_{\bar{q}})$ with opposite signs. It therefore cancels in their sum, in which the two gluons couple to the quark line like photons [5].

```
In[1]:= tA40 = BuildAntenna[A,4,0,Component->Subleading];

        col34Rules = {s34 -> lambda, s13 -> z s1P, s14 -> (1-z) s1P,
            s23 -> z sP2, s24 -> (1-z) sP2, s134 -> s1P + lambda,
```

```
            s234 -> sP2 + lambda, q2 -> s12 + s1P + sP2 + lambda};

        FullSimplify[SeriesCoefficient[Together[tA40 /. col34Rules],
            {lambda, 0, -1}]]

Out[1]:=
                                    0
```

B.8.3 B_4^0

The B_4^0 antenna is the ordinary, non-identical-flavour four-quark antenna of the process $\gamma^* \to qq'\bar{q}'\bar{q}$. This antenna has three non-vanishing limits, the single collinear limit $3_{q'} \parallel 4_{\bar{q}'}$, the double soft limit $3_{q'}, 4_{\bar{q}'} \to 0$, and the triple-collinear limit $1_q \parallel 3_{q'} \parallel 4_{\bar{q}'}$.

B.8.3.1 Single Collinear Limit

This limit is computed similarly to A_4^0's $3_g \parallel 4_g$ collinear limit. We find that,

$$B_4^0(1_q, 3_{q'}, 4_{\bar{q}'}, 2_{\bar{q}}) \xrightarrow{3_{q'} \parallel 4_{\bar{q}'}} \frac{1}{s_{34}} P^{\mu\nu}_{q\bar{q}\to g}(z, k_\perp) \left(A_3^0\right)_{\mu\nu} \Big(1, (34), 2\Big) \tag{B.66}$$

which after azimuthal averaging becomes,

$$\left\langle B_4^0 \right\rangle_\phi \xrightarrow{3_{q'} \parallel 4_{\bar{q}'}} \frac{1}{s_{34}} P_{q\bar{q}\to g}(z) A_3^0\Big(1, (34), 2\Big), \quad P_{q\bar{q}\to g}(z) = \frac{z^2 + (1-z)^2 - \epsilon}{1-\epsilon}. \tag{B.67}$$

The check below keeps ϵ symbolic throughout, directly validating the full ϵ-dependent kernel $P_{q\bar{q}\to g}(z)$ above, rather than only its $\epsilon = 0$ piece [5].

```
In[1]:= B40 = BuildAntenna[B,4,0];

        col34Rules = {s34 -> lambda, s13 -> z s1P, s14 -> (1-z) s1P,
            s23 -> z sP2, s24 -> (1-z) sP2, s134 -> s1P, s234 -> sP2,
            q2 -> s12 + s1P + sP2};
        SeriesCoefficient[Together[B40 /. col34Rules],
            {lambda, 0, -1}] // FullSimplify

Out[1]:=
```

$$\frac{2(\epsilon-1)s_{12}\left(s_{12}+s_{1P}+s_{P2}\right) - (\epsilon - 1 - 2z(z-1))\left[(\epsilon-1)\left(s_{1P}^2+s_{P2}^2\right) + 2\epsilon s_{1P}s_{P2}\right]}{(\epsilon-1)s_{1P}s_{P2}\left(s_{12}+s_{1P}+s_{P2}\right)}$$

Subtracting $P_{q\bar{q}\to g}(z)A_3^0(1_q, (34)_g, 2_{\bar{q}})$ from this result leaves exactly the spin-correlation term $4s_{12}z(z-1)/\left[(\epsilon-1)s_{1P}s_{P2}\right]$, which vanishes after azimuthal averaging.

B.8.3.2 Double-Soft Limit

A soft quark-antiquark pair is singular when both particles become soft simultaneously. To compute this limit we start scaling,

$$\begin{aligned} p_3 \to \lambda p_3, \quad p_4 \to \lambda p_4, \\ s_{i3}, s_{i4} \sim \lambda, \quad s_{34} \sim \lambda^2. \end{aligned} \tag{B.68}$$

The antenna has a leading λ^{-4} behaviour such that [5],

$$\lim_{\lambda\to 0}\lambda^4 B_4^0(1,\lambda 3,\lambda 4,2) = S_{12}(3,4),$$
$$S_{12}(3,4) = \frac{2\left(s_{12}s_{34}-s_{13}s_{24}-s_{14}s_{23}\right)}{s_{34}^2(s_{13}+s_{14})(s_{23}+s_{24})} + \frac{2}{s_{34}^2}\left[\frac{s_{13}s_{14}}{(s_{13}+s_{14})^2}+\frac{s_{23}s_{24}}{(s_{23}+s_{24})^2}\right] \tag{B.69}$$

```
In[1]:= B40 = BuildAntenna[B,4,0];

        doubleSoftRules = {s13 -> lambda s13, s14 -> lambda s14,
            s23 -> lambda s23, s24 -> lambda s24, s34 -> lambda^2 s34,
            s134 -> lambda (s13+s14) + lambda^2 s34,
            s234 -> lambda (s23+s24) + lambda^2 s34,
            q2 -> s12 + lambda (s13+s14+s23+s24) + lambda^2 s34,
            Epsilon -> 0};
        SeriesCoefficient[Together[B40 /. doubleSoftRules],
            {lambda, 0, -4}] // FullSimplify

Out[1]=
```

$$\frac{2s_{12}s_{34}(s_{13}+s_{14})(s_{23}+s_{24}) - 2\left(s_{14}s_{23}-s_{13}s_{24}\right)^2}{s_{34}^2(s_{13}+s_{14})^2(s_{23}+s_{24})^2}$$

This is the double-soft factor $\mathcal{S}_{12}(3,4)$ of the equation above, written over a common denominator.

B.8.3.3 Triple-Collinear Limit

For the $1_q \parallel 3_{q'} \parallel 4_{\bar{q}'}$ limit, all three partons become collinear together. We introduce the rescaled invariants r_{13}, r_{14}, r_{34}, of order unity as $\lambda \to 0$, via

$$s_{13} \to \lambda r_{13}, \quad s_{14} \to \lambda r_{14}, \quad s_{34} \to \lambda r_{34}, \tag{B.70}$$

and the momentum fractions w, x, and $1-w-x$ carried by partons 1, 3, and 4 respectively within the recombined cluster, via

$$s_{12} \to w\, s_{P2}, \quad s_{23} \to x\, s_{P2}, \quad s_{24} \to (1-w-x)\, s_{P2}. \tag{B.71}$$

At leading power, the four-parton antenna factorises onto the trivial LO antenna, $A_2^0 \equiv 1$, as,

$$\lim_{\lambda\to 0}\lambda^2 B_4^0(1_q,3_{q'},4_{\bar{q}'},2_{\bar{q}}) = P^{(134)}_{\text{NonIdent}}(w,x;r_{13},r_{14},r_{34})\, A_2^0(1,2), \tag{B.72}$$

where the non-identical-flavour triple-collinear splitting function is [5],

$$\begin{aligned} P^{(134)}_{\text{NonIdent}}(w,x;r_{13},r_{14},r_{34}) = &-\frac{(1-\epsilon)+\dfrac{2r_{13}}{r_{34}}}{(r_{13}+r_{14}+r_{34})^2} - \frac{2\left[x(r_{13}+r_{14}+r_{34})-(1-w)r_{13}\right]^2}{r_{34}^2(r_{13}+r_{14}+r_{34})^2(1-w)^2} \\ &+ \frac{1}{r_{34}(r_{13}+r_{14}+r_{34})}\left(\frac{1+x^2+(x+w)^2}{1-w} - \epsilon(1-w)\right). \end{aligned} \tag{B.73}$$

```
In[1]:= B40 = BuildAntenna[B,4,0];
```

```
        tripleCollinearRules134 = {s13 -> lambda r13, s14 -> lambda r14,
            s34 -> lambda r34, s12 -> w sP2, s23 -> x sP2,
            s24 -> (1-w-x) sP2, s134 -> lambda (r13+r14+r34),
            s234 -> (1-w) sP2 + lambda r34,
            q2 -> sP2 + lambda (r13+r14+r34)};
        SeriesCoefficient[Together[B40 /. tripleCollinearRules134],
            {lambda, 0, -2}] // FullSimplify
```

Out[1]:=

$$\frac{\begin{aligned}&-2r_{13}^2(w-1)^2+r_{13}r_{34}(w-1)\left(1+\epsilon(w-1)^2-w(w+2)\right)\\&+r_{14}r_{34}(w-1)\left(\epsilon(w-1)^2-1-w^2\right)+r_{34}^2 w\left(1+\epsilon(w-1)^2-w^2\right)\\&-2(r_{13}+r_{14}+r_{34})\left[(w-1)(2r_{13}+r_{34}w)x+(r_{13}+r_{14}+r_{34}w)x^2\right]\end{aligned}}{r_{34}^2(r_{13}+r_{14}+r_{34})^2(w-1)^2}$$

Since $A_2^0 = 1$, this is the splitting function $P^{(134)}_{\text{NonIdent}}$ of the equation above, written over the common denominator $r_{34}^2(r_{13}+r_{14}+r_{34})^2(1-w)^2$.

B.8.4 C_4^0

The C_4^0 antenna is the identical-flavour four-quark antenna of the process $\gamma^* \to q\bar{q}q\bar{q}$, built as the symmetrised direct–exchange interference. Unlike B_4^0, it has a single non-vanishing limit, the triple-collinear limit $2_{\bar{q}} \parallel 3_q \parallel 4_{\bar{q}}$; the direct $3 \parallel 4$ single-collinear region is regular, as the check below confirms.

B.8.4.1 Collinear $3 \parallel 4$ Limit

```
In[1]:= C40 = BuildAntenna[C,4,0];

        col34Rules = {s34 -> lambda, s13 -> z s1P, s14 -> (1-z) s1P,
            s23 -> z sP2, s24 -> (1-z) sP2, s134 -> s1P, s234 -> sP2,
            q2 -> s12 + s1P + sP2, Epsilon -> 0};

        FullSimplify[SeriesCoefficient[Together[C40 /. col34Rules],
            {lambda, 0, -1}]]
```

Out[1]:=

$$0$$

B.8.4.2 Triple-Collinear Limit

For the $2_{\bar{q}} \parallel 3_q \parallel 4_{\bar{q}}$ limit, partons 2, 3, and 4 become collinear together, leaving parton 1 as the sole hard remaining leg. As for B_4^0, we introduce the rescaled invariants r_{23}, r_{24}, r_{34}, of order unity as $\lambda \to 0$, via

$$s_{23} \to \lambda r_{23}, \quad s_{24} \to \lambda r_{24}, \quad s_{34} \to \lambda r_{34}, \tag{B.74}$$

and the momentum fractions w, x, and $y \equiv 1 - w - x$ carried by partons 2, 3, and 4 respectively within the recombined cluster, via

$$s_{12} \to w\, s_{1P}, \quad s_{13} \to x\, s_{1P}, \quad s_{14} \to y\, s_{1P}. \tag{B.75}$$

We also define $S_{234} \equiv r_{23} + r_{24} + r_{34}$.

At leading power, the four-parton antenna factorises onto the trivial LO antenna, $A_2^0 \equiv 1$, with an explicit identical-particle factor of $\frac{1}{2}$, as,

$$\lim_{\lambda\to 0} \lambda^2 C_4^0(1_q, 3_q, 4_{\bar{q}}, 2_{\bar{q}}) = \frac{1}{2} P^{(234)}_{\text{Ident}}(w, x; r_{23}, r_{24}, r_{34})\, A_2^0(1,2), \tag{B.76}$$

where the identical-flavour triple-collinear splitting function is a sum of two mirror-symmetric pieces [5],

$$\begin{aligned} P^{(234)}_{\text{Ident}}(w, x; r_{23}, r_{24}, r_{34}) = &-\frac{2r_{23}/r_{34} + 2}{S_{234}^2} - \frac{x(1+x^2)}{2r_{23}r_{34}(1-y)(1-w)} + \frac{1}{r_{34}S_{234}}\left(\frac{1+x^2}{1-y} + \frac{2x}{1-w}\right) \\ &-\frac{2r_{34}/r_{23} + 2}{S_{234}^2} - \frac{x(1+x^2)}{2r_{34}r_{23}(1-w)(1-y)} + \frac{1}{r_{23}S_{234}}\left(\frac{1+x^2}{1-w} + \frac{2x}{1-y}\right). \end{aligned} \tag{B.77}$$

```
In[1]:= C40 = BuildAntenna[C,4,0];

        tripleCollinearRules234 = {s23 -> lambda r23, s24 -> lambda r24,
            s34 -> lambda r34, s12 -> w s1P, s13 -> x s1P,
            s14 -> (1-w-x) s1P, s234 -> lambda (r23+r24+r34),
            s134 -> (1-w) s1P + lambda r34,
            q2 -> s1P + lambda (r23+r24+r34), Epsilon -> 0,
            s123 -> (w+x) s1P + lambda r23,
            s124 -> (1-x) s1P + lambda r24};
        SeriesCoefficient[Together[C40 /. tripleCollinearRules234],
            {lambda, 0, -2}] // FullSimplify
```

Out[1]:=

$$\frac{\begin{aligned} &-r_{23}^2\left(2w - (x-1)^2\right)(w+x-1) \\ &+r_{23}\left[r_{24}(x-1)\left(1 + w(x-1) + x(2x-1)\right) + r_{34}\left((x-1)^3 - 4w^2 - 4w(x-1)\right)\right] \\ &+(r_{24}x - r_{34}w)\left(r_{24} - r_{34} + 2r_{34}w + (r_{24}+r_{34})x^2\right) \end{aligned}}{2r_{23}r_{34}(w-1)(w+x)(r_{23}+r_{24}+r_{34})^2}$$

Since $A_2^0 = 1$, this is $\frac{1}{2}P^{(234)}_{\text{Ident}}$ of the equation above, written over a common denominator.

B.8.5 Massive A_3^0

For this massive antenna, described as,

$$A_3^0\left(1_Q, 3_g, 2_{\bar{Q}}\right) \tag{B.78}$$

where Q denotes a quark of mass m_q, there is one infrared singular limit, for the soft gluon, and two mass-regulated quasi-collinear limits, for $Q \parallel g$ and $\bar{Q} \parallel g$. These two collinear limits are symmetric as

$1 \leftrightarrow 2$ and thus, as before, the validation of one validates the other. Relative to the massless case, it is relevant to note that there is no ordinary collinear pole at fixed m_q.

B.8.5.1 Soft-Gluon Limit

This limit is computed by taking the gluon's momentum to zero, as $p_3 \to 0$. The antenna reduces to a product of an A_2^0 antenna and the massive eikonal factor as [26],

$$A_3^0(1_Q, 3_g, 2_{\bar{Q}}) \xrightarrow{3_g \to 0} S_{132}(m_q, m_q), \quad S_{132}(m_q, m_q) = \frac{2s_{12}}{s_{13}s_{23}} - \frac{2m_q^2}{s_{13}^2} - \frac{2m_q^2}{s_{23}^2}. \tag{B.79}$$

```
In[1]:= mA30 = BuildAntenna[A,3,0,quarkMass->mQ];

        softRules = {s13 -> lambda s13, s23 -> lambda s23,
            s123 -> s12 + lambda (s13+s23),
            q2 -> s12 + 2 mQ^2 + lambda (s13+s23), Epsilon -> 0};
        SeriesCoefficient[Together[mA30 /. softRules],
            {lambda, 0, -2}] // FullSimplify

Out[1]:=
```

$$\frac{2s_{12}s_{13}s_{23} - 2m_q^2\left(s_{13}^2 + s_{23}^2\right)}{s_{13}^2 s_{23}^2}$$

This is $\mathcal{S}_{132}(m_q, m_q)$ of the equation above, written over a common denominator.

B.8.5.2 Quasi-Collinear Limit and the Massive Splitting Function

To compute this limit, we start by replacing,

$$p_1 \parallel p_3, \quad s_{13} \to 0, \quad m_q^2 \to 0, \quad \frac{m_q^2}{s_{13}} \to \mu_{Qg}^2. \tag{B.80}$$

Here μ_{Qg} is some fixed quantity. As usual, with z representing the quark momentum fraction,

$$p_1 \to z p_{13}, \quad p_3 \to (1-z)p_{13}. \tag{B.81}$$

The antenna factorises as follows [26],

$$A_3^0(1_Q, 3_g, 2_{\bar{Q}}) \xrightarrow{1_Q \parallel 3_g} \frac{1}{s_{13}} P_{Qg\to Q}(z, \epsilon, \mu_{Qg}^2), \qquad \begin{aligned} P_{Qg\to Q}(z, \epsilon, \mu_{Qg}^2) &= \frac{1+z^2}{1-z} - \epsilon(1-z) - 2\mu_{Qg}^2 \\ &= \frac{1+z^2}{1-z} - \epsilon(1-z), \quad \mu_{Qg}^2 = 0. \end{aligned} \tag{B.82}$$

```
In[1]:= mA30 = BuildAntenna[A,3,0,quarkMass->mQ];

        quasi13Rules = {s13 -> lambda r13, s12 -> z sP2,
            s23 -> (1-z) sP2,
            mQ^2 -> lambda muQg2 r13, mQ^4 -> lambda^2 muQg2^2 r13^2};
        r13 SeriesCoefficient[Together[mA30 /. quasi13Rules],
            {lambda, 0, -1}] // FullSimplify
```

```
Out[1]:=
```

$$\frac{1+z^2}{1-z} - \epsilon(1-z) - 2\mu_{Qg}^2$$

This is $P_{Qg\to Q}(z, \epsilon, \mu_{Qg}^2)$ of the equation above, with ϵ kept symbolic.

Appendix C

Master Integrals

This appendix collects the definitions and the analytic evaluation of the master integrals that appear in the integrated antenna functions of Chapter 4, organised by antenna family: the two-loop A-type integrals of the A_2^2 set, the one-loop phase-space V-type integrals of the A_3^1 set, and the four-particle phase-space R-type integrals of the four-parton antennae. The only massless master integral treated in the main text is R_3, which is needed for the worked example of Subsection 4.1.1, and the massive master integrals are presented together with the massive A_3^0 antenna in Section 4.5, with their derivation given in Section C.5. The notation for the phase-space measures and for the normalisation factors S_Γ and $S_{\Gamma,2}$ is collected in Section C.1, and the scalar one-loop functions B_0, C_0 and D_0 are discussed in Appendix B.2. The derivation of $R_{8,a}$, the most involved of the massless master integrals, is given in full detail in Section C.4 as an example of the method used for the others.

Table C.1: Overview of the master integrals used in the build- and integration-stage by each antenna type [5, 13]. The entries marked † arise only from the A_2^1 ultraviolet counterterm attached at the public build boundary; the two-loop source itself is reduced by IBP to the A-type masters.

Antenna type(s)	Build-Side MI	Integration-Side MI
A_2^0	–	–
A_3^0	–	R_3
A_4^0, $\tilde{A}_4^0$, B_4^0, C_4^0	–	R_4, R_6, $R_{8,a}$, $R_{8,b}$
A_2^1	–	B_0, C_0
A_3^1, $\tilde{A}_3^1$, $\hat{A}_3^1$	B_0, C_0, D_0	$V_{5,a}$, $V_{5,b}$, V_8
A_2^2	$A_{22,\mathrm{LO}}$, A_3, A_4, $B_0^\dagger$, $C_0^\dagger$	$A_{22,\mathrm{LO}}$, A_3, A_4
$\tilde{A}_2^2$	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6	$A_{22,\mathrm{LO}}$, A_3, A_4, A_6
$\hat{A}_2^2$	A_4, $B_0^\dagger$, $C_0^\dagger$	A_4
$\check{A}_2^2$	$A_{22,\mathrm{LO}}$	$A_{22,\mathrm{LO}}$
$A_3^{0,(m_q)}$	–	$I_1^{(m_q,0,m_q)}$, $I_2^{(m_q,0,m_q)}$

C.1 Phase-Space Notation and Normalisation

The master integrals obtained through IBP reduction are associated with different phase spaces, depending on the final-state multiplicity of the corresponding antenna. These phase spaces can be parametrised in terms of the appropriate independent Mandelstam invariants. To simplify the notation

for the corresponding integration measures, we introduce the following shorthand definitions:

$$\begin{aligned} &s_{12} = s_{12}, \quad d\mathcal{S}_2 = ds_{12} \\ &s_{123} = s_{12} + s_{13} + s_{23}, \quad d\mathcal{S}_3 = ds_{12}ds_{13}ds_{23} \\ &s_{1234} = s_{12} + s_{13} + s_{14} + s_{23} + s_{24} + s_{34}, \quad d\mathcal{S}_4 = ds_{12}ds_{13}ds_{14}ds_{23}ds_{24}ds_{34}. \end{aligned} \tag{C.1}$$

Using this notation, the d-dimensional differential phase-space measures can be written as [13, 16],

$$\begin{aligned} d\Phi_2 &= (2\pi)^{2-d} s_{12}^{\frac{d-4}{2}} 2^{1-d} d\Omega_{d-1} d\mathcal{S}_2 \delta\Big(q^2 - s_{12}\Big) \\ d\Phi_3 &= (2\pi)^{3-2d} 2^{-1-d} \Big(q^2\Big)^{\frac{2-d}{2}} (s_{12}s_{13}s_{23})^{\frac{d-4}{2}} d\Omega_{d-1} d\Omega_{d-2} d\mathcal{S}_3 \delta\Big(q^2 - s_{123}\Big) \\ d\Phi_4 &= (2\pi)^{4-3d} \Big(q^2\Big)^{\frac{2-d}{2}} 2^{1-2d} (-\Delta_4)^{\frac{d-5}{2}} \Theta(-\Delta_4) \\ &\qquad d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} d\mathcal{S}_4 \delta\Big(q^2 - s_{1234}\Big). \end{aligned} \tag{C.2}$$

For the four-particle phase space, Δ_4 denotes the Gram determinant, which can be expressed in terms of the Källén function as [13, 16],

$$\Delta_4 = \lambda(s_{12}s_{34}, s_{13}s_{24}, s_{14}s_{23}), \quad \lambda(x, y, z) = x^2 + y^2 + z^2 - 2(xy + xz + yz). \tag{C.3}$$

The exponent of q^2 in $d\Phi_4$ is fixed by dimensional analysis: a four-particle phase space has mass dimension $3d-8$, so, with Δ_4 as defined above, the prefactor must be $\left(q^2\right)^{\frac{2-d}{2}}$, as for $d\Phi_3$. Equation (A.5) of Ref. [13] prints the exponent $3 - \frac{d}{2}$ with the same definition of Δ_4. This is a misprint: rescaling all invariants by q^2, the exponent $\frac{2-d}{2}$ gives a total phase-space volume $\propto \left(q^2\right)^{2-3\epsilon}$, in agreement with the four-particle volume P_4 quoted in Eq. (A.10) of the same reference, whereas $3 - \frac{d}{2}$ would give $\left(q^2\right)^{4-3\epsilon}$. A numerical evaluation of the $d = 4$ measure above also reproduces the normalisation of P_4. The factor $\Theta(-\Delta_4)$, where Θ is the Heaviside step function, restricts the integration to the physical region of the four-particle phase space. The angular factors in Eq. (C.2) are written in terms of $d\Omega_n$, the solid-angle element on the unit $(n-1)$-sphere, with

$$\Omega_n = \int d\Omega_n = \frac{2\pi^{\frac{n}{2}}}{\Gamma(n/2)}. \tag{C.4}$$

The master integrals are expressed using normalisation factors that collect common dimensional-regularisation and phase-space factors. We define [25, 13],

$$S_\Gamma = \left(\frac{(4\pi)^\epsilon}{16\pi^2\Gamma(1-\epsilon)}\right)^2. \tag{C.5}$$

For phase-space master integrals, it is also convenient to include the integrated two-particle phase-space volume Φ_2 in the normalisation, defining

$$S_{\Gamma,2} = \Phi_2 S_\Gamma, \quad \Phi_2 = \int d\Phi_2 = \frac{2^{-3+2\epsilon}\pi^{-1+\epsilon}\Gamma(1-\epsilon)}{\Gamma(2-2\epsilon)}\Big(q^2\Big)^{-\epsilon}. \tag{C.6}$$

C.2 A-Type Integrals of the A_2^2 Set

C.2.1 The $A_{22,\mathrm{LO}}$ Integral

The first master integral required for the A_2^2 antenna set is $A_{22,\mathrm{LO}}$, which corresponds to the product of two massless one-loop bubble integrals. In contrast to the phase-space master integrals encountered

in the integration of real-radiation antennae, $A_{22,\mathrm{LO}}$ is a purely virtual two-loop integral involving two independent loop momenta. Defining

$$q = k_1 + k_2, \qquad q^2 = s_{12},$$

where k_1 and k_2 are the momenta of the final-state quark and antiquark, respectively, the integral is given by [5],

$$\begin{aligned} A_{22,\mathrm{LO}} &= \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_2)^2 \ell^2 (\ell - k_1 - k_2)^2} \\ &= -\frac{\mu^{-4\epsilon}}{\left(16\pi^2\right)^2} \Big[B_0\Big(q^2; 0, 0\Big) \Big]^2 \\ &= S_\Gamma \Big(-q^2\Big)^{-2\epsilon} \, \frac{\Gamma^2(1+\epsilon)\Gamma^6(1-\epsilon)}{\Gamma^2(2-2\epsilon)} \Big(-\frac{1}{\epsilon^2}\Big) \\ &= S_\Gamma \Big(-q^2\Big)^{-2\epsilon} \left[-\frac{1}{\epsilon^2} - \frac{4}{\epsilon} - 12 + O(\epsilon) \right]. \end{aligned} \tag{C.7}$$

Since the propagators depending on p and ℓ are independent, the integral factorises into the product of two identical massless one-loop bubble integrals. It can therefore be written in terms of the scalar two-point function $B_0(q^2; 0, 0)$.

C.2.2 The A_3 Integral

The A_3 master integral is a massless two-loop integral containing three propagators. It is defined as [5],

$$A_3 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - \ell - k_1 - k_2)^2}. \tag{C.8}$$

Unlike $A_{22,\mathrm{LO}}$, the two loop integrations do not factorise, since the third propagator depends on both loop momenta. Performing the ℓ-integration first gives a massless one-loop bubble, so that,

$$A_3 = \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2} B_0\Big((p - k_1 - k_2)^2; 0, 0\Big). \tag{C.9}$$

Substituting the analytic expression for the massless B_0 function and carrying out the remaining p-integration gives

$$A_3 = S_\Gamma \Big(-q^2\Big)^{1-2\epsilon} \, \frac{\Gamma(1+2\epsilon)\Gamma^5(1-\epsilon)}{\Gamma(3-3\epsilon)} \Big(-\frac{1}{2(1-2\epsilon)\epsilon}\Big), \tag{C.10}$$

which can be expanded around $\epsilon = 0$ as

$$A_3 = S_\Gamma \Big(-q^2\Big)^{1-2\epsilon} \left[-\frac{1}{4\epsilon} - \frac{13}{8} + O(\epsilon) \right]. \tag{C.11}$$

C.2.3 The A_4 Integral

The A_4 master integral is a massless two-loop integral containing four propagators,

$$A_4 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2 \ell^2 (p - k_1 - k_2)^2 (p - \ell - k_1)^2}. \tag{C.12}$$

As in the A_3 integral, the two loop momenta are coupled and the integral does not factorise into a product of independent one-loop integrals. The ℓ-integration can, however, be performed first. The two

propagators depending on ℓ

$$\frac{1}{\ell^2(p-\ell-k_1)^2}, \tag{C.13}$$

form a massless one-loop bubble with external momentum $p - k_1$. Hence,

$$A_4 = \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2(p-k_1-k_2)^2} B_0\Big((p-k_1)^2; 0, 0\Big). \tag{C.14}$$

Substituting the analytic expression for the massless B_0 function and evaluating the remaining p-integration gives the following result [5], expressed as a Laurent series in ϵ:

$$\begin{aligned} A_4 &= S_\Gamma\Big(-q^2\Big)^{-2\epsilon} \frac{\Gamma(1-2\epsilon)\Gamma(1+\epsilon)\Gamma^4(1-\epsilon)\Gamma(1+2\epsilon)}{\Gamma(2-3\epsilon)} \left(-\frac{1}{2(1-2\epsilon)\epsilon^2}\right) \\ &= S_\Gamma\Big(-q^2\Big)^{-2\epsilon} \left[-\frac{1}{2\epsilon^2} - \frac{5}{2\epsilon} - \left(\frac{19}{2} + \frac{\pi^2}{6}\right) + O(\epsilon)\right]. \end{aligned} \tag{C.15}$$

C.2.4 The A_6 Integral

The final master integral required for the A_2^2 antenna set is A_6, a massless two-loop integral containing six propagators. It is defined as [5],

$$A_6 = \int \frac{d^d p}{(2\pi)^d} \int \frac{d^d \ell}{(2\pi)^d} \frac{1}{p^2\ell^2(p-k_1-k_2)^2(p-\ell)^2(p-\ell-k_2)^2(\ell-k_1)^2}. \tag{C.16}$$

In contrast to the preceding master integrals, neither loop integration reduces directly to a scalar one-loop function of the B_0, C_0, or D_0 type. The integral is the two-loop crossed triangle of the massless quark form factor. It is evaluated by expressing it as a Mellin–Barnes contour integral, evaluating the residues, and expanding as a Laurent series around $\epsilon = 0$, which gives,

$$\begin{aligned} A_6 &= \frac{S_\Gamma \left(-q^2\right)^{-2-2\epsilon}}{(2\pi i)^2} \int_{-i\infty}^{i\infty} \int_{-i\infty}^{i\infty} dz_1 dz_2 \frac{\Gamma(-z_1)\Gamma(-z_2)\Gamma(2+2\epsilon+z_1+z_2)\Gamma^2(-\epsilon-z_1)\Gamma^2(-\epsilon-z_2)}{\Gamma(-2\epsilon-z_1-z_2)\Gamma(2+\epsilon+z_1+z_2)} \\ &= S_\Gamma\Big(-q^2\Big)^{-2-2\epsilon} \left[-\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{27}{\epsilon}\zeta_3 + \frac{23\pi^4}{36} + O(\epsilon)\right]. \end{aligned} \tag{C.17}$$

This expansion agrees with that of the exact expression in ϵ given in Ref. [12], which involves generalised hypergeometric functions of unit argument,

$$\begin{aligned} A_6 = S_\Gamma\Big(-q^2\Big)^{-2-2\epsilon} \Bigg[&- \frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^4(1-2\epsilon)\Gamma^3(1+2\epsilon)}{\epsilon^4\,\Gamma^2(1-4\epsilon)\Gamma(1+4\epsilon)} \\ &+ \frac{\Gamma^4(1-\epsilon)\Gamma(1+\epsilon)\Gamma(1-2\epsilon)\Gamma(1+2\epsilon)}{2\epsilon^4\,\Gamma(1-3\epsilon)}\, {}_3F_2(1,-4\epsilon,-2\epsilon;1-3\epsilon,1-2\epsilon;1) \\ &- \frac{4\,\Gamma^4(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+2\epsilon)}{\epsilon^2(1+\epsilon)(1+2\epsilon)\,\Gamma(1-4\epsilon)}\, {}_3F_2(1,1,1+2\epsilon;2+\epsilon,2+2\epsilon;1) \\ &- \frac{\Gamma^5(1-\epsilon)\Gamma(1+2\epsilon)}{2\epsilon^4\,\Gamma(1-3\epsilon)}\, {}_4F_3(1,1-\epsilon,-4\epsilon,-2\epsilon;1-3\epsilon,1-2\epsilon,1-2\epsilon;1) \Bigg]. \end{aligned} \tag{C.18}$$

C.3 V-Type Integrals of the A_3^1 Set

The integration of the A_3^1 antenna set combines a one-loop integral with the integration over the $1 \to 3$ antenna phase space. After applying the IBP reduction described in Chapter 3, the leading- and subleading-colour contributions are expressed in terms of the master integrals $V_{5,a}$, $V_{5,b}$, and V_8. These integrals therefore combine loop propagators with the three-particle phase-space integration and constitute the master-integral basis required for the real–virtual antenna contributions considered in Subsections 4.3.1 and 4.3.2. Only the real part of each of them is needed: the one-loop antennae are built from the real part of the tree–one-loop interference, $2\,\mathrm{Re}\big(\mathcal{M}^{0\,\dagger}\mathcal{M}^1\big)$ (Chapter 2), so the imaginary parts that the loop integrals develop in the time-like region $q^2 > 0$ do not contribute.

C.3.1 The $V_{5,a}$ Integral

The master integral $V_{5,a}$ consists of a massless one-loop bubble integrated over the three-particle phase space. Denoting the momenta of the final-state quark, antiquark, and gluon by k_1, k_2, and k_3, respectively, their total momentum is

$$q = k_1 + k_2 + k_3, \qquad q^2 = s_{12} + s_{13} + s_{23}.$$

The three final-state momenta are massless, $k_i^2 = 0$, while $q^2 > 0$ is the invariant mass squared of the antenna system. The master integral is defined as

$$\begin{aligned} V_{5,a} &= \mathrm{Re}\left[-i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_2 - k_3)^2}\right] \\ &= \mathrm{Re}\left[\int d\Phi_3\, B_0\Big(q^2; 0, 0\Big)\right]. \end{aligned} \tag{C.19}$$

Since the scalar function $B_0\big(q^2; 0, 0\big)$, discussed in Subsubsection B.2.2.1, depends only on the fixed total invariant q^2, and not on the individual phase-space invariants, it can be taken outside the three-particle phase-space integration, which leaves the phase-space volume $R_3 = \int d\Phi_3$ of Eq. (4.20). Hence this integral yields [5],

$$\begin{aligned} V_{5,a} &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon} \cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right) \frac{\Gamma^6(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)} \\ &= S_{\Gamma,2}\Big(q^2\Big)^{1-2\epsilon} \left[-\frac{1}{2\epsilon} - \frac{13}{4} + O(\epsilon)\right]. \end{aligned} \tag{C.20}$$

C.3.2 The $V_{5,b}$ Integral

The master integral $V_{5,b}$ has a structure similar to $V_{5,a}$, consisting of a massless one-loop bubble integrated over the three-particle phase space. The essential difference is that the external momentum of the bubble is now $k_1 + k_3$, rather than the total antenna momentum q. It is therefore defined by,

$$\begin{aligned} V_{5,b} &= \mathrm{Re}\left[-i \int d\Phi_3 \int \frac{d^d p}{(2\pi)^d} \frac{1}{p^2 (p - k_1 - k_3)^2}\right] \\ &= \mathrm{Re}\left[\int d\Phi_3\, B_0(s_{13}; 0, 0)\right]. \end{aligned} \tag{C.21}$$

Unlike q^2, which is fixed during the phase-space integration, s_{13} varies over the three-particle phase space. Consequently, the scalar function $B_0(s_{13}; 0, 0)$ cannot be taken outside the phase-space integral,

and $V_{5,b}$ does not factorise into the product of a scalar one-loop function and the phase-space volume R_3.

Using the analytic expression for the massless scalar function $B_0(s_{13};0,0)$ and performing the remaining three-particle phase-space integration gives [5],

$$\begin{aligned} V_{5,b} &= \mathrm{Re}\left[\int d\Phi_3\, B_0(s_{13};0,0)\right] \\ &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon}\cos(\pi\epsilon)\left(-\frac{1}{\epsilon}\right)\frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)\Gamma(1+\epsilon)}{\Gamma(2-2\epsilon)\Gamma(3-4\epsilon)} \\ &= S_{\Gamma,2}\left(q^2\right)^{1-2\epsilon}\left[-\frac{1}{2\epsilon}-4+O(\epsilon)\right]. \end{aligned} \tag{C.22}$$

C.3.3 The V_8 Integral

The third master integral required for the A_3^1 antenna set is V_8. In contrast to $V_{5,a}$ and $V_{5,b}$, whose loop integrations correspond to scalar two-point functions, V_8 contains a massless one-loop scalar four-point integral integrated over the three-particle phase space. It is defined as [5],

$$\begin{aligned} V_8 &= \mathrm{Re}\left[-i\int d\Phi_3 \int \frac{d^dp}{(2\pi)^d}\frac{1}{p^2(p-k_1)^2(p-k_1-k_3)^2(p-k_1-k_2-k_3)^2}\right] \\ &= \mathrm{Re}\left[-i\int d\Phi_3\,\frac{1}{s_{12}}\,\mathrm{Box}(s_{13},s_{23},s_{12})\right]. \end{aligned} \tag{C.23}$$

The loop integral in this definition, denoted Box, is a massless scalar box integral. Using the scalar one-loop conventions introduced in Subsubsection B.2.2.3, it may be written in terms of the D_0 function as

$$V_8 = \left(\frac{(4\pi)^\epsilon\mu^{-2\epsilon}}{16\pi^2}\right)\frac{\Gamma^2(1-\epsilon)\Gamma(1+\epsilon)}{\Gamma(1-2\epsilon)}\int d\Phi_3\,\frac{1}{s_{12}}\,\mathrm{Re}\left[D_0\left(0,0,0;q^2,s_{13},s_{23};0,0,0,0\right)\right]. \tag{C.24}$$

Unlike $V_{5,a}$, the scalar box depends explicitly on the phase-space invariants s_{12}, s_{13}, and s_{23}. The loop integral therefore cannot be factorised from the three-particle phase-space integration. Performing the remaining phase-space integration and applying the normalisation introduced above gives [5],

$$V_8 = S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[-\frac{5}{2\epsilon^4}+\frac{9\pi^2}{2\epsilon^2}+\frac{89\zeta_3}{\epsilon}+\frac{13\pi^4}{180}+O(\epsilon)\right]. \tag{C.25}$$

C.4 R-Type Integrals of the Four-Parton Antennae

The integration of the tree-level four-parton antennae considered in Subsections 4.4.1–4.4.4 requires integration over the massless $1 \to 4$ antenna phase space. After mapping the antenna functions onto the corresponding integral family and applying the IBP reduction described in Chapter 3, the integrated antennae can be expressed in terms of a small set of master integrals, R_4, R_6, $R_{8,a}$, and $R_{8,b}$. Unlike the V-type master integrals encountered for the real–virtual A_3^1 antennae, these integrals contain no loop integration. They are integrals over the four-particle phase space, with additional propagator-like factors for R_6, $R_{8,a}$, and $R_{8,b}$. The simplest member, R_4, is the four-particle phase-space volume itself. The following subsections present their definitions and analytic evaluation.

C.4.1 The R_4 Integral

The simplest master integral in the four-parton basis is R_4, corresponding to the total massless four-particle phase-space volume,

$$R_4 = \int d\Phi_4.$$

In $d = 4 - 2\epsilon$ dimensions, the phase-space measure may be expressed in terms of the six two-particle invariants $s_{ij} = 2k_i \cdot k_j$, subject to momentum conservation and the constraint imposed by the four-particle Gram determinant Δ_4. This gives

$$R_4 = (2\pi)^{4-3d}\left(q^2\right)^{\frac{2-d}{2}} 2^{1-2d} \int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \left(-\Delta_4\right)^{\frac{d-5}{2}} \Theta(-\Delta_4)\delta\left(q^2 - \mathcal{S}_4\right). \tag{C.26}$$

To evaluate R_4, the four-particle phase space is factorised into a two-particle phase space and a tripole phase-space factor (the phase space of the unresolved partons clustered onto the radiator, relative to the spectator),

$$d\Phi_4 = d\Phi_2 \, d\Phi_T,$$

where $d\Phi_2$ describes the reduced two-particle configuration obtained by clustering the momenta p_1, p_3, p_4 into the mapped radiator momentum $\tilde{p}_{134}$, with $\tilde{p}_2$ denoting the corresponding mapped spectator momentum. The factor $d\Phi_T$ is the associated tripole phase-space measure and contains the remaining degrees of freedom required to resolve the original four-particle configuration. Following Ref. [13], these degrees of freedom are parametrised by five dimensionless variables with support on the unit hypercube. We start by defining $y_{ij} = s_{ij}/q^2$ and $y_{ijk} = s_{ijk}/q^2$. Our five dimensionless variables become [13]:

- The rescaled invariant mass of the three-parton cluster, $y_{134} = \frac{s_{134}}{q^2}$;
- The longitudinal momentum fraction $z_1 = \frac{p_3 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$, which parametrises the fraction associated with parton 3, measured with respect to the mapped momenta $\tilde{p}_{134}$ and $\tilde{p}_2$;
- The variable t, introduced through the second longitudinal momentum fraction $z_2 = (1 - z_1)t = \frac{p_4 \cdot \tilde{p}_2}{\tilde{p}_{134} \cdot \tilde{p}_2}$, which parametrises the momentum fraction associated with parton 4 after the fraction z_1 carried by parton 3 has been specified;
- The variable v, introduced through the rescaled two-particle invariant $y_{14} = \frac{s_{14}}{q^2} = y_{134}(1 - z_1)v$;
- The variable χ, which parametrises the allowed range of y_{13}, where $y_{13,a}$ and $y_{13,b}$ are the lower and upper boundaries determined by the Gram-determinant constraint as $y_{13} = (y_{13,b} - y_{13,a})\chi + y_{13,a} = \Delta y_{13}\chi + y_{13,a}$.

Performing the change of variables from the phase-space invariants to the five dimensionless variables y_{134}, z_1, t, v, χ, including the corresponding Jacobian, maps the physical integration region onto the unit hypercube $[0, 1]^5$ and the master integral becomes

$$\begin{aligned} R_4 = \Phi_2 \frac{16^{-2}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)}\left(q^2\right)^{2-2\epsilon} \int_{[0,1]^5} dy_{134} dz_1 dt dv d\chi \left[y_{134}(1 - y_{134})(1 - z_1)\right]^{1-2\epsilon} \\ \left[z_1 t(1-t)v(1-v)\right]^{-\epsilon}\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}. \end{aligned} \tag{C.27}$$

A useful feature of this parametrisation is that the dependence on the five integration variables factorises into powers of each variable and its complement. Each integration can therefore be evaluated using the

Euler Beta function,

$$\int_0^1 dx\, x^{a-1}(1-x)^{b-1} = B(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}.$$

Performing these integrations and simplifying the resulting Gamma functions gives [5, 13],

$$\begin{aligned} R_4 &= \Phi_2\left(q^2\right)^{2-2\epsilon} \frac{2^{-11+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}\Gamma^3(1-\epsilon)}{\Gamma(3-3\epsilon)\Gamma\left(\frac{5}{2}-2\epsilon\right)} \\ &= S_{\Gamma,2}\left(q^2\right)^{2-2\epsilon} \frac{\Gamma^5(1-\epsilon)\Gamma(2-2\epsilon)}{\Gamma(3-3\epsilon)\Gamma(4-4\epsilon)} \\ &= S_{\Gamma,2}\left(q^2\right)^{2-2\epsilon} \left[\frac{1}{12} + \frac{59}{72}\epsilon + O\left(\epsilon^2\right)\right]. \end{aligned} \tag{C.28}$$

C.4.2 The R_6 Integral

The second master integral required for the four-parton antennae is R_6, defined by [13],

$$\begin{aligned} R_6 &= \int d\Phi_4 \frac{1}{s_{134}s_{234}} \\ &= (2\pi)^{4-3d}\left(q^2\right)^{\frac{2-d}{2}} 2^{1-2d} \int d\mathcal{S}_4 d\Omega_{d-1} d\Omega_{d-2} d\Omega_{d-3} \frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{134}s_{234}} \Theta(-\Delta_4)\delta\left(q^2-\mathcal{S}_4\right). \end{aligned} \tag{C.29}$$

Unlike R_4, however, R_6 does not reduce to a product of elementary Beta-function integrals under the tripole change of variables introduced above. In particular, s_{234} has a non-linear dependence on the tripole variables, preventing the factorisation of the integrand that made the direct evaluation of R_4 straightforward. Following Ref. [13], R_6 is therefore evaluated indirectly using the optical theorem.

The optical theorem relates the imaginary part of a forward-scattering integral to the sum over its possible unitarity cuts. In the present case, the relevant three-loop vacuum-polarisation integral I_6 admits cuts that can be expressed in terms of the four-particle phase-space integral R_6 and the lower-multiplicity contribution involving the two-loop integral A_4. With the conventions of Ref. [13], this relation reads

$$2\,\mathrm{Im}\, I_6 = -2\Phi_2\,\mathrm{Re}\, A_4 + 2R_6. \tag{C.30}$$

The required expressions for I_6 and A_4 are known analytically [13]. In the conventions adopted here, they are

$$\begin{aligned} I_6 = \frac{(4\pi)^{3\epsilon}}{\left(16\pi^2\right)^3} \frac{\Gamma^6(1-\epsilon)\Gamma^3(1+\epsilon)}{3\epsilon^3\Gamma^3(2-2\epsilon)}\left(-q^2\right)^{-3\epsilon} \\ \left[1+\epsilon+\epsilon^2+\epsilon^3(14\zeta_3-7)+\epsilon^4\left(-67+14\zeta_3+\frac{21\pi^4}{90}\right)+O\left(\epsilon^5\right)\right] \end{aligned} \tag{C.31}$$

$$\begin{aligned} A_4 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2\left(-q^2\right)^{-2\epsilon}\left[-\frac{1}{2\epsilon^2}-\frac{5}{2\epsilon}-\left(\frac{19}{2}+\frac{\pi^2}{6}\right)\right. \\ \left. -\epsilon\left(\frac{65}{2}+\frac{5\pi^2}{6}-2\zeta_3\right)-\epsilon^2\left(\frac{211}{2}+\frac{19\pi^2}{6}-10\zeta_3\right)+O\left(\epsilon^3\right)\right]. \end{aligned} \tag{C.32}$$

The two-loop master integral A_4 was introduced previously in Section C.2 in the calculation of the two-parton NNLO antennae. Its appearance here illustrates that the same virtual integral enters the lower-multiplicity cut of the three-loop forward-scattering integral.

Since the forward-scattering integral is evaluated in the time-like region $q^2 > 0$, the factors involving $(-q^2)$ must be analytically continued according to the Feynman $+i0$ prescription. In the convention of Ref. [13], this gives

$$\left(-q^2 - i0\right)^{-\epsilon} = \left(q^2\right)^{-\epsilon} e^{i\pi\epsilon}.$$

The real and imaginary parts required in the relation above can then be extracted from these phases. Substituting the real part of A_4 and the imaginary part of I_6, and denoting by $\mathcal{A}_4$ and $\mathcal{I}_6$ the ϵ-series in square brackets in the expressions for A_4 and I_6 above, we solve for R_6 and obtain

$$\begin{aligned} R_6 &= S_{\Gamma,2}\left(q^2\right)^{-2\epsilon} \frac{\Gamma^6(1-\epsilon)\Gamma^2(1+\epsilon)}{6}\left(\frac{6\cos(2\pi\epsilon)}{\Gamma^2(1-2\epsilon)}\mathcal{A}_4 + \frac{\Gamma(1-\epsilon)\Gamma(1+\epsilon)\sin(3\pi\epsilon)}{\epsilon^3\pi\Gamma^2(2-2\epsilon)}\mathcal{I}_6\right) \\ &= S_{\Gamma,2}\left(q^2\right)^{-2\epsilon}\left[-1+\frac{\pi^2}{6}+\epsilon\left(-12+\frac{5\pi^2}{6}+9\zeta_3\right)+O\left(\epsilon^2\right)\right]. \end{aligned} \tag{C.33}$$

C.4.3 The $R_{8,a}$ Integral

The master integral $R_{8,a}$ is the first of the two four-particle phase-space master integrals containing four invariant denominators. It is defined as [13],

$$\begin{aligned} R_{8,a} &= \int d\Phi_4 \frac{1}{s_{13}s_{23}s_{14}s_{24}} \\ &= (2\pi)^{4-3d}\left(q^2\right)^{\frac{2-d}{2}} 2^{1-2d}\int d\mathcal{S}_4 d\Omega_{d-1}d\Omega_{d-2}d\Omega_{d-3}\frac{(-\Delta_4)^{\frac{d-5}{2}}}{s_{13}s_{23}s_{14}s_{24}}\Theta(-\Delta_4)\,\delta\left(q^2-\mathcal{S}_4\right) \\ &= \Phi_2\left(q^2\right)^{-2-2\epsilon}\frac{16^{-2+\epsilon}\pi^{-5+2\epsilon}}{\Gamma(1-2\epsilon)}\int_{[0,1]^5} dy_{134}dz_1dtdvd\chi \\ &\qquad \left[(1-y_{134})(1-z_1)\right]^{-1-2\epsilon}\left[tvz_1^{-\epsilon}(\Delta y_{13}\chi + y_{13,a})\right]^{-1}\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}. \end{aligned} \tag{C.34}$$

In contrast to R_4, the resulting integrand does not factorise into independent Beta-function integrals. Its analytic evaluation instead proceeds through a sequence of integrations involving hypergeometric functions. Following Ref. [13], we perform the integrations in the order,

$$\chi \to v \to y_{134} \to t \to z_1,$$

with an auxiliary integration variable $u \in [0,1]$ introduced at a later stage through an Euler integral representation of the hypergeometric function.

We begin with the χ integration. The dependence on χ occurs through the Gram-determinant factor and the invariant $y_{13} = y_{13,a} + \Delta y_{13}\chi$, leading to,

$$I_\chi = \int_0^1 d\chi\left[\chi(1-\chi)\right]^{-\frac{1}{2}-\epsilon}(\Delta y_{13}\chi + y_{13,a})^{-1} = \frac{\Delta y_{13}^{-2\epsilon}}{y_{13,a}}\frac{\Gamma^2\left(\frac{1}{2}-\epsilon\right)}{\Gamma(1-2\epsilon)}{}_2F_1\left(1,\frac{1}{2}-\epsilon;1-2\epsilon;-\frac{\Delta y_{13}}{y_{13,a}}\right) \tag{C.35}$$

where ${}_2F_1$ is the Gauss hypergeometric function. To simplify the argument of the hypergeometric function, we use the phase-space boundaries, $y_{13,(a,b)} = y_{134}(A \mp B)^2$, with

$$A = \sqrt{(1-t)(1-v)}, \qquad B = \sqrt{z_1 tv}, \qquad Z = -\frac{B}{A}.$$

These definitions imply,

$$-\frac{\Delta y_{13}}{y_{13,a}} = \frac{4Z}{(1+Z)^2}, \quad (1+Z)^2 = \frac{y_{13,a}}{y_{134}A^2}. \tag{C.36}$$

The quadratic transformation of the Gauss hypergeometric function [41],

$$ {}_2F_1\left(1, b; 2b; \frac{4Z}{(1+Z)^2}\right) = (1+Z)^2 {}_2F_1\left(1, \frac{3}{2} - b; b + \frac{1}{2}; Z^2\right) \tag{C.37} $$

together with

$$ 16^{\epsilon} \Delta y_{13}^{-2\epsilon} = \Big[t(1-t)v(1-v)y_{134}^2 z_1\Big]^{-\epsilon}, \tag{C.38} $$

allows the χ-integrated expression to be written as,

$$ I_\chi = \frac{\pi\Gamma(1-2\epsilon)}{y_{134}\Gamma^2(1-\epsilon)} \Big(tvy_{134}^2 z_1\Big)^{-\epsilon} \Big[(1-t)(1-v)\Big]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \tag{C.39} $$

Substituting this expression into the phase-space integral eliminates χ and reduces $R_{8,a}$ to a four-dimensional integral,

$$ \begin{aligned} R_{8,a} = \Phi_2\Big(q^2\Big)^{-2-2\epsilon} \frac{16^{-2+\epsilon}\pi^{-4+2\epsilon}}{\Gamma^2(1-\epsilon)} \int_{[0,1]^4} dy_{134} dz_1 dt dv \Big[y_{134}(1-y_{134}) z_1(1-z_1)\Big]^{-1-2\epsilon} \\ \Big[t(1-t)v(1-v)\Big]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right). \end{aligned} \tag{C.40} $$

The v integration is performed by expanding the Gauss hypergeometric function in its defining series and integrating term by term. This yields,

$$ \begin{aligned} I_v &= \int_0^1 dv \Big[v(1-v)\Big]^{-1-\epsilon} {}_2F_1\left(1, 1+\epsilon; 1-\epsilon; \frac{z_1 tv}{(1-t)(1-v)}\right) \\ &= \sum_{n\geq 0} \frac{(1+\epsilon)_n}{(1-\epsilon)_n} \Big(\frac{z_1 t}{1-t}\Big)^n \frac{\Gamma(-\epsilon-n)\Gamma(-\epsilon+n)}{\Gamma(-2\epsilon)} \\ &= -\frac{2\Gamma^2(1-\epsilon)}{\epsilon\Gamma(1-2\epsilon)} {}_2F_1\Big(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\Big). \end{aligned} \tag{C.41} $$

After substituting I_v, the y_{134} dependence factorises and can be integrated directly, leaving a two-dimensional integral over z_1 and t,

$$ \begin{aligned} R_{8,a} = -\Phi_2\Big(q^2\Big)^{-2-2\epsilon} \frac{2^{-7+8\epsilon}\pi^{-\frac{7}{2}+2\epsilon}}{\epsilon^2\Gamma\Big(\frac{1}{2}-2\epsilon\Big)} \int_{[0,1]^2} dz_1 dt\, (1-z_1)^{-1-2\epsilon} \\ \Big[z_1 t(1-t)\Big]^{-1-\epsilon} {}_2F_1\Big(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\Big). \end{aligned} \tag{C.42} $$

To perform the remaining integrations, the hypergeometric function is represented through its Euler integral,

$$ {}_2F_1\Big(1, -\epsilon; 1-\epsilon; -\frac{z_1 t}{1-t}\Big) = -\epsilon \int_0^1 du\, u^{-1-\epsilon} \Big[1 + \frac{u z_1 t}{1-t}\Big]^{-1} \tag{C.43} $$

The auxiliary variable u therefore converts the remaining hypergeometric dependence into an additional ordinary integral over the unit interval. The remaining t, u, and z_1 integrations can then be performed sequentially. After simplifying the resulting Gamma and generalised hypergeometric functions, one

obtains,

$$R_{8,a} = \Phi_2\left(q^2\right)^{-2-2\epsilon} 2^{-7+4\epsilon}\pi^{-3+2\epsilon}\csc(\pi\epsilon)\Gamma(-2\epsilon)\left[\frac{2^{8\epsilon}\pi}{\epsilon^2\Gamma^2\left(\frac{1}{2}-2\epsilon\right)}{}_3F_2\left(\begin{matrix}-2\epsilon,-2\epsilon,-2\epsilon\\ 1-2\epsilon,-4\epsilon\end{matrix};1\right)\right.$$
$$\left. - \frac{2\Gamma^3(-\epsilon)}{\Gamma(1-4\epsilon)\Gamma(1-\epsilon)\Gamma(1+\epsilon)\Gamma(-3\epsilon)}{}_4F_3\left(\begin{matrix}1,-\epsilon,-\epsilon,-\epsilon\\ 1-\epsilon,1+\epsilon,-3\epsilon\end{matrix};1\right)\right] \tag{C.44}$$

Rewriting the result in the normalisation used throughout this chapter and expanding about $\epsilon = 0$ gives [13, 30],

$$\begin{aligned} R_{8,a} =& S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[\frac{6}{\epsilon^4}\frac{\Gamma^5(1-\epsilon)\Gamma(1-2\epsilon)}{\Gamma(1-3\epsilon)\Gamma(1-4\epsilon)}{}_4F_3\left(\begin{matrix}1,-\epsilon,-\epsilon,-\epsilon\\ 1-\epsilon,1+\epsilon,-3\epsilon\end{matrix};1\right)\right. \\ &\left. - \frac{1}{\epsilon^4}\frac{\Gamma^3(1-\epsilon)\Gamma(1+\epsilon)\Gamma^3(1-2\epsilon)}{\Gamma^2(1-4\epsilon)}{}_3F_2\left(\begin{matrix}-2\epsilon,-2\epsilon,-2\epsilon\\ -4\epsilon,1-2\epsilon\end{matrix};1\right)\right] \\ =& S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[\frac{5}{\epsilon^4} - \frac{20\pi^2}{3\epsilon^2} - \frac{126\zeta_3}{\epsilon} + \frac{7\pi^4}{18} + O(\epsilon)\right]. \end{aligned} \tag{C.45}$$

C.4.4 The $R_{8,b}$ Integral

The second four-denominator master integral is $R_{8,b}$, defined as [13],

$$R_{8,b} = \int d\Phi_4 \frac{1}{s_{13}s_{134}s_{23}s_{234}}. \tag{C.46}$$

As in the case of R_6, the presence of s_{234} prevents the tripole change of variables from reducing the integral to a convenient factorised form. Following Ref. [13], $R_{8,b}$ is therefore determined indirectly by applying the optical theorem to the corresponding three-loop vacuum-polarisation integral I_8. The different unitarity cuts of I_8 generate contributions involving the two-loop virtual master integral A_6, the real–virtual master integral V_8, and the two four-particle phase-space master integrals $R_{8,a}$ and $R_{8,b}$. With the conventions of Ref. [13], their relation is

$$2\,\mathrm{Im}\,I_8 = -2\Phi_2\,\mathrm{Re}\,A_6 + 4\,\mathrm{Re}\,V_8 + R_{8,a} + 4R_{8,b}. \tag{C.47}$$

The required three-loop integral I_8 is known analytically [13], while A_6 and V_8 were introduced previously in Sections C.2 and C.3, respectively. Their relevant expressions are

$$I_8 = S + O(\epsilon), \quad \mathrm{Im}\,S = -\frac{13\pi^4}{90}, \tag{C.48}$$

where S is the ϵ^0 coefficient of I_8;

$$V_8 = -i\int d\Phi_3 \frac{1}{s_{12}}\,\mathrm{Box}(s_{13}, s_{23}, s_{12}); \tag{C.49}$$

$$A_6 = \left(\frac{(4\pi)^\epsilon}{16\pi^2}\frac{\Gamma(1+\epsilon)\Gamma^2(1-\epsilon)}{\Gamma(1-2\epsilon)}\right)^2\left(-q^2\right)^{-2-2\epsilon}\left[-\frac{1}{\epsilon^4} + \frac{5\pi^2}{6\epsilon^2} + \frac{23\zeta_3}{\epsilon} + \frac{103\pi^4}{180} + O(\epsilon)\right]. \tag{C.50}$$

Solving the optical-theorem relation for $R_{8,b}$ gives,

$$R_{8,b} = \frac{1}{4}\left[2\,\mathrm{Im}\,I_8 + 2\Phi_2\,\mathrm{Re}\,A_6 - 4\,\mathrm{Re}\,V_8 - R_{8,a}\right]. \tag{C.51}$$

Substituting the corresponding ϵ-expansions and rewriting the result in the $S_{\Gamma,2}$ normalisation used throughout this section yields [13],

$$R_{8,b} = S_{\Gamma,2}\left(q^2\right)^{-2-2\epsilon}\left[\frac{3}{4\epsilon^4} - \frac{17\pi^2}{12\epsilon^2} - \frac{44\zeta_3}{\epsilon} - \frac{61\pi^4}{60} + O(\epsilon)\right]. \tag{C.52}$$

C.5 Massive Integrals of the $A_3^{0,(m_q)}$ Antenna

In this section, we derive the two massive master integrals $I_1^{(m_q,0,m_q)}$ and $I_2^{(m_q,0,m_q)}$ used in Subsection 4.5.1. We also check the change of basis in Eq. (4.44) by evaluating the corresponding integrals numerically.

C.5.1 Massive Three-Particle Phase Space

For $\gamma^*(q) \to Q(k_1)\,\bar{Q}(k_2)\,g(k_3)$ with $k_1^2 = k_2^2 = m_q^2$ and $k_3^2 = 0$, the three-particle phase space can be written in a form analogous to the massless $d\Phi_3$ of Eq. (C.2). The product $s_{12}s_{13}s_{23}$ is replaced by four times the Gram determinant of $\{k_1, k_2, k_3\}$,

$$K = s_{12}s_{13}s_{23} - m_q^2\left(s_{13}^2 + s_{23}^2\right), \tag{C.53}$$

and $q^2 = s_{12} + s_{13} + s_{23} + 2m_q^2$. The measure is then

$$d\Phi_3^{(m_q,0,m_q)} = (2\pi)^{3-2d}2^{-1-d}\left(q^2\right)^{\frac{2-d}{2}}K^{\frac{d-4}{2}}\,\Theta(K)\,d\Omega_{d-1}d\Omega_{d-2}\,d\mathcal{S}_3\,\delta\left(q^2 - 2m_q^2 - s_{123}\right). \tag{C.54}$$

The equal-mass two-particle phase-space volume is

$$\Phi_2^{(m_q,m_q)} = (2\pi)^{2-d}\,2^{1-d}\,\Omega_{d-1}\left(q^2\right)^{\frac{d-4}{2}}r^{\frac{d-3}{2}}, \tag{C.55}$$

where r is the threshold variable defined in Eq. (4.40). For $m_q = 0$, both expressions reduce to their massless counterparts.

C.5.2 Parametrisation

To carry out the integration, we use the gluon energy fraction and the heavy-quark angle in the $Q\bar{Q}$ rest frame. Defining $u = (s_{13} + s_{23})/q^2 \in [0, r]$ and $c \in [-1, 1]$,

$$s_{13} = \frac{q^2u}{2}\,(1 - \beta_u c)\,, \qquad s_{23} = \frac{q^2u}{2}\,(1 + \beta_u c)\,, \qquad \beta_u = \sqrt{\frac{r-u}{1-u}}, \tag{C.56}$$

where β_u is the heavy-quark velocity in the $Q\bar{Q}$ rest frame, whose invariant mass is $(1-u)q^2$. In these variables the Gram factor and the Jacobian become

$$K = \left(q^2\right)^3\frac{u^2}{4}\,(r-u)\left(1-c^2\right), \qquad ds_{13}\,ds_{23} = \left(q^2\right)^2\frac{u\,\beta_u}{2}\,du\,dc. \tag{C.57}$$

The integrand therefore factorises in u and c, with the physical integration region given by $u \in [0, r]$ and $c \in [-1, 1]$.

C.5.3 Evaluation

The angular integral is elementary,

$$\int_{-1}^{1} dc \, \left(1-c^2\right)^{-\epsilon} = \frac{2^{1-2\epsilon}\,\Gamma^2(1-\epsilon)}{\Gamma(2-2\epsilon)}. \tag{C.58}$$

The integration over u can be written in the Euler form,

$$\int_0^r du\, u^{\alpha-1}(r-u)^{\gamma-\alpha-1}(1-u)^{-\frac{1}{2}} = r^{\gamma-1}B(\alpha,\gamma-\alpha)\;{}_2F_1\!\left(\tfrac{1}{2},\alpha;\gamma;r\right). \tag{C.59}$$

For I_1, we have $\alpha = 2-2\epsilon$ and $\gamma = \frac{7}{2}-3\epsilon$. For I_2, the additional factor s_{13} contributes only its part even in c, $q^2u/2$, as the term linear in c integrates to zero. This gives $\alpha = 3-2\epsilon$ and $\gamma = \frac{9}{2}-3\epsilon$. After collecting the prefactors, dividing by $\Phi_2^{(m_q,m_q)}$ and simplifying the Gamma functions, we obtain

$$\begin{aligned}
I_1^{(m_q,0,m_q)} &= \left(q^2\right)^{1-\epsilon} r^{2-2\epsilon}2^{-2\epsilon}\pi^{-2+\epsilon}\,\frac{\Gamma(2-2\epsilon)\Gamma(3-3\epsilon)}{\Gamma(6-6\epsilon)}\;{}_2F_1\!\left(\tfrac{1}{2},2-2\epsilon;\tfrac{7}{2}-3\epsilon;r\right),\\
I_2^{(m_q,0,m_q)} &= \left(q^2\right)^{2-\epsilon} r^{3-2\epsilon}2^{1-2\epsilon}\pi^{-2+\epsilon}\,\frac{\Gamma(3-2\epsilon)\Gamma(4-3\epsilon)}{\Gamma(8-6\epsilon)}\;{}_2F_1\!\left(\tfrac{1}{2},3-2\epsilon;\tfrac{9}{2}-3\epsilon;r\right),
\end{aligned} \tag{C.60}$$

in agreement with Subsection 4.5.1 and Ref. [26]. In the expression for I_1, the factor $r^{2-2\epsilon}$ comes from $r^{\frac{5}{2}-3\epsilon}$ in the Euler integral and $r^{-\frac{1}{2}+\epsilon}$ in $\Phi_2^{(m_q,m_q)}$. We can also check the massless limit: for $r\to 1$ and $\epsilon = 0$, $I_1 \to 1/(32\pi^2)$, as expected from the massless ratio R_3/Φ_2.

C.5.4 Check of the Change of Basis

The *LiteRed2* master J_2 contains a squared massive cut denominator. Using $1/D_1^2 = -\partial_{m_1^2}(1/D_1)$ for $D_1 = m_1^2 - k_1^2$, we can obtain it from J_1 evaluated with unequal masses, $k_1^2 = m_1^2$ and $k_2^2 = m_q^2$,

$$J_2^{(m_q,0,m_q)} = -\frac{\partial}{\partial m_1^2}\, J_1^{(m_1,0,m_q)}\Big|_{m_1=m_q}, \tag{C.61}$$

where the unequal-mass Gram factor is $s_{12}s_{13}s_{23} - m_1^2 s_{23}^2 - m_q^2 s_{13}^2$ and the normalisation follows from $J_1 = I_1$. At fixed s_{23}, the s_{13} integration is of Beta-function type. We then evaluate the remaining integral and the mass derivative numerically. For $m_q^2/q^2 \in \{0.05,\, 0.15\}$ and $\epsilon \in \{0,\, -0.2\}$, the combination $aJ_1 + bJ_2$, with a and b defined in Section 4.5, agrees with the expression for $I_2^{(m_q,0,m_q)}$ above to at least 12 significant digits. This provides a direct numerical check of the change of basis in Eq. (4.44).

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